Cognitive Radar-Vision Scene Analysis via False Discovery Rate Control

Kognitive Radar-Vision-Szenenanalyse mittels Kontrolle der Falscherkennungsrate
Bachelor thesis in Electrical Engineering and Information Technology
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Abstract

Fog and rain fill the road, headlights shine straight into the camera. The image turns white, yet the radar reports an object ahead. It is unclear if the unidentifiable object could be a little child or even just a ghost. One thing is clear, autonomous driving, robotics, aerospace, search and rescue and many other fields need both sensors: the radar measures distance, speed and direction through fog and darkness, and the camera reads colour and images where the radar is blind. Yet every radar report carries doubt: a detector sorts thousands of cells into object or noise, so some reports are false however carefully it decides. Detectors bound the error of one cell, not of the whole list. Controlling the false discovery rate (FDR), the expected share of false discoveries among all discoveries, bounds this doubt and distills a flood of data into truths that quench the thirst for knowledge. This thesis proposes a cognitive radar-vision framework via FDR control to test whether this guarantee survives real radar data and reaches localization, tracking, and decision making in real time.

Reconstruction of the scene from a sparse set of truths starts on the implemented ARTIFACT platform. A 77 GHz automotive radar streams raw samples into a Python framework, a camera adds colour and vision, and a motor stage of known pose moves both. First, CT-Rex and Infini-Rex estimate directions of arrival from single snapshots under FDR control. They belong to the T-Rex family of selectors that control the FDR in high dimensions with random dummies. Second, the sparse classifier M^2-SRC reads the material and the motion of an object from its echo. Third, MetricSight reads text with optical character recognition at the image scale that the radar range fixes, and a radar-anchored glare removal detects light sources and mitigates their interference. Fourth, KalmanCQR gives radar tracking prediction regions of chosen coverage, and a cognitive chirp choice lets the radar pick its next chirp from the doubt of its prediction. Finally, a world model and a planner turn both sensors into decisions on a moving platform.

From a single snapshot, CT-Rex and Infini-Rex find 89 percent of the targets without a false direction, and Infini-Rex for Radar finds 94 percent at an FDR of 0.02. In contrast, 68 percent of the directions of classical MUSIC and ESPRIT are false. M^2-SRC recognises the motion in 98.1 percent and the material in 94.6 percent of the windows of an unseen session. MetricSight reads text with a character error rate of 2.1 percent, against 13.6 percent for a larger neural network, and the glare removal lowers its error under the light by 5.7 percent. The prediction regions of KalmanCQR hold 92 percent of the next measurements at a stated level of 90 percent, and the cognitive chirp choice raises the higher order tracking accuracy on one recording from 49.9 to 58.5 percent. On the moving platform, the planner waits after entries of a person and obstacles on the track, and the scene analysis reaches a Scene Panoptic Quality of 56 percent, against 29 percent for a classical chain of detection, clustering and tracking.

The results indicate that the guarantee holds in a controlled setting and weakens where the scene adds more echoes and objects of different characteristics. The findings suggest that electromagnetic effects set these limits: multipath and coherent echoes weaken the detection guarantee, and dielectric objects, whose echo is the weakest, are the hardest class to read. Within these limits, the cognitive scene analyser quantifies its uncertainty and solves tasks from localization to decision making in real time.

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1. Introduction

1.1. Motivation

"All truths are easy to understand once they are discovered; the point is to discover them."

— Galileo Galilei

Galileo pointed a telescope at the sky and found moons around Jupiter. The truth had always been there, and it took an instrument to reveal it. Radar is such an instrument. It sends a wave into the dark and listens to what comes back, and the echo tells it how far away an object is, how fast it moves and in which direction it lies. With this simple idea, radar has discovered truths that no eye could see. An orbiter above Mars looked through about one and a half kilometres of polar ice and found a bright reflection at its base, which the dielectric properties of the echo identify as liquid water (Figure 1.1) [1]. Radars in orbit map the mountains and the slow deformation of the Earth [2], [3], they follow many objects in space at once [4], and in networks, or as passive receivers that borrow the signals of existing transmitters, they listen to the world with more than one ear [5], [6].

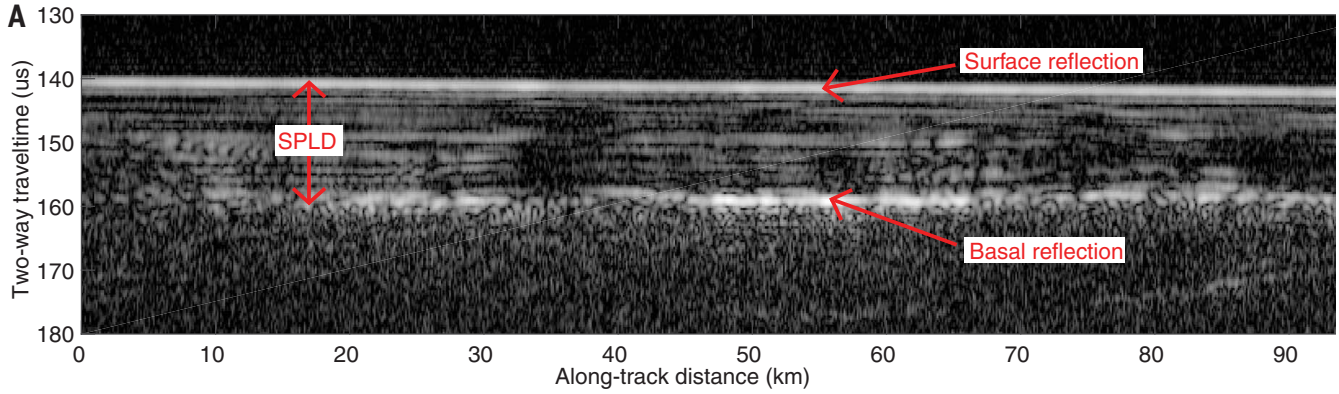


Figure 1.1.: Radar echoes from beneath the ice of Mars. The radargram of the MARSIS radar over the south polar layered deposit shows the surface reflection and, below it, a bright basal reflection that is interpreted as liquid water. Reproduced from Orosei et al. [1].

Closer to home, the same instrument watches over our roads. A radar does not tire, is not dazzled by headlights and does not care whether it is day or night. Darkness means nothing to it, and fog and rain disturb it far less than they disturb a camera. This robustness has made radar a core sensor of driver assistance and automated driving [7]. Safety depends on this knowledge everywhere: an airborne radar must see through

jamming that hides targets in noise or invents false ones [8], [9], and when a child steps from behind a parked car in the fog, a road radar must know that something is there, how far away it is and how fast it approaches.

Yet a radar is half-blind in its own way. It knows where things are, but not what they look like. A camera is its natural partner. It sees colour, shape and writing, and it gives a system a pair of eyes. It also has weaknesses that radar does not share: it cannot measure distance directly, it fails in fog and darkness, and a bright lamp in its lens washes out the whole image. The two sensors can complete each other. The radar can tell the camera where to look and how large an object will appear, and the camera can tell the radar what it is looking at. Together they promise more than scene detection, namely scene analysis: a picture of what is there, what it is made of, how it moves and where it will be next [10], [11].

Galileo's sentence, however, hides a second difficulty. Understanding a truth is easy once it has been discovered, but a sensor never delivers truths alone. Every echo carries noise, and every discovery carries a doubt. A radar frame holds thousands of cells, and each cell is a small question: is there an object, or only noise? However good the test for a single cell, some of the thousands of answers will be wrong. Physicists know this as the look-elsewhere effect. A peak that looks striking at one place in a spectrum loses much of its meaning once the search over all possible places is counted, as it was when the Higgs boson was announced [12], [13]. The consequence is severe when real objects are rare: even a precise test then reports mostly false findings [14]. Fields that search large maps for faint signals therefore stopped asking how likely a single report is to be false and asked how many of all reports are false. They controlled this share, the false discovery rate, and nowhere is the need greater than in genome-wide studies, which test an entire genome at once [15]. Selectors that bound the false discovery rate there now handle millions of genetic variants [16], [17], and the same idea tamed the images of the brain [18], of the sky [19] and of the stock market [20]. A radar needs the same answer. A false object on the road can trigger an emergency brake, and no test drive can prove that a vehicle is reliable, since hundreds of millions of miles would be needed [21]. A perception system must therefore carry its error bound by construction and not find it by experiment.

Nature solved the problem of sensing in the dark long before engineers did. A bat hunts insects at night by echolocation, with sound alone. While it searches, it sends out long calls at a slow rate. Once it has found a prey, it shortens the calls and speeds them up until it closes in on the insect [22]. The bat does not merely listen to the echo; it decides what to send next, and it pays for every call in time and energy [23]. With this loop it tracks a moving insect and steers through the dark at the same time. Engineers have taken up the idea under the name active perception, in which the interpretation of the data controls the next measurement [24]. Neuroscientists call the loop the perception-action cycle [25], and Haykin brought it to radar as cognitive radar, a radar that feeds what it has understood back to what it transmits [26], [27]. The research community lists such cognitive, learning-based and multi-sensor perception among the grand challenges of radar [28].

This thesis follows that path. It builds a radar that looks at the scene, a camera that adds its eyes, and a cognition that decides where and how to look next, and it ties them together by a guarantee on the share of false discoveries. If the point is to discover truths, as Galileo said, then this thesis asks how a radar and a camera can discover them together and how they can know which of their discoveries deserve to be called true.

1.2. Research Rationale and Objectives

A system that discovers truths with a radar, a camera and a measure of its own doubt must unite scientific worlds that rarely meet. Statistical signal processing asks how many of its discoveries are false [29], and it has built selectors that control this share for problems with millions of variables [16], [30], [31], [32]. Array signal processing and radar engineering ask how a wave at a very high frequency carries the position and the motion of an object into an antenna array [33], [34], and they answer with beamforming [35], [36], subspace methods [37], [38] and thresholds on a single cell [39], [40]. Computer vision reads and recognises what an image shows [41], [42], [43], and the study of intelligence asks how a system chooses its own measurements in a loop of perception and action [24], [26], [27]. Each toolbox rests on assumptions about its own data. The statistical selectors assume a design matrix with convenient properties [30], [31], the array methods assume a number of sources and a clean noise model [37], [44], and the vision methods assume that an image is all there is [45]. The assumptions of one field rarely hold in the data of another, and this interdisciplinary gap explains why the pieces exist and the whole does not.

Consequently, the processing chain fractures into isolated stages: each field guards its own domain, leaving unmapped gaps between them that a rigorous statistical framework must bridge. Selectors that control the false discovery rate were conceived for genomics and finance. For radar, they were proposed as a detection framework [46] and tested for direction finding in simulation [47], but their guarantee on measured data is untested. Radar practice bounds the false alarm probability of a single cell [39], [40], so the share of false objects in its output stays uncontrolled. Material classifiers with millimetre waves mostly work on contact [48], and micro-Doppler methods concentrate on motion [49]. Kalman trackers state their uncertainty by a covariance that is exact only for linear models with Gaussian noise [50], [51], whereas conformal prediction guarantees coverage without a model of the data [52], [53]. Networks fuse radar and camera without an error bound [11], [54]. Cognitive radar closes the loop on real hardware [55], [56], but not yet on a moving platform with the guarantees of the earlier stages.

These gaps matter because a perception system is only as reliable as the weakest guarantee in its chain. A detector with controlled errors is of little use if the tracker that follows it is overconfident, and a classifier that explains nothing cannot be trusted in a decision that involves a person. Since safety cannot be demonstrated by driving alone [21], the guarantees must be part of the method at every stage. The objective of this thesis is to build such a chain on one sensor platform with one statistical framework, and to test every link on measured data. It serves one central question.

Central research question: Can a system distill overwhelming information into a minimal set of bounded truths, leveraging its evolving beliefs and cognitive decisions to reconstruct a true scene?

Three observations motivate the question. First, a radar frame holds thousands of cells and a camera image millions of pixels, yet a decision needs only a handful of statements. A system must therefore reduce the data to a few reports. Second, every report can be wrong. A report becomes a truth for others to rely on only if it comes with a guarantee on how many of its fellow reports may be wrong [29]. Third, a system that moves chooses its next measurement from what it already believes [24], [26], so its belief changes with every look. A belief that changes without a guarantee carries its errors into every later decision [21]. The question joins the three: reduce the data, bound the truths and steer the next measurement. No single work settles it. Five research questions pursue it, one for each stage of the chain. They mark directions, and the experiments settle each one in part.

RQ1: How much of what a radar reports can be certified as real, that is, can the share of false discoveries in detection and direction-of-arrival estimation be bounded by a guarantee that survives the step from

theory to real measurements? Classical estimators need many snapshots and a covariance [37], [44]. Selectors that control the false discovery rate work with a single snapshot [47]. Their guarantee rests on assumptions about noise and geometry, and it was tested only in simulation. Measured data must show whether the guarantee belongs to the radar or only to its model.

RQ2: How much of the nature of an object can a millimetre-wave radar read from its echo alone, in real time, and how many of its properties can it read at once? The echo carries the strength of the reflection, which depends on the material [57], [58], and its phase over time, which depends on the motion [59]. A walking person and falling rain both consist of water, yet a vehicle must react to them differently [60]. Millimetre-wave classifiers mostly read objects on contact or only their motion [48], [49]. How much a single echo reveals at once, and how fast that revelation is, has much to be investigated.

RQ3: What can a radar and a camera know together that neither knows alone, and what must the two sensors exchange to know it? The camera supplies colour, shape and writing, and the radar supplies distance, speed and robustness against light [61], [62]. Networks merge both in a learned representation without an error bound [11], [54]. The sensors can instead exchange knowledge. The radar gives the direction and distance of an object, so the camera knows where to look and at what scale [24]. For instance, images could show whether identified objects optically interfere the images, whose glare spreads over the picture [63], which can be filtered via the location of the object. The tasks that profit from the radar-vision exchange remain to be identified.

RQ4: Can a system state honestly how little it knows about the future of what it tracks, without assuming a model of how the object moves, and can this statement improve its next measurement? A prediction is useful only with a statement of its uncertainty. Kalman trackers state it by a covariance that is exact only for linear models with Gaussian noise [50], [51]. Conformal prediction offers regions with coverage under weak assumptions [52], [53], [64]. A test on moving objects must show how such regions behave when the motion changes, and whether they can steer the next measurement [65], [66].

RQ5: Can cognition, the choice of what to measure next, turn guarantees about perception into guarantees about action when the sensor itself moves? A cognitive radar chooses what to measure next from what it has understood [26], [66], [67], and a vehicle must act on the same understanding [21]. When the sensor moves, the static world must be separated from the movers [8], [68], and the time of perception becomes part of the reaction time. Two difficulties remain: scoring such a scene [10], [69], and finding where the guarantees of the components stop holding for the whole.

The five questions describe one processing chain, and the literature answers each link alone. Selectors that control false discoveries were tested for direction finding only in simulation [47]. Material classifiers read objects on contact [48], and fusion networks give no error bound [54]. Trackers state their uncertainty only under a motion model [51], and cognitive radar closes its loop without a guarantee on a moving platform [56]. No work joins these links under one guarantee and tests the whole chain on measured data. This gap defines the contribution of the thesis.

1.3. Contributions and Overview

One idea runs throughout this thesis: whatever a radar reports, one should be able to ask how much of it is true. Chapter 2 collects the fundamentals of radar, signal processing, statistics and vision, and Chapter 3 states, for each experiment, the conventional method, the recent work and the question that this work leaves open.

Chapter 4 introduces **ARTIFACT**, the Adaptive Radar Tracking Inference of False Alarm Controlled Targets: an automotive radar whose **raw samples stream into an own implementation**, behind a **modular printed radome**, with a camera and a motor stage. It records the **measured data sets** of all experiments.

Chapter 5 detects and locates reflectors and shows how many of the reports of a radar can be trusted. It tests the **selectors that control false discoveries** against classical estimators on measured data and derives **Infini-Rex**, which replaces many random experiments by one closed-form step.

Chapter 6 shows what a single echo reveals about an object, such as what it is made of and how it moves, so that a person can be told from rain. The classifier **M^2-SRC** does this in real time and shows the examples behind each decision.

Chapter 7 shows what radar and camera can do together that neither can do alone, such as reading text through the glare of a lamp. **MetricSight** reads at a scale that the radar range fixes, and a **radar-anchored glare removal** clears the view when the lamp shines into the lens.

Chapter 8 follows moving objects over time and shows how sure a system can be about where they will be next. **Track-guided detection** lets predictions steer the search, **KalmanCQR** gives a Kalman filter prediction regions of stated coverage, and a **cognitive chirp choice** lets the radar decide what to measure next.

Chapter 9 shows how all of this works on a platform that moves and decides, for example to wait for a person or to step around an obstacle. A **world model** and a **planner** turn the scene into decisions, and a **spatial envelope** describes the scene that radar and camera see together.

Chapter 10 summarises the results and shows what remains open. Appendices A to D hold the implementation, the proofs, the data sets and the code base. Table 1.1 lists the contributions.

Table 1.1.: Contributions of the thesis, with the location in which each is illustrated.

Contribution	What it does	Where
ARTIFACT radar platform ★	Raw samples of a radar in own Python code	Chapter 4
Measured data sets ★	Recordings of five experiments with ground truth	Chapters 5 to 9, Appendix C
Modular radome and housing ★	Printed radome with measured pattern	Sections A.2, A.4 and 4.2
FDR-controlled selectors in radar ★	Benchmarked on measured radar data	Section 5.2.3
Infini-Rex (for Radar) ★	Closed-form e-values; continuous angles	Sections B.1.3 and 5.2.4
M^2-SRC ★	Material and motion from one echo	Section 6.2.2
MetricSight ★	Reads text at a radar-fixed scale	Section 7.2.2
Radar-anchored glare removal ★	Removes the veil of a located lamp	Section 7.2.3
Track-guided detection ★	Tracks weight the detection	Section 8.2.2
KalmanCQR ★	Conformal regions for a Kalman filter	Section 8.2.1
Cognitive chirp choice ★	Chirps chosen by predicted entropy	Section 8.2.3
World model and planner ★	Measured reach turned into decisions	Sections 9.2.1 and 9.2.2
Spatial envelope of radar and camera ★	One scene description for both sensors	Section 9.1.1
Radar tutorial and code base ★	Start-up guide and open source code	Section A.5, Appendix D

Original contributions of this thesis are marked with a green star (★) in the table and in the chapter headings.

2. Fundamentals

Every radar measurement begins in the same place: a wave leaves an antenna and returns changed. This chapter follows that wave and gathers what the later chapters take for granted, namely the fundamentals. Section 2.1 follows the wave from the antenna to an object and back, and explains how a millimetre-wave radar turns the echo into a range, a speed and a direction, how antennas form a virtual array, and what the material of an object and the chirp configuration add. Section 2.2 shows how echoes become decisions: how the radar cube is built, how a single cell is tested, how directions are estimated, and how objects are tracked and their next positions predicted with a stated uncertainty. Section 2.3 widens the view from one decision to thousands. It explains how many hypotheses are tested at once, how the false discovery rate bounds the share of false reports among them and how a few relevant variables are picked from many candidates. Section 2.4 turns to the second sensor and describes how an image represents the scene, how operations on neighbouring pixels separate ink from its background and how convolutional networks learn their own features. Each concept appears once, where it is first needed, so that the experiment chapters can refer back instead of repeating it.

2.1. mmWave Radar

A radar perceives distance, speed and direction through darkness, fog and rain, because it listens to the echo of its own radio wave and needs no daylight. The name abbreviates *radio aircraft detection and ranging* and recalls the origin of the method: a ground station transmitted a radio wave to detect aircraft, the aircraft scattered part of it back, and the delay of the echo gave the range [9], [70]. Modern radar keeps the principle and drops the aircraft: it denotes any sensor that transmits an electromagnetic wave and reads the echo to learn whether an object is present, how far it is, how fast it moves and in which direction it lies. A millimetre-wave (mmWave) radar operates at wavelengths from 1 to 10 mm, which correspond to frequencies from 30 to 300 GHz. Automotive and industrial sensors use the band from 76 to 81 GHz [7], where the wavelength spans 3.7 to 3.9 mm. This short wavelength admits small antennas with narrow beams, and the wide band admits a fine range resolution.

2.1.1. Electromagnetic Wave Propagation and the Radar Equation

A radar knows an object only through the way the object changes its wave. Like a shout in a canyon, the wave leaves its source, strikes a surface and returns delayed and weakened, and the delay and the weakening reveal where the surface lies and how it reflects. Electrodynamics supplies the exact description: Maxwell's equations define the wave, the boundary conditions define the reflection, and the radar equation defines the power of the echo.

From Maxwell's equations to the plane wave. Maxwell's equations [71] couple the electric field $\mathbf{E}$ (in V/m) and the magnetic field $\mathbf{H}$ (in A/m). Far from any source, the two fields sustain each other, and this self-sustaining pair is a wave. Consider a region without free charges and currents, filled with a linear, isotropic and homogeneous medium of permittivity ε and permeability μ. Faraday's and Ampère's laws read

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}, \qquad \nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t}, \tag{2.1}$$

and Gauss's law requires $\nabla \cdot \mathbf{E} = 0$. Take the curl of the first equation. Schwarz's theorem permits exchanging the order of the spatial and the temporal derivative, and Ampère's law then replaces $\nabla \times \mathbf{H}$. With the identity $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$, the result is the wave equation

$$\nabla^2 \mathbf{E} = \mu\varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}, \tag{2.2}$$

which $\mathbf{H}$ obeys as well [72]. Its solutions travel at the speed $v = 1/\sqrt{\mu\varepsilon}$, which equals $c = 1/\sqrt{\mu_0\varepsilon_0} \approx 2.998 \cdot 10^8$ m/s in vacuum.

The simplest solution is the time-harmonic plane wave

$$\mathbf{E}(\mathbf{r}, t) = \mathrm{Re}\{\mathbf{E}_0\, e^{\mathrm{j}(\omega t - \mathbf{k}\cdot\mathbf{r})}\}, \qquad |\mathbf{k}| = k = \frac{\omega}{v} = \frac{2\pi}{\lambda}, \tag{2.3}$$

where j is the imaginary unit, $\mathbf{r}$ the position, $\omega = 2\pi f$ the angular frequency, $\mathbf{k}$ the wave vector, $\mathbf{E}_0$ the complex amplitude vector and $\lambda = v/f$ the wavelength. The field repeats along $\mathbf{k}$ with the period λ and in time with the period $1/f$. Gauss's law demands $\mathbf{k} \cdot \mathbf{E}_0 = 0$, so $\mathbf{E}$ is transverse, and Faraday's law yields $\mathbf{H} = \hat{\mathbf{k}} \times \mathbf{E}/\eta$ with the wave impedance $\eta = \sqrt{\mu/\varepsilon}$, which is $\eta_0 \approx 377\,\Omega$ in vacuum. Power flows along the Poynting vector $\mathbf{E} \times \mathbf{H}$, and its time average in a lossless medium is the power density

$$\Pi = \frac{|\mathbf{E}_0|^2}{2\eta} \tag{2.4}$$

in W/m^2. Doubling the field amplitude quadruples the power density.

Real scenes consist of layered, anisotropic materials, whereas the derivation assumes a homogeneous medium. The analysis cuts the scene into regions of constant material and joins the solutions at the boundaries. Air and common materials respond linearly at radar power densities, so waves superpose [72], and a chirp splits into one plane wave per frequency. A 4 GHz chirp spans 5 per cent of a 79 GHz carrier, so the permittivity at the centre frequency describes every component.

Waves in matter and at boundaries. A real medium absorbs part of the wave. Conduction currents and rotating molecular dipoles extract its energy, which the complex relative permittivity $\varepsilon_\mathrm{r} = \varepsilon' - \mathrm{j}\varepsilon''$ summarises. The real part ε' slows the wave to $c/\sqrt{\varepsilon'}$ and shortens its wavelength to $\lambda_0/\sqrt{\varepsilon'}$, where λ_0 is the wavelength in vacuum. The imaginary part ε'' damps the amplitude along the path z as $e^{-\alpha z}$ [72]. For weak loss, the attenuation constant α follows from ε'', and in a good conductor of conductivity σ_c, the skin depth δ_s marks the depth at which the field has fallen to $1/e$ of its surface value:

$$\alpha \approx \frac{\pi\varepsilon''}{\lambda_0\sqrt{\varepsilon'}}, \qquad \delta_\mathrm{s} = \frac{1}{\sqrt{\pi f \mu \sigma_\mathrm{c}}}. \tag{2.5}$$

For copper at 77 GHz, $\delta_\mathrm{s} = 0.24\,\mu$m, so the field of a metal fills a layer 16 000 times thinner than the wavelength, and the metal reflects the wave in full.

At a boundary between two media, the tangential components of **E** and **H** stay continuous, which fixes the reflected wave. At normal incidence from air onto a medium of relative permittivity ε_r and wave impedance η, the reflection coefficient Γ, the ratio of the reflected to the incident field amplitude, is

$$\Gamma = \frac{\eta - \eta_0}{\eta + \eta_0} = \frac{1 - \sqrt{\varepsilon_\mathrm{r}}}{1 + \sqrt{\varepsilon_\mathrm{r}}} \tag{2.6}$$

[72]. The share $|\Gamma|^2$ of the power returns, and a perfect conductor reflects with $\Gamma = -1$.

Near field, far field and beams. A real antenna of finite size D radiates a spherical wave (Figure 2.1a). Close to the antenna, the wave front is curved across the aperture, and far away, a small piece of the sphere looks flat. The wave counts as plane where the front deviates by less than $\lambda/16$ across the aperture, which geometry converts into the far-field distance

$$r_\mathrm{ff} = \frac{2D^2}{\lambda} \tag{2.7}$$

[9], [73]. An aperture of $D = 20\,$mm at 77 GHz reaches its far field at 0.21 m.

The aperture also shapes the beam (Figure 2.1b). Huygens' principle treats every point of a wave front as the source of a spherical wavelet, so an aperture of width a acts as many sources. For $a \gg \lambda$, their waves add in phase in the forward direction and cancel elsewhere, which forms a narrow beam. For $a \ll \lambda$, a single source remains, and it radiates in all directions. A uniformly illuminated aperture radiates a beam of half-power width θ_3, the angle between the two directions in which the power has fallen to half of its peak, and a gain G, the peak power density relative to an isotropic radiator of equal power, an ideal antenna that radiates equally in all directions:

$$\theta_3 \approx 0.89\,\frac{\lambda}{a}, \qquad G = \frac{4\pi A_\mathrm{e}}{\lambda^2}. \tag{2.8}$$

Here A_e is the effective area, the area over which the aperture collects power, and θ_3 is measured in radians [9], [73], [74].

(a) Spherical to Plane Waves

(b) Diffraction at an Aperture

Figure 2.1.: A wave leaves an antenna. (a) An isotropic source radiates spherical wave fronts, which are plane to within $\lambda/16$ over an aperture D beyond the far-field distance. (b) A wave passing an opening of width a keeps its direction if $a \gg \lambda$ and spreads in all directions if $a \ll \lambda$. Adapted from Richards et al. [9].

Echo power and the radar equation. The radar equation follows the power to an object and back. An isotropic source of power P_t spreads it over a sphere, so the power density at range r is $P_\mathrm{t}/(4\pi r^2)$, and an antenna of gain G_t raises it by the factor G_t. The object intercepts part of this power and scatters part of it back. The radar cross section σ quantifies the result through the field $\mathbf{E}_\mathrm{s}$ scattered to the radar and the field $\mathbf{E}_\mathrm{i}$ incident at the object [9]:

$$\sigma = \lim_{r\to\infty} 4\pi r^2 \frac{|\mathbf{E}_\mathrm{s}|^2}{|\mathbf{E}_\mathrm{i}|^2}. \tag{2.9}$$

The echo spreads over a sphere again, and a receiving antenna of gain G_r captures it with its effective area $G_\mathrm{r}\lambda^2/(4\pi)$. Together, these factors yield the received power

$$P_\mathrm{r} = \frac{P_\mathrm{t}\, G_\mathrm{t}\, G_\mathrm{r}\, \lambda^2\, \sigma}{(4\pi)^3\, r^4\, L_\mathrm{s}}, \tag{2.10}$$

where $L_\mathrm{s} \geq 1$ summarises the system losses [9], [74]. The echo competes with thermal noise,

$$P_\mathrm{n} = k_\mathrm{B} T_0 F B, \tag{2.11}$$

with the Boltzmann constant k_B, the reference temperature $T_0 = 290\,\mathrm{K}$, the noise figure F of the receiver and its bandwidth B. The ratio $P_\mathrm{r}/P_\mathrm{n}$ is the Signal-to-Noise Ratio (SNR).

Three consequences follow. Power ratios are stated in decibel (dB), ten times the base-10 logarithm of the ratio, so that 3 dB halve the power. First, $P_\mathrm{r} \propto r^{-4}$, because the wave spreads on the outbound and on the inbound path, and doubling the range costs 12 dB. Second, the detection range grows only as $\sigma^{1/4}$, so a 16-fold smaller cross section halves the range. Third, the noise power grows with the bandwidth, whereas a fine range resolution requires a large bandwidth (Section 2.1.2), so resolution and SNR compete.

Effects of the real scene. The radar equation assumes a lossless path, a single path and one number σ per object. Four effects of the real scene relax these assumptions.

Absorption by the atmosphere weakens the echo. Oxygen and water-vapour molecules resonate near 60 and 22 GHz and absorb strongly there [9]. Between 76 and 81 GHz, the one-way attenuation, the power lost per kilometre of path, is ≈ 0.4 dB/km at sea level, so an object at 300 m behind a round trip of 600 m costs 0.24 dB. Rain and fog absorb and scatter the wave at their droplets. Fog droplets span tens of micrometres at most, far below the wavelength of 3.9 mm, and leave the wave nearly intact, whereas they block optical and infrared radiation. Raindrops of millimetre size match the wavelength and attenuate it more strongly [9].

Scattering fixes the cross section of each object. At 3.9 mm, vehicles, people and walls span many wavelengths and behave as a set of scattering centres, points on the surface that return the wave independently. The echo is the complex sum of their contributions [9]. Specular reflection, in which the reflected ray leaves at the angle of incidence, forms the strongest centres. A flat plate of area A that faces the radar returns $\sigma = 4\pi A^2/\lambda^2$, and a sphere of radius a returns $\sigma = \pi a^2$ from every view. The complex sum reacts sensitively to the geometry: a relative shift of 0.97 mm between two centres changes the round-trip path by half a wavelength, 1.95 mm, and turns reinforcement into cancellation. The echo power of a complex object fluctuates from look to look, so σ in the radar equation is a mean value.

Roughness decides whether a surface mirrors the wave or scatters it. The root-mean-square (rms) height h is the standard deviation of the surface elevation about its mean level. Two points that differ in elevation by h reflect the wave over paths that differ by $2h\cos\theta$, where θ is the angle of incidence measured from the

surface normal. The reflected waves add coherently while this difference stays below a quarter wavelength, which is the Rayleigh criterion

$$h \cos\theta < \frac{\lambda}{8} \tag{2.12}$$

[9]. At 77 GHz and normal incidence, the limit is $h < 0.49$ mm. Below the limit, a surface mirrors the wave and returns an echo only while it faces the radar, and above it, the surface scatters diffusely.

Multipath adds echoes that no object produces. A smooth surface such as a road or a wall offers a second path to the object and back [9]. The longer path places the echo at a longer range, and the mirror image puts it behind the surface as a ghost, an echo without an object at its apparent position. The paths also interfere, and a change of their difference by half a wavelength, 1.95 mm, turns reinforcement into cancellation.

Phase and the Doppler effect. The phase of the echo encodes the path length. A wave that travels to range r and back covers the distance $2r$ and accumulates the phase ϕ. If the object moves with the radial speed v_r, positive away from the radar and small against c, the phase changes in time. The Doppler effect is this change, and the Doppler frequency f_D is the phase rate, $f_\mathrm{D} = (1/2\pi)\,\mathrm{d}\phi/\mathrm{d}t$:

$$\phi = \frac{4\pi r}{\lambda}, \qquad f_\mathrm{D} = \frac{2v_\mathrm{r}}{\lambda}. \tag{2.13}$$

A displacement of half a wavelength rotates the phase by a full 2π, and a speed of 1 m/s shifts the frequency by 514 Hz at 77 GHz [74]. A phase error of 1° corresponds to a displacement of 5.4 µm at 77 GHz, so the radar senses displacements far below the wavelength.

The receiver stores the echo as a complex number V with the amplitude $A \geq 0$ and the phase ϕ. Mixing the antenna voltage, which is proportional to $\mathbf{E}$, with the transmitted signal splits it into the in-phase component I and the quadrature component Q:

$$V = Ae^{\mathrm{j}\phi} = I + \mathrm{j}Q, \qquad I = A\cos\phi, \quad Q = A\sin\phi. \tag{2.14}$$

The squared amplitude A^2 is proportional to the received power of Eq. (2.10), whereas the phase ϕ carries the path length in fractions of a wavelength.

The shout in the canyon now has an exact counterpart. The echo returns weakened by the r^{-4} law and by the reflectivity of the surface, delayed by the path, and shifted in phase by the path length in fractions of a wavelength. Its power reports size and material through σ and Γ, its delay and its phase report the range, and the rate of its phase reports the speed. How a receiver reads a delay of nanoseconds and a phase of a fraction of a wavelength at a modest sampling rate remains open.

2.1.2. FMCW Principle and Chirps

Range hides in a delay of nanoseconds. An echo from range r returns after $\tau = 2r/c$, so an object at 10 m answers after 67 ns, and a range resolution of 4.4 cm demands a timing of 0.29 ns. A pulse radar timestamps this delay, but a fine resolution requires a short pulse with a large bandwidth, a high peak power and fast electronics [70], [74]. A Frequency-Modulated Continuous Wave (FMCW) radar converts the delay into a frequency instead, as two close piano strings convert a minute pitch difference into a slow, audible beat.

It transmits a chirp, a signal whose frequency rises linearly as $f(t) = f_0 + St$ with the slope S during the chirp duration T, so that the chirp sweeps the bandwidth $B = ST$. The echo is the same chirp delayed by τ

(Figure 2.2). The receiver mixes the echo with the transmitted chirp: it multiplies both signals and retains their phase difference, the beat signal

$$s_{\mathrm{b}}(t) \propto \exp\{\mathrm{j}\,(2\pi S\tau\, t + 2\pi f_0 \tau - \pi S\tau^2)\}, \tag{2.15}$$

where $\propto$ hides a constant amplitude. The first term oscillates with the constant frequency

$$f_{\mathrm{b}} = S\tau = \frac{2Sr}{c}, \tag{2.16}$$

because the delayed chirp is a copy of the transmitted chirp shifted along the time axis, and this shift appears as the frequency difference $S\tau$. Range becomes a frequency, which is low and easy to sample [33], [75]. Each object contributes one tone, and a Fast Fourier Transform (FFT) of the sampled beat signal, the range FFT, evaluates the spectrum at discrete frequency steps called bins and exhibits one peak per object.

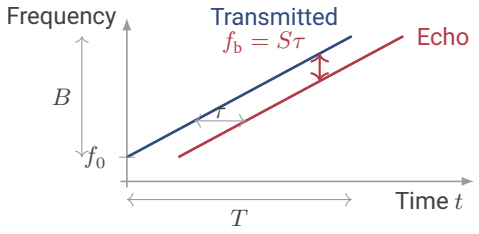


Figure 2.2.: Chirp, echo and beat frequency. The echo is the transmitted chirp delayed by $\tau = 2r/c$. At every instant, the two frequencies differ by the constant beat frequency $f_{\mathrm{b}} = S\tau$, which the receiver extracts by mixing. The chirp sweeps the bandwidth B in the duration T.

Two objects form separate peaks if their beat frequencies differ by more than $1/T$, the frequency resolution of the FFT. Since the beat frequency grows by $2S/c$ per metre, this condition defines the range resolution

$$\Delta r = \frac{c}{2ST} = \frac{c}{2B}. \tag{2.17}$$

Only the swept bandwidth sets the resolution [70]. The maximum range follows from the sampling. The Analog-to-Digital Converter (ADC) samples the beat signal at the rate f_{s} and must capture the highest beat frequency. Complex samples cover the bandwidth f_{s}, whereas real samples represent only frequencies below $f_{\mathrm{s}}/2$ and halve the maximum range [33]:

$$r_{\max} = \frac{f_{\mathrm{s}}\, c}{2S} \quad \text{(complex samples)}. \tag{2.18}$$

Speed hides in the phase. The constant phase in Eq. (2.15) is dominated by $2\pi f_0 \tau = 4\pi r/\lambda$ with $\lambda = c/f_0$, because the bandwidth is small against f_0. An object that moves by vT_{c} between two chirps that start T_{c} apart shifts this phase by

$$\Delta\phi = \frac{4\pi v T_{\mathrm{c}}}{\lambda}. \tag{2.19}$$

The radar transmits a sequence of K chirps and applies a second FFT across the chirps for each range bin, the Doppler FFT. This transform converts the steady phase step into a speed [33], [75]. A phase is known only modulo 2π, so faster speeds wrap around. The maximum speed and the speed resolution follow:

$$v_{\max} = \frac{\lambda}{4T_{\mathrm{c}}}, \qquad \Delta v = \frac{\lambda}{2KT_{\mathrm{c}}}. \tag{2.20}$$

The beat of the piano strings has become a ruler. The bandwidth alone sets the range resolution, the slope and the sampling rate set the maximum range, and the chirp interval and the sequence length set the maximum speed and the speed resolution. The two transforms arrange the echoes in a two-dimensional spectrum of range and speed, in which every object sounds its own tone and occupies one cell. A single antenna cannot tell from where in space a tone arrives, and the missing angle calls for more than one antenna.

2.1.3. MIMO and Virtual Arrays

A single antenna measures range and speed, but it cannot tell from which direction an echo arrives. Direction requires at least two antennas, because the echo of an object reaches them at different times. A row of antennas converts these time differences into an angle, and a Multiple-Input Multiple-Output (MIMO) radar builds a long row from two short ones, since a long row separates neighbouring objects more finely.

An array is a row of identical receiving antennas with the spacing d. The angle θ of an object is measured from the array normal, the line perpendicular to the row, so $\theta = 0$ means straight ahead. The wave of a distant object arrives as a plane wave (Section 2.1.1) and reaches the antennas one after another. The extra path between two neighbours is $d \sin\theta$ (Figure 2.3a) and appears as the phase difference

$$\Delta\phi = \frac{2\pi d \sin\theta}{\lambda}. \tag{2.21}$$

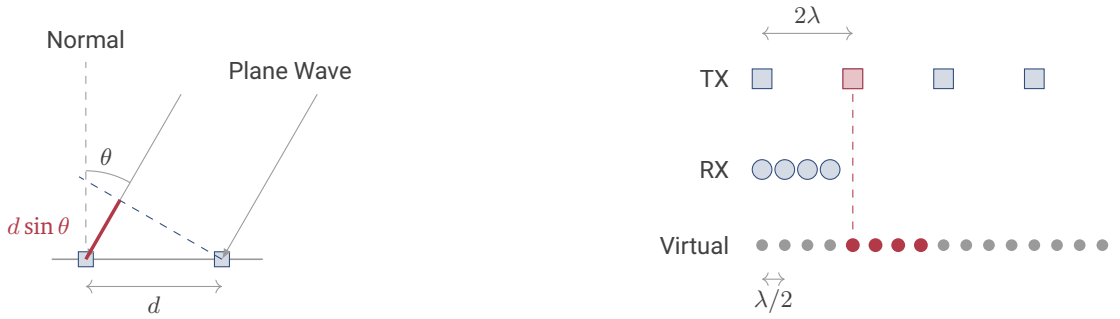
An array of N receivers samples this phase progression in space, just as K chirps sample the phase progression in time, and an FFT across the antennas, the angle FFT, converts the progression into the direction sine $u = \sin\theta$ [33]. The spacing d corresponds to the chirp interval and bounds the unambiguous field of view, which is $\pm 90^\circ$ for $d \leq \lambda/2$. The array length Nd corresponds to the frame duration and sets the resolution: two directions are resolved if their sines differ by more than $\lambda/(Nd)$.

A long array needs many receivers, and a MIMO radar avoids this by adding transmitters. It uses N_{tx} transmitting and N_{rx} receiving antennas. For a distant object, the echo of transmitter i at receiver l carries the phase that a single one-way antenna at the position $x_i + x_l$ would measure [34]. Each transmitter-receiver pair acts as one element of a virtual array (Figure 2.3b). A virtual element exists only in the calculation, since no antenna sits at its position, yet its phase equals that of a real antenna there. Four receivers spaced $\lambda/2$ and four transmitters spaced 2λ yield 16 virtual elements spaced $\lambda/2$.

The receivers must separate the echoes of the transmitters. In Time-Division Multiplexing (TDM), the transmitters send one after another, and the time slot identifies the sender [76]. Each transmitter then repeats only after N_{tx} chirp slots, which lowers the maximum speed by the factor N_{tx}. A moving object also advances its phase between the slots, which shifts its apparent direction unless the radar compensates it [77]. Two short rows now act as one long row, and the price is paid in speed: each transmitter repeats only after N_{tx} slots, so a MIMO radar trades maximum speed for angular resolution. Range, speed and direction place an object in space, yet the echo carries more than position: it also records how the object reflects, absorbs and moves.

2.1.4. Material Sensing

Material sensing infers what an object consists of from the way it changes the echo. Every material answers the wave in its own way, and four answers leave a trace in the echo: reflection, absorption, surface scattering and motion.



(a) Phase Step Across Two Antennas

(b) Four Transmitters and Four Receivers

Figure 2.3.: Direction from an antenna array. (a) A plane wave from the angle θ reaches the near antenna $d \sin\theta$ earlier than the far one, which produces the phase step of Eq. (2.21). (b) Each transmitter-receiver pair acts as one virtual element at the sum of the two positions. Four receivers spaced $\lambda/2$ and four transmitters spaced 2λ yield 16 virtual elements spaced $\lambda/2$; the four elements of the second transmitter are highlighted.

A boundary returns the share $|\Gamma|^2$ of the incident power, and Eq. (2.6) shows that this share grows with the relative permittivity ε_r [78]. A metal returns all of it. Polycarbonate ($\varepsilon_\mathrm{r} = 2.9$) returns 7 % [79], whereas water returns a larger share, because its polar molecules rotate with the field and produce a high permittivity at millimetre waves [57]. The part of the wave that enters loses power at a rate that grows with ε'' (Eq. (2.5)): plastics pass the wave with little loss, and water absorbs it within a fraction of a millimetre.

The surface and the motion add two further traces. A smooth surface mirrors the wave away from the radar, and a rough surface scatters it in all directions (Rayleigh criterion, Eq. (2.12)), so the echo depends as strongly on the viewing angle as on the material. Vibrating or rotating parts add sidebands around the Doppler frequency of the whole object, the micro-Doppler effect [59], and the spectrum reveals whether an object moves as a rigid body or with moving parts.

The four answers of a material survive in the echo: the amplitude, corrected for the r^{-4} law of Eq. (2.10), carries reflection and absorption, and the phase over time carries the motion. Material sensing ranks materials by reflection, penetration and motion, because no single trace names a material: the echo strength also depends on size and viewing angle. How many of these traces a measurement resolves depends on the chirp, since bandwidth, sampling and chirp interval decide which differences between echoes the radar still sees.

2.1.5. Radar Configuration

A radar configuration decides what the radar can see. Four quantities limit its view: the finest distance and the finest speed it resolves, and the largest distance and the largest speed it measures without ambiguity. The previous subsections derived these quantities from the chirp, from parameters to measurement. A designer needs the opposite direction. The application states the four quantities, for example a pedestrian at 10 m or a car at 30 m/s, and the chirp parameters must follow from them. This inversion is the design problem, and its solution fixes everything that the radar can and cannot distinguish.

The inversion proceeds in a fixed order [33]. The maximum speed of Eq. (2.20) comes first, because it fixes the time T_c between two chirps of the same transmitter. The range resolution of Eq. (2.17) fixes the sampled bandwidth $B = SN_\mathrm{s}/f_\mathrm{s}$ of slope S, sample count N_s and sampling rate f_s. The maximum range of Eq. (2.18)

fixes the ratio $f_s/S = 2r_{max}/c$, so that bandwidth and maximum range together determine the slope and the sample count for a given ADC rate. The speed resolution of Eq. (2.20) finally fixes the number K of chirps per frame.

Table 2.1.: Example chirp configuration of a 77 GHz radar with four transmitters in TDM and the quantities it fixes. The band centre of 79.2 GHz fixes the wavelength $\lambda = 3.785$ mm.

Parameter	Fixes	Value
Slope S	Beat frequency per metre, Eq. (2.16)	70 MHz/µs
Samples N_s, rate f_s	Sampled bandwidth $B = SN_s/f_s$	656, 13.35 MS/s, $B = 3.44$ GHz
Bandwidth B	Range resolution Δr, Eq. (2.17)	4.36 cm
Sampling rate f_s	Maximum range $f_s c/(4S)$, real samples	14.3 m
Idle time, ramp time	Chirp period $T_c = 4\,(T_{idle} + T_{ramp})$	186 µs, 57.14 µs, $T_c = 972.6$ µs
Chirp period T_c	Maximum speed $\lambda/(4T_c)$	± 0.97 m/s
Chirps per transmitter K	Speed resolution $\lambda/(2KT_c)$	16, 0.122 m/s
Transmitters $\times$ receivers	Virtual channels, 12 in the azimuth row	$4 \times 4 = 16$

The four requirements compete for the same parameters. At a given ADC rate, a steeper slope sharpens the range resolution but shortens the maximum range, because the beat frequency of a distant object rises with S. A shorter chirp period raises the maximum speed, and a longer frame sharpens the speed resolution at the price of a longer measurement time. Every parameter serves one requirement and limits another. Table 2.1 shows the compromise: the four transmitters share the chirp period, so each repeats only after four slots, and a range resolution of 4.4 cm comes with an unambiguous speed of only ± 0.97 m/s. The configuration fixes what the radar can tell apart, yet what it delivers is still a stream of numbers.

2.2. Signal Processing

A sensor delivers numbers, not knowledge. A microphone reports pressure, a camera brightness and a radar voltages, and each series mixes the quantity of interest with noise and with other sources. Signal processing recovers the quantity: it models how information and disturbance enter the signal and applies the operation that separates them. A listener who follows one voice in a loud room solves the same task. The ear separates the voice by its pitch, a frequency, and by its direction, the delay between the two ears. These two cues mark the classic tools of the discipline. The Fourier transform sorts a signal by frequency, and an array of sensors extends the same transform from time to space, where it sorts waves by direction (Eq. (2.21)). Radar signal processing combines both: a radar samples time, chirps and antennas together. It adds a decision to the result, because noise is random and every statement about an object is a statement about probabilities. Figure 2.4 traces the path from samples to forecast, and the red marks locate the points where an error can enter.

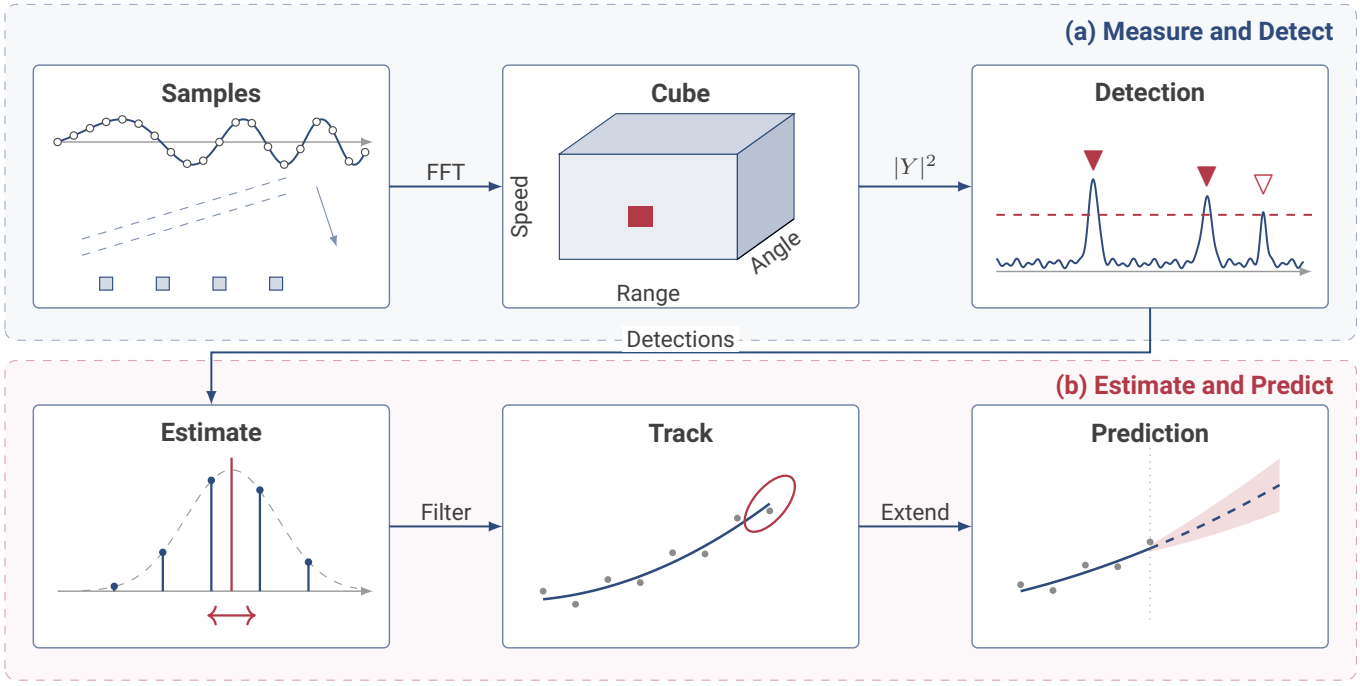


Figure 2.4.: From samples to forecast. (a) Samples in time and across an antenna row fill the cube, whose power a threshold tests; filled markers show objects, the hollow marker a noise spike. (b) Estimation, tracking and prediction refine the detections; red marks where an error enters.

2.2.1. Radar Signal Processing

A radar echo hides its object in a stream of numbers. Radar signal processing is the sequence of operations that turns the digitised echoes of a transmit-receive array into estimates of range, speed and direction, and it begins with a question that every sensor poses: what does one measurement contain, and how do many measurements fit together? A camera answers with a grid of brightness values, one for every position. A radar answers with a grid of values for every instant, chirp and antenna, and each value carries two facts: its amplitude measures the echo power, and its phase measures the path length in fractions of a wavelength [9]. The arrangement of these values decides what later operations can ask of them.

The receiver mixes the echo with the transmitted chirp (Section 2.1.2). A receiver with in-phase and quadrature outputs delivers the complex sample $y = I + \mathrm{j}Q$ of Eq. (2.14). A receiver with one real output keeps only the real part, and the range FFT then restores the quadrature: an object contributes one tone at the positive beat frequency and a mirror image at the negative one, so the lower half of the spectrum carries all information and halves the maximum range of Eq. (2.18) [33]. After the range FFT, every sample is complex, and the following transforms keep the sign of the Doppler frequency.

A frame arranges the samples along three axes. The index $s \in \{0, \ldots, N_\mathrm{s} - 1\}$ counts the samples within one chirp and forms the fast-time axis, because they follow each other within microseconds. The index $k \in \{0, \ldots, K - 1\}$ counts the chirps of one transmitter and forms the slow-time axis, because the chirps repeat only after the chirp period T_c. The index $c \in \{0, \ldots, n - 1\}$ labels the n channels of the virtual array

(Section 2.1.3). The radar cube, also called the datacube, is the three-way array

$$y \in \mathbb{F}^{N_\mathrm{s} \times K \times n}, \qquad y[s, k, c], \tag{2.22}$$

with $\mathbb{F} = \mathbb{R}$ for real and $\mathbb{F} = \mathbb{C}$ for complex samples [9], [75]. The example configuration of Table 2.1 fills one frame with $656 \times 16 \times 16 = 167\,936$ values.

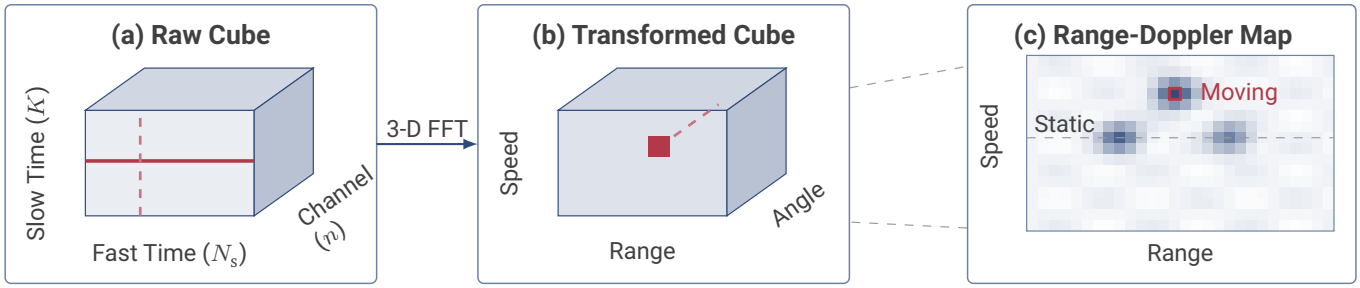


Figure 2.5.: The radar cube. (a) The solid line marks the fast-time fibre of one chirp, the dashed line the slow-time fibre of one sample over all chirps. (b) Three Fourier transforms turn the axes into range, speed and angle; the dashed line marks the angle fibre of one cell. (c) The slice at one angle bin is the range-Doppler map, here schematic. The highlighted cell is the cell of (b).

Three Fourier transforms read the cube along its axes. The transform over fast time converts the beat frequency into range (Eq. (2.16)), the transform over slow time converts the phase step into speed (Eq. (2.19)), and the transform over the channels converts the phase progression into direction (Eq. (2.21)). Each transform acts on samples that a window w has tapered. With the range, Doppler and angle bins ℓ, m and q, the three transforms combine into

$$Y[\ell, m, q] = \sum_{c=0}^{n-1} \sum_{k=0}^{K-1} \sum_{s=0}^{N_\mathrm{s}-1} w_\mathrm{r}[s]\, w_\mathrm{d}[k]\, w_\mathrm{a}[c]\; y[s, k, c]\; e^{-\mathrm{j}2\pi\left(\frac{\ell s}{N_\mathrm{s}} + \frac{mk}{K} + \frac{qc}{n}\right)}. \tag{2.23}$$

The transformed cube $Y \in \mathbb{C}^{N_\mathrm{r} \times K \times n}$ holds one complex value per bin, with $N_\mathrm{r} = N_\mathrm{s}$ range bins for complex and $N_\mathrm{r} = N_\mathrm{s}/2$ for real samples.

Every bin names a physical quantity. The resolutions of Eqs. (2.17) and (2.20) and the array of Eq. (2.21) with the element spacing d convert the indices into

$$r_\ell = \ell\, \Delta r, \qquad v_m = m\, \Delta v, \qquad u_q = \frac{\lambda\, q}{n\, d}, \tag{2.24}$$

where $u_q = \sin\theta_q$ is the direction sine, and the cyclic shift of the usual implementation restricts $m \in \{-K/2, \ldots, K/2 - 1\}$ and $q \in \{-n/2, \ldots, n/2 - 1\}$, which places zero speed and the array normal in the centre.

Reading the cube means choosing a fibre or a slice. A fibre fixes two indices and varies the third: along range it is a range profile, along speed a Doppler spectrum, and along angle the angular spectrum of one cell. A slice fixes one index: at one angle bin, it is the range-Doppler map of Figure 2.5c. An object appears there as a compact peak: a static object lies on the zero-speed line, a moving object in its own speed row, and two objects at equal range and speed separate along the angle axis [76].

The window controls the sidelobes that a strong echo spills into the neighbouring bins and that can bury a weak echo. The Hann window tapers the samples smoothly to zero at both ends. It lowers the sidelobes

at the price of a wider main peak, which grows from one bin to between 1.5 and 2 bins [9]. The squared magnitude $|Y|^2$ is the power map over range, speed and direction.

The size of the cube is not its dimension in the statistical sense. Statistics counts the n measurements and the p unknown candidates of an inference problem. The problem is classical for $n > p$, where the data determine the unknowns. It is high-dimensional for $p > n$, where infinitely many solutions fit the data and an additional assumption, such as few objects, must select one. In the cube, the snapshot of the n channels in one range-Doppler cell supplies the measurements, and a grid of candidate directions finer than n supplies the unknowns, so p exceeds n. The $N_\mathrm{r}Kn$ values of the whole cube do not enter this count. They form many separate problems, one per cell.

The stream of numbers has become a map. As a pixel of a camera frame names a position, a bin of the cube names a quantity (Eq. (2.24)), and each cell is one resolution cell of the scene: two echoes share it only if they agree in range, speed and direction, so a scene leaves most of the $N_\mathrm{r}Kn$ cells empty and each peak stands for a scatterer. This reading supports three tasks: a decision whether a cell holds an object, an estimate of the parameters of an object from the neighbouring bins and the snapshot, and a comparison of maps from frame to frame. The power map keeps the cell count of the cube, $328 \times 16 \times 16 = 83\,968$ for the example configuration, and nearly all of these cells hold noise alone. Which of them hold an object is the question the map poses.

2.2.2. Detection

Detection decides for each cell of the power map whether an object stands there, that is, a reflector whose echo rises above the interference of receiver noise, clutter and jamming [9]. The example configuration of Section 2.2.1 yields 83 968 cells, and every one needs an answer. A smoke alarm faces the same decision: it listens to one quantity, compares it with a threshold and accepts two mistakes, a missed fire and a false ring for toast.

Statistics casts the decision as a hypothesis test between two statements. The null hypothesis H_0 states that the cell holds interference only, and the alternative H_1 states that it holds interference and an object echo. The complex value of one cell is $y = w$ under H_0 and $y = a + w$ under H_1, with the echo a and complex Gaussian noise w of power σ^2. The detector forms the test statistic $T = |y|^2$, the power of the cell, and reports an object if T exceeds the threshold η (Figure 2.6a). A false alarm reports an object in a cell of interference only, and a miss overlooks an object. Their probabilities define the quality of the detector:

$$P_\mathrm{fa} = \mathbb{P}(T > \eta \mid H_0), \qquad P_\mathrm{d} = \mathbb{P}(T > \eta \mid H_1). \tag{2.25}$$

The probability of detection P_d is the power of the test, and a miss has probability $1 - P_\mathrm{d}$.

The exponential distribution makes the trade explicit. Under H_0, the power T has the mean σ^2. If the echo amplitude fluctuates as a complex Gaussian variable of power S, the Swerling I model [9], the power under H_1 has the mean $\sigma^2(1 + \rho)$ with the SNR $\rho = S/\sigma^2$. The exponential tails give

$$P_\mathrm{fa} = e^{-\eta/\sigma^2}, \qquad P_\mathrm{d} = e^{-\eta/(\sigma^2(1+\rho))} = P_\mathrm{fa}^{1/(1+\rho)}. \tag{2.26}$$

For $P_\mathrm{fa} = 10^{-6}$, an SNR of 13 dB detects every second object ($P_\mathrm{d} = 0.52$), and 20 dB detects 87 %. A higher threshold lowers both probabilities, so a fixed detector cannot improve one without worsening the other [9]. The receiver operating characteristic, P_d against P_fa over all thresholds, displays this trade [80].

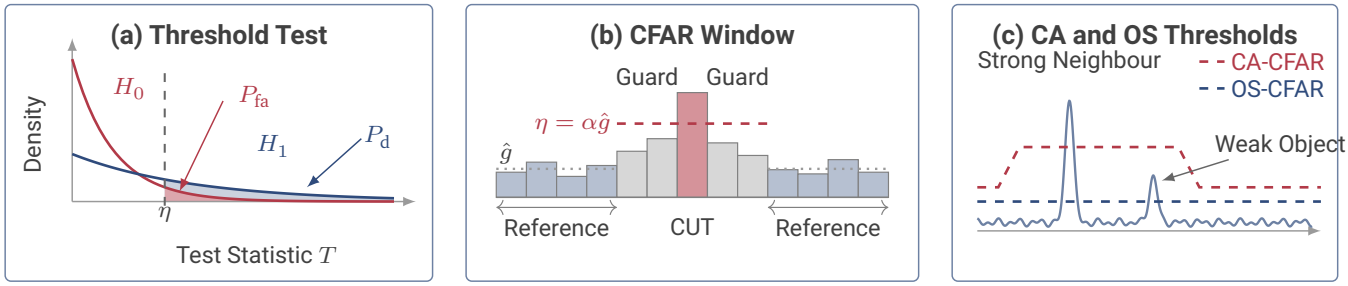


Figure 2.6.: Threshold test and CFAR detection, schematic. (a) The threshold η cuts the tail P_{fa} of the density under H_0 and the tail P_{d} of the density under H_1. (b) A CFAR detector scales the mean $\hat{g}$ of the reference cells to the threshold of the cell under test (CUT). (c) A strong neighbour lifts the CA-CFAR threshold above a weak object, whereas the OS-CFAR threshold stays flat.

The Neyman-Pearson criterion fixes P_{fa} and maximises P_{d} under this constraint [9]. Its lemma states that the likelihood-ratio test, which compares $p(\boldsymbol{y} \mid H_1)/p(\boldsymbol{y} \mid H_0)$ with a threshold, reaches the optimum [75]. In the model above, this ratio grows with T, so the optimal test is a threshold on the power, and a chosen P_{fa} fixes it through Eq. (2.26):

$$\eta = -\sigma^2 \ln P_{\text{fa}}. \tag{2.27}$$

This threshold needs the noise power σ^2, and the radar does not know it, because the interference level changes with range, direction and scene.

The Constant False Alarm Rate (CFAR) detector replaces σ^2 with an estimate from the surroundings of the cell. It scales the threshold to the local interference, $\eta = \alpha\hat{g}$, with the interference estimate $\hat{g}$ and the CFAR constant α [9]. A fixed α keeps P_{fa} constant, because $\hat{g}$ grows with σ^2 as T does, so the ratio $T/\hat{g}$ no longer depends on the noise level. The window around the cell under test (CUT) has two parts (Figure 2.6b). The guard cells next to the CUT stay out of the estimate, because the echo of an object spills into them, and the reference cells beyond them feed $\hat{g}$. The detector slides this window across the map and tests every cell in turn.

The Cell-Averaging Constant False Alarm Rate (CA-CFAR) detector averages the N_{T} reference cells T_i [39]. For exponential noise, the mean of N_{T} cells follows a gamma distribution, and $P_{\text{fa}} = (1 + \alpha/N_{\text{T}})^{-N_{\text{T}}}$ yields the constant

$$\eta = \alpha_{\text{CA}} \frac{1}{N_{\text{T}}} \sum_{i=1}^{N_{\text{T}}} T_i, \qquad \alpha_{\text{CA}} = N_{\text{T}} \left(P_{\text{fa}}^{-1/N_{\text{T}}} - 1 \right). \tag{2.28}$$

The mean estimates σ^2 most efficiently in homogeneous noise, the home ground of CA-CFAR. Its weakness is masking: a strong neighbour inflates the average, raises the threshold and hides a weak object (Figure 2.6c).

The Ordered-Statistic Constant False Alarm Rate (OS-CFAR) detector sorts the reference cells and takes the k-th smallest value $T_{(k)}$ as the estimate, with $k \approx 3N_{\text{T}}/4$ as a common choice [40]. Strong neighbours sit among the largest cells and leave $T_{(k)}$ untouched, as long as at most $N_{\text{T}} - k$ of them lie in the window. For exponential noise, the gaps between ordered cells are independent exponential variables, and the false alarm probability becomes

$$\eta = \alpha_{\text{OS}}\, T_{(k)}, \qquad P_{\text{fa}} = \prod_{i=0}^{k-1} \frac{N_{\text{T}} - i}{N_{\text{T}} - i + \alpha_{\text{OS}}}, \tag{2.29}$$

which fixes α_{OS} numerically. It serves scenes with several close objects and pays with a sort in every window and a weaker estimate in homogeneous noise.

The alarm of the introduction now sets its own threshold from the room, and it pays for this calm with sensitivity. CFAR lowers P_{d} and costs SNR, so it can discard true but weak echoes [70]. The threshold must follow the interference and lies above the optimum of Eq. (2.27), and the extra SNR that restores P_{d}, the CFAR loss, amounts to 1 to 2 dB under standard conditions [9]. The detector condenses the power map into a short list of exceedances, yet it judges each cell alone with the same P_{fa}, and m cells produce mP_{fa} false alarms on average, whatever the number of objects: at $P_{\mathrm{fa}} = 10^{-4}$, the 83 968 cells of the example add 8.4 false alarms to every frame. The guarantee concerns one cell, not the list, and the list names cells only. It says that an object stands in a cell, and it leaves open where in the cell, how fast exactly and from which direction the echo comes.

2.2.3. Parameter Estimation and Direction of Arrival

Detection raises a flag in a cell and stops there. Parameter estimation supplies the numbers that follow: it infers the unknown parameters θ of a signal model from noisy data $\boldsymbol{y}$, for example the range, the speed or the direction of an object. A lighthouse keeper who spots a ship has detected it, and the harbour master still needs its bearing. An estimator $\hat{\theta}(\boldsymbol{y})$ maps the data to a value of θ. Its bias $\mathbb{E}[\hat{\theta}] - \theta$ and its variance describe its quality, and the square root of the mean squared error, the Root-Mean-Square Error (RMSE), states the typical error in the unit of the parameter.

Limit of Every Estimator. The Cramér-Rao bound states how small the variance of an unbiased estimator can become:

$$\operatorname{Var}(\hat{\theta}) \;\geq\; \frac{1}{\mathbb{E}\Big[\big(\partial \ln p(\boldsymbol{y};\theta)/\partial\theta\big)^{2}\Big]}. \tag{2.30}$$

The denominator, the Fisher information, measures how sharply the likelihood $p(\boldsymbol{y};\theta)$ peaks at the true parameter, and a sharp peak lets the data single out θ. The maximum likelihood estimate selects the θ under which the data are most probable and, in Gaussian noise, attains the bound [9]. For range, speed and angle, the variance falls inversely with the SNR. Robust variants resist the outliers of real noise [81].

Direction of Arrival. Direction of Arrival (DoA) estimation recovers the direction sines $u_i = \sin\theta_i$ of k objects from the phases that one wave leaves across the n channels of an array, as two ears recover the direction of a sound from its delay. The problem is parametric, because a few directions and amplitudes describe the data, non-linear, because the data depend on the directions through phases, and spatial, because the information sits in the variation across the antennas. Its goal is the smallest RMSE of the angle. The data are the snapshot $\boldsymbol{y} \in \mathbb{C}^{n}$, the complex values of the n channels in one range-Doppler cell, and the array turns each direction into a phase pattern, the steering vector $\boldsymbol{a}(u)$, that the snapshot superposes:

$$\boldsymbol{y} = \sum_{i=1}^{k} \beta_i\, \boldsymbol{a}(u_i) + \boldsymbol{\varepsilon}, \qquad [\boldsymbol{a}(u)]_m = e^{\mathrm{j}2\pi d_m u}. \tag{2.31}$$

Here d_m is the position of channel m in wavelengths, β_i the complex amplitude of object i and $\boldsymbol{\varepsilon}$ the noise. The covariance $\boldsymbol{R} = \mathbb{E}[\boldsymbol{y}\boldsymbol{y}^{\mathrm{H}}]$ of the snapshot carries the directions, and the sample covariance of L snapshots estimates it.

Model. For one object, the model fixes what the data look like at every candidate direction: $\boldsymbol{y}$ scatters around $\beta \boldsymbol{a}(u)$. Complex Gaussian noise of variance σ^2 makes values near this centre likely and values far from it unlikely, which defines the density

$$p(\boldsymbol{y}; u) \;\propto\; \exp\!\left(-\frac{\left\| \boldsymbol{y} - \beta \boldsymbol{a}(u) \right\|^2}{\sigma^2} \right). \tag{2.32}$$

Estimate. The measured $\boldsymbol{y}$ is fixed, and the candidate u varies. The maximum likelihood estimate selects the u whose prediction lies closest to $\boldsymbol{y}$. Eliminating the unknown β turns this minimisation into the maximisation of $|\boldsymbol{a}^{\mathrm{H}}(u) \boldsymbol{y}|^2$, the Bartlett power of Eq. (2.34) for one snapshot. The likelihood curve over u is a spectrum, and its peak is the direction.

Bound. For one object, n channels at half-wavelength spacing, L snapshots and the per-channel SNR ρ, the Cramér-Rao bound on the direction sine, derived for arrays by Stoica and Nehorai [82], reads

$$\mathrm{Var}(\hat{u}) \;\geq\; \frac{6}{\pi^2 \, L \, n \, (n^2 - 1) \, \rho}. \tag{2.33}$$

The error shrinks with the SNR, with the snapshots and, through n^3, steeply with the aperture. Twelve channels at 20 dB reach 0.11° at broadside from a single snapshot.

Four Methods. One peak can hide two objects. Four classical methods turn $\boldsymbol{R}$ into directions that separate them: spectral beamforming (Bartlett), adaptive beamforming (Capon), the subspace method Multiple Signal Classification (MUSIC) (Schmidt) and the rotational subspace method Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT) (Roy and Kailath). Each repairs a weakness of its predecessor (Figure 2.7, Table 2.2).

Table 2.2.: Classical direction-of-arrival methods, each repairing a weakness of the one before.

Method	Core Process	Gain	Limit
Spectral Beamforming, Bartlett [35]	Align the phases and sum	Robust and simple	Resolution fixed by the aperture, strong sidelobes
Adaptive Beamforming, Capon [36]	Minimise the power at unit gain	Narrower peaks, lower sidelobes	Many snapshots, correlated objects cancel
Subspace Method, Schmidt [37]	Orthogonality to the noise subspace	Peaks beyond the aperture limit	Known k, grid scan, no coherent objects
Rotational Subspace Method, Roy and Kailath [38]	Rotation between two subarrays	Directions in closed form	Known k, shift-invariant array

Spectral Beamforming. The spatial form of Bartlett's periodogram [35] scans the direction u, aligns the phases of the channels with $\boldsymbol{a}(u)$ and sums them:

$$P_{\mathrm{B}}(u) = \frac{\boldsymbol{a}^{\mathrm{H}}(u) \, \boldsymbol{R} \, \boldsymbol{a}(u)}{n^2}. \tag{2.34}$$

The peak width equals the Rayleigh limit $\lambda/(Nd)$ of Section 2.1.3, so closer objects merge.

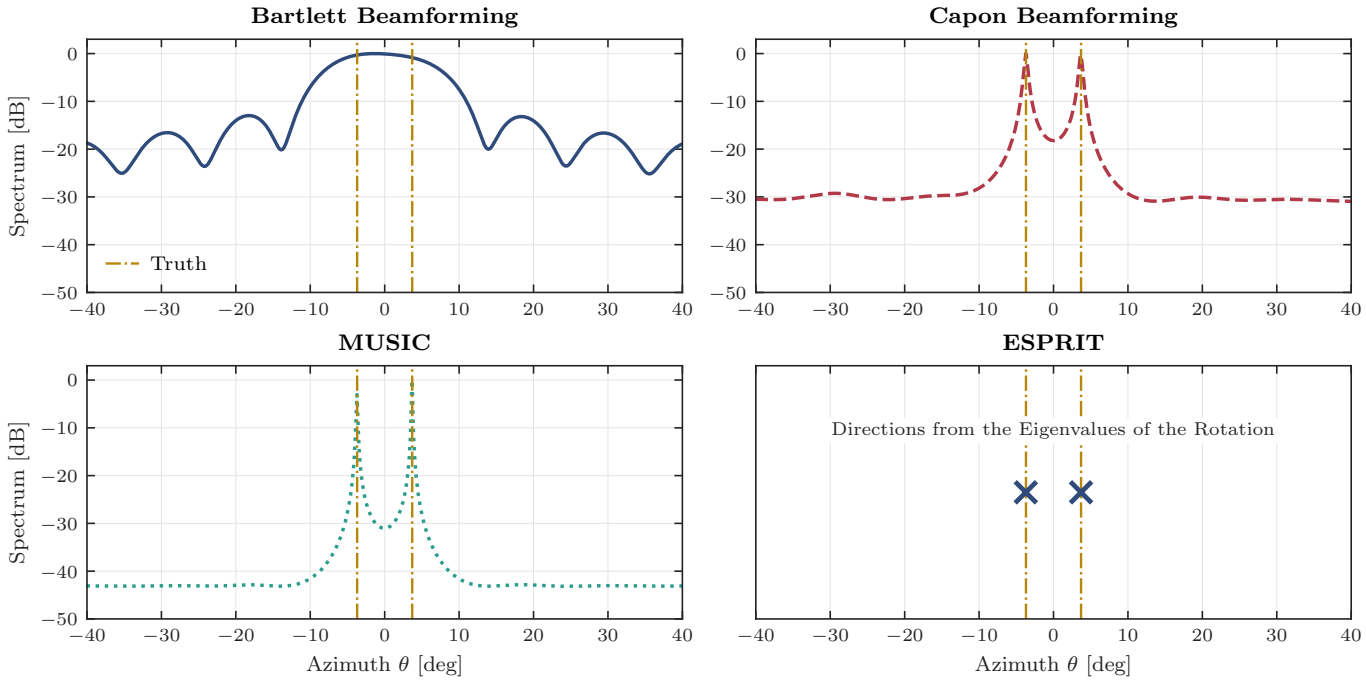


Figure 2.7.: Four direction-of-arrival methods on the same data: 100 simulated snapshots of two objects at $\pm 3.7^\circ$ on 12 channels at half-wavelength spacing with an SNR of 20 dB. Bartlett merges the objects into one peak, Capon separates them, MUSIC sharpens the peaks, and ESPRIT returns the directions without a spectrum. Dash-dotted lines mark the true directions.

Adaptive Beamforming. Capon [36] let the weights follow the data. The weights $\boldsymbol{w}$ minimise the output power $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{R}\boldsymbol{w}$ under the constraint $\boldsymbol{w}^{\mathrm{H}}\boldsymbol{a}(u) = 1$, which passes the look direction with unit gain and suppresses all others. The solution $\boldsymbol{w} = \boldsymbol{R}^{-1}\boldsymbol{a}/(\boldsymbol{a}^{\mathrm{H}}\boldsymbol{R}^{-1}\boldsymbol{a})$ yields

$$P_{\mathrm{C}}(u) = \frac{1}{\boldsymbol{a}^{\mathrm{H}}(u)\,\boldsymbol{R}^{-1}\boldsymbol{a}(u)}. \tag{2.35}$$

Subspace Method. Schmidt's MUSIC [37] replaced the power scan by the structure of $\boldsymbol{R}$. For k uncorrelated objects, its eigenvectors split into a signal subspace of k dimensions and a noise subspace $\boldsymbol{E}_{\mathrm{n}}$ of $n-k$ dimensions with equal eigenvalues. The steering vectors of the true directions lie in the signal subspace and stand orthogonal to $\boldsymbol{E}_{\mathrm{n}}$:

$$P_{\mathrm{MUSIC}}(u) = \frac{1}{\left\|\boldsymbol{E}_{\mathrm{n}}^{\mathrm{H}}\boldsymbol{a}(u)\right\|^2}. \tag{2.36}$$

Coherent objects, whose snapshots are scaled copies of each other, collapse the signal subspace, and spatial smoothing over subarrays, introduced by Shan, Wax and Kailath and extended by Pillai and Kwon, restores its rank [83], [84].

Rotational Subspace Method. Roy and Kailath's ESPRIT [38] removed the scan. A uniform array shifted by one element sees the same wave front with one phase rotation per object. If $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$ hold the first and the last $n-1$ rows of a signal-subspace basis, then $\boldsymbol{E}_2 = \boldsymbol{E}_1\boldsymbol{\Psi}$, and the eigenvalues ϕ_i of $\boldsymbol{\Psi}$ equal $e^{\mathrm{j}2\pi d u_i}$ for the element spacing d in wavelengths:

$$\hat{u}_i = \frac{\arg \phi_i}{2\pi d}. \tag{2.37}$$

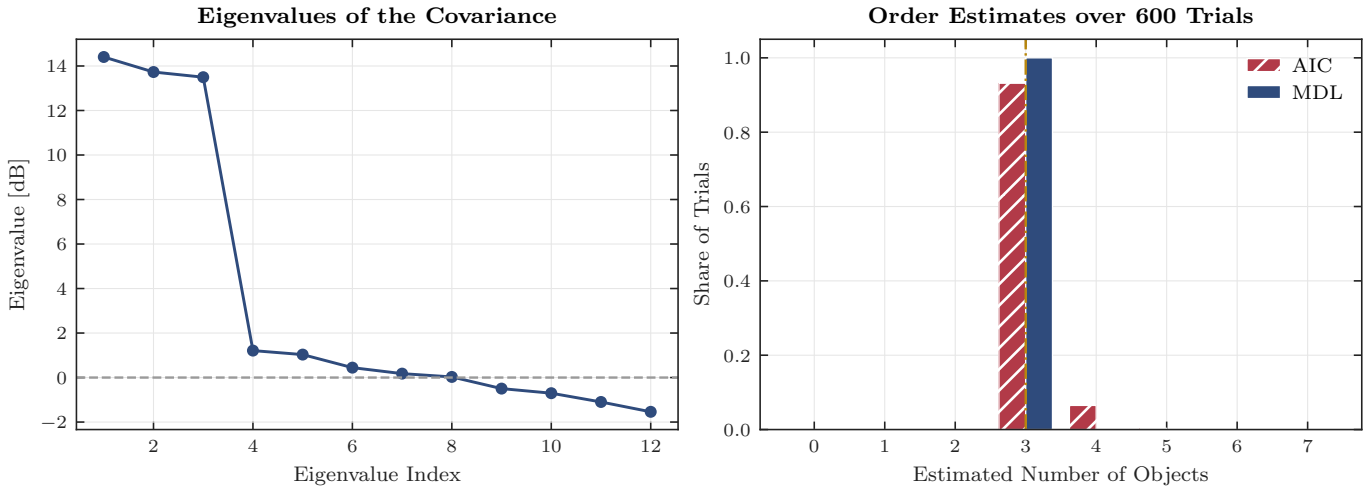


Figure 2.8.: Model order by AIC and MDL for three objects on 12 channels, 200 snapshots and an SNR of 3 dB per object. (a) Three eigenvalues of one sample covariance stand above the noise level. (b) MDL finds the true order in all 600 trials, AIC overestimates it in 7 %.

Number of Objects. MUSIC and ESPRIT need k. Information criteria select it from the eigenvalues $l_1 \geq \cdots \geq l_n$ of the sample covariance. The noise eigenvalues are equal, and the geometric mean g_k of the $n-k$ smallest eigenvalues equals their arithmetic mean a_k exactly in that case. The term $-L(n-k)\ln(g_k/a_k)$ measures the deviation from pure noise and falls as k grows. Akaike [85] added a penalty on the free parameters, the Akaike Information Criterion (AIC). Rissanen [86] measured the model by its description length, which raises the penalty by the factor $\ln L$, the Minimum Description Length (MDL), and Wax and Kailath [44] applied both to arrays:

$$\mathrm{AIC}(k) = -2L(n-k)\ln\frac{g_k}{a_k} + 2k(2n-k), \tag{2.38}$$

$$\mathrm{MDL}(k) = -L(n-k)\ln\frac{g_k}{a_k} + \tfrac{1}{2}k(2n-k)\ln L. \tag{2.39}$$

The estimate $\hat{k}$ minimises the criterion. MDL is consistent: its estimate converges to the true k as the snapshots grow, whereas AIC overestimates it with a fixed probability [44]. In 600 simulated trials, MDL found the three objects every time and AIC overestimated their number in 7 % (Figure 2.8).

A direction completes the position of a detected cell: range, speed and angle now fix where an object stands and how it moves, and objects at equal range and speed separate by angle alone. The angle also gives the radar a common axis with a camera, since a pixel column corresponds to an azimuth. A grid of candidate directions turns Eq. (2.31) into a sparse regression that needs no covariance and works with one snapshot, although a direction between two candidates leaves a remainder [87], and a grid with more candidates than channels leaves the data unable to decide alone. The ship has a bearing now, but each frame measures it once and says nothing about the course the ship will steer.

2.2.4. State Estimation and Tracking

State estimation infers the hidden state of a changing system from noisy measurements over time. Parameter estimation fixes one unknown from one data set, whereas state estimation follows an unknown that moves,

so it needs a model of how the state evolves between two measurements. Tracking is state estimation applied to detections: it combines the detections of a moving object in successive frames with a motion model and estimates the state of the object, for example its position and velocity [9], [51]. A detection says where an object was seen, and a track says where it is and how it moves. With several objects, tracking adds a second task, data association, which decides which detection belongs to which track.

State-Space Model. A state-space model describes the hidden state and its measurement with two equations:

$$\boldsymbol{s}_t = \boldsymbol{F}\boldsymbol{s}_{t-1} + \boldsymbol{w}_t, \qquad \boldsymbol{z}_t = \boldsymbol{H}\boldsymbol{s}_t + \boldsymbol{\nu}_t. \tag{2.40}$$

The state $\boldsymbol{s}_t \in \mathbb{R}^d$ collects the quantities that fix the future of the object, and the measurement $\boldsymbol{z}_t \in \mathbb{R}^m$ is what the sensor reports in frame t. The transition matrix $\boldsymbol{F}$ moves the state from one frame to the next, and the observation matrix $\boldsymbol{H}$ selects what the sensor sees. The process noise $\boldsymbol{w}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q})$ accounts for unknown manoeuvres, and the measurement noise $\boldsymbol{\nu}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_\nu)$ for the error of the sensor. Both are white, that is, uncorrelated in time, and independent of each other. The present state holds everything that the past says about the future. For an object of nearly constant velocity, the state holds position and velocity on each axis, the radar measures the position, and an unknown acceleration with variance σ_a^2 in one frame of length Δt fixes $\boldsymbol{Q}$ [51]:

$$\boldsymbol{s}_t = \begin{pmatrix} p_t \\ v_t \end{pmatrix}, \quad \boldsymbol{F} = \begin{pmatrix} 1 & \Delta t \\ 0 & 1 \end{pmatrix}, \quad \boldsymbol{H} = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad \boldsymbol{Q} = \sigma_a^2 \begin{pmatrix} \Delta t^4/4 & \Delta t^3/2 \\ \Delta t^3/2 & \Delta t^2 \end{pmatrix}. \tag{2.41}$$

Kalman Filter. Kalman [50] derived the recursion that estimates the state of this model from all measurements up to the present. For linear models with Gaussian noise, it returns the conditional mean and the covariance of the state exactly, so the pair of estimate and covariance describes the belief about the state completely [88]. The notation $\hat{\boldsymbol{s}}_{t|j}$ and $\boldsymbol{P}_{t|j}$ denotes the estimate and its covariance at time t given the measurements up to time j. Each frame passes through three stages (Figure 2.9a), and each stage ends in the equations that implement it.

Predict. The model carries the last estimate forward, and the process noise $\boldsymbol{Q}$ widens the covariance: without a measurement, the uncertainty grows with every frame [89].

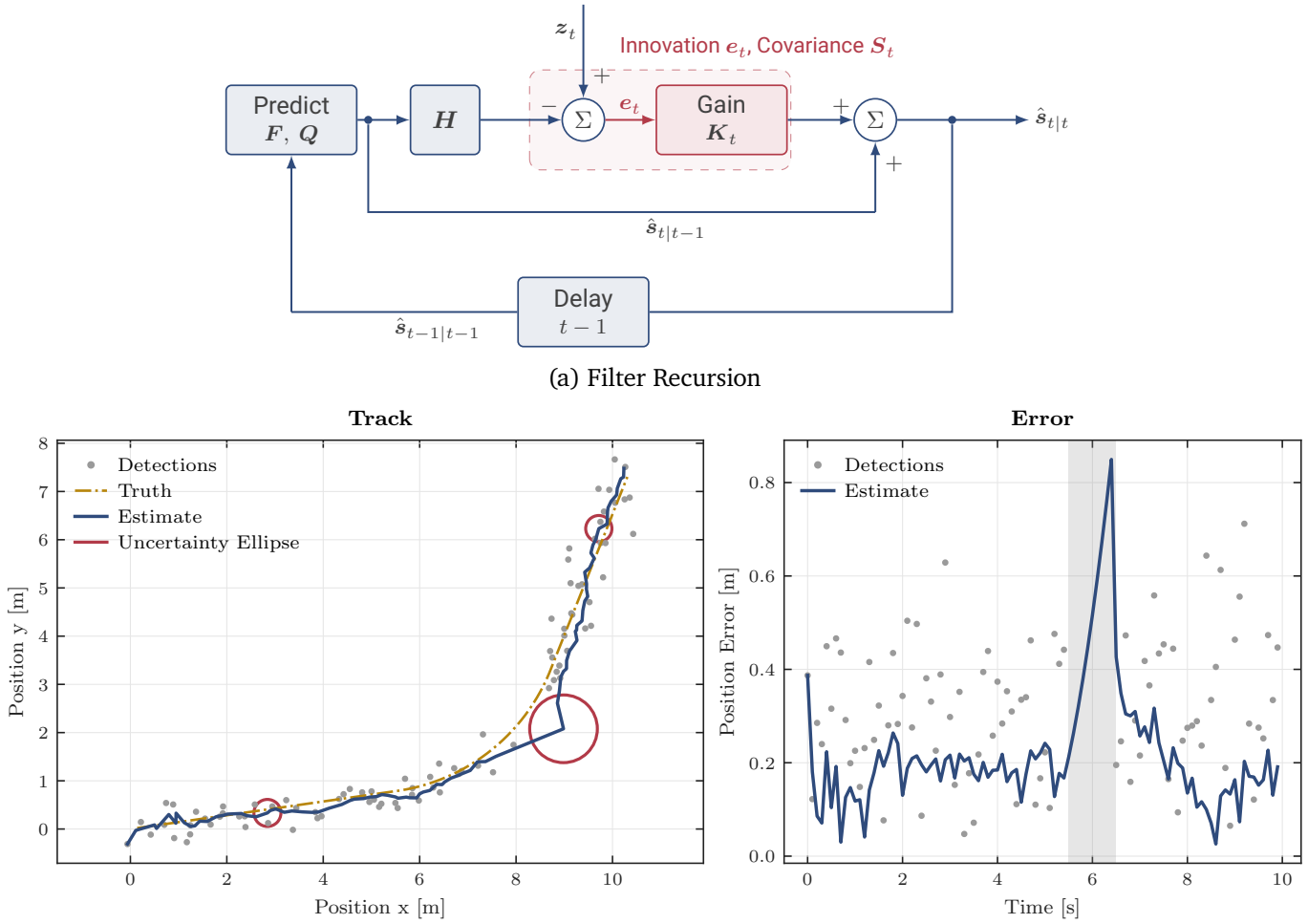
$$\hat{\boldsymbol{s}}_{t|t-1} = \boldsymbol{F}\hat{\boldsymbol{s}}_{t-1|t-1}, \qquad \boldsymbol{P}_{t|t-1} = \boldsymbol{F}\boldsymbol{P}_{t-1|t-1}\boldsymbol{F}^{\mathrm{T}} + \boldsymbol{Q}. \tag{2.42}$$

Innovation. The prediction of the measurement is $\boldsymbol{H}\hat{\boldsymbol{s}}_{t|t-1}$. The innovation $\boldsymbol{e}_t$ is the part of the measurement that this prediction did not foresee, and its covariance $\boldsymbol{S}_t$ adds the uncertainty of the prediction and of the sensor:

$$\boldsymbol{e}_t = \boldsymbol{z}_t - \boldsymbol{H}\hat{\boldsymbol{s}}_{t|t-1}, \qquad \boldsymbol{S}_t = \boldsymbol{H}\boldsymbol{P}_{t|t-1}\boldsymbol{H}^{\mathrm{T}} + \boldsymbol{R}_\nu. \tag{2.43}$$

If the model is correct, the innovations are zero-mean, white and Gaussian with covariance $\boldsymbol{S}_t$, so the normalised innovation squared $\boldsymbol{e}_t^{\mathrm{T}}\boldsymbol{S}_t^{-1}\boldsymbol{e}_t$ follows a chi-squared distribution with m degrees of freedom [51]. The innovations are the evidence of how well the model fits the data.

Gain and Update. The Kalman gain $\boldsymbol{K}_t$ weighs the innovation by the two uncertainties. It minimises the trace of the updated covariance [50], and for one scalar state observed directly it reduces to $K_t = P_{t|t-1}/(P_{t|t-1} + R_\nu)$: the gain approaches one when the sensor is precise and zero when the prediction is,



(a) Filter Recursion

(b) Simulated Track and Position Error

Figure 2.9.: The Kalman filter and a tracking example. (a) Each frame, the filter predicts the state, forms the innovation from the measurement and the predicted measurement, and corrects the prediction with the gain-weighted innovation; the red layer marks the innovation. (b) A simulated object manoeuvres, the detections carry a position error of 0.25 m, and ten frames bring no detection. Outside the gap, the position error falls from 0.35 m (detections) to 0.20 m (estimate), and the uncertainty ellipse grows while detections are missing.

so the filter trusts the source with the smaller covariance. The update adds the weighted innovation to the prediction and shrinks the covariance:

$$\boldsymbol{K}_t = \boldsymbol{P}_{t|t-1}\boldsymbol{H}^{\mathrm{T}}\boldsymbol{S}_t^{-1}, \tag{2.44}$$

$$\hat{\boldsymbol{s}}_{t|t} = \hat{\boldsymbol{s}}_{t|t-1} + \boldsymbol{K}_t\boldsymbol{e}_t, \tag{2.45}$$

$$\boldsymbol{P}_{t|t} = (\boldsymbol{I} - \boldsymbol{K}_t\boldsymbol{H})\boldsymbol{P}_{t|t-1}. \tag{2.46}$$

The covariance depends only on $\boldsymbol{F}$, $\boldsymbol{H}$, $\boldsymbol{Q}$ and $\boldsymbol{R}_\nu$, never on the measured values.

Tracking Example. Figure 2.9b shows the recursion on one object with a constant-velocity model. The estimate smooths the detections, bridges the gap by prediction and keeps its ellipse honest: the ellipse grows

during the gap and shrinks again when detections return. The mean normalised innovation squared of this simulation is 1.96, close to the value $m = 2$ that a consistent filter produces.

Several Objects. The tracker validates a detection for a track only inside a gate around the predicted measurement, assigns the detections to the tracks, and starts tentative tracks from the unassigned ones. It confirms a track after enough updates and deletes it after a period without them [51].

Uncertainty in Bits. The covariance $\boldsymbol{P}$ measures the uncertainty of an estimate, and the differential entropy of a Gaussian vector in d dimensions expresses it in bits [90], [91]:

$$h = \frac{1}{2} \log_2\Big((2\pi e)^d \det \boldsymbol{P} \Big) . \tag{2.47}$$

In two dimensions, halving the area of the uncertainty ellipse lowers the entropy by one bit.

The detections of the introduction have become tracks: an estimate and a covariance for every object, refreshed every frame, with gaps bridged by prediction at the price of a growing ellipse. Radar trackers run this recursion for every object in every frame, because it needs only the last estimate and stays exact for linear Gaussian models [9], [51]. The ellipse is only as honest as the model behind it: an unforeseen manoeuvre or a non-Gaussian error makes it wrong, and the innovations are the first witness. What the track says about the next frames, and how far to trust it, remains open.

2.2.5. Time Series Prediction

Every decision about a moving object acts on its future: a vehicle brakes for the place where a pedestrian will stand, and a radar points its beam to where an object will appear. Prediction estimates a future value of a time series from its past. A time series is a sequence of values $y_1, y_2, \ldots, y_t$ ordered in time, and the order matters, because the next value depends on the earlier ones. A predictor maps the past to the value h steps ahead, $\hat{y}_{t+h} = f(y_1, \ldots, y_t)$. Simple predictors need no model of the world: persistence repeats the last value, and linear extrapolation continues the last slope. Model-based predictors such as the Kalman prediction of Eq. (2.42) use the motion model. Each returns one value, a point prediction.

One value does not say how wrong it may be. The usual remedy assumes a distribution for the error: the Kalman filter predicts the next measurement as a Gaussian with the mean $\boldsymbol{H}\hat{\boldsymbol{s}}_{t|t-1}$ and the covariance $\boldsymbol{S}_t$ of Eq. (2.43), and a Gaussian interval around the point follows.

What can uncertainty mean when the predictor selects one point? A set $\mathcal{C}$ that contains the true value y with a stated probability. It has the coverage $1 - \epsilon$ if

$$\mathbb{P}\big(y \in \mathcal{C}\big) \geq 1 - \epsilon. \tag{2.48}$$

The Gaussian interval has this coverage only if the Gaussian assumption holds, and real data break it: a person turns, the noise has heavy tails, the behaviour drifts, and the interval fails without warning.

Vovk, Gammerman and Shafer [92] developed conformal prediction to guarantee the coverage of Eq. (2.48) for any point predictor [64], [93]. The guarantee is distribution-free: it holds for every data distribution, every predictor and every sample size, with no Gaussian assumption. It needs one weak assumption about how the data arrive.

The assumption is exchangeability. A sequence $Z_1, \ldots, Z_{n+1}$ is exchangeable if its joint distribution stays the same under every reordering of its entries. Independent and identically distributed data are exchangeable,

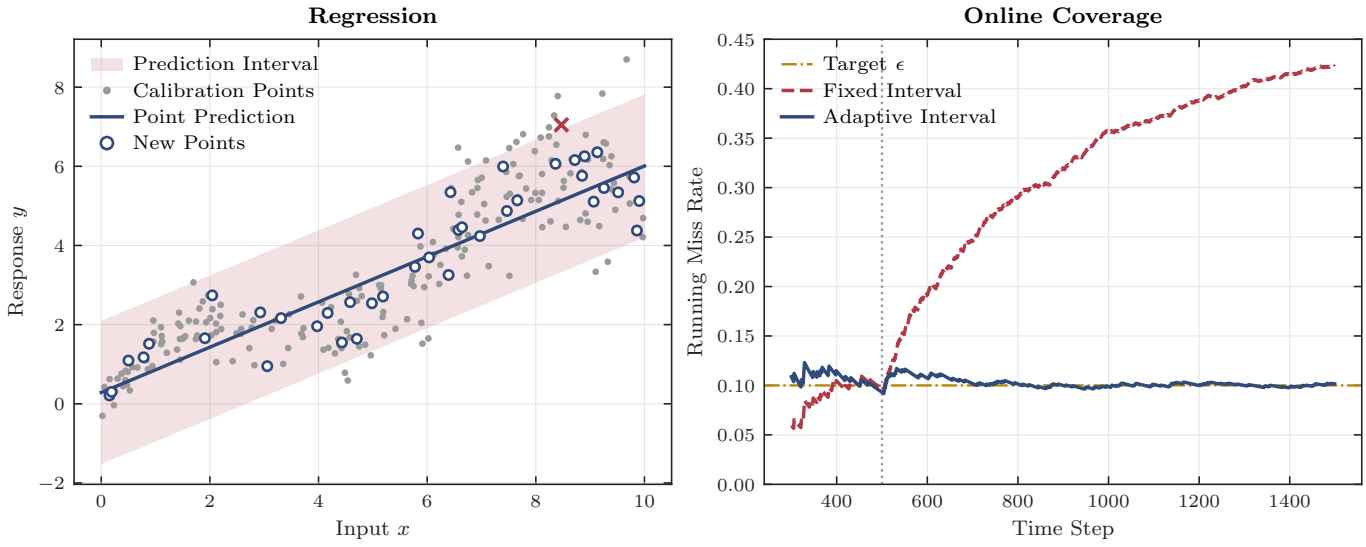


Figure 2.10.: Conformal prediction on simulated data. (a) A line fitted to 200 training points predicts the response, and 200 calibration points set the interval $\hat{y} \pm \hat{q}$ for $\epsilon = 0.1$. This calibration set covers 92 % of 20 000 new points (99.9 % for $x < 3$, 85 % for $x > 7$); averaged over 300 calibration sets, the coverage is 90.2 %. (b) After the noise level jumps at step 500, the running miss rate of a fixed interval climbs to 43 %, whereas adaptive conformal inference with $\gamma = 0.02$ holds 10.2 %.

whereas a trend or a jump in the noise level breaks it. Under exchangeability, the score of the new point is equally likely to take any rank among the $n + 1$ scores, and the guarantee rests on exactly this fact.

Split conformal prediction needs a point predictor $\hat{y}$ and a score that measures its error, for example the absolute residual. (i) Compute the scores $s_i = |y_i - \hat{y}_i|$ on n calibration points that the predictor has not seen. (ii) Take the $\lceil (1-\epsilon)(n+1) \rceil$-th smallest score as the quantile $\hat{q}$. (iii) Widen the point prediction by it (Figure 2.10a):

$$\mathcal{C} = \big[\hat{y} - \hat{q},\ \hat{y} + \hat{q}\big], \qquad 1 - \epsilon \ \le\ \mathbb{P}\big(y \in \mathcal{C}\big) \ \le\ 1 - \epsilon + \frac{1}{n+1}. \tag{2.49}$$

The interval misses only if the score of the new point ranks above $\lceil (1-\epsilon)(n+1) \rceil$ among the $n + 1$ scores, which has probability at most ϵ under exchangeability.

The coverage of Eq. (2.49) is marginal: it averages over the random calibration set and the new point [64]. A single calibration set scatters around $1 - \epsilon$, and the coverage at one particular input can differ from the average.

A time series violates exchangeability, because the future depends on the past and the behaviour can change. Adaptive conformal inference of Gibbs and Candès [53] restores a long-run guarantee without any assumption. At every step t, it (i) builds the interval $\mathcal{C}_t$ from the recent scores at the level $1 - \epsilon_t$, (ii) observes y_t and records the miss $\mathrm{err}_t = \mathbb{1}\{y_t \notin \mathcal{C}_t\}$, and (iii) updates the level with the step size γ:

$$\epsilon_{t+1} = \epsilon_t + \gamma\,(\epsilon - \mathrm{err}_t)\,. \tag{2.50}$$

A miss lowers ϵ_t and widens the next interval, and a hit narrows it. For every T, the share of misses differs from ϵ by at most $(\max(\epsilon_1, 1 - \epsilon_1) + \gamma)/(\gamma T)$ [53]. After a jump in the noise level, a fixed interval loses its coverage and the adaptive one keeps it (Figure 2.10b).

A track that predicts a position and states how far to trust it can steer a beam or a brake. Conformal prediction supplies the statement: a coverage guarantee that survives a wrong model, needs one assumption, and adapts online when the behaviour changes. A Kalman filter supplies the motion and the point prediction, together with the innovations of Section 2.2.4, scores that measure how wrong each prediction was and that conformal prediction takes as calibration material. Whether the innovations of a moving object are exchangeable, and how the interval can widen where predictions are harder, are the questions that remain.

2.3. Statistical Inference

Statistical inference draws conclusions about an unknown truth from data that noise has touched, and it states how far the conclusions can be trusted. Detection (Section 2.2.2) and estimation (Section 2.2.3) are its two oldest tasks: a test decides whether an effect exists, and an estimator reports how large it is. Each comes with an error statement for one decision, a false alarm probability or a variance. A radar frame, however, poses thousands of decisions at once, and the errors of single decisions add up: m cells judged at the same false alarm probability P_{fa} produce mP_{fa} false alarms, whatever the number of objects. A laboratory that screens 20 000 genes for a disease meets the same arithmetic. Inference at scale therefore needs an error statement about the whole set of decisions. Three steps build it. Multiple hypothesis testing counts the errors of many tests together, false discovery rate control bounds the share of false reports among the reported discoveries, and variable selection finds the few relevant variables among many candidates while it keeps the same share in check.

2.3.1. Multiple Hypothesis Testing

A hypothesis test turns data into one decision about a null hypothesis H_0, and the p-value measures how surprising the data are under it. The p-value is the probability, under H_0, of a statistic at least as extreme as the one observed. Ten heads in ten tosses of a coin have $p = 2^{-10} \approx 10^{-3}$ under the null of a fair coin, which makes the null hard to keep. A p-value is valid if it never falls below a level more often than that level allows:

$$\mathbb{P}(p \le a \mid H_0) \le a \qquad \text{for every } a \in [0,1]. \tag{2.51}$$

Rejecting H_0 when $p \le \alpha$ then commits a type I error, the rejection of a true null, with probability at most α:

$$\mathbb{P}(\text{reject } H_0 \mid H_0) = \mathbb{P}(p \le \alpha \mid H_0) \le \alpha. \tag{2.52}$$

The false alarm probability of Section 2.2.2 is this error for one cell.

One test answers whether something is there, but a radar must also say where, and an object can stand in any of thousands of cells. Every cell, every direction and every speed needs its own test. Each test keeps its own error at α, yet the errors accumulate. For m independent true nulls, the probability of at least one false rejection is

$$1 - (1-\alpha)^m, \tag{2.53}$$

which reaches 99.4 % for $m = 100$ at $\alpha = 0.05$. The radar example is harsher: at $P_{\text{fa}} = 10^{-4}$, the 83 968 cells of Section 2.2.1 contain a false alarm in 99.98 % of all frames if they hold noise only. The guarantee that held for one test collapses for many.

Table 2.3.: Outcomes of m tests, of which m_0 are true nulls.

	Not rejected	Rejected	Total
True null	$m_0 - V$	V (false discoveries)	m_0
False null	$m - m_0 - S$	S (true discoveries)	$m - m_0$
Total	$m - R$	R	m

Counting the outcomes shows what a family of m tests must control (Table 2.3): of R rejections, V are false discoveries and S true discoveries. The Family-Wise Error Rate (FWER) redefines the type I error for the whole family as the probability of at least one false discovery:

$$\text{FWER} = \mathbb{P}(V \geq 1). \tag{2.54}$$

The Bonferroni correction controls it by testing each hypothesis at the level α/m. The union bound over the m_0 true nulls gives

$$\mathbb{P}(V \geq 1) \leq \sum_{i \in \mathcal{H}_0} \mathbb{P}\Big(p_i \leq \frac{\alpha}{m}\Big) \leq \frac{m_0\,\alpha}{m} \leq \alpha, \tag{2.55}$$

whatever the dependence between the tests. The price is power: with 10^4 tests at $\alpha = 0.1$, each p-value must reach 10^{-5}, so only the strongest echoes pass and weak objects are lost [29].

The FWER bounds the probability of one event, the event of at least one false discovery. A family of tests invites counting instead. The number of false discoveries has the expectation

$$\mathbb{E}[V] = \sum_{i \in \mathcal{H}_0} \mathbb{P}(\text{reject } i), \tag{2.56}$$

and expectations add up whatever the dependence between the tests. The evidence suited to expectations is the e-value [94]:

$$e \geq 0, \qquad \mathbb{E}[e \mid H_0] \leq 1. \tag{2.57}$$

It is the payoff of a bet against H_0 that costs one unit. Under H_0, the bet is fair or unfavourable, so a large payoff is evidence against it. The likelihood ratio of a simple alternative against a simple null is an e-value, because its expectation under H_0 equals one.

An e-value connects to the p-value and goes beyond it. Markov's inequality bounds the chance of a large e-value,

$$\mathbb{P}\left(e \geq \frac{1}{\alpha} \,\middle|\, H_0\right) \leq \alpha, \tag{2.58}$$

so $1/e$ behaves like a p-value, and the rule $e \geq 1/\alpha$ keeps the type I error of one test. Unlike p-values, e-values can be averaged: the mean of valid e-values is again a valid e-value, whatever their dependence. This matters for radar data, because neighbouring cells share the sidelobes of the same echo, and their tests depend on each other. Rules that read e-values then bound the expected share of false discoveries without any assumption on this dependence [95].

One test keeps its type I error, and many tests keep it only after the error is redefined: the FWER restores the guarantee at the price of power, and the e-value replaces the probability of an event by an expectation that counts well under dependence. A single false discovery is intolerable when one wrong report is costly, as in a certified list of obstacles. A frame with many objects can accept a few wrong reports if they stay a small share of all reports. How large that share may be, and what bounds it, remains to be asked.

2.3.2. False Discovery Rate Control

Controlling the FWER demands a list without a single false entry, and this demand no longer fits high-throughput data. A genomic screen reports hundreds of genes and a radar frame hundreds of cells, and the user accepts a few wrong entries as the price of finding the right ones. The FWER charges that price in missed discoveries, the type II errors, because the Bonferroni threshold falls with m. The false discovery rate asks what a list needs: of all reported discoveries, how many are false? With the counts of Table 2.3, the false discovery proportion is the share realised in one list, and the False Discovery Rate (FDR) is its expectation [29]:

$$\mathrm{FDP} = \frac{V}{\max(R, 1)}, \qquad \mathrm{FDR} = \mathbb{E}[\mathrm{FDP}]. \tag{2.59}$$

Since $\mathrm{FDP} \leq \mathbb{1}\{V \geq 1\}$, the FDR never exceeds the FWER, and both coincide when all nulls are true. The weaker condition leaves room for power. For a radar, the FDR is the share of reported objects that do not exist.

An empty list has an FDR of zero, so the error rate alone says little about usefulness. The power is the share of the true effects that a list contains, the true positive proportion, and the True Positive Rate (TPR) is its expectation:

$$\mathrm{TPP} = \frac{S}{\max(m - m_0, 1)}, \qquad \mathrm{TPR} = \mathbb{E}[\mathrm{TPP}]. \tag{2.60}$$

The FDR plays for a list the role of the type I error of one test, and the TPR the role of the detection probability P_d of Eq. (2.25). A list of everything has $\mathrm{TPR} = 1$ and $\mathrm{FDR} = m_0/m$, and an empty list has both at zero. The design task mirrors the Neyman-Pearson criterion: maximise the TPR subject to $\mathrm{FDR} \leq \alpha$. Three procedures solve it, and each repairs a limit of its predecessor.

BH and BY Procedures. The Benjamini–Hochberg Procedure (BH) [29] sorts the m p-values and compares the k-th smallest with a threshold that grows with k. It rejects the $\hat{k}$ smallest:

$$\hat{k} = \max\left\{k \colon p_{(k)} \leq \frac{k\,\alpha}{m}\right\}. \tag{2.61}$$

The more discoveries a list holds, the more lenient its threshold: with $m = 10^4$ and $\alpha = 0.1$, the hundredth p-value may reach 10^{-3}, a hundred times the Bonferroni limit. For independent p-values [29] and for positively dependent ones [96], the procedure controls the FDR at

$$\mathrm{FDR} \leq \frac{m_0}{m}\,\alpha \leq \alpha. \tag{2.62}$$

Three steps explain why. Fix a true null i and set its p-value to zero, and let the other p-values fix the number $R^{(i)}$ of rejections. The null i is rejected exactly when $p_i \leq \alpha R^{(i)}/m$, and then $R = R^{(i)}$. Conditioning on the other p-values and using Eq. (2.51) for the independent null p_i gives

$$\mathrm{FDR} = \sum_{i \in \mathcal{H}_0} \mathbb{E}\left[\frac{\mathbb{1}\{p_i \leq \alpha R^{(i)}/m\}}{R^{(i)}}\right] \leq \sum_{i \in \mathcal{H}_0} \mathbb{E}\left[\frac{\alpha R^{(i)}/m}{R^{(i)}}\right] = \frac{m_0}{m}\,\alpha. \tag{2.63}$$

Benjamini and Yekutieli obtained control under any dependence by replacing α in Eq. (2.61) with

$$\alpha_{\mathrm{BY}} = \frac{\alpha}{\sum_{i=1}^{m} 1/i}, \tag{2.64}$$

which defines the Benjamini–Yekutieli Procedure (BY) [96], at the price of a threshold smaller by a factor of about $\ln m$. The full proofs of Eq. (2.62) and of the BY guarantee are Proposition B.3 and Proposition B.4. The Benjamini–Hochberg Procedure for E-Values (e-BH) of Wang and Ramdas applies the BH rule to e-values, $e_{[1]} \geq \cdots \geq e_{[m]}$, and rejects the $\hat{k}$ largest with

$$\hat{k} = \max\left\{k \colon e_{[k]} \geq \frac{m}{\alpha k}\right\}. \tag{2.65}$$

A rejected hypothesis satisfies $e_i \geq m/(\alpha R)$, hence $1/R \leq \alpha e_i/m$, and $\mathbb{E}[e_i] \leq 1$ for true nulls gives control under any dependence [95]:

$$\mathrm{FDP} \leq \frac{\alpha}{m} \sum_{i \in \mathcal{H}_0} e_i, \qquad \mathrm{FDR} \leq \frac{\alpha m_0}{m}. \tag{2.66}$$

Model-X Knockoffs. A regression with more variables than observations offers no p-values for its coefficients. Barber and Candès supply known nulls instead [30]: for every variable $\boldsymbol{x}_j$, the knockoff filter builds a copy $\tilde{\boldsymbol{x}}_j$ that is unrelated to the response and exchangeable with the original, that is, swapping any subset of variables with their copies leaves the joint distribution unchanged. Candès, Fan, Janson and Lv extended the filter to Model-X knockoffs, which allow any dependence of the response on the variables if their distribution is known [31]. A statistic W_j contrasts the importance of $\boldsymbol{x}_j$ with that of its copy, for example the difference of their lasso coefficients. For a null variable, the sign of W_j is a fair coin, so it acts as a one-bit p-value. The threshold

$$\tau = \min\left\{t \colon \frac{1 + \#\{j \colon W_j \leq -t\}}{\max(\#\{j \colon W_j \geq t\}, 1)} \leq \alpha\right\} \tag{2.67}$$

selects the variables with $W_j \geq \tau$. Again three steps explain the control. Let $V^+(t)$ and $V^-(t)$ count the null variables with $W_j \geq t$ and $W_j \leq -t$; fair signs give them the same distribution. At τ, the rule Eq. (2.67) gives at least $(1 + V^-(\tau))/\alpha$ selections, because the negative statistics of the nulls are among all negative statistics. The ratio $V^+(t)/(1 + V^-(t))$ is a supermartingale when the threshold is lowered, and τ is a stopping time, so the optional stopping theorem bounds its expectation:

$$\mathrm{FDP} \leq \alpha \, \frac{V^+(\tau)}{1 + V^-(\tau)}, \qquad \mathbb{E}\left[\frac{V^+(\tau)}{1 + V^-(\tau)}\right] \leq 1 \quad \Longrightarrow \quad \mathrm{FDR} \leq \alpha. \tag{2.68}$$

Two limits remain. The distribution of the variables must be known or estimated, and the $+1$ in Eq. (2.67) demands at least $1/\alpha$ positive statistics before anything is selected. The full proofs are Proposition B.7 for the fixed design and Proposition B.8 for Model-X knockoffs, both resting on Proposition B.6.

T-Rex Selector. The Terminating-Random Experiments Selector (T-Rex) of Machkour, Muma and Palomar needs neither a model of the variables nor a constructed copy [32], [97]. It appends to $\boldsymbol{X}$ a set of L dummies, independent standard normal columns that are null by construction, and repeats this in K random experiments (Figure 2.11). Each experiment runs a forward selection, which adds the variable that best explains the current residual, on the enlarged matrix and stops after T dummies have entered. The candidate set $\mathcal{C}_k(T)$ keeps the original variables that entered before. The relative occurrence of a variable is the share of experiments that selected it, and a voting level $v \in [0.5, 1)$ fuses the candidate sets:

$$\Phi_{T,L}(j) = \frac{1}{K} \sum_{k=1}^{K} \mathbb{1}\{j \in \mathcal{C}_k(T)\}, \qquad \mathcal{A}(v, T) = \{j \colon \Phi_{T,L}(j) > v\}. \tag{2.69}$$

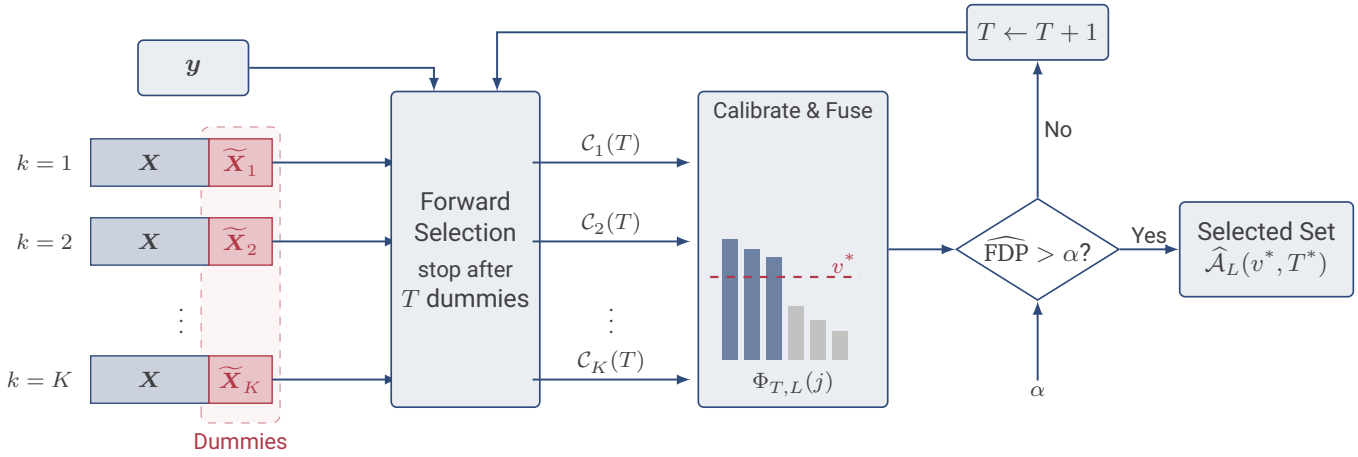


Figure 2.11.: Schematic of the T-Rex selector. Each of the K random experiments appends dummies (red) to the predictors, and a forward selection stops after T dummies. The candidate sets are fused by their relative occurrences and a voting level v. The calibration raises T until the estimated false discovery proportion exceeds the target level α, and the output uses the last feasible values v^* and T^*. Adapted from the overview of the T-Rex framework in Machkour [97].

Dummies replace the signs of W_j as the count of nulls: they enter the paths like null variables, so the number of dummies that have entered measures the nulls that have entered. The calibration raises T and picks, for each T, the smallest v whose estimated false discovery proportion $\widehat{\mathrm{FDP}} = \widehat{V}/R$ stays below α, where $\widehat{V}$ estimates the number of selected nulls from the dummies. Three steps explain the control. The calibration gives $\mathrm{FDP} \le \alpha\, V(v^*)/\widehat{V}(v^*)$. The ratio $V(v)/\widehat{V}(v)$ is a supermartingale when v is lowered, and v^* is a stopping time. At $v = 0.5$, the expected number of selected nulls does not exceed the estimate, because the nulls are stochastically dominated by a negative hypergeometric count with mean $Tp_0/(L+1)$, so for a large number K of experiments

$$\mathrm{FDP} \le \alpha \frac{V(v^*)}{\widehat{V}(v^*)}, \qquad \mathbb{E}\left[\frac{V(v^*)}{\widehat{V}(v^*)}\right] \le \mathbb{E}\left[\frac{V(0.5)}{\widehat{V}(0.5)}\right] \le 1 \quad \Longrightarrow \quad \mathrm{FDR} \le \alpha. \tag{2.70}$$

The full proofs are Proposition B.10 for the T-Rex selector and Proposition B.11 for its complex-valued form.

The three procedures share one recipe: bound the false proportion by a ratio of counts whose expectation is at most one. BH needs independent p-values, BY pays a factor of $\ln m$ for dependence, e-BH changes the currency, knockoffs drop the p-values but need the distribution of the variables, and the T-Rex selector replaces the constructed copies by dummies. The guarantee fixes the share of false reports and leaves open how the candidates are ordered: the forward selection, the lasso coefficients and the statistics W_j all stem from a linear model, and how such a model ranks many variables remains to be built.

2.3.3. Statistical Learning and Variable Selection

Statistical learning infers a relation between a response and many candidate variables from data. The linear model writes this relation as

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{2.71}$$

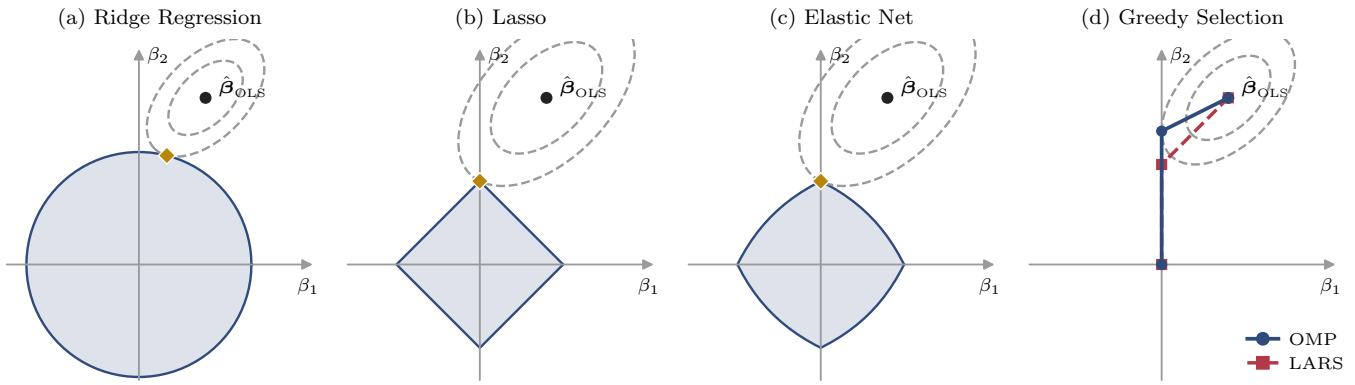


Figure 2.12.: How regularisation and greedy selection come up with variables, simulated. (a–c) Regularisation: the contours of the squared error around the least-squares solution touch the constraint region of the penalty, and the contact point (diamond) has a zero coordinate only where the region has a corner. (d) Greedy selection in the plane of two coefficients: OMP jumps along the axis of the most correlated variable to the least-squares point, LARS stops where a second variable ties in correlation, and both then reach the full solution.

where the response $\boldsymbol{y}$ holds n measurements, the design matrix $\boldsymbol{X}$ has one column for each of p candidate variables, the coefficients $\boldsymbol{\beta}$ weight the variables, and $\boldsymbol{\varepsilon}$ is the noise. Least squares minimises $\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}\|_2^2$ and has the unique solution $(\boldsymbol{X}^H\boldsymbol{X})^{-1}\boldsymbol{X}^H\boldsymbol{y}$ if $\boldsymbol{X}$ has full column rank. The problem is high-dimensional when $p > n$: infinitely many $\boldsymbol{\beta}$ fit the data equally well, and the data alone cannot choose among them. Radar data are high-dimensional in this sense, because a grid of candidate directions has many more entries than the array has channels (Section 2.2.3). Sparsity, the assumption that only a few entries of $\boldsymbol{\beta}$ are nonzero, restores a unique answer, and finding the nonzero entries is variable selection. Its direct formulation counts them:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \frac{1}{2}\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_0 \le k. \tag{2.72}$$

This problem is non-convex, because the count $\|\boldsymbol{\beta}\|_0$ is not a convex function, and it is NP-hard [98]: the search covers all $\binom{p}{k}$ subsets of k variables, a number that grows combinatorially with p. Two families of selectors avoid this search, regularisation and greedy forward selection, and they differ in how they come up with the variables (Figure 2.12).

Regularisation relaxes the count into a penalty and turns the search into a convex program. Three penalties form one family,

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \frac{1}{2}\|\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda_1\|\boldsymbol{\beta}\|_1 + \lambda_2\|\boldsymbol{\beta}\|_2^2, \tag{2.73}$$

with the weights $\lambda_1, \lambda_2 \ge 0$. The selected variables are those with nonzero coefficients, and the weights tune how many remain. A penalty equals a constraint region around the origin, and the solution is the point where the contours of the squared error first touch it. The circle of ridge regression ($\lambda_1 = 0$) has no corners, so the contact point has no zero coordinate and ridge regression shrinks all coefficients without selecting [99]. The diamond of the lasso ($\lambda_2 = 0$) has corners on the axes, where the contours touch for typical data, so coefficients become exactly zero [100]. The elastic net keeps these corners, rounds the sides and keeps correlated variables together [101]. The ℓ_1 norm is the convex relaxation of the count: every local minimum

is global, and proximal gradient methods such as Fast Iterative Shrinkage-Thresholding Algorithm (FISTA) solve the problem with a soft-threshold step [102], [103], [104].

Greedy forward selection attacks the count directly and builds the support one variable at a time. Orthogonal Matching Pursuit (OMP) starts from an empty set and adds in every step the variable most correlated with the residual $\boldsymbol{r}_{t-1}$ [105]. With unit-norm columns, it then refits all selected variables by least squares and updates the residual,

$$j_t = \arg\max_j \left| \boldsymbol{x}_j^H \boldsymbol{r}_{t-1} \right|, \qquad \mathcal{S}_t = \mathcal{S}_{t-1} \cup \{j_t\}, \qquad \boldsymbol{r}_t = \boldsymbol{y} - \boldsymbol{X}_{\mathcal{S}_t} \hat{\boldsymbol{\beta}}_{\mathcal{S}_t}, \tag{2.74}$$

so that the residual stays orthogonal to the selected columns, and the variables come out in their order of entry. Random measurements suffice for OMP to recover sparse signals [106]. Least Angle Regression (LARS) moves the coefficients continuously: it follows the direction that keeps all active variables equally correlated with the residual until another variable reaches the same correlation, and with a small modification it traces the lasso path [107], so the greedy and the penalised route meet. Newtonized Orthogonal Matching Pursuit (NOMP) adapts the greedy idea to the continuum, with Newton steps that refine the parameters of all detected components after each detection [108].

Every path orders the variables, but it does not say where to stop: a selection that stops too late includes noise, and one that stops too early misses variables. The selectors of Section 2.3.2 supply the rule with a guarantee on the share of false selections. Knockoffs compare every variable with a control copy, and the T-Rex selector wraps around a forward selection, runs the path on a matrix enlarged by dummies and stops it after T dummies have entered [32]: the path supplies the order and the dummies the stopping rule. FDR-controlled sparse regression handles five million variables on a laptop [16] and tracks a stock index with a few stocks [109]. Sparse coding without such a guarantee recognises a face by coding it over a dictionary of training images, the idea of Sparse Representation Classification (SRC) [110]. An underdetermined problem has become solvable through sparsity, relaxation, greedy paths and control copies. All these matrices came from the radar, whose entries are complex numbers with a physical meaning. A second sensor delivers a matrix of another kind, and how its entries carry meaning remains to be read.

2.4. Computer Vision

Computer vision infers structure and meaning from images. It complements the radar: the radar measures range and speed through darkness and fog but senses an object only as a set of echoes, whereas a camera records the appearance of the object, its colour, its texture and its outline, on a grid of pixels. An image is a matrix of brightness values, like the design matrices of Section 2.3.3, but its entries carry a structure that those matrices lack: neighbouring pixels belong to the same surface and vary together. Computer vision exploits this structure in three stages. A camera forms the image by projecting the scene through a lens onto a sensor and sampling the light in colour. Image processing enhances and separates the content with operations that act on neighbourhoods of pixels. Learned models map the processed image to a decision, with features that the data rather than the designer define. The subsections follow these stages.

2.4.1. Image Representations and Color Spaces

An image representation encodes what a camera records as numbers that a computer can store and compare. It consists of a function that assigns to every position on a plane the intensity of light in a few colour channels,

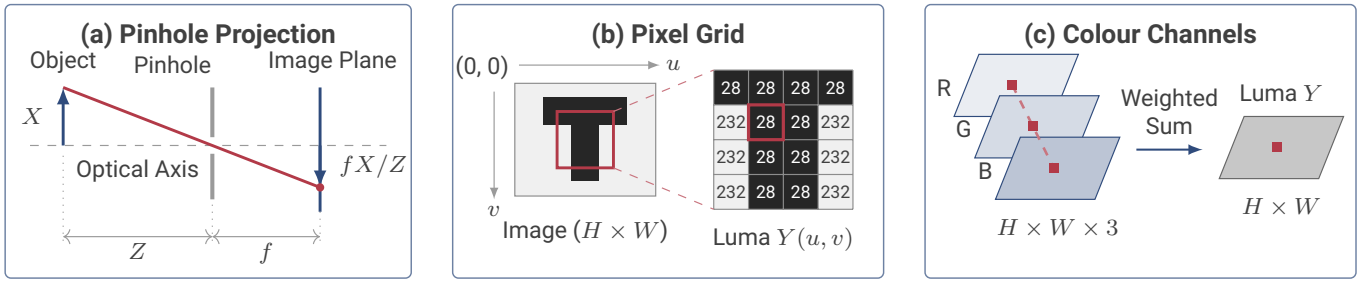


Figure 2.13.: Image representation. (a) The pinhole model maps a point at lateral position X and depth Z to the offset fX/Z on the image plane; all points on the ray through the pinhole share this pixel. (b) The pixel grid with column u and row v, counted from the upper left corner; the enlarged patch shows schematic luma values. (c) The colour tensor consists of three channel planes, and the luma is their weighted sum.

and of a geometry that ties these positions to points in the scene. A mosaic shows the principle: it breaks a picture into tiles of uniform colour, and the finer the tiles, the more detail the mosaic keeps. Three choices define the representation, namely the grid of positions, the values stored at each position and the projection that links the grid to space. The following blocks treat these choices in turn.

A sensor samples the light on a regular grid. A pixel is one sample, and its position is the pair (u, v) of column and row, counted from the upper left corner of the image with v growing downwards. The camera reports for every pixel the intensity in a red, a green and a blue channel and quantises each intensity to 256 levels. An image of height H and width W is therefore the tensor

$$\boldsymbol{I} \in \{0, \dots, 255\}^{H \times W \times 3}, \qquad \boldsymbol{I}(u, v) = \begin{bmatrix} I_{\mathrm{R}}(u, v) \\ I_{\mathrm{G}}(u, v) \\ I_{\mathrm{B}}(u, v) \end{bmatrix}, \quad u \in \{0, \dots, W-1\},\ v \in \{0, \dots, H-1\}, \tag{2.75}$$

whose entry $\boldsymbol{I}(u, v)$ lists the three channel values of the pixel in column u and row v. Like the radar cube of Section 2.2.1, the tensor is an indexed array. Its first two axes are already positions in the picture, so that no transform has to reveal them, and neighbouring entries describe neighbouring points of the same surface and vary together.

Brightness and colour are separable. The luma condenses the three channels into the single value

$$Y(u, v) = 0.299\, I_{\mathrm{R}}(u, v) + 0.587\, I_{\mathrm{G}}(u, v) + 0.114\, I_{\mathrm{B}}(u, v), \tag{2.76}$$

the first coordinate of a linear change of colour space whose other two coordinates, the colour differences, carry the colour [111]. Many tasks work on Y alone, since the contrast between ink and paper or between a lamp and its surroundings lives in brightness, and one matrix is cheaper to process than three. The range of the sensor ends at 255. Light above this limit records as the same value, so that a clipped pixel, whose luma reaches the limit, reports that the light was at least this bright and not how bright. A bright source leaves traces outside its core as well: light scattered inside the lens adds a veil of brightness over the whole image and lowers the contrast of everything else [63].

The pixel grid relates to space through the camera geometry. A sensor with its own frame, such as the radar, reports a point $\boldsymbol{p}$ in that frame, whereas the camera sees the point from its own position and orientation. The

change of frame is a rigid transform, a shift followed by a rotation,

$$\boldsymbol{p}_\mathrm{c} = \boldsymbol{R}_\mathrm{c}\,(\boldsymbol{p} - \boldsymbol{c}) = \begin{bmatrix} X_\mathrm{c} & Y_\mathrm{c} & Z_\mathrm{c} \end{bmatrix}^\top, \tag{2.77}$$

where $\boldsymbol{c}$ is the position of the camera in the first frame and the rotation matrix $\boldsymbol{R}_\mathrm{c}$, with $\boldsymbol{R}_\mathrm{c}^\top \boldsymbol{R}_\mathrm{c} = \boldsymbol{I}$ and $\det \boldsymbol{R}_\mathrm{c} = 1$, preserves lengths and angles. The Z_c-axis of the camera frame points along the optical axis. The pinhole model projects the point $\boldsymbol{p}_\mathrm{c}$ to the pixel

$$u = u_0 + f\,\frac{X_\mathrm{c}}{Z_\mathrm{c}}, \qquad v = v_0 + f\,\frac{Y_\mathrm{c}}{Z_\mathrm{c}}, \tag{2.78}$$

where f is the focal length in pixels and (u_0, v_0) the principal point, at which the optical axis meets the image [45]. A real lens bends the rays further. A radial distortion with the coefficient k_1 scales the offset from the principal point by $1 + k_1 r^2$, where $r^2 = (X_\mathrm{c}/Z_\mathrm{c})^2 + (Y_\mathrm{c}/Z_\mathrm{c})^2$ [45]. Two consequences of Eq. (2.78) matter. An object of width w at depth Z_c covers

$$n_w = \frac{f\,w}{Z_\mathrm{c}} \tag{2.79}$$

pixels, so that its image shrinks in proportion to its distance. In addition, the division by Z_c maps all points on a ray through the pinhole to the same pixel, so that a pixel fixes a direction but not a distance.

The parameters of this chain, namely $\boldsymbol{R}_\mathrm{c}$, $\boldsymbol{c}$, f, (u_0, v_0) and k_1, are unknown for a given setup. They follow from correspondences, points $\boldsymbol{p}_i$ whose pixels $\boldsymbol{q}_i$ are known. The best parameters $\boldsymbol{\vartheta}$ minimise the squared pixel distances

$$\hat{\boldsymbol{\vartheta}} = \arg\min_{\boldsymbol{\vartheta}} \sum_i \left\| \boldsymbol{q}_i - \pi(\boldsymbol{p}_i; \boldsymbol{\vartheta}) \right\|_2^2, \tag{2.80}$$

where π chains Eq. (2.77), Eq. (2.78) and the distortion. The projection divides by Z_c and contains rotations, so the problem is nonlinear and has no closed-form solution. The Levenberg-Marquardt method iterates instead. With the residuals $\boldsymbol{e}$ and their Jacobian $\boldsymbol{J}$ at the current parameters, every iteration solves

$$\left(\boldsymbol{J}^\top \boldsymbol{J} + \mu\,\mathrm{diag}(\boldsymbol{J}^\top \boldsymbol{J})\right) \boldsymbol{\Delta} = -\boldsymbol{J}^\top \boldsymbol{e}, \qquad \boldsymbol{\vartheta} \leftarrow \boldsymbol{\vartheta} + \boldsymbol{\Delta}, \tag{2.81}$$

and the damping μ moves the step between a Gauss-Newton step for small μ and a gradient step for large μ [112], [113]. Camera calibration uses the same refinement for the intrinsic parameters [45].

A camera image is thus a grid of positions, three intensities at every position and a projection that preserves the direction of a point and discards its distance. Like the mosaic of the introduction, the representation keeps what it records and nothing beyond: no tile knows whether it belongs to ink or paper, to a lamp or to the reflection of one. The radar of Section 2.1 measures the distance that a pixel lacks. Only the way a tile differs from its neighbours can say what it means.

2.4.2. Image Processing

A raw image mixes the property of interest with everything else: the lighting, the noise of the sensor, the size of the object. Image processing separates them. It transforms an image into another image or into a few numbers that expose the content a task depends on, and it is signal processing on a two-dimensional grid, with pixels in the place of time samples. Like the equaliser of an audio system, which leaves the signal where it is and lifts the bands that matter, these operations leave the picture in place and emphasise what a decision needs. Three kinds cover most of the practice: operations on single pixels, operations on neighbourhoods and operations that condense a region into a feature vector.

The simplest operation acts on one pixel at a time. A point operation maps the luma $Y(u,v)$ to a new value that depends on this value alone, as in the scaling and the offset of brightness. Thresholding is the point operation that matters most for text. It separates dark ink from a bright background by comparing every pixel with a level τ,

$$B(u,v) = \mathbb{1}\{Y(u,v) < \tau\}, \tag{2.82}$$

where the binary image B marks the ink with one. A single level fails as soon as the lighting varies over the image, because the paper in shadow is darker than the ink in the light. The decision needs the surroundings of the pixel.

Operations on neighbourhoods supply them. A linear filter replaces every pixel by a weighted sum of its neighbourhood, the convolution

$$(\boldsymbol{I} * \boldsymbol{k})(u,v) = \sum_{i} \sum_{j} \boldsymbol{I}(u-i, v-j)\, \boldsymbol{k}(i,j), \tag{2.83}$$

applied to each channel, with the kernel $\boldsymbol{k}$ holding the weights. An averaging kernel smooths the image. Since neighbouring pixels belong to the same surface and vary together, the average keeps the surface and cancels the noise, which differs from pixel to pixel. Averaging over a window $\mathcal{W}_{u,v}$ around (u,v) also yields the local mean m, and the same window yields the local standard deviation s,

$$m(u,v) = \frac{1}{|\mathcal{W}_{u,v}|} \sum_{(i,j)\in\mathcal{W}_{u,v}} Y(i,j), \qquad s(u,v) = \sqrt{\frac{1}{|\mathcal{W}_{u,v}|} \sum_{(i,j)\in\mathcal{W}_{u,v}} \big(Y(i,j) - m(u,v)\big)^2}. \tag{2.84}$$

The pair describes the brightness and the contrast around every pixel, and it turns Eq. (2.82) into an adaptive rule: a pixel counts as ink if it is darker than its surroundings by $\kappa \geq 0$ standard deviations, which sets the level to

$$\tau(u,v) = m(u,v) - \kappa\, s(u,v). \tag{2.85}$$

Document binarisation uses such local thresholds, and Sauvola's rule additionally scales the threshold with the contrast, so that a smooth background is not mistaken for ink [114]. A binary image of ink is then cut into letters at the gaps between them or along its connected components [115].

A difference kernel estimates the change of brightness instead of its level. The central differences of the two neighbours along each axis,

$$G_u = \tfrac{1}{2}\big(Y(u+1,v) - Y(u-1,v)\big), \qquad G_v = \tfrac{1}{2}\big(Y(u,v+1) - Y(u,v-1)\big), \tag{2.86}$$

form the gradient of the luma. Its magnitude and orientation,

$$g = \sqrt{G_u^2 + G_v^2}, \qquad \theta = \operatorname{atan2}(G_v, G_u), \tag{2.87}$$

are large where ink meets paper, with an orientation perpendicular to the contour, and near zero on flat areas. The gradient therefore describes the outline of a shape and is blind to the level of brightness.

A decision such as which letter a region shows needs a description that survives small shifts and changes of lighting. The Histogram of Oriented Gradients (HOG) divides the region into cells of a few pixels and collects, in every cell c, the gradient magnitudes of its pixels in orientation bins,

$$h_c(b) = \sum_{(u,v)\in c} g(u,v)\, \mathbb{1}\{\theta(u,v) \in \text{bin } b\}, \qquad \boldsymbol{\phi} = \frac{\boldsymbol{h}}{\|\boldsymbol{h}\|_2}, \tag{2.88}$$

where $\boldsymbol{h}$ stacks the histograms of all cells and the normalisation fixes the length of the feature vector $\boldsymbol{\phi}$ [116]. A shift of a few pixels moves a gradient within its cell and leaves the histogram unchanged, and a change of brightness scales all magnitudes alike, so that the normalisation cancels it. Dalal and Triggs normalise in overlapping groups of cells and use the descriptor to find pedestrians.

Each operation thus removes one nuisance. The adaptive threshold ignores the lighting, the average removes the noise and the normalisation removes the contrast, and together they separate ink from paper and describe outlines well enough to find a pedestrian in a street. Like the equaliser of the introduction, none of them decides anything: they hand over a vector of numbers. Which of the possible letters this vector describes depends on weights that someone must choose.

2.4.3. Deep Learning and Convolutional Networks

A designer chooses the operations of image processing: the kernel, the threshold, the number of orientation bins. This succeeds for a defined task such as reading a printed letter, but every new task asks for a new design. Deep learning reverses the order. It writes the operations as a function with free parameters and leaves it to labelled examples to fix them. A deep network composes many simple parametrised functions, the layers, and training adjusts the parameters until the output matches the labels [41]. The network is a filter bank that tunes itself.

A neuron computes $y = \varphi(\boldsymbol{w}^\top \boldsymbol{x} + b)$, a weighted sum of its inputs $\boldsymbol{x}$ passed through a nonlinear activation φ, for example the rectified linear unit $\varphi(a) = \max(0, a)$ [42]. A layer holds many neurons, and a network stacks layers, so that every layer describes the output of the one before. For classification, the last layer outputs one score z_c per class. The softmax function turns the scores into probabilities, and training minimises their cross-entropy with the one-hot label y_c,

$$p_c = \frac{e^{z_c}}{\sum_j e^{z_j}}, \qquad \mathcal{L} = -\sum_c y_c \log p_c. \tag{2.89}$$

The chain rule yields the gradient of $\mathcal{L}$ with respect to every weight, a procedure called backpropagation, and an optimiser such as Adam updates the weights with step sizes that adapt to the history of each gradient [117].

A layer in which every neuron sees every pixel suits images poorly. A letter of 32×32 pixels offers 1024 inputs, every neuron would own 1024 weights, and a stroke at a new position would have to be learned again. Images provide the structure of the previous subsections instead: neighbouring pixels vary together, and the same pattern, an edge or a corner, appears anywhere. A convolutional layer exploits both facts. It slides small kernels over the image, as in Eq. (2.83), with the same weights at every position,

$$\boldsymbol{F}_k(u, v) = \varphi\big((\boldsymbol{I} * \boldsymbol{k}_k)(u, v) + b_k\big), \qquad k \in \{1, \dots, K\}, \tag{2.90}$$

where every kernel $\boldsymbol{k}_k$ spans all channels of the input and is learned together with its bias b_k [41]. The feature map $\boldsymbol{F}_k$ responds where the pattern of its kernel appears. With kernels of size $s \times s$ on C channels, the layer has $K(s^2 C + 1)$ parameters, independent of the size of the image. A difference kernel as in Eq. (2.86) is one possible kernel, and the network finds its kernels itself.

Pooling replaces every small window, usually of 2×2 pixels, by its maximum or its mean. It halves the resolution and makes the response tolerant to a shift of one pixel, the role that the cells of Eq. (2.88) play. Stacking convolution and pooling enlarges the region of the input that one value sees. The first layers see

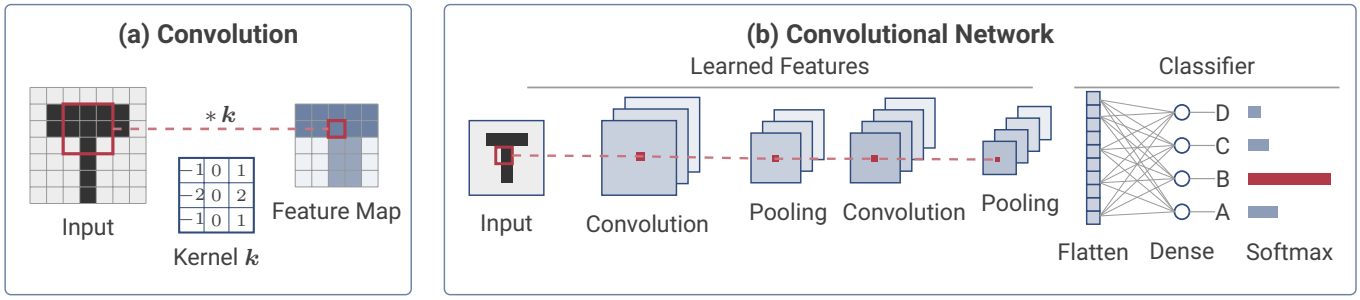


Figure 2.14.: A convolutional network. (a) A kernel slides over the input, and every position yields one value of the feature map; the kernel shown is a gradient-like example, whereas a network learns its kernels. (b) Convolution and pooling extract features, and dense layers with a softmax classify them. The highlighted position traces one value back to an ever larger region of the input.

few pixels and respond to local patterns such as edges, whereas deeper layers see larger regions and combine the patterns into strokes and shapes. The last feature maps are flattened into a vector, which dense layers and the softmax of Eq. (2.89) turn into class probabilities (Figure 2.14). The chain is the one of the previous subsection, a descriptor followed by a classifier, with one difference: the kernels and the classifier are learned together.

Such networks classify images at a level that made them the standard approach once large labelled data sets existed [42], and combined with a recurrent network they read whole lines of text without cutting letters [43]. The self-tuning filter bank spares the designer a new construction for every task, and it asks for data, a loss and an optimiser in return. Its output is a score: a softmax probability states how strongly the network prefers a class, and not how often such a preference is wrong.

3. State of the Art

The central objective of this thesis is a radar and camera system that reports only findings with a bounded share of errors and reconstructs a scene from them. Researchers have already built many parts of such a system. Their work gives this thesis a reference to build on and shows which questions still need an answer. The sections below follow the order of the experiments. Each section presents the conventional method, the recent work that represents the field and the point where that work stops. The topics are the directions of arrival that an array reports (Section 3.1), the material of an object (Section 3.2), the text that a camera reads (Section 3.3), a radar that chooses its next measurement (Section 3.4), a radar that moves (Section 3.5) and a scene that several sensors observe together (Section 3.6).

3.1. Detection and Direction of Arrival

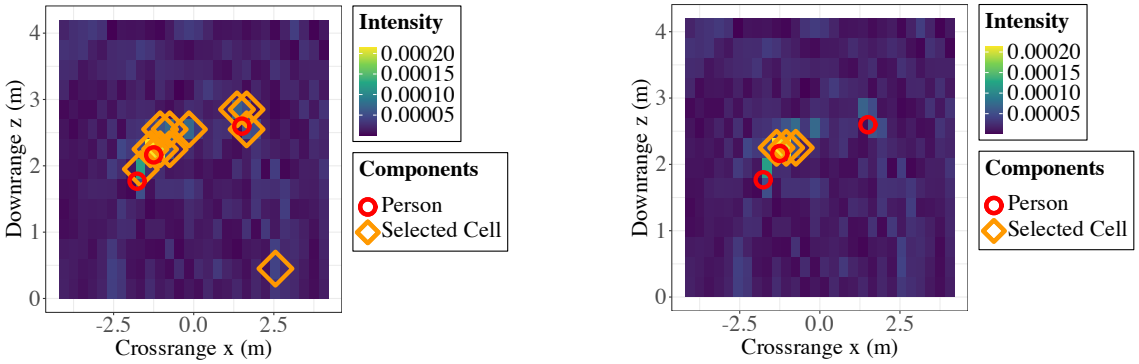
An antenna array answers two questions in every range cell: how many objects are there, and in which directions do they lie? A CFAR detector answers the first. It keeps the false alarm probability of a single cell constant [9]. Spectral methods answer the second. Beamforming scans the directions with a spatial filter [35], [36], and subspace methods such as MUSIC and ESPRIT resolve closely spaced sources [37], [38]. Both steps rest on a model of the array. Neither bounds the share of false objects among those that the radar reports.

Liu et al. replace the array model by a neural network [118]. The network learns from simulated data to decide for every direction of a preset grid whether a signal lies near it, and its error stays below that of MUSIC when the array is imperfect. The price is a large amount of labelled data, and the paper reports the error of the directions but not the share of false ones.

Löser et al. bring a guarantee on the share of false reports to the localisation of people with a rescue robot [119]. Smoke, dust and walls blind a camera there, so the robot carries a radar that steps through 137 frequencies between 6.3 and 8.0 GHz and sees through a wooden door. The authors cut the room into cells of 0.3 m and explain a single snapshot of the radar by a few occupied cells (Figure 3.1). Selecting these cells is the detection of the persons. They select them with Complex-Valued Terminating-Random Experiments Selector (CT-Rex) [47], which adds artificial cells that hold only noise to the grid and runs a forward selection until these dummies enter. Many such random experiments are fused, and a calibration step sets the vote threshold so that the estimated share of false cells meets the target level. The method needs neither a tuned sparsity weight nor the number of persons.

The test covers simulation and measurement. In simulation, three occupied cells hide among 392 on a virtual array of eight channels. Over 100 trials with signal-to-noise ratios from -18 to 6 dB, the share of false cells respects the target of 5 percent, and the detection rate of CT-Rex is slightly higher than that of three competing sparse-recovery methods, also under impulsive noise. The measurement uses real radar data: a benchmark

set of 62 scenarios, recorded with the radar of the rescue robot, with up to five persons in free space or behind obstacles. In scenario 28, three persons sit 2.5 to 3 m from the robot. CT-Rex finds all three in free space and behind a wooden door, while the competing method SAEN finds one in free space and misses persons behind the door.



(a) CT-Rex (Target FDR $\alpha = 5\%$).

(b) SAEN.

Figure 3.1.: Localisation of three seated persons from a single radar snapshot in free space. Red circles mark the persons and orange diamonds the selected cells. (a) CT-Rex at a target false discovery rate of 5 percent, (b) the sequential adaptive elastic net, a competing sparse-recovery method. Adapted from Löser et al. [119].

This evidence shows that a guarantee on false cells fits into a radar localisation. On real data from the rescue robot, the selector finds all three persons, also behind a door, where attenuation and multipath disturb the echoes. The share of false cells, however, appears only in the simulation, where the truth is exact. In the measured scenario, the ground truth marks one nominal cell per person, while a person covers several cells. The measurement therefore confirms the detections but reports no share of false cells. The paper also shows one of the 62 scenarios and a single snapshot, and its signal model leaves out wall returns and multipath. Its authors name a larger evaluation on real data, multiple snapshots, wall returns and real-time operation on the robot as future work.

Chapter 5 addresses these points. Point-like reflectors at taped positions give a known truth for every report, so the false share becomes measurable for one and for 128 snapshots in a room with its own echoes. The selector also reaches directions between the grid points.

3.2. Radar Material Classification

Most radar classification reads what an object does. Parts that vibrate or rotate modulate the Doppler frequency of the echo. This adds sidebands around the bulk Doppler line. A time-frequency analysis turns the sidebands into features [59]. Surveys list classifiers for human gait, small drones and concealed targets [49]. The material of the object is a second source of information, because reflection and absorption of the wave depend on it. Micro-Doppler classifiers leave this source unused.

RadarCat of Yeo et al. classifies the material itself [48]. It uses the Google Soli chip, which Lien et al. built for gesture sensing [120]. The chip is an FMCW radar at 57 to 64 GHz with two transmit and four receive antennas. The object lies on the chip or touches it. A random forest assigns the 661 features of the echo to

one of 26 everyday materials and objects (Figure 3.2), to one of 16 transparent plastics or to one of 10 body parts.

Figure 3.2.: RadarCat. Adapted from Yeo et al. [48]. (a) The object lies on top of the Soli sensor, and a machine learning classifier on a computer assigns the raw radar signals to a material or an object. (b) The 26 materials and objects of the first study and the 16 transparent plastics of the second.

A forest trained on five sessions and tested on the sixth reaches 96.0 percent on the 26 objects. The 16 transparent plastics reach 98.7 percent. A classifier for body parts trained per user reaches 90.4 percent in real time. Trained on other users, it reaches 62.2 percent. Confusions occur between objects that share a material. An aluminium laptop cover and a scale with aluminium parts are an example.

RadarCat shows that the echo alone identifies a material when the geometry is fixed. The object touches the sensor, so range and angle drop out of the signal. The price is the contact. The authors report accuracies over sessions and users. They give no error statement for a single decision.

Chapter 6 removes the contact. It classifies material and motion together at a distance. Each decision is explained by the training examples that rebuild it.

3.3. Computer Vision for Text Reading

Classical text recognition cuts first and classifies afterwards. The engine Tesseract builds on a connected component analysis. It finds the text lines, splits them into words and characters, and classifies every character. An adaptive classifier trains on the words that it has recognised well [121]. Touching characters make the cutting the weak step. Dissection, projection profiles and recognition-driven cuts exist to repair it [115].

The first learned readers dropped the cutting. The Convolutional Recurrent Neural Network (CRNN) of Shi et al. passes the word image through convolutional layers, a bidirectional Long Short-Term Memory (LSTM) and a transcription layer. It trains end to end on synthetic words [43]. Some commercial recognisers still use this design [122].

Du et al. drop the recurrent layers as well [122]. Their SVTR cuts the word image into small patches, which they call character components. Three stages of mixing blocks process the patches. The blocks apply self-attention at two ranges: within a neighbourhood the size of a stroke, and across all components. A merging step halves the height between the stages (Figure 3.3). A linear layer reads the characters from the result.

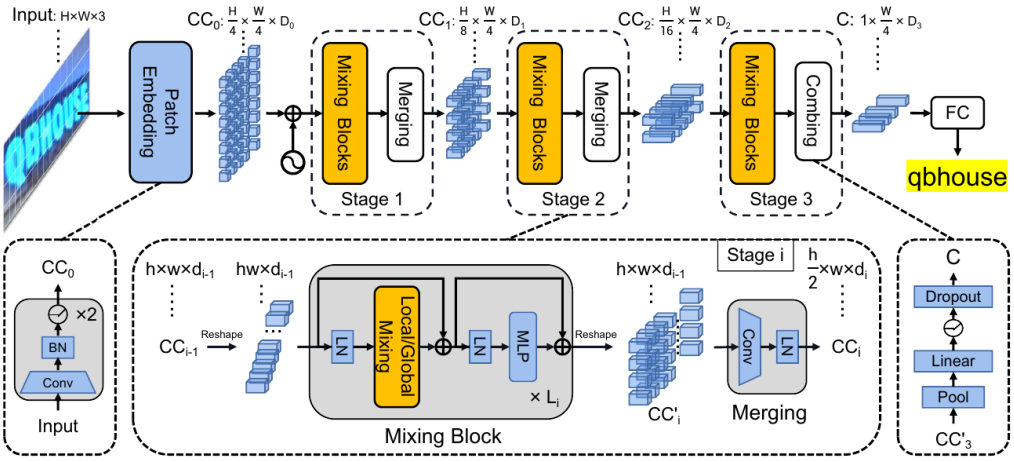


Figure 3.3.: A single visual model for text. Adapted from Du et al. [122]. Patch embedding turns the word image into character components. Three stages of mixing blocks with merging refine them, and a linear layer returns the word.

In the comparison table of the paper, the tiny model SVTR-T has 6.03 million parameters. It needs 4.5 ms per word image on a GPU. It reaches 94.4 percent on IIIT5k and 91.6 percent on SVT. The CRNN of the same table reaches 82.9 and 81.6 percent with 8.3 million parameters and 6.3 ms.

SVTR shows that recognition without cutting has become more accurate and cheaper. The training data must still show every font and size. The image alone fixes no metric scale, so the network learns the variation of size from data. A bright source causes a second problem. Light scattered inside the lens adds a veil that lowers the contrast of the whole image [63]. An image-only network cannot tell the lamp from a bright reflection.

What the image lacks, the radar can supply. Chapter 7 lets the radar fix the metric scale of the text. It also marks the lamp, so that the glare can be modelled.

3.4. Cognitive and Adaptive Radar

Haykin names three ingredients of a cognitive radar. The first is intelligent signal processing that learns from the interaction with the environment. The second is feedback from the receiver to the transmitter. The third is the preservation of the information in the returns through Bayesian detection by tracking [26]. The echolocation system of a bat realises all three. Gürbüz and colleagues order the stages. An adaptive radar changes the processing of the received data. A fully adaptive radar adapts its transmission as well. A cognitive radar adds what Fuster found in the brain, among it the perception-action cycle, attention and memory [27].

The memory of such a radar can be a map. The knowledge-aided radar of the DARPA programme KASSPER feeds high-fidelity knowledge of the environment into the adaptive space-time processor [123]. The knowledge helps where interference varies too fast for traditional adaptation. A tutorial of the same year widens the idea to knowledge-based signal and data processing. There, the knowledge also steers the positioning, the waveform selection and the modes of the radar [68]. Knowledge-aided processing thus ranks above plain adaptation, because it carries information into new situations [27].

The action side is older than the word cognitive. Kershaw and Evans embed the waveform in the Kalman update. They select the waveform with the smallest predicted error covariance [65]. Bell and colleagues extend the rule to a framework with track initiation and termination. A cost built from the predicted error covariance of the tracker drives the next illumination. The authors report gains over a feed-forward radar [66]. An experimental workspace confirms the lower tracking error on hardware [55].

Time keeps cognitive radar from practice. Martone and colleagues name the system response time as one of three factors of operationalising cognitive radar. It is the latency of all algorithms and hardware in the loop. On a software-defined radar, they close the loop for spectrum sharing [56].

Learning replaces the model. Thornton et al. train a deep reinforcement-learning policy [124]. A Q-network maps the sensed interference to the bandwidth and the centre frequency of the next waveform. A double Q-network with a recurrent LSTM layer adds stability and a memory over several time steps. A software-defined radar with 100 MHz of instantaneous bandwidth carries the policy. The trained network is frozen into a look-up table. The selection thus fits into the time slot of the interference.

In simulated coexistence scenarios, the recurrent variant collides least with the interference. On the hardware, the learned radar reaches a higher signal-to-interference-and-noise ratio on recorded interference of communication systems. Its detector also finds more targets at the same false alarm rate (Figure 3.4).

Stephan et al. adapt the tracker itself [125]. A reinforcement-learning agent chooses the tracking parameters from the scene, for example the noise variances of the filter. Engineers would fix these parameters once. On two reward formulations, the choice comes 226 and 93 percent closer to the optimal return than the baseline parameters. The normalised innovation squared tests the settings. A value above the 95-percent bound marks noise variances that are too low. A value below the 5-percent bound marks variances that are too high.

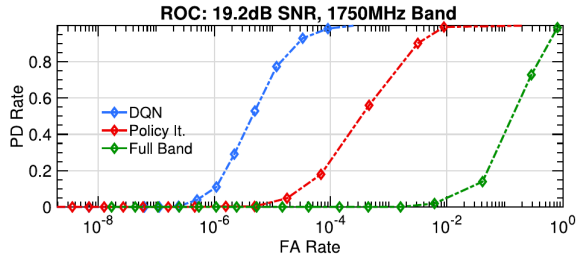


Figure 3.4.: A learned policy on a software-defined radar. Adapted from Thornton et al. [124]. Receiver operating characteristics of the CFAR detector on recorded interference. The learned policy (DQN) detects more targets at a given false alarm rate than policy iteration and a full-band radar.

Both lines leave a gap. The model-based policies reason with tracks. Their uncertainty comes from a Gaussian covariance, which can misjudge the error, as the consistency test of Stephan et al. shows. The learned policies adapt to data. They need training on recorded scenes, and their reward measures the spectrum or the return, not the uncertainty of the tracks. Neither line states how reliable the reported tracks are.

Chapter 8 keeps the model-based decision and repairs its weak point. A conformal calibration replaces the Gaussian covariance. The tracks come from detections with a guaranteed error budget, so the perception that the decision acts on is controlled as well. The choice also counts the processing latency, as the response time of Martone and colleagues demands. It runs on an unmodified automotive radar chip and needs no training.

3.5. Radar in Motion

A radar that moves sees the world move past it. The echo of the static scene then follows the motion of the platform. Airborne radar uses this fact: the Doppler shift of the ground echo is set by the platform, so the processing can separate the targets that move on their own [8]. Radar odometry reverses the argument. It reads the motion of the platform from the echo of the surroundings, provided that the scene is mostly static [126]. Both methods sort the world into a static and a moving part. Each rests on one assumption: the motion of the platform is known, or the scene is static.

Ding et al. ask the radar itself which points move [127]. Their network takes two consecutive point clouds of a 4D radar. For every point, it estimates a motion vector and a probability of moving. It also recovers the ego-motion between the two scans, which alone explains the flow of the static points. Nobody labels the data. The odometry, the lidar and the camera of the vehicle supply the training cues instead.

On the View-of-Delft data, the scene-flow error is 0.13 m. The strongest baseline reaches 0.23 m. The masks of moving points follow the ground truth (Figure 3.5). The by-product ego-motion also follows long trajectories better than an iterative-closest-point baseline.

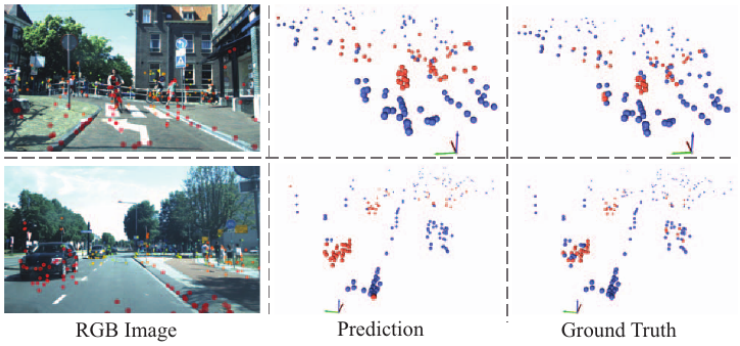


Figure 3.5.: Moving and static radar points. Adapted from Ding et al. [127]. The network marks the radar points that move (orange) and those that stand still (blue) from two consecutive scans of a moving radar. The prediction follows the ground truth.

The result has a price. The radar learns motion from the sensors that it is meant to relieve in the dark. The odometer and the lidar supply the labels of the moving points. The network also scores every point without stating how many of the points marked as moving are wrong.

Chapter 9 poses the same question from a rail. The pose of the platform is known there, and a reference run of the empty path takes the place of the learned labels. A statistical test then decides which detections move. It keeps the error budget of the detection intact.

3.6. Multi-Modal Scene Analysis

A scene states what is where and how it moves. A driving scene is read by many sensors at once. The benchmark that fixed this setting for radar and camera is nuScenes [128]. It is the first dataset with the

full sensor suite of a vehicle: six cameras, five radars and one lidar. It holds 1000 scenes of 20 s with three-dimensional boxes of 23 classes. New metrics score detection and tracking.

Fusion is the established way to use such data. Surveys sort the networks by the place where the streams meet: at the input, at the decision or in between [11], [61]. Radar enters in two ways. Its points are projected into the image and appended as extra channels [54]. Alternatively, a network segments the raw radar tensor directly [129].

Fog breaks the redundancy on which these designs rely. Bijelic et al. drove 10 000 km through northern Europe. They labelled lidar, camera, radar and gated near-infrared data. Their single-shot fusion network trains on clean data only. It fuses its features adaptively, driven by the entropy of each measurement. In hard scenarios, it beats the compared fusion methods by more than 8 percent in average precision [62].

CAFuser carries this idea from boxes to whole scenes [130]. The camera names the condition. An RGB camera classifies the weather and the lighting into a Condition Token. A contrastive loss aligns this token with verbal descriptions of the scene. The token steers the fusion of camera, lidar, radar and event camera. One shared pre-trained backbone maps all sensors into a common feature space, with a light adapter for every modality. A panoptic head returns the segmented scene (Figure 3.6).

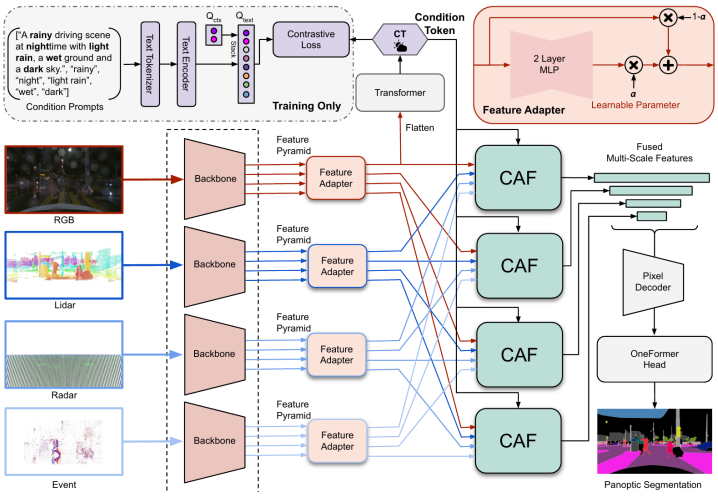


Figure 3.6.: Condition-aware fusion for panoptic scene analysis. Adapted from Brödermann et al. [130]. A shared backbone and one adapter per sensor feed the fusion modules. The Condition Token from the camera steers them, and a pixel decoder and a panoptic head return the scene.

On the MUSES benchmark, CAFuser ranks first. The benchmark records camera, lidar, radar and event camera in adverse weather and at night. The model reaches a panoptic quality of 59.7 and an intersection over union of 78.2 percent for semantic segmentation. It also sets a new state of the art on DeLiVER. The panoptic quality scores every segment by its overlap. It counts missed and false segments alike [69].

Cooperation can also run the other way. In the system of Wang et al., a camera localises the person. The radar then forms its beams on this position and measures the vital signs [131].

The rescue robot Scout shows such a system in the field. It carries lidar, an inertial unit, a 360-degree camera, a zoom camera, a thermal camera and a through-wall radar. The radar chain of Schroth et al. localises up to five persons and measures their breathing (Figure 3.7). In most scenarios, the localisation error stays below

15 cm. The chain senses at rest and almost fails behind a wall. The authors suggest using the camera to steer the robot to a door [132].

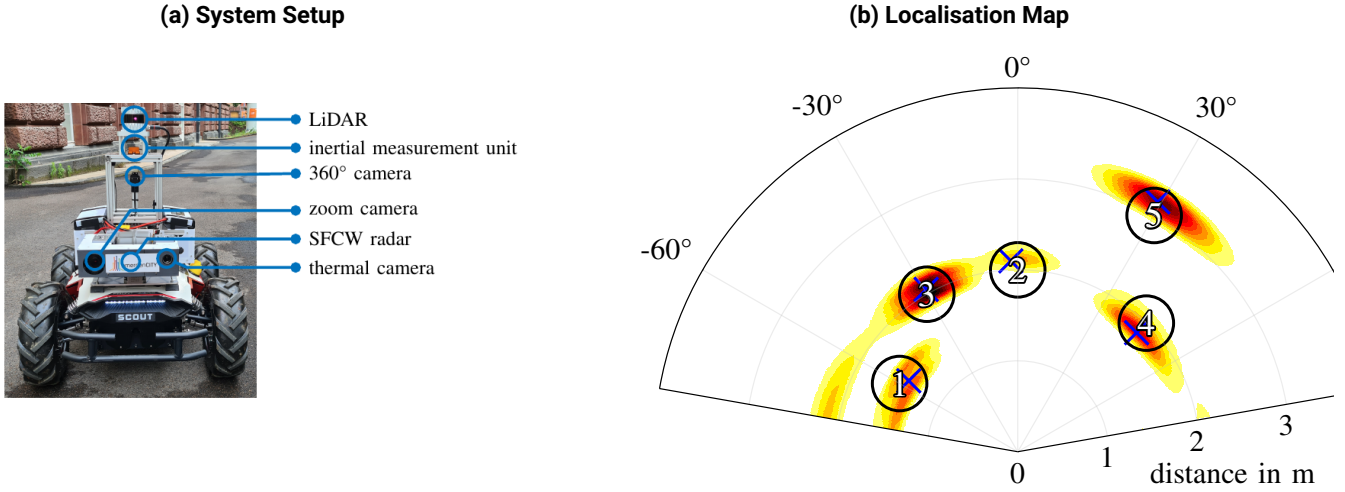


Figure 3.7.: The rescue robot Scout and its localisation of persons. Adapted from Schroth et al. [132]. (a) The robot with its sensors: lidar, inertial measurement unit, 360-degree camera, zoom camera, stepped-frequency radar and thermal camera. (b) Two-dimensional MUSIC spectrum of a scene with five persons. Circles mark the reference positions and crosses the estimated positions.

Three points stand out. The strongest systems learn from large labelled data sets. They report one score over a data set and give no statement about how many reported objects are false. The radar is one input among several. Where it does what only a radar can do, the camera steers and the radar measures. The radar never fixes what the camera reads. Finally, none of these systems carries the errors of its perception to the action of its platform.

Chapter 9 chooses another path. Each sensor keeps its own interpretable output. The radar selects the object and its scale, and the camera reads it. One scene panoptic quality, the measure on which CAFuser leads MUSES, scores the understanding. The platform acts on detections that carry an error budget.

4. Implementation ★

A radar that decides what to measure next needs a body. ARTIFACT, the Adaptive Radar Tracking Inference of False Alarm Controlled Targets, is that body: one platform that senses, moves and thinks. It must show the raw signal of a radar, keep a camera in step with it and let the whole system move. Only then can a claim about false discoveries meet real recordings.

An automotive radar is the heart of ARTIFACT. Its 16 virtual channels hand their raw samples to a computer, without any vendor processing (Section 4.1). A printed chassis and a modular radome protect the electronics, and an exchangeable plate changes what the radar looks through (Section 4.2). A webcam adds what the radar cannot see, namely colour, texture and printed letters (Section 4.3). Radar and camera ride on a three-axis motor stage, which turns a static sensor into a platform that drives, stops and looks again (Section 4.4). A Python framework on a computer configures, records and processes everything, live or from a recording with the same code (Section 4.5). Figure 4.1 shows the parts and their cables, and Appendix A adds their schematics, the design of the housing, the laboratory and the start-up.

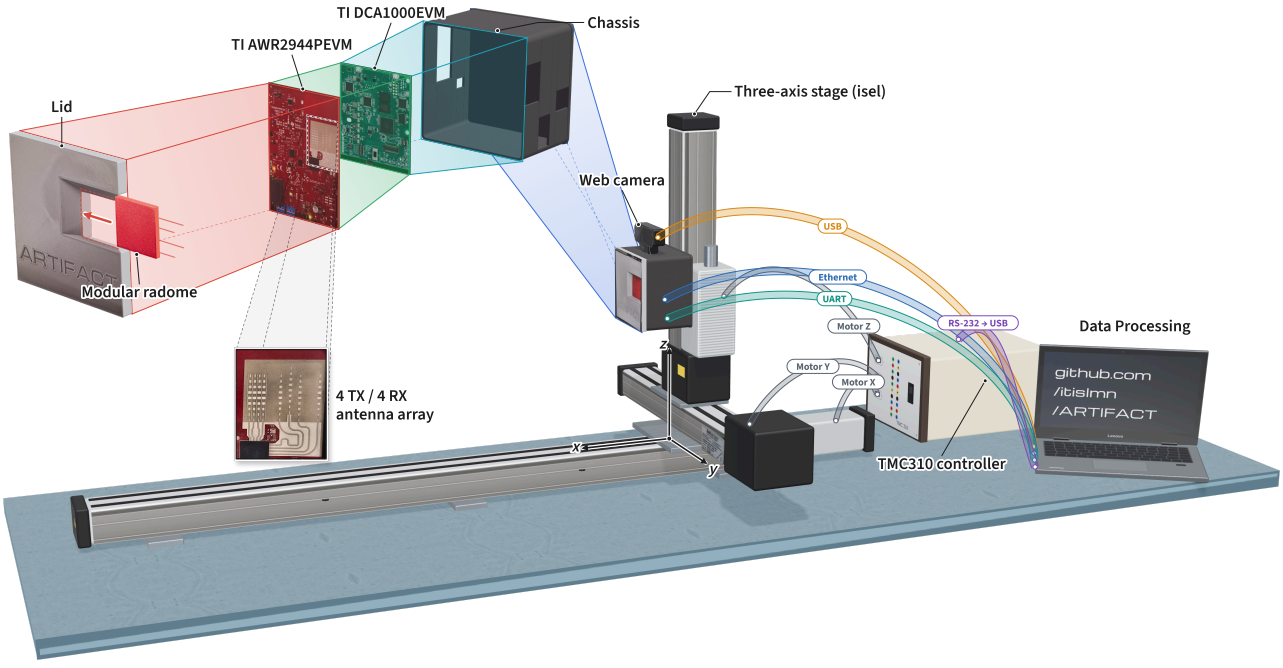


Figure 4.1.: The ARTIFACT platform in an exploded view. The radar and the capture card sit in the chassis behind the modular radome. The webcam and the radar ride on a three-axis stage with three motors. USB, Ethernet, a serial port and RS-232 connect the platform to the computer.

4.1. Radar Sensorics and Evaluation Modules

A radar that decides what to measure next must first sense the scene and then hand over what it sensed. Sensing turns echoes into digital samples. Delivery carries these samples, unaltered, to a computer. Two boards share this work in ARTIFACT. The AWR2944PEVM senses, and the DCA1000EVM acquires the data. Appendix A.1 gives the schematics and the specifications of all four hardware blocks.

4.1.1. AWR2944P Evaluation Module

Every claim about false discoveries begins with a measurement that nobody has edited. ARTIFACT therefore needs a radar that separates objects standing centimetres apart and exposes its raw signal. A millimetre-wave radar meets the first demand, because its short wavelength resolves fine detail with a compact antenna array. An automotive design adds a form made for vehicles that move, which matches a platform that drives.

The AWR2944P evaluation module of Texas Instruments is such an automotive radar for the band from 76 to 81 GHz [133]. A single chip generates the FMCW chirps, receives the echoes and converts them into samples (Figure 4.2). The antenna array of four transmit and four receive antennas sits on the module. At a wavelength of about 3.8 mm (Section 2.1), the array stays compact.

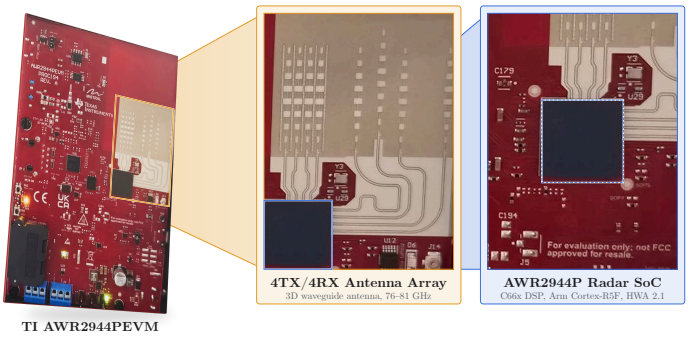


Figure 4.2.: The AWR2944P evaluation module with its four-transmit, four-receive antenna array and the radar chip.

Range. The bandwidth decides how finely the radar separates two objects in distance (Section 2.1). The chip sweeps more than 4 GHz, which promises a range cell of 3.7 cm [134]. The sampling rate decides how far the radar sees. A distant object produces a high beat frequency, and the sampler must follow it.

Speed. The radar measures speed from the phase change between chirps of the same transmitter. In TDM, the four transmitters take turns, so each one repeats only after four slots. This cuts the unambiguous speed to a quarter of the single-transmitter value. Consequently, the chirp period becomes a design choice of every experiment.

Angle. The four transmit and four receive antennas form 16 virtual channels (Section 2.1.3). Twelve of them lie in one horizontal row and carry the azimuth. One transmitter sits higher, and its four channels add the elevation. The receive antennas stand 1.900 mm apart, half a wavelength at 78.9 GHz, whereas the transmit antennas stand 7.837 mm apart, about two wavelengths. A wave from the angle θ reaches each channel with a different phase (Eq. (2.21)). The row of twelve resolves about 9.6° at broadside, and two closer directions

blur into one peak. Because the spacing is not exactly uniform, the steering vectors of Chapter 5 use the antenna positions of the board layout. This choice lowers the angle error from about 0.9° to 0.3 to 0.4° (Appendix C.1.1).

Field of View. Each antenna has a peak gain of 13 dBi. In azimuth, the gain falls by 3 dB at $\pm 30^\circ$ and by 6 dB at $\pm 45^\circ$. In elevation, the beam is thinner, with 3 dB at $\pm 3^\circ$ and 6 dB at $\pm 5^\circ$ [133]. The radar thus sees a wide horizontal sector through a thin slice. The angle grid of Chapter 5 spans $\pm 60^\circ$, where the gain lies about 10 dB below the peak. The ideal region of the experiments stays within $\pm 30^\circ$.

Signal Strength. An echo must rise above the noise (Section 2.1.1). A transmit power of 13.5 dBm and a receiver noise figure of 12 dB set the budget [134]. Combining the 16 virtual channels coherently adds up to 12 dB of SNR for a point echo.

Raw Output. Behind the antennas, the chip holds a 16-bit ADC with up to 37.5 MS/s, a C66x Digital Signal Processor (DSP), an Arm Cortex-R5F controller and a hardware accelerator for detection [134]. The chip could therefore deliver detections on its own. ARTIFACT leaves this function unused. A CFAR bounds the false alarms of one cell, not the share of false objects. Instead, the chip runs the out-of-box demonstration of the software development kit, version 04.07.02.01, which only configures and starts the chirps [135]. The ADC delivers real samples, and the Fourier transform in software forms the complex values (Section 2.2.1).

Engineers start from the requirement and derive the chirp parameters, as Section 2.1.5 describes. Table 4.1 applies this inversion to the search chirp that all experiments share. The scenes lie within a few metres, so a maximum range of 10.7 m suffices. Walking persons stay far below the unambiguous speed of 3.8 m/s. A sampled bandwidth of 1.8 GHz, less than the chip offers, keeps the frame at 1 MB (Eq. (4.6)).

Table 4.1.: Search chirp of all experiments and the quantities it fixes. The wavelength $\lambda \approx 3.8$ mm.

Parameter	Fixes	Value
Idle Time, Ramp Time	Chirp period $T_\mathrm{c} = 4\,(T_\mathrm{idle} + T_\mathrm{ramp})$	26.5 µs, 36 µs, $T_\mathrm{c} = 250$ µs
Chirp Period T_c	Maximum speed $\lambda/(4T_\mathrm{c})$	± 3.8 m/s
Slope S	Beat frequency per metre	70 MHz/µs
Samples N_s, Rate f_s	Sampled bandwidth $B = SN_\mathrm{s}/f_\mathrm{s}$	256, 10 MS/s, $B = 1.79$ GHz
Bandwidth B	Range resolution $c/(2B)$	8.4 cm
Sampling Rate f_s	Maximum range $f_\mathrm{s}c/(4S)$, real samples	10.7 m
Snapshots K	Speed resolution $\lambda/(2KT_\mathrm{c})$	128, 0.059 m/s
Transmitters $\times$ Receivers	Virtual channels, 12 in the azimuth row	$4 \times 4 = 16$

In operation, the module draws power from a 12 V jack that supplies more than 2.5 A. A text file of commands configures the chirps over the serial line at 115 200 baud, through the micro-USB port. Once started, the module streams its samples over four Low-Voltage Differential Signaling (LVDS) lanes to the capture card.

4.1.2. DCA1000EVM Data Capture Card

A radar chip streams its samples without pause, yet the evaluation module has no memory to keep them. Every raw sample matters here, since the methods of this thesis work on the unprocessed signal. A capture card must therefore catch the stream and pass it to a computer.

The DCA1000EVM of Texas Instruments serves this purpose [136]. A flat cable joins the 60-pin connector of the radar board to the connector of the card (Figure 4.3). The cable carries the four LVDS lanes, the clock and the serial interfaces of the radar. The lanes run at up to 600 Mbit/s.

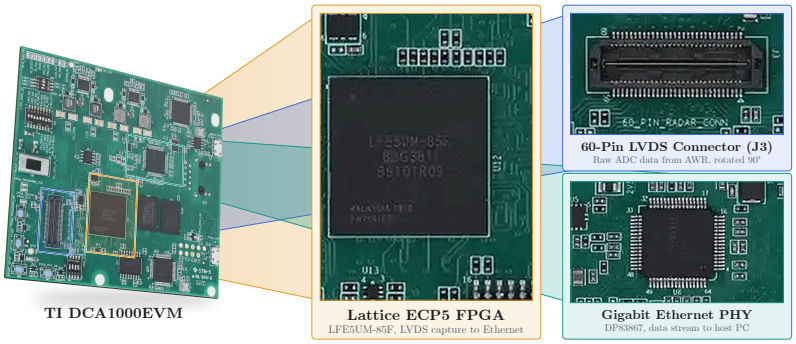


Figure 4.3.: The DCA1000EVM capture card: the 60-pin connector for the radar, the FPGA and the Gigabit Ethernet transceiver.

A Lattice ECP5 Field-Programmable Gate Array (FPGA) of the type LFE5UM-85F does the work. It offers 84 000 look-up tables and 2 GB of DDR3L memory. Its program, stored in a flash memory on the card, runs in real time without a processor. First, the FPGA receives the lanes in raw mode and buffers the samples. Next, it cuts them into packets and sends them as User Datagram Protocol (UDP) datagrams over a Gigabit Ethernet port, served by the DP83867 transceiver. Every packet carries a sequence number and a byte count, so the receiver detects a loss. Commands to start and stop a recording travel back on a separate port.

The Ethernet link, not the LVDS interface, limits the data rate. The delay between packets sets the throughput. A delay of 5 µs yields about 706 Mbit/s, the default of 25 µs yields 325 Mbit/s, and 50 µs yields 193 Mbit/s. One frame of the search chirp carries about 1 MB of payload and 2 MB on the wire, because every real sample travels in a 32-bit word. At 10 Hz, this needs 168 Mbit/s (Eq. (4.6)), half of the default limit. The cognitive experiments stream two or three such sub-frames in a frame of 105 ms, which needs 320 or 480 Mbit/s. Hence the delay must fall to 10 µs, which offers 545 Mbit/s (Section 8.2.3).

Integration imposes two conditions. The Ethernet adapter must sit on a USB 3 port, since packets drop on a USB 2 port (Appendix A.5). A 5 V supply with at most 700 mA powers the card [136]. Section 4.5 shows how the laptop rebuilds the frames.

4.2. Modular Radome Design ★

A cover protects a radar from dust and touch, yet it stands directly in the beam. Every plastic slows the wave, reflects a part of it and absorbs another part. A fixed cover would therefore tie all experiments to one compromise. ARTIFACT instead uses a modular radome (Figure 4.4). A front cover with a window and a funnel fixes the geometry, and an exchangeable plate of 44×44 mm closes the window. Materials thus compare on identical geometry. Moreover, a printed plate costs little to replace, and the housing holds every plate at the same distance from the antenna (dimensions and printing in Appendix A.2).

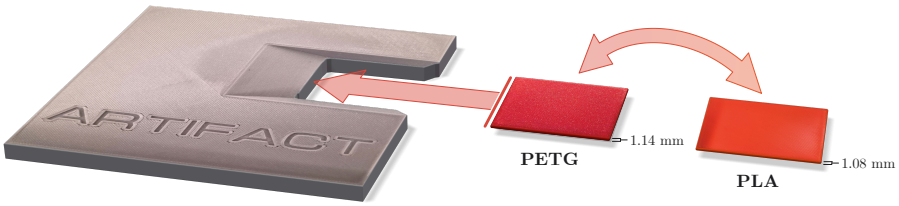


Figure 4.4.: The modular radome. An exchangeable plate of PETG (1.14 mm) or PLA (1.08 mm) closes the window of the front cover, which has a funnel-shaped recess.

Physics dictates what the cover may do. The plastic slows the wave to $c/\sqrt{\varepsilon_\mathrm{r}}$ and shortens it to $\lambda_0/\sqrt{\varepsilon_\mathrm{r}}$ (Section 2.1.1). Each boundary returns the share $|\Gamma|^2$ of the power (Eq. (2.6)), and the material absorbs a further share (Eq. (2.5)). A careless cover also distorts the antenna pattern, which shifts the angles that the steering vectors of Chapter 5 assume.

The two reflections of a plate cancel when its thickness is a multiple of half the wavelength in the material [79]. For a carrier of $f = 78.9\,\mathrm{GHz}$, with $\lambda_0 = 3.80\,\mathrm{mm}$, the design reads

$$\lambda_\mathrm{m} = \frac{\lambda_0}{\sqrt{\varepsilon_\mathrm{r}}}, \qquad t = \frac{n\lambda_\mathrm{m}}{2}, \quad n \in \mathbb{N}. \tag{4.1}$$

The first multiple, $n = 1$, yields the thinnest plate, which absorbs the least. PETG with $\varepsilon_\mathrm{r} = 2.8$ asks for $t = 1.14\,\mathrm{mm}$, and PLA with $\varepsilon_\mathrm{r} = 3.1$ asks for $t = 1.08\,\mathrm{mm}$. Both permittivities are design values from the range of 2 to 4 that the guide lists for polymers. A single boundary reflects 6.3 percent (PETG) or 7.6 percent (PLA) of the power, and the half-wave plate largely removes this share. An error of 0.05 mm detunes the round-trip phase by $4\pi\,\Delta t/\lambda_\mathrm{m} \approx 16^\circ$. Because the tolerance of printed plates is large against the wavelength of 3.8 mm [137], the printed plates test the design as well as the material.

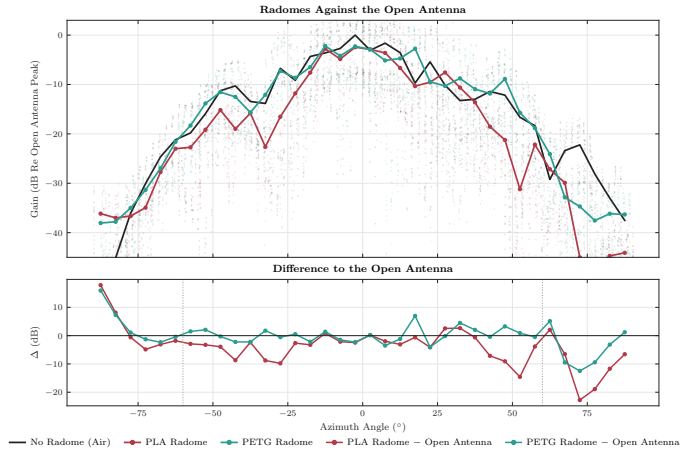
The same logic fixes the gap between antenna and plate. The reflected wave returns in phase with the transmitted wave when the gap is a multiple of half the wavelength in air [79],

$$D = \frac{n\lambda_0}{2} = 1.90\,\mathrm{mm}, \qquad n = 1. \tag{4.2}$$

The housing holds the plate at this distance. A flat plate is tuned only at boresight, since a wave at an angle crosses a longer path in it. The funnel therefore widens outward and keeps the cover out of the beam where the pattern is still strong [79].

The measurement tests the design. A corner reflector stood at 0.17 m while the radar turned through $\pm 90^\circ$ in front of it. The energy of each of the 16 channels in the range bin of the reflector was averaged into bins of 5°. The open antenna sets the reference of 0 dB at its maximum, and the loss of a radome is the difference to this pattern (Appendix A.4).

Figure 4.5 shows the result. Both plastics lower the peak by about 2.3 dB. Behind PETG, the pattern stays within 0.2 dB of the open antenna on average over $\pm 60^\circ$, whereas behind PLA it falls by 3.7 dB. This order agrees with the theory, since PETG has the lower permittivity and reflects less. The reflected shares differ by only 1.3 percentage points. Absorption or the deviation of the printed thickness may explain the rest, and the measurement cannot separate the two. A plate of half a wavelength thus keeps the pattern within 0.2 dB over the sector of the angle grid of Chapter 5, at a cost of 2.2 dB at the peak. The radar looks through PETG almost as through air, so the measurement favours PETG as the radome material.



	Peak	$\leq 30^\circ$	$\leq 60^\circ$
Open	0.0	0.0	0.0
PLA	-2.4	-2.2	-3.7
PETG	-2.2	-0.5	$+0.2$

Figure 4.5.: Antenna pattern with and without radome. Left: gain over the azimuth angle for the open antenna and behind PLA and PETG, in decibels relative to the maximum of the open antenna, and the difference to it. Right: the peak and the mean difference to the open antenna within 30° and 60° of the boresight, in decibels.

4.3. Camera

A radar knows where an echo lies and how strong it is, but it cannot tell what the object looks like. Shape, texture and colour separate a wall from a person, and only a camera records them. ARTIFACT therefore carries a camera next to the radar.

The ISY IW-1000-1 is a webcam on top of the radar housing (Figure 4.6). As a USB Video Class (UVC) device, it lists its pixel formats, frame sizes and controls through standard requests. The computer needs no vendor driver, and a program selects the format directly [138], [139], [140], [141]. The specification promises 1280×720 pixels at 30 frames per second and a focus range from 30 cm to infinity.

Quantity	Value
Resolution, Frame Rate	1280×720, 30 fps
Focus Range	30 cm to infinity
Signal-to-Noise Ratio	> 48 dB (30 fps, 220 lux)
Dynamic Range	> 72 dB
Supply	5 V, 500 mA over USB

Figure 4.6.: The camera of the platform and its specification [139].

The pixel format decides whether the link carries the images. A frame of $W \times H$ pixels with $b = 16$ bit per pixel (4:2:2, Section 2.4.1) at f frames per second needs

$$R = W\,H\,b\,f = 1280 \cdot 720 \cdot 16 \cdot 30\,\text{bit/s} \approx 442\,\text{Mbit/s}. \tag{4.3}$$

This almost fills the 480 Mbit/s of a high-speed USB 2.0 link, as the supply of 500 mA suggests. Consequently, the camera sends Motion JPEG (MJPEG). Every frame is a complete Joint Photographic Experts Group (JPEG)

image. Blocks of 8×8 pixels pass through a discrete cosine transform, and the rounded coefficients are coded without further loss [142]. Frames therefore stay independent, and a lost frame damages only itself.

The program selects MJPEG at 1280×720 and aborts if the device does not offer this format. The exposure control takes an integer e and sets the exposure time to 2^{e} s. A fixed value keeps the brightness of a pixel comparable between frames, which the glare detection needs, whereas the automatic mode serves scenes that change. The focus range covers the radar range of 10.7 m (Table 4.1). Each frame receives its time stamp on arrival from the high-resolution counter of the computer, so that camera and radar share one time base (Section 7.2.1). One cable carries power and data.

4.4. Three-Axis Motor Stage

A radar that moves must know where it is. A pose error of a few centimetres enters every range and angle that the planner derives, and it would blur the very evidence of the experiments. An isel motor stage removes this doubt, because it moves the sensor head along three axes of known position. The forward axis changes range and closing speed. The sideways axis changes the bearing. The vertical axis changes the elevation. Each axis thus shifts the scene along a direction that the array resolves (Section 4.1.1). Together, the axes reproduce the climb-and-approach path of a drone.

The forward axis is an LES 6 ball screw unit of 1 390 mm length and 4 mm pitch. The sideways axis is an LES 4 of 290 mm length and 2.5 mm pitch. A stepper module MS135HT drives the vertical axis [143]. A ball screw moves the carriage by one pitch per revolution with rolling friction, and the catalogue lists a repeatability of ± 0.02 mm.

A stepper motor turns by a fixed angle per pulse, so the position needs no sensor. The usual NEMA 23 motors turn by 1.8°, which gives $N_{\mathrm{fs}} = 200$ full steps per revolution. The catalogue does not state this value. With $N_{\mu} = 16$ microsteps, the forward axis moves by

$$\Delta x = \frac{p}{N_{\mathrm{fs}}\, N_{\mu}} = \frac{4\,\mathrm{mm}}{200 \cdot 16} = 1.25\,\mu\mathrm{m}, \tag{4.4}$$

which is sixteen times finer than the repeatability. The spindle permits $n_{\max} = 1500$ revolutions per minute, so the axis reaches $v_{\max} = p\, n_{\max}/60 = 100$ mm/s. This speed lies far below the unambiguous radar speed of ± 3.8 m/s (Table 4.1).

The Trinamic TMCM-310 drives the three motors [144]. It runs on 8 to 34 V, delivers 0.3 to 1.1 A Root Mean Square (RMS) per coil and microsteps up to a factor of 16. The host sends nine-byte Trinamic Motion Control Language (TMCL) commands over RS-232 at 9 600 baud. Per motor, these commands set the maximum speed and the acceleration (0 to 2047 each), the coil current (0 to 1500 mA) and the microstep resolution. Since the torque of a stepper grows in proportion to the coil current, the current matches what the load needs. The position counter of $\pm 2^{23}$ microsteps, 10.5 m at $1.25\,\mu$m, returns the pose.

Without an encoder, the counter holds commanded microsteps. A move of N microsteps over a reference length L sets the scale

$$k = \frac{N}{L}, \qquad k_{\mathrm{ideal}} = \frac{N_{\mathrm{fs}} N_{\mu}}{p} = 800\,\mathrm{steps/mm}, \tag{4.5}$$

and its relative error equals that of the length. A reading uncertainty of 0.5 mm over 1 m fixes the scale to 0.05 percent. The pose error then stays below 0.7 mm over 1.39 m, far below the range resolution of 8.4 cm

(Table 4.1). The pose thus serves as ground truth for every radar frame, and the experiments of Chapter 9 rest on it.

4.5. Data Acquisition and Processing ★

ARTIFACT works as one recording and processing chain. Three sensors feed it, namely the radar, the camera and the motor stage. A laptop receives their three streams, labels every item with a time stamp from one clock and stores the items as one record. A method then reads this record, live or from a stored take, with the same code.

The platform divides into five groups of devices (Figure 4.7). The sensor head carries the radar and the camera. The data capture card receives the radar samples. The motor stage moves the sensor head, and the motion controller drives its three motors. The laptop hosts one process per stream and the scene method.

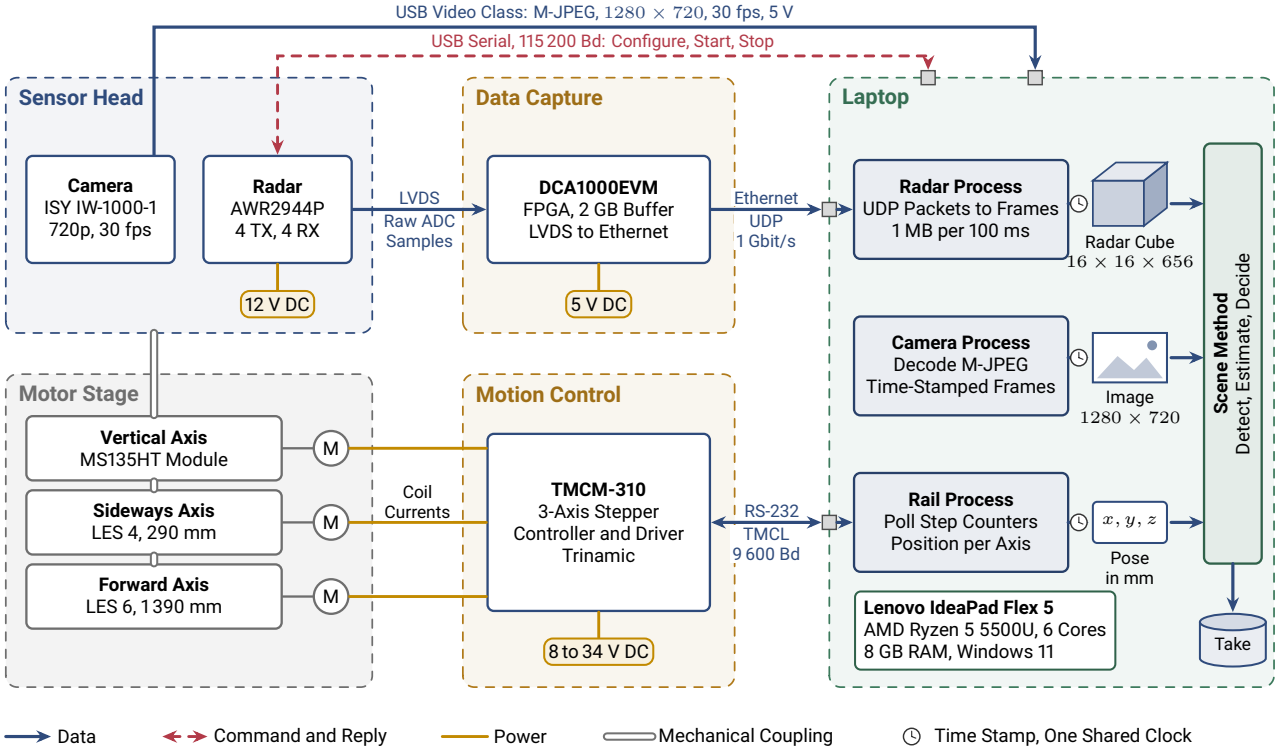


Figure 4.7.: System overview of the platform. Ethernet and RS-232 reach the laptop through USB adapters. Three processes receive the streams, stamp them with one clock and hand them to the method.

Five transfers connect the groups. First, the radar passes raw ADC samples to the capture card over the LVDS lanes. Second, the capture card sends UDP packets of 1 466 bytes to the laptop over Ethernet. There, the radar process orders them by sequence number and cuts them into frames. Third, the camera sends M-JPEG frames over USB. The camera process decodes them. Fourth, the rail process sends TMCL commands to the motion controller over RS-232. The controller replies with the step counters of the three axes. Finally, the motion controller feeds the coil currents to the three motors, whose axes carry the sensor head. The step counters therefore give the pose of the radar and the camera.

Three data formats result. A radar frame holds the raw samples of one frame period of 100 ms. It consists of 128 snapshots, 16 virtual channels and 256 real samples of 16 bit each. The payload rate is

$$R = \frac{128 \cdot 16 \cdot 256 \cdot 2\,\mathrm{B}}{100\,\mathrm{ms}} = \frac{1\,048\,576\,\mathrm{B}}{0.1\,\mathrm{s}} \approx 84\,\mathrm{Mbit/s}. \tag{4.6}$$

The wire carries twice this rate, 168 Mbit/s, because the card sends every 16-bit sample in a 32-bit word. Both values lie below the 325 Mbit/s of the link (Section 4.1.2). A snapshot is one pass of the four transmitters and lasts 0.25 ms. A camera frame is an image of 1280×720 pixels with a time stamp. A pose is the position of the three axes in millimetres.

The samples are real, and the complex values arise in the Fourier transform. A Hann window weights the 256 samples of every chirp. The transform of a real signal mirrors its spectrum, so only half of the bins carry information. This leaves 128 complex range bins, each holding the amplitude and the phase of the beat signal. The radar cube thus has 128 range bins, 128 snapshots and 16 virtual channels (Section 2.2.1), and every method reads this cube. One snapshot at one range bin is the vector of the 16 channels.

One process per stream keeps a slow stream from delaying another. Each process stamps its data on arrival with the same high-resolution clock, so the cube, the image and the pose of one moment can be paired. A Lenovo IdeaPad Flex 5 with an AMD Ryzen 5 5500U (six cores) and 8 GB of memory runs the processes and the method in Python [145], [146]. A logged take stores the raw frames, the images and the poses. The method runs again on a recording with identical results (Appendices D and A.5).

With these parts, ARTIFACT meets what the experiments demand. The radar delivers its raw signal, the camera adds appearance and the motor stage supplies a known pose. One clock ties the three streams into one record. Consequently, every method reads the same data, live or from a recording, and a claim about false discoveries meets real measurements. The first experiment (Chapter 5) asks the most basic question: where are the reflectors?

5. Detection and Estimation

Every list that a radar reports is a wager. Each entry bets that an object exists, and the detector settles the bets with a threshold that fixes the odds of a false alarm in one cell and ignores the list as a whole. Ten reports at a false alarm probability of 10^{-3} per cell reveal nothing about how many of the ten are wrong, because the number of wrong entries depends on how many cells were tested and how many objects exist. The false discovery rate (FDR) prices the list instead of the cell: $\mathbb{E}[V/\max(R,1)] \leq \alpha = 0.1$ promises that, on average, at most one entry in ten is false, whatever the scene holds, where V of the R reports are false. Statistics has proven this promise for models that a radar violates. A steering matrix correlates every column with its neighbours, reflectors lie between grid points, and a room echoes uninvited. The research question concerns whether the promise survives these violations on a real radar with twelve channels. The scene (Section 5.1) sets one table with three reflectors and a known truth, the methodology (Section 5.2) builds the methods from the classical estimators to Infini-Rex, a new FDR method, the results (Section 5.3) test the bound on measured and simulated data, and the discussion (Section 5.4) weighs what it is worth.

5.1. Scene: Detecting and Locating Reflectors

Radar practice controls a different quantity. A constant false alarm rate detector fixes the probability of a false alarm in one range cell (Section 2.2.2), which says little about the share of wrong reports in the list. Selectors that keep a chosen level α exist for direction finding, but only in simulation, with 80 sensors and directions on a grid [47] (Section 3.1). The chapter tests classical estimators, FDR-controlled selectors and Infini-Rex, the selector of this thesis, by four metrics: the achieved FDR, which must not exceed α, the TPR of the reflectors found, the angle error and the computation time. Simulated and measured data are compared, because a real array offers twelve channels at slightly irregular positions, reflectors between grid points and room echoes.

A bound needs an answer key, so the scenes have a known truth. A laboratory table provides it (Figure 5.1a). The radar stands at the far end of a linear stage and looks along the table with the twelve virtual channels of its azimuth row (Section 2.1.3). Trihedral corner reflectors stand at taped positions between 1.25 and 2.05 m. Three metal plates meet at right angles in each of them, so that several bounces return a strong echo over a wide range of angles [9]. The session owns three reflectors, one big and two small. The Radar Cross Section (RCS) of a trihedral grows with the fourth power of its edge length (Eq. (2.9)), and after the correction for range the big reflector returns 17 to 19 dB more than a small one, which is 50 to 80 times the power. The big reflector stands at an elevation of -0.3° and the two small ones at $+1.5^\circ$ and -4.2°. The elevation beam of the antenna is narrow (Section 4.1.1), so the radar sees a thin horizontal slice, and the problem is one of range and azimuth. A wall on the right side of the table and a metal rail on the left return echoes that the answer key does not list. The platform of Chapter 4 provides the measurement: the AWR2944P (Section 4.1.1) records the twelve virtual channels, the motor stage (Section 4.4) holds the radar

at a known position along the table, and the data acquisition (Section 4.5) delivers the complex range bins to the methods. The code of this experiment is in the folder `estimation/` of the code base (Appendix D).

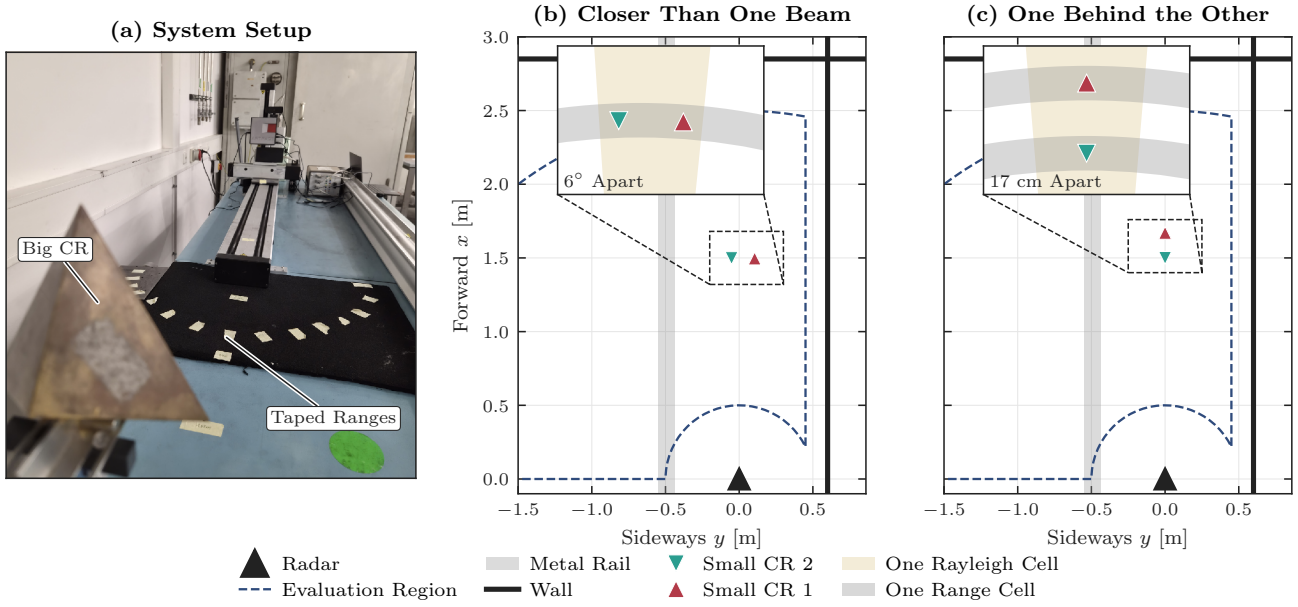


Figure 5.1.: The measured scene. (a) The big corner reflector faces the radar on its linear stage; the photograph looks towards the radar, so the wall appears on the left. (b) Two small reflectors in one range cell, 6° apart. (c) Two small reflectors at one angle, 17 cm apart. The top views show the taped positions and the evaluation region, and the insets zoom into each pair.

The takes grow harder. A single-target take places one small reflector at $+10^\circ$, -30° or -45°, and every report beyond it is false. A multi-target take places two or three reflectors. In Figure 5.1b, two reflectors share one range cell and stand 6° apart, which is 0.6 of the Rayleigh cell of 9.6°. In Figure 5.1c, two reflectors share one angle and stand 17 cm apart, two range cells of 8.37 cm. A big reflector beside a small one and two takes with reflectors spread in angle and range complete the set, and one take without any reflector shows what the room reports alone.

The session holds twelve takes of 20 frames. A calibration take with the big reflector at boresight fixes the phase of every channel, and an empty take records the room, which every other take subtracts. The remaining ten takes are evaluated: three with one reflector, six with two or three, and a null take. These 200 frames contain 320 reflector observations (Appendix C.1.1). A frame holds 128 snapshots, and one snapshot is a pass of the four transmitters in time-division multiplexing, which delivers all channels within 0.25 ms, a time in which a static scene does not change. The evaluation reads the first $K \in \{1, 4, 16, 128\}$ snapshots of each frame. A simulated dataset serves as the control. It renders raw frames of point reflectors with exact truth and without a room, at an SNR from 0 to 60 dB and with 300 trials per setting, and it switches the violations of the assumptions on one at a time: reflectors on the grid, between grid points, and between grid points with random phase errors of 5° per channel. The difference between simulated and measured results measures what the room adds.

Given the $n = 12$ channels of K snapshots from a scene with an unknown number k of reflectors at ranges r_i and angles θ_i, $i \in \{1, \ldots, k\}$, find k, every r_i and every θ_i from as few snapshots as possible, subject to a target level $\alpha = 0.1$ on the FDR. A report counts as a *true discovery* if it is matched one-to-one to a real

reflector within two range cells and half a Rayleigh cell; all other reports are *false discoveries*. With R reports, V of them false and $S = R - V$ true, the false discovery proportion and its expectation are

$$\mathrm{FDP} = \frac{V}{\max(R, 1)}, \qquad \mathrm{FDR} = \mathbb{E}\left[\mathrm{FDP}\right], \tag{5.1}$$

and the true positive rate is the expected share of real reflectors that are found,

$$\mathrm{TPR} = \mathbb{E}\left[\frac{S}{\max(k, 1)}\right]. \tag{5.2}$$

The RMSE of the N matched pairs, $\mathrm{RMSE}_\theta = \left(\frac{1}{N}\sum_{i=1}^{N}(\hat{\theta}_i - \theta_i)^2\right)^{1/2}$ and the same expression for the range, measures accuracy, and the computation time per range gate completes the four metrics. A method keeps the bound if FDR $\leq \alpha$, and among those that keep it, the one with the highest TPR and the lowest error is best. Averages over frames estimate the expectations.

Three obstacles stand in the way: twelve channels, reflectors between grid points and a room that echoes. One fact gives hope. Even a single snapshot lifts every reflector by 18 to 43 dB over the median of the range profile, so the directions are already in the data. What the data lack is a yardstick that separates a direction with a real echo from a direction that merely fits well, and this yardstick must work on one real snapshot and finish before the next frame arrives.

5.2. Methodology

The radar enters the scene with twelve complex numbers per range gate and no key to the truth. The methods answer the same task with growing ambition. The classical estimators ignore the error, the test of Benjamini and Hochberg and the knockoffs promise a bound under assumptions that a steering matrix breaks, CT-Rex keeps the bound with random dummies at a high computational price, and Infini-Rex replaces the dummies by their probability (Appendix B.1.2 lists all methods).

5.2.1. Signal Model

Every method receives the same complex numbers. The board samples each chirp as 256 real values, and a Hann-windowed range FFT turns them into complex range bins (Section 2.2.1). The result is the cube of Figure 5.2a: range bins, snapshots and virtual channels. An OS-CFAR detector on the range profile marks the bins that hold an echo [40]. A marked bin, the range gate, yields one vector $\boldsymbol{y}_\kappa \in \mathbb{C}^{12}$ per snapshot κ, and the K snapshots are averaged, $\bar{\boldsymbol{y}} = \frac{1}{K}\sum_{\kappa=1}^{K}\boldsymbol{y}_\kappa$.

A plane wave from the direction sine $u = \sin\theta$ reaches virtual channel m with the phase $2\pi x_m u$, where x_m is the position of the channel in wavelengths: the sum of a transmit and a receive position. The receivers sit $0.4968\,\lambda$ apart and the three transmitters $2.0491\,\lambda$ apart,

$$x_{(i,l)} = (i-1)\cdot 2.0491 + (l-1)\cdot 0.4968, \qquad i \in \{1,2,3\},\ l \in \{1,\ldots,4\}, \tag{5.3}$$

which gives twelve positions from 0 to $5.589\,\lambda$ with a spacing that deviates from half a wavelength by up to a tenth. The fourth transmitter is mounted higher, and its four channels carry the elevation and stay out of the azimuth problem, so $n = 12$ of the 16 virtual channels are used. The steering vector uses these exact

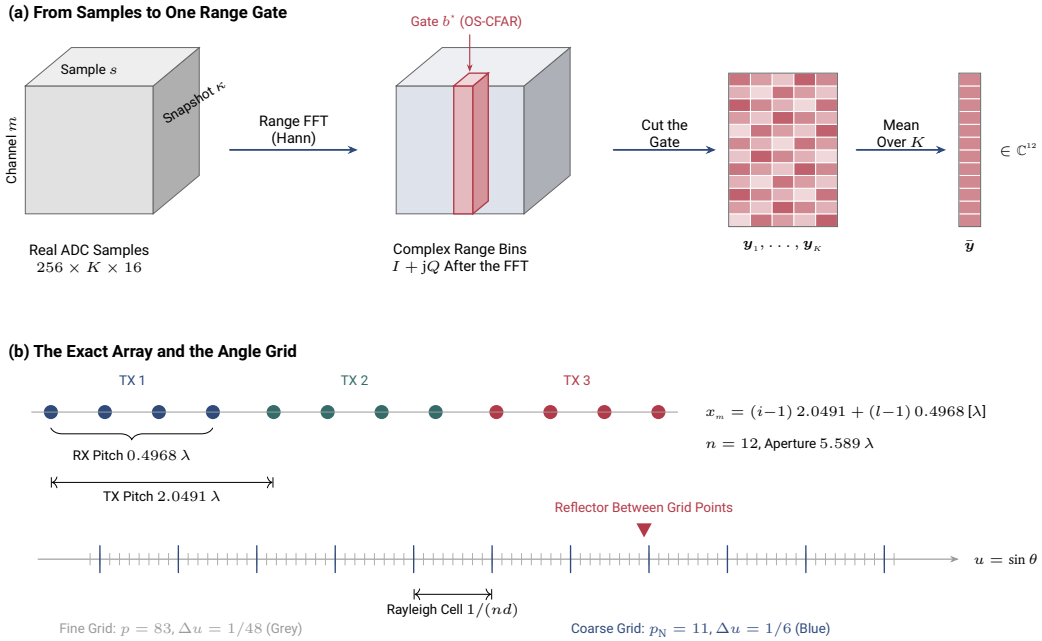


Figure 5.2.: From the radar cube to the angle problem. (a) The range FFT turns real samples into complex range bins; a gate cuts one vector per snapshot, and the mean over K snapshots gives $\bar{\boldsymbol{y}}$. (b) Twelve virtual channels from three transmitters (colours) and four receivers, at exact positions in wavelengths. The directions lie on a fine grid of 83 and a coarse grid of 11 points; one reflector lies between grid points.

positions, $\boldsymbol{a}(u) = [e^{\mathrm{j}2\pi x_1 u}, \ldots, e^{\mathrm{j}2\pi x_{12} u}]^\top$. With $n = 12$ and the nominal spacing $d = 1/2$, the Rayleigh cell is $1/(nd) = 1/6$ in u, or 9.6° at broadside. The fine grid $u_j = j/(8nd) = j/48$ oversamples it eight times and covers $\pm 60^\circ$ with $p = 83$ points; the coarse grid keeps every eighth point, $p_{\mathrm{N}} = 11$. The steering matrix with unit-norm columns and the model of one gate are

$$\boldsymbol{X} = \frac{1}{\sqrt{n}} \left[\boldsymbol{a}(u_1), \ldots, \boldsymbol{a}(u_p)\right] \in \mathbb{C}^{n \times p}, \qquad \bar{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{5.4}$$

with the sparse amplitude vector $\boldsymbol{\beta} \in \mathbb{C}^p$ and complex Gaussian noise of variance σ^2/K. The coarse matrix is $\boldsymbol{X}_{\mathrm{N}} \in \mathbb{C}^{n \times p_{\mathrm{N}}}$. The number of snapshots enters only through the noise level, so every method that uses $\bar{\boldsymbol{y}}$ runs unchanged for $K = 1$ and $K = 128$. For static reflectors, the amplitude vector is the same in every snapshot, so the average is exactly the single-snapshot model with the noise reduced by the factor K. This extends the single-snapshot selectors of Scheidt and colleagues [47] to K snapshots without changing them. A snapshot is one TDM loop of four chirps, one per transmitter, which delivers all virtual channels at one instant (0.25 ms). The methods use K snapshots in two ways. The selectors average them into one vector $\bar{\boldsymbol{y}}$ and stay single-measurement methods with less noise. The classical estimators form the sample covariance of the K snapshots (Section 5.2.2), which has rank one for $K = 1$ and holds no noise subspace; forward-backward smoothing restores the rank at the price of aperture. A selector that treats the K snapshots as one joint measurement does not exist yet: the authors of CT-Rex name multiple snapshots as future work, and they tested their selector in simulation with 80 sensors, whereas this experiment uses 12 channels. Since $p > n$, Eq. (5.4) has infinitely many solutions, and only sparsity makes it solvable: direction finding becomes variable selection (Section 2.3.3), and each direction poses a test $H_{0,j}\colon \beta_j = 0$ (Section 2.3.1). A reflector between grid points is not a column of $\boldsymbol{X}$; its neighbours explain it in part and leave a deterministic remainder

that a selector can mistake for further reflectors [87]. Sparse recovery on a grid goes back to Malioutov and colleagues [147], who set the penalty by hand; here the level α replaces it.

5.2.2. Classical Estimators

The reference chain works on the sample covariance $\hat{\boldsymbol{R}}$ of the snapshots, a second-order statistic, whereas the selectors below use only their mean. Static reflectors are coherent, and forward-backward smoothing over subarrays of $m = 8$ channels breaks the coherence [83], [84]. The MDL criterion (Eq. (2.39)) with $N_{\text{eff}} = 2(n-m+1)K$ estimates the number of reflectors [44]. Given $\hat{k}$, the Bartlett beamformer (Eq. (2.34)), the Capon beamformer (Eq. (2.35)), MUSIC (Eq. (2.36)) and ESPRIT [38] report the $\hat{k}$ highest peaks of their spectra. CA-CFAR and OS-CFAR compare the Bartlett power of each direction with a threshold from its neighbours at $P_{\text{fa}} = 10^{-3}$, with the sixth of eight ordered cells for OS-CFAR [39], [40]. For $K = 1$, the smoothing is the only source of a noise subspace. It trades the aperture of 12 channels for subarrays of 8, which widens the Rayleigh cell by half and leaves the order estimate fragile.

Figure 5.3 shows the classical methods on one measured gate of the close pair. Bartlett merges the two reflectors into one wide peak, because both lie within one Rayleigh cell. Capon and MUSIC resolve one reflector sharply and the other as a weaker, shifted peak. MDL counts the echoes of the room as sources and reaches its upper limit $\hat{k} = 7$ for two reflectors. The first two eigenvalues of the smoothed covariance stand 15 dB above the third, which fits two reflectors, whereas the rest fall to -50 dB without a gap, and the profile at one snapshot equals that at 128 within 1 dB, because the scene is static and its echoes are coherent. None of these methods states how many of its reports are wrong.

A learned baseline, a multilayer perceptron after Liu and colleagues, is described in Appendix B.1.2.

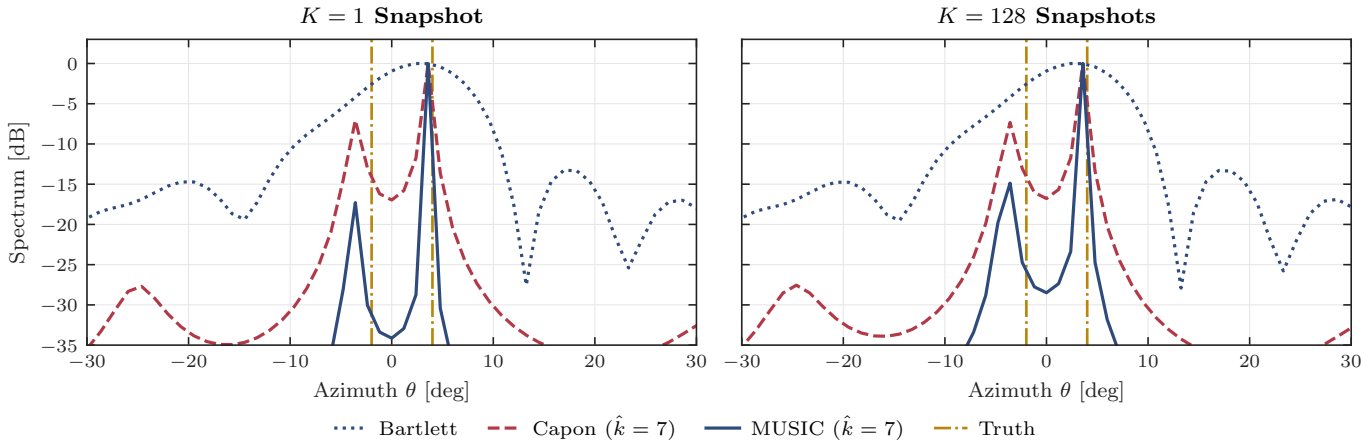


Figure 5.3.: Classical estimators on one measured gate of the close pair, with one snapshot (left) and with 128 snapshots (right). The truth (dash-dotted) is two reflectors at -2° and $+4^\circ$. MDL reaches its upper limit $\hat{k} = 7$ in both cases.

5.2.3. FDR-Controlled Selectors in Radar ★

The selectors of this section guarantee FDR $\leq \alpha$, each under its own assumptions (Section 2.3.2), and the experiment applies all of them to the same measured data.

Benjamini–Hochberg and Benjamini–Yekutieli. The BH test needs one p-value per direction (Eq. (2.61)). A p-value needs an estimate of every β_j, which least squares provides only if the unknowns do not outnumber the channels, so the test works on the coarse grid, with

$$\hat{\boldsymbol{\beta}} = \left(\boldsymbol{X}_{\mathrm{N}}^H \boldsymbol{X}_{\mathrm{N}}\right)^{-1} \boldsymbol{X}_{\mathrm{N}}^H \bar{\boldsymbol{y}}, \qquad \mathrm{Var}\left(\hat{\beta}_j\right) = \frac{\sigma^2}{K}\left[\left(\boldsymbol{X}_{\mathrm{N}}^H \boldsymbol{X}_{\mathrm{N}}\right)^{-1}\right]_{jj}. \tag{5.5}$$

Under $H_{0,j}$, $\hat{\beta}_j$ is circularly symmetric complex Gaussian, so $T_j = |\hat{\beta}_j|^2 / \mathrm{Var}(\hat{\beta}_j)$ is exponential with mean one, and $p_j = e^{-T_j}$ is a valid p-value. Appendix B.1.2 lists the steps of the test. The BY variant divides the level by a harmonic sum and holds for any dependence [96]. Both are exact only if the reflectors lie on the coarse grid.

Knockoffs. The knockoff filter of Section 2.3.2 compares each direction with a constructed copy [30]. Here, one lasso path runs on the steering matrix and its copies, the statistic is $W_j = |\hat{\beta}_j| - |\hat{\beta}_{\tilde{j}}|$, and the filter selects the directions with $W_j \geq \tau$ by Eq. (2.67). *Fixed-X knockoffs* treat the steering matrix as fixed, which fits the physics, and need at least as many rows as unknowns; this holds only for the coarse grid written as a real problem with $2n = 24$ rows and $2p_{\mathrm{N}} = 22$ unknowns, the low-dimensional case. *Model-X knockoffs* [31] assume random rows from a known distribution and work on the fine grid, but row m of a steering matrix is the deterministic vector $[e^{\mathrm{j}2\pi x_m u_1}, \ldots, e^{\mathrm{j}2\pi x_m u_p}]$. Both follow the reference implementation [148]. Eq. (2.67) adds a limit for radar: without negative statistics the ratio falls below α only if at least $1/\alpha = 10$ directions are selected, while a gate rarely holds three reflectors. The variant without the $+1$ (offset 0) escapes this limit, but it controls only a modified FDR [30].

T-Rex and CT-Rex. The T-Rex selector replaces the knockoff copy by random dummies (Figure 2.11): it appends L columns of independent noise to $\boldsymbol{X}$ and forward-selects until T dummies have entered, because the entry of a dummy shows when the selection starts to pick noise [32]. CT-Rex carries this to complex data [47]: complex Gaussian dummies and complex arithmetic keep the real and imaginary part of an amplitude together, whereas the real-valued T-Rex splits them into $2n = 24$ real rows and loses this link. Figure 5.4 shows the method, and Appendix B.1.2 states it as Algorithm 4. Twenty runs with independent dummies give each direction its relative occurrence $\Phi_{T,L}(j) = \frac{1}{K_{\mathrm{rex}}} \sum_k \mathbb{1}\{j \in \mathcal{C}_k(T, L)\}$, where $\mathcal{C}_k(T, L)$ holds the real directions that run k selected before it stopped. The dummies also estimate the share of false selections among the directions with $\Phi_{T,L}(j) > v$, and the calibration takes the largest selection whose estimate stays at or below α; the martingale proof of Machkour and colleagues bounds the FDR (Proposition B.10 in Appendix B.1.3). The experiment compares several forward solvers (Appendix C.1.2); the price is the computation of 20 runs with up to $10p$ dummies for every gate.

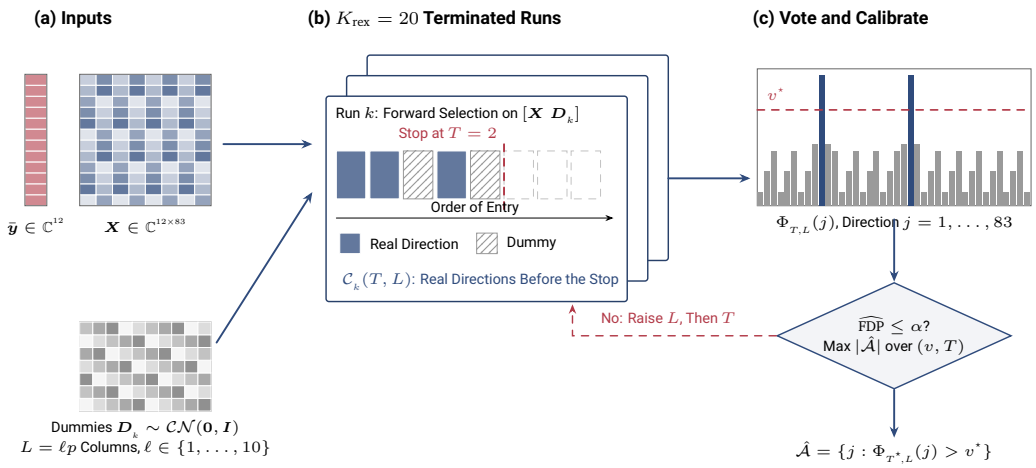


Figure 5.4.: The complex T-Rex selector. (a) The measurement, the steering matrix and complex Gaussian dummies. (b) Twenty runs forward-select on the steering matrix extended by their own dummies and stop after T dummies. (c) The relative occurrence of each direction is compared with the voting level $v^\star$; the calibration raises the number of dummies L and the stop T until the estimated false discovery proportion stays below α.

5.2.4. Infini-Rex ★

CT-Rex tests every direction against one criterion: whether it entered the selection before the dummies. Twenty runs estimate the answer. Virtual dummies already avoid storing the dummies and sample only their projections onto the residual [17], [149]. Infini-Rex goes one step further. It asks the same question in the limit of infinitely many runs and answers it in closed form, because the distribution of such a projection is known (Figure 5.5). It follows one forward selection and computes at every step the probability that a random dummy would have entered instead.

One step of the race. Infini-Rex runs OMP on $\boldsymbol{X}$ [150]. At step t, the residual $\boldsymbol{r}_{t-1}$ is what the first $t-1$ selected columns cannot explain, and the column with the largest correlation enters, $j_t = \arg\max_j |\boldsymbol{x}_j^H \boldsymbol{r}_{t-1}|^2$, with the explained share

$$\rho_t = \frac{\left|\boldsymbol{x}_{j_t}^H \boldsymbol{r}_{t-1}\right|^2}{\|\boldsymbol{r}_{t-1}\|_2^2} \in [0, 1]. \tag{5.6}$$

A dummy $\boldsymbol{d} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{I}_n)$ would have entered instead if its own share were larger. Its direction is uniform on the unit sphere of $\mathbb{C}^n$, so for any fixed $\boldsymbol{r}$ the share follows a $\text{Beta}(1, n-1)$ distribution (Appendix B.1.3), and the probability that the dummy beats step t is

$$P_t = \mathbb{P}\left(\frac{\left|\boldsymbol{d}^H \boldsymbol{r}_{t-1}\right|^2}{\|\boldsymbol{d}\|_2^2 \, \|\boldsymbol{r}_{t-1}\|_2^2} \ge \rho_t \right) = (1 - \rho_t)^{n-1}. \tag{5.7}$$

A strong step, with ρ_t near one, is almost never beaten; a step that picks noise explains about $1/n$ and is beaten with probability near e^{-1}.

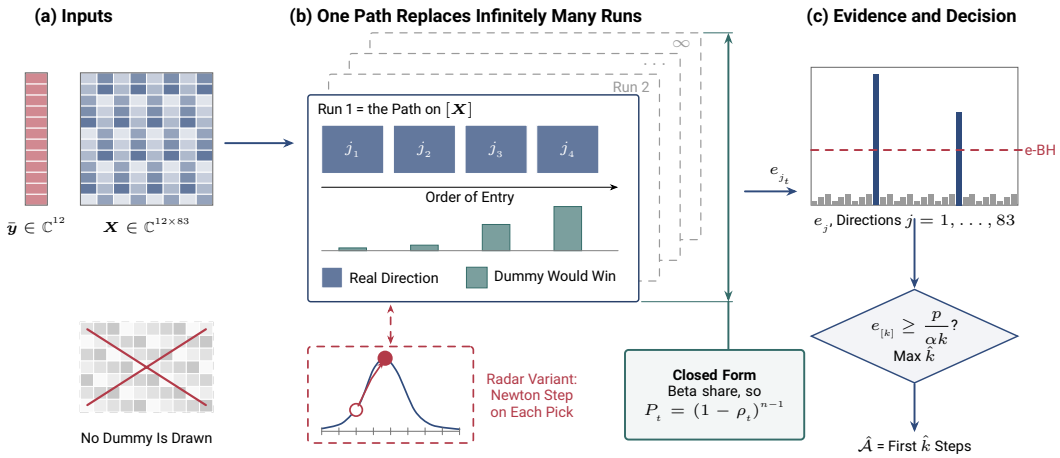


Figure 5.5.: Infini-Rex and its radar variant, drawn like the CT-Rex selector of Figure 5.4. (a) The inputs are the same, but no dummy is drawn, following the idea of virtual dummies [17]. (b) One path replaces infinitely many runs: the real directions enter in order, and the green bar shows the chance that a dummy would have won at this step. The two pointers mark that the closed form covers all runs at once. The radar variant moves each pick by a Newton step to the peak of the spectrum. (c) The e-values are compared with the e-BH line, and the first $\hat{k}$ steps are reported.

From the race to evidence. A run of T-Rex with L dummies and $T = 1$ follows this path until the first dummy enters. Column j_t is selected before every dummy if and only if no dummy beats any of the steps $1, \ldots, t$. One dummy beats at least one of them with probability at most $S_t = \sum_{\tau \leq t} P_\tau$, and the L dummies are independent, so given the measurement column j_t wins with probability at least $(1 - S_t)_+^L$, the outcome of infinitely many runs. An e-value is a nonnegative statistic with $\mathbb{E}[e] \leq 1$ under the null, the payoff of a fair bet against it, and Markov's inequality gives $\mathbb{P}(e \geq 1/\alpha) \leq \alpha$ [94]. A null direction that races on equal terms against L dummies wins with probability at most $1/(L+1)$, so $(L+1)$ times the winning probability is an e-value. Averaging over several L avoids choosing one and keeps validity,

$$e_{j_t} = \frac{1}{|\mathcal{L}|} \sum_{L \in \mathcal{L}} (L+1)\,(1 - S_t)_+^L\,, \qquad \mathcal{L} = \{p, 2p, 5p, 10p, 20p, 50p, 100p\}, \tag{5.8}$$

with $e_j = 0$ for every direction that never enters. A large L is needed because a term never exceeds $L+1$ and the first report must reach p/α. Yet a large L nearly cancels the term when S_t is not tiny, since $(L+1)(1-S_t)^L$ peaks near $L = 1/S_t$. No single L suits every step, so the set spans two decades from p to $100p$ in steps of 1, 2 and 5. Any other set gives valid e-values; the thesis does not test the choice. The e-BH rule (Eq. (2.65)) selects, with the e-values sorted in decreasing order,

$$\hat{k} = \max\left\{k \colon e_{[k]} \geq \frac{p}{\alpha k}\right\}. \tag{5.9}$$

Since S_t grows with every step, the e-values fall along the path, and the selection is the first $\hat{k}$ steps. The path ends as soon as $e_{j_t} < p/(\alpha s_{\max})$ with $s_{\max} = \lfloor n/2 \rfloor = 6$, because no later step can pass Eq. (5.9). One path of at most six steps replaces twenty runs.

Assumption 5.1 (Fair race)**.** For every null direction j and every $L \in \mathcal{L}$, column j is selected before all of L independent dummies with probability at most $1/(L+1)$.

Proposition 5.2 (FDR control of Infini-Rex)**.** *Under Assumption 5.1, every e-value of Eq.* (5.8) *with* $\beta_j = 0$ *satisfies* $\mathbb{E}[e_j] \leq 1$*, and the selection of Eq.* (5.9) *has* FDR $\leq \alpha\, p_0/p \leq \alpha$*, where* p_0 *is the number of null directions.*

The proof applies the e-BH theorem to the e-values of the race (Appendix B.1.3).

Assumption 5.1 fixes the dummy design. Each dummy is a complex Gaussian vector, $\boldsymbol{d} \sim \mathcal{CN}(\mathbf{0}, \boldsymbol{I}_n)$, so its direction is uniform on the unit sphere. The race is fair if the residual left by the true reflectors is white noise, because its direction is then uniform as well, and a fixed unit-norm column and a dummy share the same Beta$(1, n-1)$ law of the explained share (Appendix B.1.3). The proposition needs no independence between neighbouring directions, which the BH test needs and a steering matrix does not provide.

Two cases break the fairness. A reflector on the grid equals one column of $\boldsymbol{X}$, so the model is exact and only noise remains. A reflector between grid points equals no column: the nearest columns explain part of its echo and leave a fixed remainder in the residual, which the neighbouring columns correlate with and keep winning against the dummies. The plain selector then has no guarantee; in simulation between grid points it still stays below the level (Appendix C.1.2). The radar variant removes the remainder by refinement, so its guarantee holds if the refined residual is noise-like, which the data support but do not prove. An extended object returns echoes from several scattering centres and fills several columns at once. These columns are no nulls, because they carry real echoes, and a report beside the nominal position counts as false although it is physical. The guarantee covers only directions without any echo, so it says nothing about the reports on an extended object. The experiment uses point-like corner reflectors for this reason, and Chapter 9 reports the FDR on extended objects.

Infini-Rex for Radar. The radar variant removes the remainder before it can win. Following NOMP [108], it selects each new direction on the fine grid but moves every direction found so far to the best nearby value of u on the continuous steering manifold before it forms the next residual. For direction i, it maximises the correlation with the residual $\boldsymbol{r}_{-i}$ of all other directions,

$$\hat{u}_i = \underset{|u-u_{j_i}|\leq 1/(2nd)}{\arg\max} \left| \boldsymbol{a}(u)^H \boldsymbol{r}_{-i} \right|^2 , \tag{5.10}$$

by Newton steps within half a Rayleigh cell of its grid direction, and sweeps until no direction moves. Two further changes keep the residual noise-like, as the dummies require: the channels are divided by their noise level, estimated per channel as the median power over all range bins beyond the minimum range and all snapshots, divided by $\ln 2$ (the ratio of median to mean of an exponential law), so that the noise is white, and a new column enters the race after the removal of its projection onto the refined steering vectors. Appendix B.1.2 states both variants as Algorithm 3.

The guarantee of the radar variant carries over with one correction. At step t, the residual lies in the complement of the $t-1$ refined steering vectors, which has dimension $n-t+1$, and a dummy orthogonalised in the same way has a share that follows Beta$(1, n-t)$. The exact race probability is therefore $(1-\rho_t)^{n-t}$, and the proof of Proposition 5.2 applies with this exponent under Assumption 5.1 for the refined path. The implementation uses the exponent $n-1$, which is exact at the first step and smaller than the exact probability at later steps by the factor $(1-\rho_t)^{t-1}$. For a step that picks noise, the e-value is nearly zero with either exponent, and for a true reflector P_t is negligible with either exponent, so the choice changes the e-values only at the borderline. Proposition B.2 in Appendix B.1.3 states the guarantee for the implemented race.

5.3. Results

5.3.1. Performance on Measured Data

Table 5.1 compares all methods on the measured session, which holds ten takes with up to three reflectors each. Every method reports directions of reflectors for every range gate. The table lists the metrics of Eqs. (5.1) and (5.2), FDR and TPR, and the angle error RMSE_θ in degrees. The selectors aim at an FDR of at most 0.1. The classical estimators trade one error for the other. MUSIC and ESPRIT find every reflector but report mostly false ones (68 % of their reports), whereas OS-CFAR reports no false reflector and misses 22 % of the real ones. The BH test exceeds its level: its FDR is 0.23 at $\alpha = 0.1$.

The knockoff selectors are too cautious at one snapshot. Model-X knockoff+ selects nothing, and Fixed-X knockoff+ finds 7 % of the reflectors. Without the $+1$ in the threshold, both find 91 and 89 % of the reflectors, at an FDR of 0.12 and 0.22.

CT-Rex and Infini-Rex report no false reflector. Infini-Rex for Radar finds 94 % of the reflectors at an FDR of 0.02 and an angle error of 0.45°, and it needs 1.4 ms per gate, 48 times less than CT-Rex with OMP (67.5 ms). Simulation between grid points shows the same order: the BH test exceeds its level, the selectors of the T-Rex family stay below it, and Infini-Rex for Radar finds almost all reflectors from 10 dB on (Appendix C.1.2).

The Cramér-Rao bound of Eq. (2.33) marks the smallest angle error that an unbiased estimator can reach for one reflector. For twelve channels and one snapshot, it is 0.14° for the weakest reflector of the session, which exceeds the median of the range profile by 18 dB, and 0.008° for the strongest, at 43 dB; the ratio to the median stands for the SNR per channel. With 128 snapshots, the bound for the weakest reflector falls to 0.012°. The measured angle errors are larger. The best method, Infini-Rex for Radar, reaches 0.45° at one snapshot and 0.44° at 128, three times the bound of the weakest reflector.

	$K = 1$			$K = 128$			
Method	FDR	TPR	RMSE_θ	FDR	TPR	RMSE_θ	ms/gate
Bartlett + MDL	0.72	0.89	0.50	0.81	0.89	0.50	0.3
MUSIC + MDL	0.68	1.00	0.94	0.77	1.00	1.12	0.3
ESPRIT + MDL	0.68	1.00	1.23	0.76	1.00	1.32	0.2
OS-CFAR	0.00	0.78	0.46	0.07	0.78	0.46	0.9
Benjamini-Hochberg (OLS)	0.23	0.89	0.50	0.59	0.89	0.50	0.3
Model-X knockoff+	0.00	0.00	–	0.00	0.00	–	589.6
Model-X knockoff (offset 0)	0.12	0.91	1.31	0.23	0.89	1.35	723.0
Fixed-X knockoff+ (low-dim.)	0.08	0.07	0.45	0.07	0.06	0.45	52.3
Fixed-X knockoff (offset 0, low-dim.)	0.22	0.89	0.50	0.26	0.89	0.50	49.6
CT-Rex t-LARS	0.00	0.84	0.45	0.09	0.84	0.49	92.9
CT-Rex t-OMP	0.00	0.89	1.05	0.04	0.89	0.95	67.5
T-Rex t-LARS (real split)	0.00	0.84	0.77	0.09	0.84	0.78	119.2
Infini-Rex	0.00	0.89	1.05	0.10	0.89	0.95	1.0
Infini-Rex for Radar	0.02	0.94	0.45	0.13	0.94	0.44	1.4
DNN (MLP, 3 hidden layers)	0.05	0.94	1.44	0.32	0.89	0.97	0.3

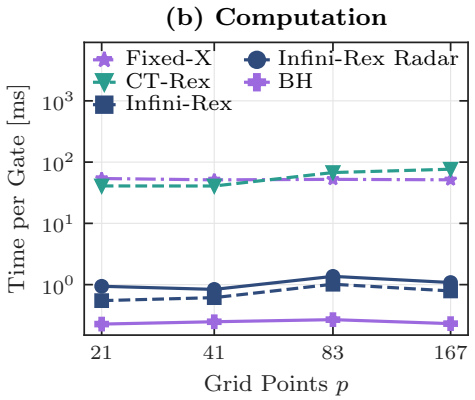


Table 5.1.: Measured session at $\alpha = 0.1$. (a) Achieved FDR, TPR and angle RMSE in degrees with one snapshot ($K = 1$) and with all 128 snapshots of a frame, and the computation time per range gate. (b) Computation time per gate against the number of grid directions p.

5.3.2. Computation Time

The last column of Table 5.1 lists the time per range gate, and panel (b) of the same table plots it against the number of grid directions. The classical estimators, the BH test and the network finish a gate in 0.2 to 0.3 ms, and OS-CFAR in 0.9 ms. The knockoffs need 50 ms (Fixed-X) and 590 to 720 ms (Model-X). CT-Rex needs 68 to 93 ms and the real-valued T-Rex 119 ms. Infini-Rex needs 1.0 ms and its radar variant 1.4 ms, which is 68 and 48 times less than CT-Rex with OMP.

The time depends little on the grid size. From 21 to 167 directions, the time of CT-Rex doubles, whereas the time of Fixed-X, of the BH test and of Infini-Rex barely changes.

5.3.3. Closely Spaced Reflectors

Figure 5.6 shows the two hardest takes as the methods report them. In the close pair, MUSIC reports six reflectors per frame on average, OS-CFAR, BH, CT-Rex and Infini-Rex report one, and only Infini-Rex for Radar reports two, one on each side of the pair. In the pair that differs only in range, every method reports a single reflector, 2 cm in front of the rear reflector, and none reports the front one. The reports of the close pair lie 4 cm behind the reflectors, the range bias of the gate (Appendix C.1.2).

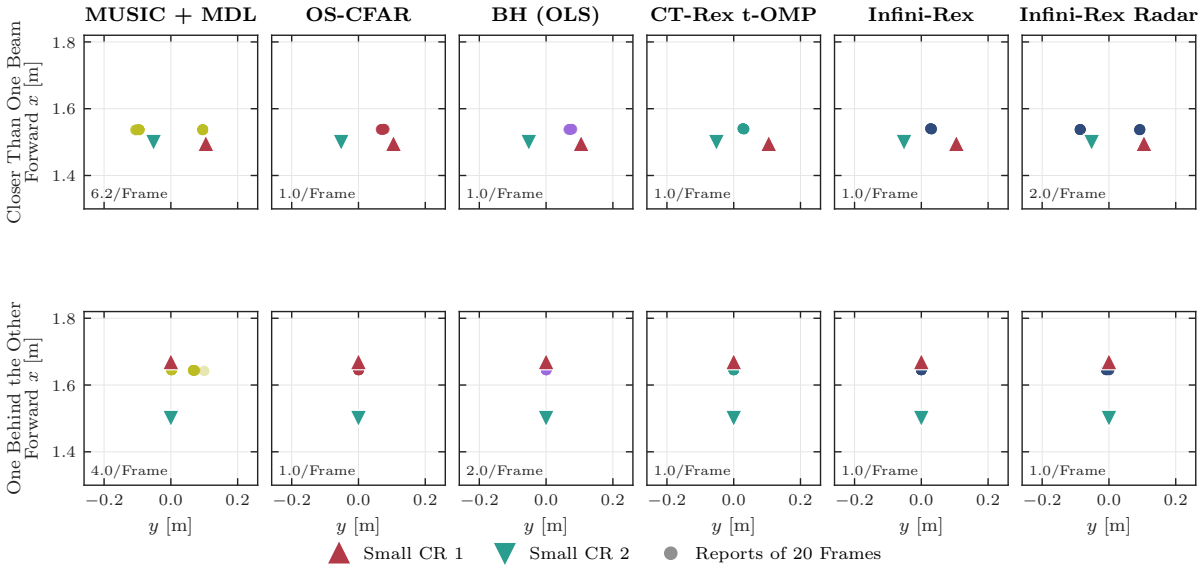


Figure 5.6.: The two hardest takes as the methods report them, in top view. Each panel shows the reflectors (triangles) and the reports of one method over the 20 frames of the take, from one snapshot. The numbers give the mean count of reports per frame. The ranges are projected onto the table plane with the elevation of the gate ($+2.0^\circ$ and $+0.5^\circ$, estimated from the elevation row); the triangles use the height of their stands.

Figure 5.7 shows the close pair as a measurement. Both reflectors form one blob of power near 1.5 m, and the azimuth-elevation map shows that their directions differ in azimuth and in elevation. CT-Rex and Infini-Rex for Radar place their reports on the blob, OS-CFAR reports one reflector between the two, and MUSIC adds a second group at 1.07 m, where the truth lists no reflector. In elevation, the reports of three methods lie near

$+2^\circ$, between the true heights of -4.2° and $+1.5^\circ$, and the reports of MUSIC spread along the ridge of the map.

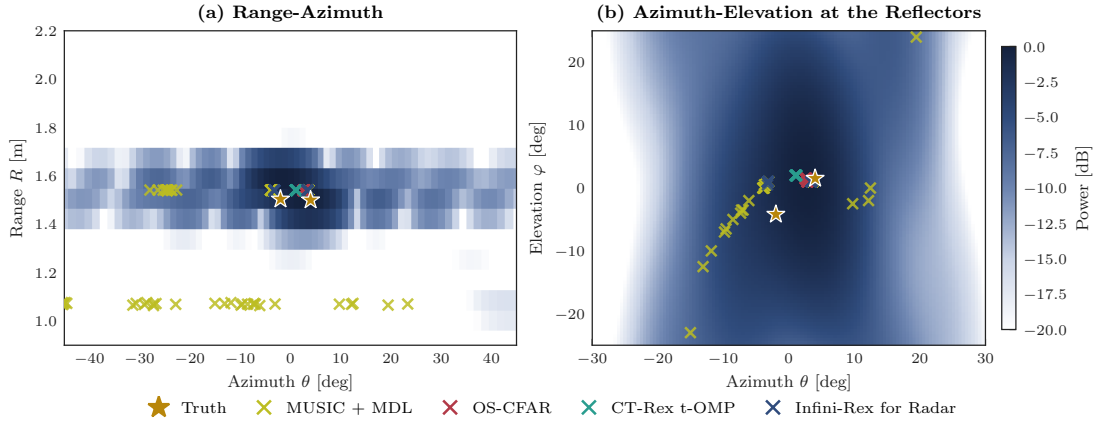


Figure 5.7.: The close pair as a measurement: range-azimuth power with 128 snapshots and the reports of four methods from one snapshot (crosses, 20 frames) (a), and azimuth-elevation power at the two reflectors (b). Stars mark the truth, crosses the reports; in (b) each report takes the elevation of the map maximum at its azimuth.

5.3.4. Dependence on the Nominal Level

Figure 5.8 varies the nominal level. The achieved FDR of the BH test hardly follows α and stays between 0.17 and 0.23. The power of the other methods rises with α: Fixed-X knockoff+ finds 7 % of the reflectors at $\alpha = 0.1$ and 67 % at $\alpha = 0.5$, CT-Rex with OMP selects nothing at $\alpha = 0.02$ and 89 % from $\alpha = 0.1$ on, and Infini-Rex and its radar variant find 28 and 33 % at the strictest level and about 90 % or more from $\alpha = 0.05$ on. The three selectors of the T-Rex family stay below the nominal level at every α.

5.3.5. Dependence on the Number of Snapshots

Figure 5.9 compares simulation and measurement over the number of snapshots K. In simulation, more snapshots help: the FDR of MUSIC falls from 0.74 to 0.04, and the selectors with a guarantee stay below the level at every K. On measured data, the trend reverses for the methods without a guarantee, since the FDR of MUSIC and of the BH test rises with K (from 0.68 to 0.77 and from 0.23 to 0.59).

The selectors with a guarantee also rise on measured data, from near zero at $K = 1$ to 0.04 (CT-Rex), 0.10 (Infini-Rex) and 0.13 (Infini-Rex for Radar) at $K = 128$; only the last exceeds α. The TPR stays constant over K.

5.4. Discussion

The results point in two directions. The bound fails where the room adds echoes that the answer key leaves out: at $K = 128$ the FDR of the radar variant rises to 0.13, above the level of 0.1, and two reflectors in one

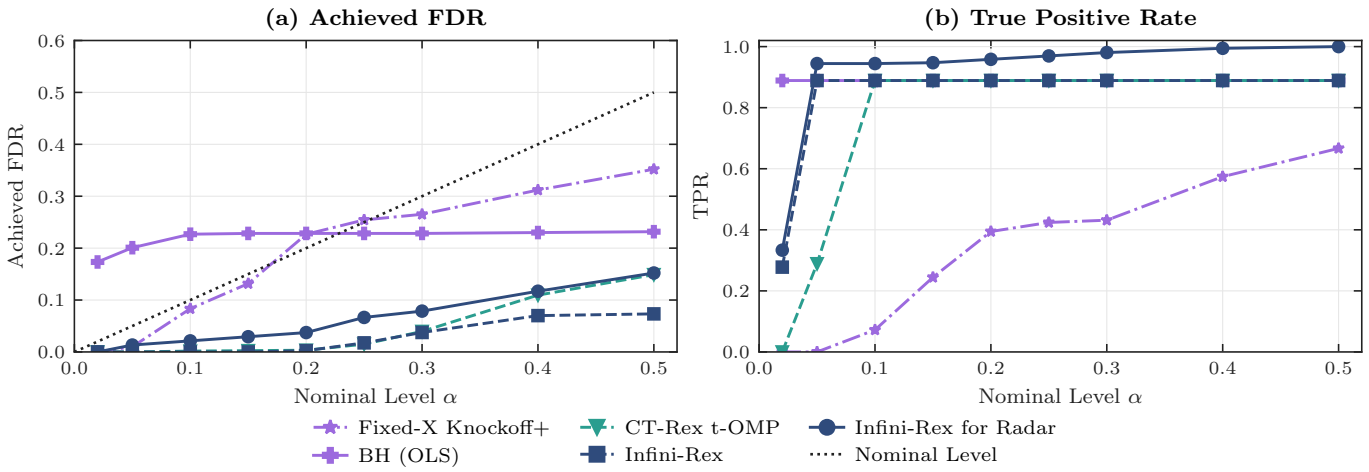


Figure 5.8.: Achieved FDR (a) and TPR (b) against the nominal level α on the measured session with one snapshot. Filled markers denote FDR-controlled methods, hollow markers methods without a guarantee.

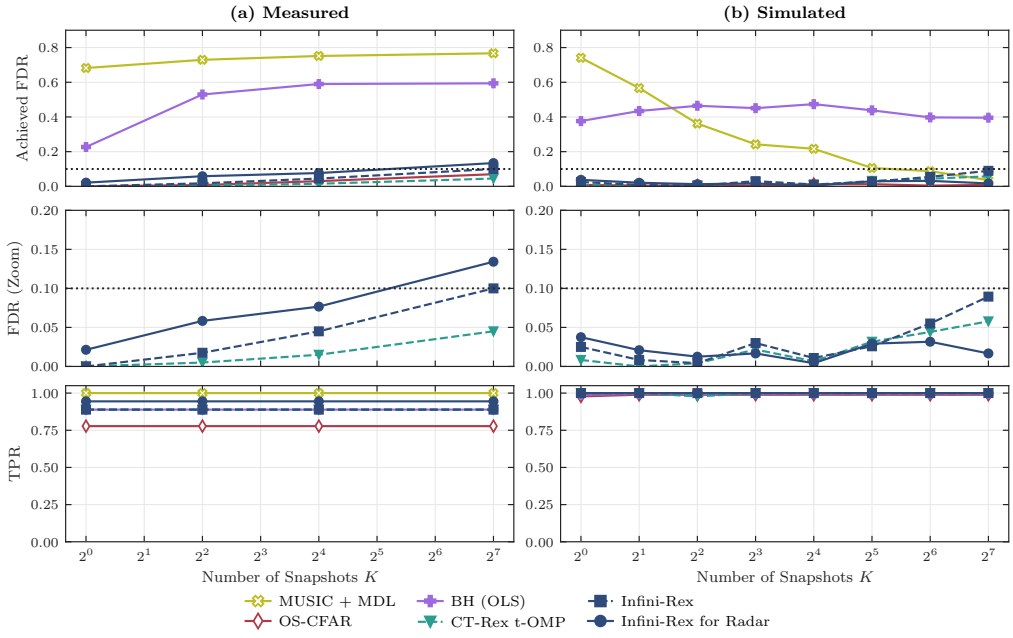


Figure 5.9.: Achieved FDR and TPR over the number of snapshots K at $\alpha = 0.1$ (dotted) on the measured session (a) and on simulated data at 15 dB per snapshot (b). The middle row zooms into the FDR of the selectors with a guarantee.

range gate stay unresolved. Under every other condition, however, CT-Rex and Infini-Rex stay at or under the level, and Infini-Rex for Radar finds 94 % of the reflectors with an angle error of 0.45° in 1.4 ms per gate. The question is whether the bound of Infini-Rex survives a real room, or whether it survives only where the room is quiet.

5.4.1. Performance on Measured Data

The selectors with a guarantee find most reflectors without reporting false ones, and the test they use explains why. Infini-Rex adds random extra directions, called dummies, to the problem. A dummy cannot be a reflector, so it shows how large a score can become by chance. A real direction counts only if its score beats these chance scores by a margin that grows as the level α shrinks. The bound holds if what remains after the reflectors are removed behaves like noise, so that every wrong direction has the same chance as a dummy to look strong (Assumption 5.1). The measured session is a hard test of this condition. Twelve channels sit up to a tenth of a wavelength off their ideal positions, some reflectors lie between grid points, and walls and a rail add echoes. The plane-wave model of Eq. (5.4) is the one assumption that holds without doubt: the nearest reflector, at 1.25 m, lies five times beyond the far-field distance of 0.24 m of the aperture (Section 2.1.1).

The reflectors between grid points threaten the fair test, because the part of their echo that the grid cannot represent is a fixed signal and not noise [87]. Infini-Rex for Radar removes this part by refining the direction of every selected reflector. The measurements agree with this reasoning: the radar variant finds 94 against 89 % of the reflectors with less than half the angle error (0.45° against 1.05°). An angle error of 0.45° is one twentieth of the Rayleigh cell of 9.6°, the width within which a standard beamformer cannot separate two reflectors.

The angle errors exceed the Cramér-Rao bound for two reasons, and the second reason hides the first. First, the model of Eq. (5.4) assumes exact channel positions and reflectors on the grid, but the real array carries phase errors, and an off-grid reflector leaves a remainder. Both raise the error above the bound that holds for the true model. Second, the ground truth rests on tape marks with an offset of up to 1.2° in azimuth. This offset exceeds the bound of the weakest reflector (0.14°) eightfold, so the measured error cannot show how close an estimator comes to the bound. An error below the bound would raise suspicion instead: it would point to a wrong ground truth. The measured 0.45° therefore limits the true error of the radar variant from above and does not measure its distance to the limit.

The caution of the selectors costs reflectors. An FDR of 0.00 lies far below the level of 0.1, and the takes show what is lost. CT-Rex and Infini-Rex find every reflector in seven of the nine takes with reflectors, the radar variant in eight, and the missed 11 % sit in two takes, the close pair and the pair that differs only in range (Figure 5.6). The evidence is narrow. The 20 frames of a take repeat the noise, but ten takes are ten scenes of one array in one room. The ground truth rests on tape marks, which lie a few millimetres from the phase centre of a reflector, so the reports of a take share an offset of 1 to 10 cm in range and up to 1.2° in azimuth. This offset is the same for all methods and stays inside the matching tolerance, so FDR and TPR do not change. It enters the angle error, however, and the 0.45° of the radar variant is an upper bound of its true error.

The methods without a bound fail where their assumptions fail. MDL, which counts the reflectors, assumes equal noise eigenvalues [44]. The covariance of a real gate holds the echoes of wall and rail, and MDL reaches its limit of seven sources for two reflectors (Figure 5.3). MUSIC inherits this count and reaches an FDR between 0.68 and 0.77. The BH test needs p-values from a least-squares fit on the coarse grid, and an off-grid reflector leaks into its neighbours there [87]. Knockoff+ can certify a selection only if it holds at least $1/\alpha$ directions [30] (Eq. (2.67)), while a gate holds one to three reflectors, and without the $+1$ the knockoffs find 89 to 91 % of the reflectors at an FDR of 0.12 and 0.22, above α. These failures do not condemn the methods. They show that a radar gate breaks the assumptions for which the methods were built. In short, the selectors with a guarantee succeed because their assumption survives the room, and the other methods fail because theirs does not.

5.4.2. Computation Time

The table sorts the methods into three groups, and each group is fast or slow for its own reason. The classical estimators, the BH test and the network compute one closed-form step: a decomposition of a 12×12 matrix, one least-squares fit on eleven directions or one forward pass. They finish within a third of a millisecond, because they never ask whether a report is right. They deliver a number and no error statement.

The methods with a guarantee ask that question, and T-Rex answers it by repetition. It asks whether a direction enters the selection before a random dummy. One experiment cannot answer this, because a dummy is random. The selector therefore runs $K_{\text{rex}} = 20$ experiments with independent dummies and counts how often each direction wins. Every experiment is a complete forward selection that stops only when T dummies have entered, and the calibration repeats the experiments with more dummies and a larger T until the estimated share of false selections stays below α. The costs multiply: 20 runs, each as long as T dummies need to enter, each over p real and up to $10p$ dummy columns. The 20 runs belong to the method. They differ from the Monte Carlo trials with which a paper estimates the FDR. The curve of CT-Rex in panel (b) of Table 5.1 suggests that overhead dominates: the time doubles while the grid grows eightfold, so the arithmetic of one step does not set the cost. The start of 20 runs, the draw of the dummies and the calibration loop set it.

Infini-Rex asks the same question for the smallest stop, $T = 1$, and increases the number of runs without limit. For $T = 1$, a run ends at the first dummy, and the chance that a dummy enters before a given step follows from the Beta law of Eq. (5.7). Infinitely many runs therefore reduce to a formula, and one forward path of at most six steps over p columns replaces the 20 runs and all dummy columns. The factor of 48 to 68 agrees in order of magnitude with this reduction: 20 runs, and for each run more columns and more steps than the single path. The code was not profiled, so this is an estimate. The choice $T = 1$ enables the closed form. A larger T allows a run to select more directions before it stops, but the joint law of several dummy entries is not known in closed form here. Whether a larger T pays off in power or in speed remains open.

The knockoffs differ for a different reason. Fixed-X knockoffs treat the steering matrix as fixed and work on the coarse grid of 22 unknowns and 24 rows: one decomposition of a small matrix and one lasso path cost 50 ms. Model-X knockoffs work on the fine grid of 83 directions and need a model for the distribution of the columns. The columns of a steering matrix are deterministic, so Model-X must fit a covariance of 83×83 before it draws the knockoffs and then runs the lasso path over twice as many columns. This takes 12 times longer than Fixed-X. The size of the problem is consistent with this ratio, but the code was not profiled either.

The timing decides what a radar can use. A look of the tracking experiment proposes up to 24 cells. At 1.4 ms per cell, Infini-Rex needs about 34 ms and fits into a frame of 105 ms. CT-Rex with OMP needs about 1.6 s for the same cells. The absolute times depend on the Python implementation, but the ratios between the methods do not.

5.4.3. Closely Spaced Reflectors

The close pair stands at 0.63 of the Rayleigh cell. Two reflectors this close produce steering vectors that correlate with

$$\mu(\Delta u) = \left| \frac{\sin(\pi n d\, \Delta u)}{n \, \sin(\pi d\, \Delta u)} \right|, \tag{5.11}$$

where n is the number of channels, d their spacing in wavelengths and Δu the difference of the directions in the sine domain. The value is 0.47 for $n = 12$ and $d = 1/2$, so the two columns of $\boldsymbol{X}$ share 22 % of their

energy, and a beamformer blurs the pair into one peak. A sparse model can still separate the pair [147] if the signal-to-noise ratio is high enough, and it is here, since every reflector exceeds the median of the range profile by 18 to 43 dB at one snapshot. CT-Rex and Infini-Rex stop at one report per frame, and only the radar variant reports two, one on each side of the pair (Figure 5.6). The refinement is the difference: it moves each selected direction to its best position between grid points, which supplies the precision to place two reflectors inside one cell.

The pair that differs only in range cannot be separated by angle, because both reflectors have the same steering vector. Their distance of 17 cm equals two range cells of 8.37 cm [9], so the two lobes overlap in one gate and add with the phase

$$\Delta\varphi = \frac{4\pi\,\Delta R}{\lambda}, \tag{5.12}$$

which is 550 rad for $\Delta R = 17$ cm and $\lambda = 3.9$ mm. In a static scene, this phase is fixed and arbitrary: a shift of 1 mm turns the sum from constructive to destructive. The single report at 1.65 m, 15 cm behind the front reflector and 2 cm in front of the rear one, shows this effect. The pair at -25° and -1° keeps the distance but yields two gates, because the echoes add with another phase in every channel and the cross term averages out over the array. A selector cannot recover a reflector that no gate contains, so this loss belongs to the shared detector and not to the selectors.

The elevation row does not rescue the pair (Figure 5.7b). The reflectors differ in elevation by 5.6° (-4.2° and $+1.4^\circ$), but the elevation map at their gate shows one peak, at $+0.5^\circ$ and 35° wide, because the row has only four channels. The separation is 0.3 of its Rayleigh cell. The elevation places the merged echo at the right height, and a selector in elevation would need more channels to repeat in height what the radar variant achieves in azimuth.

5.4.4. Dependence on the Nominal Level

The level α sets how much evidence a method asks for before it reports a reflector, and each method pays for the promise in its own way (Figure 5.8). The BH test responds little to α. A flat response is the sign of a bias that does not shrink when the level shrinks, here the leakage of off-grid reflectors. Fixed-X knockoff+ needs at least $1/\alpha$ selections, ten at $\alpha = 0.1$ and two at $\alpha = 0.5$, so its power rises with the level. Infini-Rex pays in evidence. By Eq. (5.9), the first reflector needs an e-value of p/α, which is 830 at $\alpha = 0.1$ and 4150 at $\alpha = 0.02$ for $p = 83$ directions. One snapshot carries limited evidence, so the power falls at the strictest level. This is the price of a guarantee: a stricter promise needs more evidence than one snapshot can give.

5.4.5. Dependence on the Number of Snapshots

Averaging K snapshots lowers the noise power to σ^2/K, a gain of 21 dB at $K = 128$. In simulation, noise is the only impurity, so more snapshots help every method, and the selectors keep their bound at every K (Figure 5.9). In the room, a second impurity remains that is identical in every snapshot: the echoes that the subtraction of the empty take leaves, and the remainder of off-grid reflectors. At $K = 1$ the noise hides this impurity, and the null directions race with the dummies on equal terms. At $K = 128$ the impurity stands out. A null direction next to an unlisted echo correlates with it by construction, whereas a dummy correlates only by chance, so the test becomes unfair and Assumption 5.1 fails, as the proposition anticipates.

The location of the excess supports this account. At $K = 128$, 20 of the 45 false reports of the radar variant sit at one fixed position, 1.07 m and -52°, in two takes, and an inspection places them at a real echo of the

room that the answer key does not list; the empty take adds 25 reports at changing positions. By the protocol the reports are false, and by physics the 20 at the fixed position are true. The TPR stays constant over K, because every reflector that the array can separate already exceeds the median of the range profile by 18 to 43 dB at $K = 1$. The lesson is that snapshots and aperture are different resources. Snapshots buy certainty about what is found, and only aperture and bandwidth buy the power to separate.

The selectors keep one advantage that the averaged numbers hide: they are valid at a single snapshot, and the classical estimators are not. A subspace method needs the sample covariance of several snapshots, and one snapshot yields a covariance of rank one, which smoothing repairs only by giving up aperture. In simulation, this shows. The FDR of MUSIC falls from 0.74 with one snapshot to 0.04 with 128, whereas the selectors keep their bound at every K. In the room, more snapshots help no method, so the selectors gain nothing from averaging and lose nothing without it. One snapshot also suits a moving scene. The phase of a moving reflector changes from snapshot to snapshot, so an average blurs it, and the tracking experiment accordingly detects on the cell vector of one snapshot (Chapter 8).

5.4.6. Findings and Scope

A bound on the share of false discoveries survives the step from simulation to a real room as long as what the reflectors leave behind stays noise-like. CT-Rex and Infini-Rex stay at or under α for up to 16 snapshots and at every level of Figure 5.8, and the room breaks the bound at 128 snapshots only in the way the proposition predicts. The experiment extends the single-snapshot selection of Scheidt and colleagues, which they tested in simulation with 80 sensors on a grid [47], to a real board with twelve channels and to directions between grid points. One closed-form path replaces twenty random runs and costs 1.4 ms per gate against 67.5 ms for CT-Rex.

The claim has three limits. Proposition 5.2 depends on Assumption 5.1, which the refinement supports but cannot prove for every scene, and the implementation of the radar variant uses an exponent that is exact only at the first step. The data are ten takes of one array in one room, and all selectors share the gates of one detector. A guarantee is a statement about a model, and the room is the test of that model. In this room, the guarantee held wherever the room left only noise, and it moved only where the room added echoes that the answer key omits.

6. Material Classification

A metal pole and a curtain of rain can return the same power from the same range. A vehicle brakes for the pole and may drive through the rain, so the decision rests on what an echo says beyond its strength. It says a great deal. The material sets how much of the wave is reflected, because the reflected share grows with the permittivity, and the motion sets how the phase of the echo evolves, because every moving scatterer adds its own Doppler shift. A pole returns one steady phase, and a curtain of drops returns thousands of phases that disagree. Both facts are physics, and neither is a label. The research question concerns how much of the nature of an object a millimetre-wave radar can read from its echo alone, in real time: what the object is made of and how it moves. The scene (Section 6.1) sets rain, plastic, metal and a walking person in front of the radar and defines the task. The methodology (Section 6.2) builds Material-Motion Sparse Representation Classification (M^2-SRC), a classifier whose decisions follow from physical features and name the training windows they rest on. The results (Section 6.3) show the features, the dictionary, one decision, the optimiser and the performance on an unseen session, and the discussion (Section 6.4) holds them against the physics of the scene.

6.1. Scene: Matter in Motion

Target detection is hard because an echo contains no definition of a target. A peak above the noise says that something reflects, not what it is. The first handle on identity is the material. Plastics, water and metals answer a millimetre wave in different ways (Section 2.1.4): a boundary returns the share $|\Gamma|^2$ of the incident power (Eq. (2.6)), a metal returns all of it, polycarbonate with $\varepsilon_\mathrm{r} = 2.9$ returns 7 % [79], and water returns a far larger share, because its permittivity is high and lossy at millimetre waves [57]. The material alone does not complete the picture, because a scene may move: rain and a person both consist of water, one as a cloud of independent drops and the other as limbs that swing against a torso. The phase of the echo over time carries this second answer, the micro-Doppler effect.

The objective is therefore twofold. From one window of echo, the radar decides whether an object is present and then names its material and its motion. The truth is known by construction: one object at a time stands or moves in front of the radar, between 0.3 and 1.6 m and within $\pm 25^\circ$. Three materials and four motions cover the objects. *Metal* reflects almost all incident power. *Dielectric*, such as plastic or cardboard, reflects little and lets part of the wave pass. *Waterlike* covers water and the human body, which reflect strongly. A *static* object stands still, a *rigid* object moves as one body, like a swinging ball, an *articulated* object moves its parts against each other, like the limbs of a walking person, and a *distributed* object consists of many small scatterers that move independently, like falling drops. The metal objects are aluminium and brass trihedral corner reflectors and a steel ball, the dielectric objects are plastic boxes, a bag and a radome, and the waterlike objects are water bottles, a person standing, walking or gesturing, and water poured or dripping. Appendix C.2.1 shows the objects and the data set.

Rain is the hardest case and opens the scene (Figure 6.1). A bottle with small holes in its cap stands in front of the radar, and water poured into it falls as drops through the beam (Figure 6.1a). Against the empty room, the echo of the window rises by 13 dB at 0.67 m (Figure 6.1b), so a detector finds the object, but this map says nothing about what falls. The micro-Doppler of the window answers (Figure 6.1c): the echo fills a broad band of radial speeds around zero, because thousands of drops move independently, while a rigid object traces one curve.

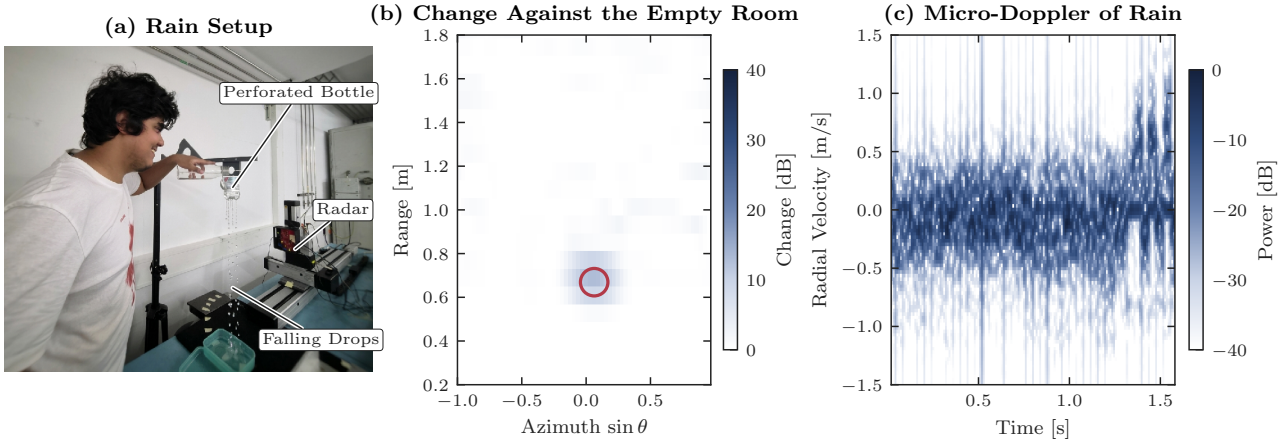


Figure 6.1.: The rain scene. (a) Water poured into a perforated bottle falls as drops in front of the radar. (b) The change of the echo against the empty room locates the object (circle). (c) The micro-Doppler of the window, radial speed against time, fills a broad band.

A window spans 1.6 s, which is 40 frames of 40 ms with 128 snapshots each; a snapshot is one pass of the four transmitters and takes 0.25 ms (Section 2.1.3). A new window starts every 0.4 s, the pace that the classifier must keep. The platform of Chapter 4 makes this pace possible: the AWR2944P (Section 4.1.1) records the frames of each window, and the data acquisition (Section 4.5) delivers them to the classifier during the recording. The code of this experiment is in the folder `material_classification/` of the code base (Appendix D). Three sessions yield 56 takes and 1 551 windows, 314 of them without an object. Sessions A and A2 (42 takes) train the classifier, and session B (14 takes), recorded separately, tests it; in leave-one-take-out, a model trained on all other takes of the training sessions classifies each take.

The task reads as follows. Given the 40 frames of a window and a recording of the empty room, find whether an object is present, its material (metal, dielectric or waterlike) and its motion (static, rigid, articulated or distributed), subject to one decision per 0.4 s, measured by the accuracy, the macro-F1 score and the detection and false alarm rates. A window is correct if the predicted label equals the label of its take. The macro-F1 score averages the F1 score over the C classes of a head,

$$\mathrm{F}_1^{\text{macro}} = \frac{1}{C} \sum_{c=1}^{C} \frac{2\,\mathrm{P}_c\,\mathrm{R}_c}{\mathrm{P}_c + \mathrm{R}_c}, \tag{6.1}$$

where the precision P_c is the share of windows predicted as class c that belong to c, and the recall R_c is the share of windows of class c that are predicted as c. The detection rate is the share of windows with an object in which the object is found, and the false alarm rate is the share of empty windows in which an object is reported.

No competing classifier stands against M^2-SRC in this chapter. RadarCat classifies objects that touch the sensor, and micro-Doppler classifiers leave the material unused (Section 3.2), so neither solves this task.

Physics is the baseline. The permittivity and the geometry of the objects predict which feature separates which class, and a classifier earns trust where it reproduces these predictions and loses it where it contradicts them. One fact gives hope: in Figure 6.1c the rain fills a band that no single swinging object produces. The classes differ in the data, and what remains is a method that reads the difference, states how it decided and finishes before the next window arrives.

6.2. Methodology

The methodology starts from the physics of the echo and ends at a class label. Features turn the window into numbers whose meaning follows from the wave, a dictionary of labelled training windows turns the numbers into evidence, and a sparse nonnegative code weighs it. Sparse representation classification, the lasso and FISTA are established methods. The fourteen features, the two-head nonnegative classifier and the explanation of each decision are contributions of this thesis.

6.2.1. Echo Window and Physical Features ★

The object is found by comparing the window with the empty room. Let $P(b, a)$ be the range-azimuth power of the window, averaged over its frames, in range bin b and direction a, let $P_0(b, a)$ be the same map of the empty room and N the noise floor of the window. The change map

$$C(b, a) = 10 \log_{10} \frac{P(b, a) + N}{P_0(b, a) + N} \tag{6.2}$$

measures in decibels how much the window differs from the empty room. The object lies at the maximum (b_0, a_0) of C within the region of the scene and counts as detected if $C(b_0, a_0) \geq 8\,\text{dB}$. In every frame f, the method follows the strongest of the five bins around b_0, so that a moving object keeps its echo, and forms the slow-time signal $x_f[\kappa]$ of the object, one complex value per snapshot $\kappa \in \{0, \ldots, 127\}$.

Motion shows in the phase of the echo. A reflector with radial speed v advances the phase by $4\pi v T_\text{s}/\lambda$ from one snapshot to the next, where $T_\text{s} = 0.25\,\text{ms}$ is the snapshot interval. The pulse-pair method of weather radar [60] estimates the phase step from the lag-one autocorrelation $R_f = \frac{1}{127} \sum_{\kappa=0}^{126} x_f[\kappa + 1]\, x_f^*[\kappa]$, and the spread of speeds from the drop of correlation between lag zero and lag one,

$$v_f = \frac{\lambda}{4\pi T_\text{s}} \angle R_f, \qquad \sigma_{v,f} = \frac{\lambda}{2\sqrt{2}\,\pi T_\text{s}} \sqrt{\ln \frac{P_f}{|R_f|}}, \tag{6.3}$$

where P_f is the mean power of x_f. A rigid object has one speed at a time, so σ_v stays small while v changes. Parts that move against each other spread the speeds (Figure C.6) [59], and falling drops spread them further. Seven motion features summarise these effects over the window: speed, speed change, spread σ_v, loss of correlation between frames, fluctuation of the amplitude, and movement in range and across range.

Material shows in the strength and the shape of the echo. The reflected power grows with the permittivity (Eq. (2.6)), and the human body behaves like water [58]. Since the received power falls with r^4 (Eq. (2.10)), the reflectivity compensates the range,

$$\rho = 10 \log_{10} P(b_0) + 40 \log_{10} r_0, \tag{6.4}$$

where $P(b_0)$ is the power of the range bin of the object and r_0 its range. A material that passes the wave shows an echo behind it, and a material that absorbs the wave casts a shadow. The seven material features therefore measure the change against the empty room, the reflectivity, the back echo, the shadow, the extent of the echo in range, and the slope and ripple of the reflectivity over four sub-bands of the chirp bandwidth (Appendix B.2.1 defines all fourteen). The physics predicts an order of the classes in two of them, and these orders are the baseline against which the results are read:

$$\rho_{\text{metal}} > \rho_{\text{waterlike}} > \rho_{\text{dielectric}}, \qquad \sigma_{v,\text{static}} < \sigma_{v,\text{rigid}} < \sigma_{v,\text{articulated}} < \sigma_{v,\text{distributed}}. \tag{6.5}$$

The first order follows from the permittivity, and the second from the number of independent scatterers: none, one, a few and many.

6.2.2. Sparse Representation Classification and M^2-SRC ★

SRC of Wright and colleagues rests on one assumption: a new sample lies near the span of the training samples of its own class [110]. It codes the sample sparsely over all training samples with an ℓ_1 penalty and assigns the class whose samples rebuild it with the smallest residual. However, the classifiers of the state of the art either read objects that touch the sensor or leave the material unused (Section 3.2), so the joint reading of material and motion at a distance stays open. M^2-SRC closes this gap with three changes, and each change answers a property of radar echoes. First, two heads replace one joint classifier. The motion head reads all fourteen features, because the shape of the echo also separates a person from rain. The material head reads only the seven material features, so that the material label cannot depend on how the object moves, and every training window serves both heads. Second, the code is nonnegative: a coefficient of either sign lets one window cancel another, while a nonnegative code forces every window to vote for its class, turns the span into a cone and makes the share of the code a confidence. Third, every decision is explained, by the training windows that carry the code and by the features that carry the margin between the winning and the second class.

The dictionary is the training set. Every feature vector is standardised with the mean and the standard deviation of the training windows, clipped to ± 5 to limit outliers, and scaled to unit length, so that class residuals compare directions and not magnitudes. The $N = 867$ standardised windows with an object form the columns of $\boldsymbol{D} \in \mathbb{R}^{q \times N}$, where $q = 14$ for the motion head and $q = 7$ for the material head. A learned dictionary such as K-SVD designs compact atoms without a label [151], whereas M^2-SRC keeps every window as a labelled atom, because the class residual and the explanation by named windows need the label. A new window $\boldsymbol{y} \in \mathbb{R}^q$ is coded as

$$\boldsymbol{a}^\star = \arg\min_{\boldsymbol{a} \geq \boldsymbol{0}} \; f(\boldsymbol{a}), \qquad f(\boldsymbol{a}) = \frac{1}{2} \left\| \boldsymbol{y} - \boldsymbol{D}\boldsymbol{a} \right\|_2^2 + \lambda \, \boldsymbol{1}^\top \boldsymbol{a}, \tag{6.6}$$

with $\lambda = 0.05$. The first term is the squared reconstruction error, and the second term penalises the number of windows used. The problem is the lasso of Eq. (2.73) with $\lambda_2 = 0$ [100] and the added constraint $\boldsymbol{a} \geq \boldsymbol{0}$. For a nonnegative vector, the ℓ_1 norm equals the sum of its entries, so the penalty is $\lambda \, \boldsymbol{1}^\top \boldsymbol{a}$. FISTA solves this nonnegative lasso (Section 6.2.3). Let $\boldsymbol{a}_c^\star$ keep the entries of the windows of class c and set all others to zero. The window is assigned to the class that rebuilds it best, and the share of the code that this class carries is the confidence,

$$\hat{c} = \arg\min_c \left\| \boldsymbol{y} - \boldsymbol{D}\boldsymbol{a}_c^\star \right\|_2, \qquad \text{conf} = \frac{\boldsymbol{1}^\top \boldsymbol{a}_{\hat{c}}^\star}{\boldsymbol{1}^\top \boldsymbol{a}^\star}. \tag{6.7}$$

The rule works under one condition. Let $\mathcal{K}_c = \{ \boldsymbol{D}_c \boldsymbol{a} : \boldsymbol{a} \geq \boldsymbol{0} \}$ be the cone of the training windows of class c.

Assumption 6.1 (Class cone)**.** A window of class c lies closer to $\mathcal{K}_c$ than to the cone of every other class.

The physics supports this condition. Windows of one class share a signature, such as a high reflectivity for metal or a broad spread of speeds for rain, so their vectors point in similar directions, while windows of different classes point apart. Under Assumption 6.1, the class residual of Eq. (6.7) is smallest for the true class. The decision is explainable by construction. The nonzero entries of $\boldsymbol{a}^\star$ name the training windows it rests on, and the squared margin between the winning class 1 and the runner-up class 2 splits exactly over the features, with $\boldsymbol{e}_c = \boldsymbol{y} - \boldsymbol{D}\boldsymbol{a}_c^\star$,

$$\|\boldsymbol{e}_2\|_2^2 - \|\boldsymbol{e}_1\|_2^2 = \sum_{k=1}^{q} \left(e_{2,k}^2 - e_{1,k}^2 \right), \tag{6.8}$$

so that every term names the share of one feature in the decision. Figure 6.2 summarises the method, and Algorithm 5 in Appendix B.2.1 states it as an algorithm.

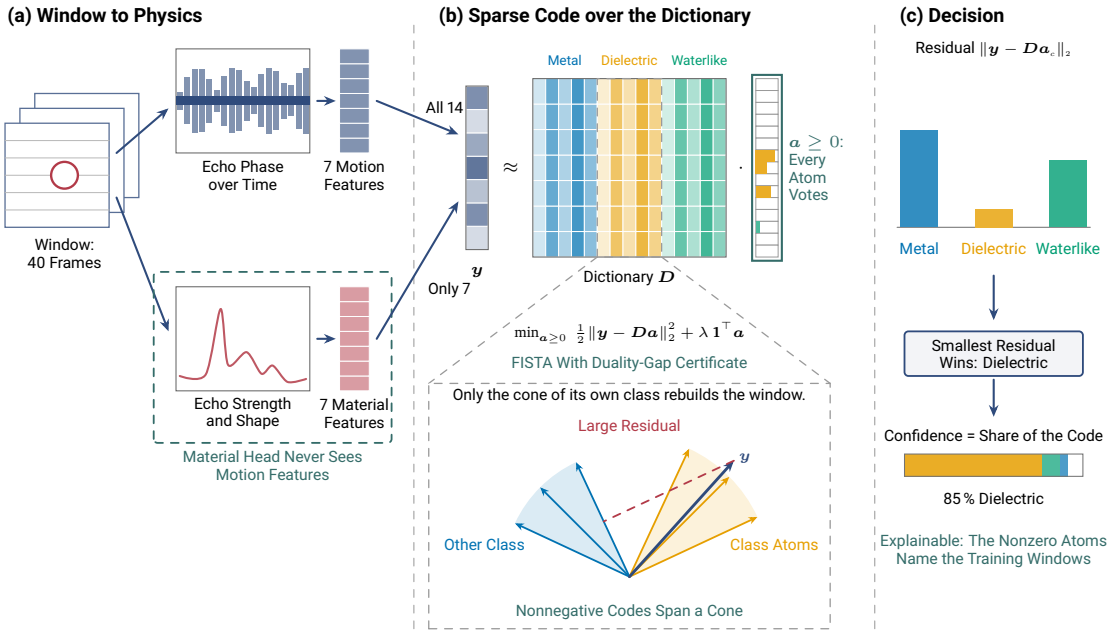


Figure 6.2.: M^2-SRC. (a) The window yields physical features of the echo phase and of the echo strength and shape; the material head never sees the motion features. (b) The window is coded as a sparse, nonnegative combination of the labelled training windows of the dictionary, solved by FISTA with a duality-gap certificate; the zoom shows why it works, because the atoms of one class span a cone. (c) The class with the smallest residual wins, the share of the code is the confidence, and the nonzero atoms explain the decision. Green marks the contributions of this thesis.

6.2.3. Solving the M^2-SRC Code

Eq. (6.6) is a convex problem, so its optimum can be computed and certified. FISTA solves it with the projected gradient step

$$\boldsymbol{a}_{k+1} = \max\left(\boldsymbol{z}_k - \frac{1}{L_D}\left(\boldsymbol{D}^\top(\boldsymbol{D}\boldsymbol{z}_k - \boldsymbol{y}) + \lambda \mathbf{1} \right),\ \mathbf{0} \right), \qquad \boldsymbol{z}_{k+1} = \boldsymbol{a}_{k+1} + \frac{\tau_k - 1}{\tau_{k+1}}\left(\boldsymbol{a}_{k+1} - \boldsymbol{a}_k \right), \tag{6.9}$$

where $L_D = \|\boldsymbol{D}\|_2^2$ is the Lipschitz constant of the gradient, the maximum acts entrywise, and $\tau_{k+1} = (1 + \sqrt{1 + 4\tau_k^2})/2$ with $\tau_1 = 1$ [104]. The momentum term makes the objective converge to its minimum at the rate $\mathcal{O}(1/k^2)$. M^2-SRC runs 300 iterations. As an exact reference, the interior-point solver Clarabel [152] solves the same problem, formulated with CVXPY [153]. The following statement shows how far an iterate lies from the optimum without knowing the optimum.

Proposition 6.2 (Duality gap). *For any $\boldsymbol{a} \geq \mathbf{0}$, let $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{D}\boldsymbol{a}$ and $\boldsymbol{\theta} = \boldsymbol{r}\,\min\bigl(1, \lambda / \max_i (\boldsymbol{D}^\top \boldsymbol{r})_i\bigr)$, with the ratio read as infinity if the maximum is not positive. Then*

$$0 \leq f(\boldsymbol{a}) - f(\boldsymbol{a}^\star) \leq G(\boldsymbol{a}) = f(\boldsymbol{a}) - \boldsymbol{\theta}^\top \boldsymbol{y} + \frac{1}{2}\|\boldsymbol{\theta}\|_2^2 . \tag{6.10}$$

The proof applies weak duality at the scaled residual (Appendix B.2.1).

6.3. Results

6.3.1. Physical Features

Figure 6.3 shows all fourteen features per class. The motion features order the classes as predicted (Eq. (6.5)): the pattern change separates the static echo from every moving class, and the speed spread grows from static through rigid and articulated to distributed objects, from 0.01 to 0.14 m/s.

The material features also follow the prediction. The reflectivity orders the materials metal, waterlike and dielectric (medians of 106.1, 99.5 and 89.1 dB), the spectral slope separates metal from the other two, and the back echo is strongest for dielectric objects.

The shadow is the exception. Metal casts none, whereas waterlike objects darken the room behind them by 4.6 dB.

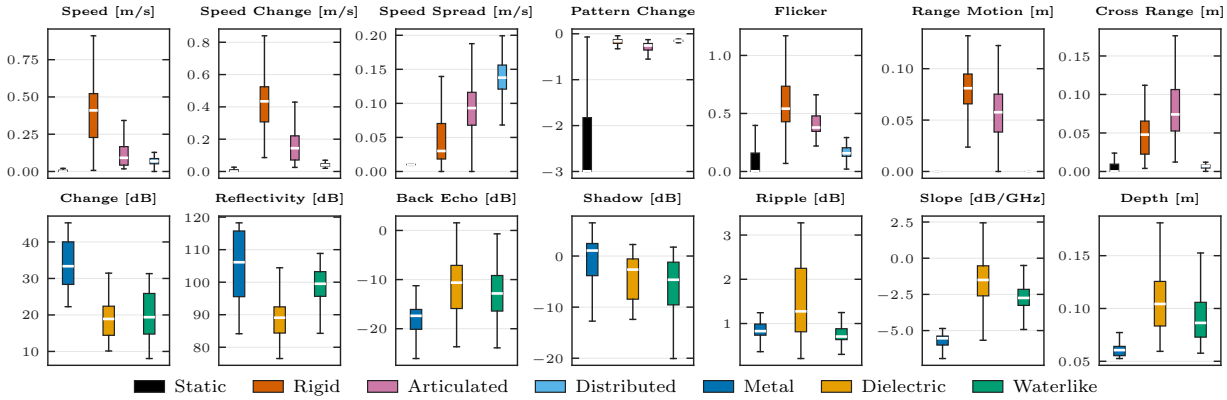


Figure 6.3.: The fourteen physical features per class over the training windows: seven motion features by motion class (top) and seven material features by material class (bottom).

6.3.2. Dictionary and Decision Maps

Figure 6.4 reads the dictionary of $N = 867$ windows. The atoms of one class resemble each other and differ from the atoms of other classes (Figure 6.4a,b), which is Assumption 6.1 in numbers. Distributed atoms resemble each other most and static atoms least. Metal atoms form a compact group that points against the other two materials, whereas dielectric and waterlike atoms are diffuse and almost orthogonal to each other.

The decision maps (Figure 6.4c,d) colour the plane of the first two principal components by the class that M^2-SRC assigns to each cell. The two components carry 52 and 59 % of the variance of the two heads, so a map is a slice of the decision and not the whole.

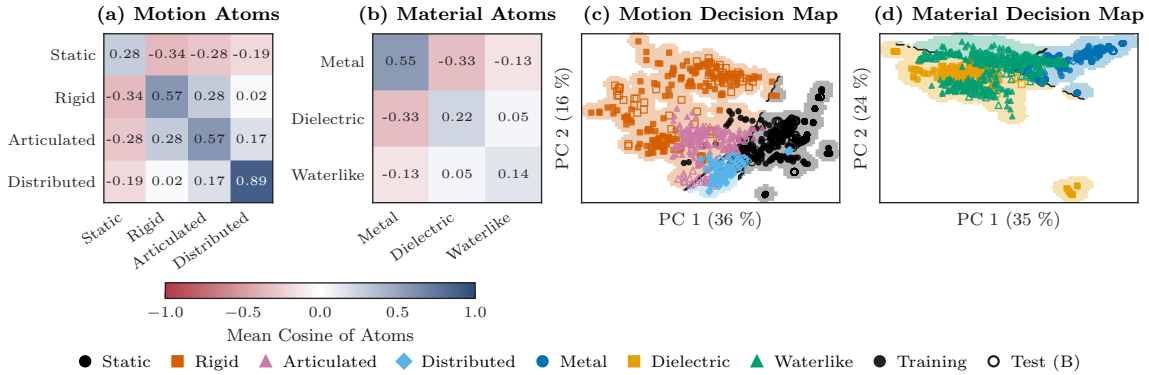


Figure 6.4.: What the dictionary says. (a,b) Mean cosine between the atoms of each pair of classes for the motion and the material head. (c,d) Training windows (filled) and session B (hollow) in the plane of the first two principal components (PCA), with the classes that M^2-SRC assigns to the plane; the other components are fixed at their mean, so the map is a slice of the decision.

6.3.3. Sparse Code of One Window

Figure 6.5 follows one window of rain from session B through the classifier. The change map locates the object at 0.67 m, and the micro-Doppler fills a broad band with short gaps in which it narrows to a line. The motion code spreads over 17 atoms, with 65 % of its weight on distributed and 35 % on static windows, and the material code uses 21 atoms, all of them waterlike. The window is classified as waterlike and distributed, both correctly.

6.3.4. Optimisation and Classification Performance

Figure 6.6a shows how FISTA reaches a code. In the plane of two atoms, the iterates move from the origin to the optimum of Clarabel and stop on the boundary where the metal weight is zero. After 300 iterations the duality gap of FISTA remains larger than that of Clarabel ($3.7 \cdot 10^{-2}$ against $2.6 \cdot 10^{-8}$), yet both solvers agree on 99.9 % of the motion labels and 97.5 % of the material labels, and FISTA is faster (15 ms against 27 ms per window). With the feature extraction, a window takes 330 ms of the 400 ms between two windows, so the classifier runs in real time (Appendix C.2.2).

Figure 6.7 shows the performance on the unseen session B and in leave-one-take-out. On session B, M^2-SRC detects every object without a false alarm and classifies the motion of 98.1 % and the material of 94.6 % of

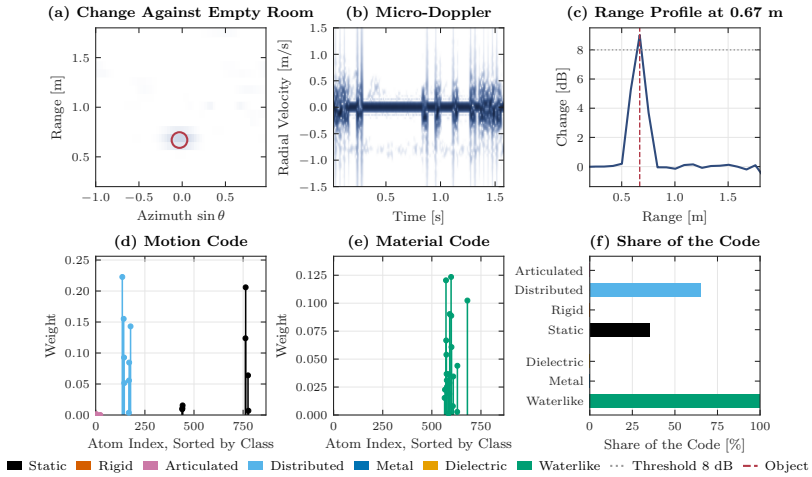


Figure 6.5.: One window of rain from the unseen session B, read by the classifier: (a) change against the empty room, (b) micro-Doppler, (c) range profile of the change, (d,e) sparse codes of the motion and the material head over the training windows, ordered by class, and (f) share of the code per class.

the windows. Metal and waterlike windows are classified without error; dielectric windows are the weak class (79 %) and are mostly called waterlike.

In leave-one-take-out, the motion stays at 95.2 %, and the material falls to 73.1 %. Dielectric windows fall to 46 %, and 54 % of them are called waterlike.

The Receiver Operating Characteristic (ROC) curves of Figure 6.6b,c show that the ranking of the classes is right in most cases: the area is 0.99 or more for every motion class in both tests, and for dielectric windows it is 0.99 on session B and 0.86 in leave-one-take-out.

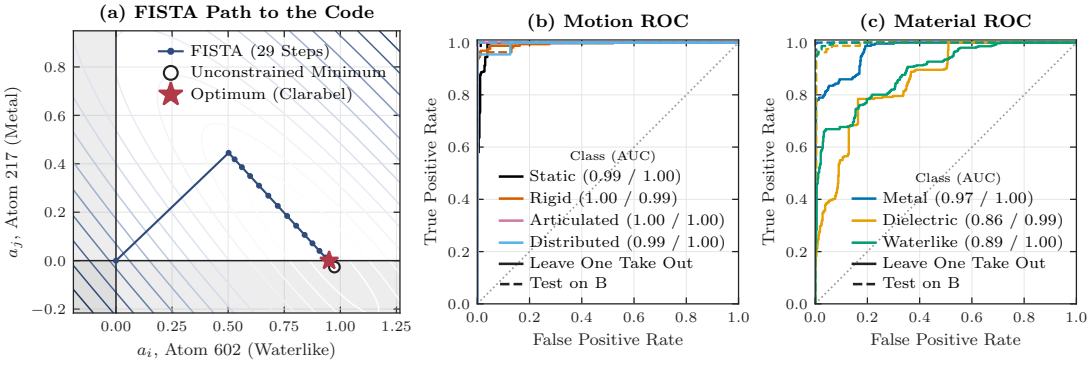


Figure 6.6.: (a) Level sets of Eq. (6.6) in the plane of two atoms, with the FISTA iterates, the unconstrained minimum and the optimum of Clarabel; the shaded region violates the nonnegativity. (b,c) ROC curves of the motion and the material head for each class against the rest, with the area under the curve in leave-one-take-out and on session B.

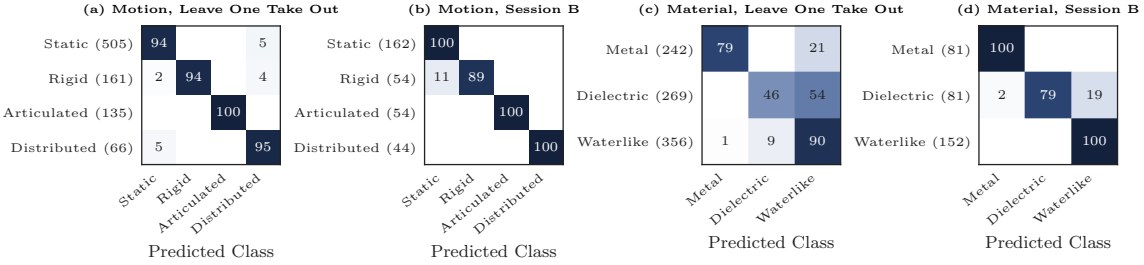


Figure 6.7.: Confusion matrices of the motion head (a,b) and the material head (c,d) in leave-one-take-out (a,c) and on the unseen session B (b,d). Each row shows, in percent, how the windows of one true class were classified; the number of windows is given in brackets.

6.4. Discussion

The results point in two directions. The material head reaches 94.6 % on the unseen session but only 73.1 % when one take is held out at a time, and dielectric windows are its weak class, with a recall of 79 and 46 %. The motion head reaches 98.1 %, metal and waterlike windows reach a recall of 100 % on session B, and a window is classified in 330 ms within the 400 ms between two windows. The question is whether M^2-SRC reads the material of an object or only recognises objects that it has seen before.

6.4.1. Physical Features

The physics fixed two orders before the first measurement (Eq. (6.5)), and the data keep both (Figure 6.3). The reflectivity ranks metal above water and water above plastic, which follows the permittivity: the larger the permittivity, the more of the wave an object reflects. The speed spread ranks the motions by their number of independent scatterers.

One prediction fails. A metal object should shadow the room behind it more than water does, yet the data show the opposite. The metal objects here are small corner reflectors and a ball, which block little of the room, whereas a bottle and a person block much more. The shadow therefore measures the size of an object as much as its absorption, so the material head rests partly on a size cue that a large metal plate would break.

6.4.2. Dictionary and Decision Maps

The dictionary shows which classes lie close together. Metal atoms point away from the other classes, whereas the regions of dielectric and waterlike windows touch in the decision maps (Figure 6.4), and the errors follow this geometry. On session B, 15 of the 17 misclassified dielectric windows are called waterlike (Figure 6.7), and in leave-one-take-out more than half of the dielectric windows are, while metal is never called dielectric. The errors cluster in few takes. Four static dielectric takes fail completely when held out, a swinging aluminium reflector and a steel ball inside a box are mostly called waterlike, and on session B one take, a swinging plastic box, holds all 17 material errors.

Dielectric objects resemble both neighbours for a physical reason. A dielectric reflects by the same law as water, only with a lower permittivity (Eq. (2.6)), so its echo shares features with water. A large, smooth plastic face mirrors the wave, and the echo then depends on its angle, so a swinging box can look like water

at one instant and like metal at another. These explanations are untested. Reflectivity, back echo and spectral slope keep metal apart on session B (recall 100 %, precision 97.6 %), and they fail only where the geometry of a dielectric object dominates its echo.

The labels may add to this confusion. The three names group objects by what they are called, whereas the echo follows the effective permittivity, thickness and content of an object. A water bottle is a plastic shell around water, a box may hold a steel ball, and a person is water under clothes, so the border between dielectric and waterlike runs through the objects. Labels that follow the echo, such as bands of the effective permittivity or a class for composite objects, might separate better; this is untested, because every window carries the name of its object.

The gap between 73.1 and 94.6 % marks the reach of the dictionary. The plastic boxes fail when one of their two takes is held out, and they succeed on session B when both takes are in the dictionary. One take samples one position, so the gap measures the coverage of the dictionary and not a limit of the physics.

6.4.3. Sparse Code of One Window

A decision of the classifier can be read from the inside. The rain window of Figure 6.5 is classified correctly, and a third of its motion code sits on static atoms. This is plausible, because a pour is interrupted and the echo of the bottle stays static in the gaps. A hard label would hide these gaps, whereas the share of the code shows them and allows a confidence threshold that rejects uncertain windows.

6.4.4. Optimisation and Classification Performance

The solver can be coarse without harming the decision. After 300 iterations the duality gap of FISTA is much larger than that of the exact solver, yet the labels agree almost always, because the decision needs the order of the class residuals and not the optimum. The ROC curves separate two kinds of error (Figure 6.6). On session B, the ranking of the classes is right even for dielectric windows, and the decision rule loses only the windows of one take. In leave-one-take-out, the ranking itself fails for dielectric windows, which shows that the dictionary lacks the object.

6.4.5. Findings and Scope

RadarCat reaches 96.0 % on 26 objects that lie on the sensor [48]. The tasks differ, so the numbers do not compare, but the failures do: RadarCat confuses objects that share a material, and M^2-SRC confuses materials that share a shape or a cover.

The second research question receives a graded and encouraging answer. A millimetre-wave radar reads the motion of an object from a single window almost completely (98.1 % on an unseen session) and the material of metal and waterlike objects reliably. It reads the material of a plastic object where its dictionary holds objects like it. The classifier finishes within the 400 ms of a window. The claim reaches as far as its evidence: one object in front of a static radar, 867 windows from one room and no label for composite objects. These limits point to a dictionary that grows, a feature for object size and labels that follow the echo. One radar window, fourteen physical features and 867 examples named the motion of nearly every window and the material of metal and water within the time of the next window, and the result repeats to the last digit (Appendix D).

7. Computer Vision

Among the sensors of the modern age, the camera is the cheapest, the densest and the richest in semantic content. For a few euros and a few watts, it delivers millions of pixels per frame, from which software can read signs, faces and labels. Its strength is also its limit. A pixel carries brightness and colour but no distance, so a camera cannot know how large a letter is, and it cannot tell a light source from the reflection of a light source. A radar closes exactly these gaps: it measures range to the centimetre and detects a light source as an object by its radar echo, but it cannot read a single letter. The research question concerns the knowledge that radar and camera hold jointly and neither holds alone. The scene (Section 7.1) sets the board, the light source and the truth, the methodology (Section 7.2) builds the alignment of both sensors, the text-recognition method MetricSight and the removal of glare, the results (Section 7.3) follow one word through alignment, reading and light, and the discussion (Section 7.4) weighs what the fusion adds and where it ends.

7.1. Scene: Reading Against the Light

Fusion earns its place where the blind spot of one sensor is the strength of the other. Modern neural networks recognise text with high accuracy (Section 3.3), but they must learn every font and letter size from training examples, and they cannot distinguish a light source from the reflection of a light source. The objective is to read a known word in every camera frame while the radar supplies what the image lacks: the position, the metric scale and the light source.

The truth is fixed by construction. A board with the printed word ARTIFACT stands in front of the radar, and the camera sits on top of the radar. The platform of Chapter 4 supplies both streams: the AWR2944P (Section 4.1.1) delivers the radar objects, the camera (Section 4.3) delivers frames of 1280×720 pixels at 30 frames per second, and the data acquisition (Section 4.5) pairs them by one common clock. The code of this experiment is in the folder `computer_vision/` of the code base (Appendix D). Nine corner reflectors anchor the alignment: their apices are marked by mouse click in the image and detected by the radar. Two further targets test the alignment independently: the printed word on a swinging ball, whose heights come from a tape measure, and the flash of a phone. The same phone torch is the light source: it shines into the camera, is switched on and off, and its state is logged. Five takes yield 5 601 evaluated frames, two without a light source (text_085b, text_085c) and three with a switched light source (light_1 to light_3), which contain 1 087 light-on frames with a fitted veil (Appendix C.3.1). Only take 3 puts the light source into the radar view: the radar names it in 566 of 603 light-on frames (93.9 %), against 1.8 % in take 1 and none in take 2. Take 3 therefore carries the test of the radar-anchored glare removal.

Four problems follow: the position of a radar point in the image, the reading at the radar scale, the detection of the light source and the reading under its light. Given the radar objects with range r, azimuth θ and elevation φ and camera frames of 1280×720 pixels, find the image position of every object, the word on the board and, when the light source is on, its position and veil, subject to a processing time of at most 33 ms

per camera frame, measured by the alignment error in pixels and centimetres, the reading error, the share of light-on frames in which the radar names the light source and the run time per frame. The edit distance [154] counts the substituted (N_S), deleted (N_D) and inserted (N_I) characters that turn the read string into the true string of N characters, and the character error rate is

$$\mathrm{CER} = \frac{N_S + N_D + N_I}{N}. \tag{7.1}$$

The word accuracy is the share of frames in which the whole word is read without error, and the stable reading is the character-wise majority vote over the last 15 frames.

One observation gives hope. A letter at a known range has a known size in pixels, and the radar knows the range to the centimetre. If the metric scale is known, a small classifier may do what a large network must learn from letters of thousands of sizes, and a light source that the radar detects is a light source whose glare a simple physical model can describe.

7.2. Methodology

The methodology follows the signal from the radar point to the letter (Figure 7.1). The pinhole camera, least squares, gradient histograms, ridge regression and convolutional networks come from Section 2.4, so only their use is stated.

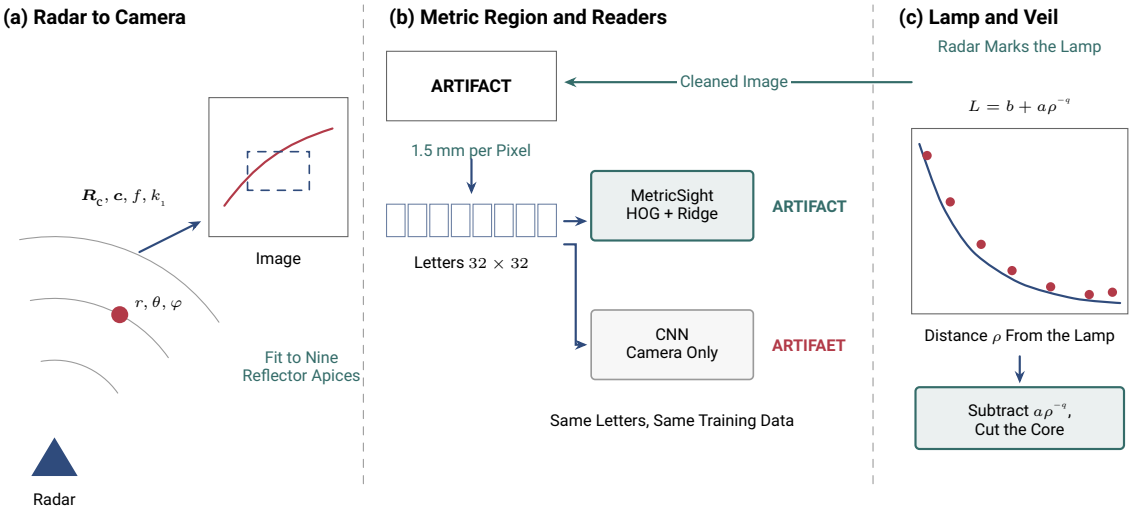


Figure 7.1.: Radar-guided reading. (a) A radar point is projected into the image. (b) The region around it is cut at 1.5 mm per pixel into letters, which two methods classify. (c) The radar marks the light source, its veil is fitted and subtracted, and the cleaned image enters the reading. Green marks the contributions of this thesis.

7.2.1. Radar-Camera Alignment

A radar point at range r, azimuth θ and elevation φ lies at $\boldsymbol{p} = r\,[\cos\varphi\cos\theta,\ \cos\varphi\sin\theta,\ \sin\varphi]^\top$ in the radar frame. The pinhole model with one radial distortion term (Section 2.4.1) moves it into the camera frame,

$\boldsymbol{p}_\mathrm{c} = \boldsymbol{R}_\mathrm{c}(\boldsymbol{p} - \boldsymbol{c}) = [X_\mathrm{c}, Y_\mathrm{c}, Z_\mathrm{c}]^\top$, and projects it to

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} + f\left(1 + k_1\left(x^2 + y^2\right)\right)\begin{bmatrix} x \\ y \end{bmatrix}, \qquad x = \frac{X_\mathrm{c}}{Z_\mathrm{c}}, \quad y = \frac{Y_\mathrm{c}}{Z_\mathrm{c}}, \tag{7.2}$$

with the rotation $\boldsymbol{R}_\mathrm{c}$ from radar to camera, the camera position $\boldsymbol{c}$ from a ruler, the focal length f in pixels, the distortion k_1 and the image centre (u_0, v_0). The three rotation angles, f and optionally k_1 follow from the least-squares problem of Eq. (2.80), solved by the Levenberg-Marquardt iteration of Eq. (2.81), with the clicked apices as pixels and the detected apices as points. With nine points, a model with k_1 may fit them and fail elsewhere, so the model with the smaller leave-one-out error is kept. A radar range without a direction is a sphere, and its image is a curved arc, so the method searches along the arc.

7.2.2. MetricSight and the Camera-Only Baseline ★

Published text-recognition networks learn the variation of letter size from training data, because an image alone fixes no metric scale (Section 3.3). MetricSight removes this variation by geometry. The radar finds the board with Infini-Rex for Radar (Section 5.2.4), and an object of width w at range r appears with fw/r pixels. The region of 0.45×0.20 m around the projected board is therefore cut out and resampled to 1.5 mm per pixel, so that a letter of a given font size covers the same number of pixels at any distance.

In this metric image, a pixel counts as ink if it is more than one standard deviation darker than the mean of its 40 mm neighbourhood. A horizontal band of ink between 10 and 200 mm high forms the text line, which is cut into letters at the gaps and, where letters touch, at the cut that makes both parts most readable [115]. Each letter is scaled to 32×32 pixels and described by a histogram of oriented gradients (Eq. (2.88)) and the ink density of its cells, a feature vector $\boldsymbol{\phi} \in \mathbb{R}^{576}$ of unit length. A linear classifier assigns one of 36 characters: with the features of N training letters in the rows of $\boldsymbol{F}$ and the true characters in $\boldsymbol{Y} \in \{-1, 1\}^{N \times 36}$, ridge regression [99] gives the weights in closed form,

$$\boldsymbol{W} = \left(\boldsymbol{F}^\top \boldsymbol{F} + \lambda \boldsymbol{I}\right)^{-1} \boldsymbol{F}^\top \boldsymbol{Y}. \tag{7.3}$$

The character with the highest score wins, and the gap to the second score is the confidence. The training set holds 18 000 distorted letters from 27 fonts, ten of them handwriting fonts [155], and 3 600 letters of three unseen fonts serve as a test. MetricSight has 20 800 parameters and needs no gradient descent. The contribution is the use of the radar range: it turns the size of a letter into a constant, so that a closed-form classifier suffices.

The camera-only baseline recognises the same letters without the metric scale of the radar. It finds ink with the Sauvola threshold in image pixels [114], uses the same letter cutting and classifies each letter with a Convolutional Neural Network (CNN) of the kind in Section 2.4.3 (Figure 2.14) with 52 400 parameters, trained with Adam on the same 18 000 letters. The two methods differ in two things only: the metric scale and the classifier.

7.2.3. Radar-Anchored Glare Removal ★

A light source in the field of view saturates the sensor at its core, and light scattered inside the lens spreads a veil over the image that lowers the contrast of the text [63]. A light event is reported when an area of light source size exceeds the running mean of its brightness by more than five standard deviations. Only the light

source is also a radar object, so the method keeps the saturated area that lies on the projected arc of a radar detection and has the strongest veil. Around its centre, the median brightness of every colour channel on rings of logarithmically spaced radius ρ, with saturated pixels excluded, follows

$$L(\rho) = b + a\,\rho^{-q}, \qquad a \geq 0, \tag{7.4}$$

where b is the brightness of the scene, a the strength of the veil and q its decay. For every exponent q on a grid from 0.5 to 4, linear least squares gives a and b (Appendix B.3.1), and the exponent with the smallest error is kept. Subtracting $a\rho^{-q}$ from every channel removes the veil, and the saturated core is cut out because it holds no information. Glare models exist in lens physics; the contribution is the anchor, because the radar names the source and the fit needs no learned labels.

7.3. Results

7.3.1. Radar-Camera Alignment

Figure 7.2 shows the alignment. The projection fits the nine calibration points to 13 pixels, and a point left out of the fit is predicted to 19 pixels (Figure 7.2b). The error grows on targets that did not enter the calibration: 39 pixels on the word of the swinging ball, which lies 0.9 cm to the left and 2.6 cm above its true position, and 71 pixels on the phone flash, which lies 3.2 cm off horizontally.

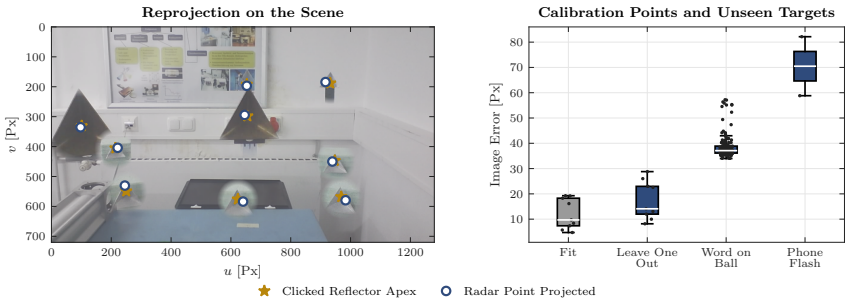


Figure 7.2.: Radar-camera alignment. (a) Projected radar points (circles) and clicked reflector apices (stars) on a composite image of the calibration scene. (b) Projection error in pixels for the fit, leave-one-out and the two independent targets.

7.3.2. Reading Without Light

Table 7.1 and Figure 7.3 compare the two text-recognition methods without a light source. MetricSight errs less per frame in both conditions (5.1 and 10.1 %), and the majority vote over the last 15 frames lowers it to 2 %. The CNN shows a Character Error Rate (CER) of 13.7 % in the text-only takes, and the vote leaves this value unchanged (13.6 %). Only MetricSight reads the whole word, in 85 % of its stable readings; the CNN never does.

The accuracy per letter (Figure 7.3b) is similar for both methods, 82 to 93 % for MetricSight and 78 to 79 % for the CNN, except for the letter C. MetricSight reads the C correctly in 84 % of the frames, whereas the

CNN never does and reads an E instead in 95.5 % of them. The confusions of MetricSight, F with G and C with D, occur in 7 % of the frames.

One frame takes a median of 30 ms with MetricSight (20.8 k parameters) and 112 ms with the CNN (52.4 k parameters); the radar detection takes 31 ms.

Table 7.1.: Reading results in percent by condition: CER and word accuracy per frame and for the stable reading (subscript s).

		MetricSight				CNN			
Condition	Frames	CER	Word	CER_s	Word_s	CER	Word	CER_s	Word_s
Text Only	3251	5.1	64.6	2.1	85.6	13.7	0.1	13.6	0.0
Lamp Off	1027	10.1	72.1	2.0	84.9	32.1	0.3	13.2	0.0
Lamp On	1087	32.2	36.6	7.0	55.5	85.1	0.0	82.4	0.0
Lamp On, Glare Removed	1087	26.5	43.9	5.3	68.4	86.4	0.1	82.1	0.0

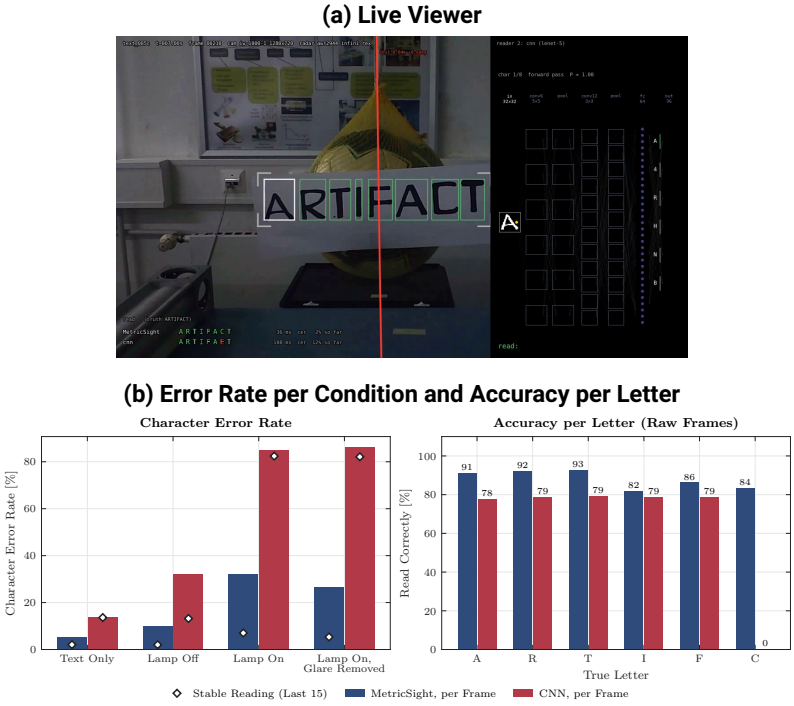


Figure 7.3.: (a) One frame of the live viewer: the boxes mark the letters cut from the radar-scaled region, MetricSight reads ARTIFACT, the CNN reads ARTIFAET, and the right panel shows the forward pass of the CNN for the letter A. (b) Character error rate per frame (bars) and for the stable reading (markers) per condition, and the share of raw frames in which each letter is read correctly.

7.3.3. Reading Under the Light Source

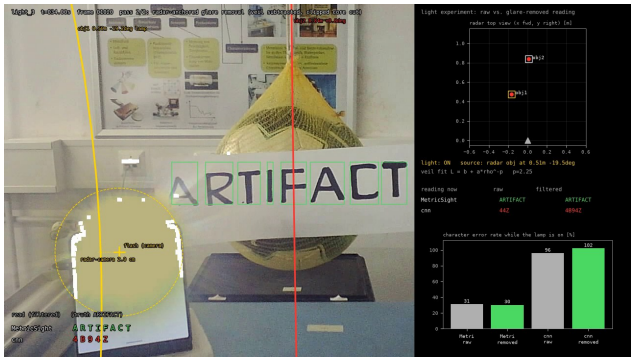
In take 3, the light source is on in about half of the 1 188 frames. The radar names it in 566 of the 603 light-on frames, and the veil fit succeeds in 544 of them. The three colour channels decay with similar exponents

between 1.9 and 2.25 (Figure 7.4b), and in the frame of Figure 7.4a the radar object lies 2 cm from the light source in the image.

The light source raises the CER of MetricSight from 8.7 % without the light source to 31.1 %. The removal of the veil lowers it only slightly, to 30.3 %, and the word accuracy stays at 55.9 %. The CNN shows a CER of 96.0 % with the light-on and 99.0 % after the removal, and it never reads the whole word.

Over all three takes, the removal lowers the CER of MetricSight from 32.2 to 26.5 %. It lowers it strongly in take 1, from 41.4 to 24.9 %, and raises it in take 2, from 10.5 to 16.0 % (Appendix C.3.2).

(a) Glare Removed, Take 3



(b) Veil Fit

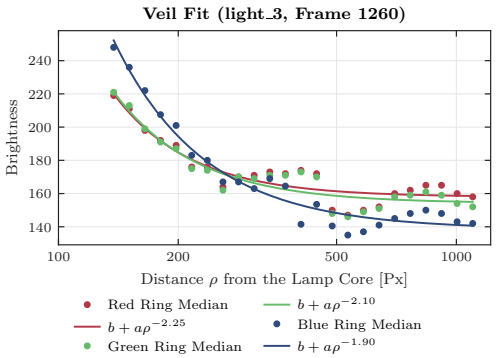


Figure 7.4.: Radar-anchored glare removal in take 3. (a) Live viewer in the cleaned pass: the yellow circle marks the light source core that the radar named, the veil is subtracted, and both methods read the cleaned frame (frame 1 020). (b) Median brightness of the three colour channels on rings around the light source against the distance ρ, with the fit of Eq. (7.4) (frame 1 260).

7.4. Discussion

The results point in two directions. Over the three light-source takes, the removal of the glare lowers the character error rate of MetricSight from 32.2 to 26.5 %, yet it raises the rate in take 2 and gains less than one percentage point in take 3 (31.1 to 30.3 %), and the alignment leaves an offset of centimetres. In the other direction, the radar named the light source in 94 % of the light-on frames, and MetricSight, with 20.8 k parameters, read the word with 2.1 % error after the vote, at the speed of the camera and with less than half the error of a network with 52.4 k parameters. The question is whether the radar helps the camera enough to justify the fusion, or whether the camera reads well on its own wherever the light is kind.

7.4.1. Radar-Camera Alignment

The alignment error grows with the distance from the calibration points (Figure 7.2b), because every step moves further from the points that the fit has seen. The remaining offset of a few centimetres stems from systematic causes. The camera position enters Eq. (7.2) directly and comes from a ruler, so an error in it shifts all points in one direction, as the vertical offset of 2.6 cm on the ball does. The clicked apex of a corner reflector is also not its radar phase centre, and nine points constrain four free parameters. The offset is small where it matters: 3 cm is less than a tenth of the width of the reading region of 45×20 cm, so the cut still

holds the word. The alignment is therefore good enough for reading, and it would limit any task that needs centimetres.

7.4.2. Reading Without Light

The majority vote lowers the error of MetricSight but not that of the CNN, because the two methods make different kinds of error. The CNN reads the C of the board as an E in 95.5 % of the frames (Figure 7.3b). The C of the board is a square bracket. Without its right-hand stroke it resembles an E whose middle bar is missing, a shape that the 27 training fonts do not contain. The error repeats in every frame, so the vote cannot remove it, and a repeated error is the dangerous kind, because it passes every filter that averages over time. The errors of MetricSight, F read as G and C read as D, change from frame to frame, so the vote lowers its error (Table 7.1). The metric scale explains the difference: with letters of fixed size, the remaining errors of MetricSight are random and not systematic.

The speed follows from the same choice. The ridge regression of Eq. (7.3) is one matrix product per letter, whereas the CNN of Figure 2.14 applies its convolutions to every letter, so only MetricSight keeps pace with the camera frame period of 33 ms. The literature is far ahead on open-world text: SVTR reaches 94.4 % on IIIT5k with 4.5 ms per word image on a graphics card (Section 3.3), and this chapter does not compete with it. It measures what a known scale buys. An image alone fixes no scale, whereas a radar supplies it directly, and methods that use only the image cannot close this gap.

7.4.3. Reading Under the Light Source

In take 3, the glare removal works and its gain is small (Figure 7.4). The three colour channels decay with similar exponents, and the radar names the source in 94 % of the frames, yet the removal lowers the error by less than one percentage point, and the stable reading improves from 3.9 to 2.8 %. The small gain stems from a visible cause. In take 3 the light source sits at the lower left and the word at the centre (Figure 7.4a), so the veil at the word is weak. The CNN gains nothing, because its errors are errors of shape and not of contrast. In take 1, where the veil covers the word, the removal lowers the error strongly (Appendix C.3.2). The two takes leave one question open: how much of the glare error is veil, and how much is lost core. A light source beside the word, named by the radar, would answer it.

7.4.4. Findings and Scope

Multimodal scene analysis in the literature learns how to weight radar and camera from large data sets, as in fusion networks that survive fog and in CAFuser, which names the lighting by the camera itself (Section 3.6). This chapter takes the opposite route: the radar enters as geometry, as a scale and a name for the light source, and the camera reads. The approach is compact and explainable. Every step follows from an equation, and no step needs more data than a closed-form fit.

For one board, radar and camera together read the word with 2.1 % error after the vote, at the speed of the camera, and the radar names the light source whose glare the camera cannot place. The limits are as clear: one board in one room, an alignment of nine points and a ruler, and one take that tests the light source. The radar does not have to read for the camera. It has to say where the word is, how big it is and which light to distrust, and then even a small reader can read.

8. Radar Tracking

A goalkeeper who watches only where the ball is concedes the goal. The save depends on where the ball will be, and on how sure the goalkeeper is of that guess. A tracking radar plays the same game ten times per second. Its detections say where an object was, its filter predicts where the object will be, and the next measurement tests the prediction. The loop has a second layer in a cognitive radar: the prediction also decides which chirp measures the object best, so that the prediction shapes the measurement that corrects it, and an error in one spreads into the other. The research question concerns whether a radar can state honestly how little it knows about the future position of a tracked object, even when its motion model is wrong, and whether it can use this statement to improve its next measurement. The scene (Section 8.1) sets a swinging ball and a walking person in front of a radar that either keeps one chirp or chooses its chirp. The methodology (Section 8.2) closes the perception-action cycle from detection with error control through the prediction regions and the choice of the chirp to the tracker. The results (Section 8.3) compare the two radars by tracking performance, statistical guarantees and scene understanding, and the discussion (Section 8.4) weighs what the loop achieves.

8.1. Scene: The Pendulum and the Passer-By

The previous chapters solved their problems one frame at a time: detection with a bound on the false discovery rate (Chapter 5), the classification of material and motion (Chapter 6) and the reading of text (Chapter 7). A moving scene adds a demand that none of them meets: the result of one frame must inform the next. The objective of this chapter is to join the abilities in one loop that tracks objects over time, states how far its predictions can be trusted, and adapts its own measurement.

A ball with the printed word ARTIFACT hangs on a string 1.55 m in front of the radar and swings across the line of sight (Figure 8.1). Behind it, a person walks back and forth at 2 to 3 m, often half outside the field of view of the camera. The camera supplies the ground truth: it detects the ball in the image and takes the moving region outside the ball as the person. The platform of Chapter 4 provides the whole loop: the AWR2944P (Section 4.1.1) transmits preloaded chirp sub-frames, the data acquisition (Section 4.5) streams them to the laptop in real time, and the camera (Section 4.3) delivers the images for the ground truth and for the reading. The code of this experiment is in the folder `tracking/` of the code base (Appendix D).

Two radars are compared. A *fixed chirp* means that the radar uses one chirp configuration, the search chirp with a maximum unambiguous speed of 3.8 m/s, in every frame and never changes it. The *cognitive chirp choice* transmits three sub-frames in every frame of 105 ms: the search chirp, a fast chirp with a maximum unambiguous speed of 5.2 m/s, and the search chirp again later in the frame. It decides in every frame which of them to process. One recording per radar forms the data (Appendix C.4.1). A second comparison on a single recording replays the cognitive recording with different look policies and removes the differences between scenes.

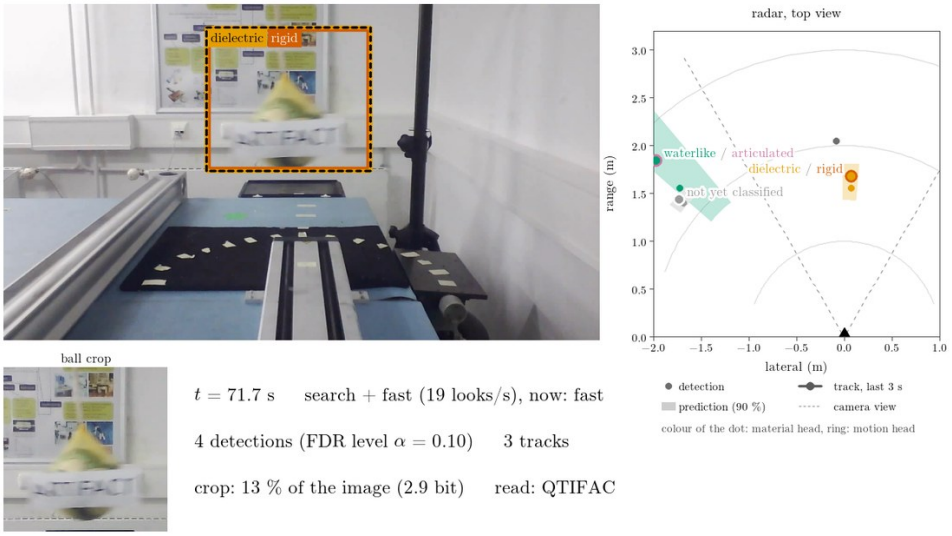


Figure 8.1.: The tracking scene in the live viewer of the cognitive run at 71.7 s. The camera image carries the ball track with its labels dielectric and rigid, the crop of the ball with the reading QTIFAC of the word ARTIFACT, and the status line with the sub-frame in use; the radar top view shows the tracks, their 90 % prediction regions and the person as a waterlike, articulated track at the left edge of the view.

Five tasks follow. The detection must keep its error budget while it reports objects. The tracker must keep one identity per object. The prediction region must hold the next measurement with a stated probability. The classifier labels the material and the motion of each track, and the camera reads the word ARTIFACT on the ball in parallel, so that a track that sits on the ball also checks itself. The chirp choice must pay for its processing time with a better track. Given the radar frames and the camera images, find the tracks, their prediction regions, their labels, the reading and the chirp of the next frame, subject to a false discovery rate of at most $\alpha = 0.1$ per look, a coverage of at least $1 - \epsilon = 0.9$ and a processing time within the frame, measured by the scores below.

A track and a true object are matched in the image column of the projected track, and the ball must also agree with its taped range of 1.55 m within 0.3 m. The scores are computed *per look*, at the time of each radar measurement, and *live*, on a fixed grid of 10 Hz with the track that was available at that moment, which includes the delay of the processing. Four scores measure the tracks. The Higher Order Tracking Accuracy (HOTA) score [156] balances finding the objects and keeping their identities, the Multiple Object Tracking Accuracy (MOTA) [157] counts all errors, including the identity switches (IDSW), the Identification F1 Score (IDF1) score [158] is the F1 score of the identity matching, and the Generalized Optimal Sub-Pattern Assignment (GOSPA) distance [159] compares the set of tracks with the set of true objects,

$$\mathrm{HOTA} = \sqrt{\mathrm{DetA} \cdot \mathrm{AssA}}, \qquad \mathrm{DetA} = \frac{|\mathrm{TP}|}{|\mathrm{TP}| + |\mathrm{FN}| + |\mathrm{FP}|}, \tag{8.1}$$

$$\mathrm{MOTA} = 1 - \frac{|\mathrm{FN}| + |\mathrm{FP}| + |\mathrm{IDSW}|}{|\mathrm{GT}|}, \qquad \mathrm{IDF1} = \frac{2\,\mathrm{IDTP}}{2\,\mathrm{IDTP} + \mathrm{IDFP} + \mathrm{IDFN}}, \tag{8.2}$$

$$\mathrm{GOSPA} = \Big(\sum_{(i,j)\in\mathcal{M}} d_{ij}^2 + \frac{c^2}{2}\big(|\mathrm{FN}| + |\mathrm{FP}|\big) \Big)^{1/2}. \tag{8.3}$$

Here DetA counts true positive (TP), missed (FN) and false (FP) track points, AssA is the average, over the

true positives, of the overlap between the history of a track and the history of its true object, and the score is averaged over several matching thresholds; |GT| is the number of true object points; IDTP, IDFP and IDFN count the points of the best one-to-one assignment between tracks and objects over the whole recording and the points outside it; $\mathcal{M}$ is the optimal assignment of the tracks within the cut-off $c = 5^\circ$ to the true objects, d_{ij} the azimuth difference of a matched pair, and every miss and every false track costs $c^2/2$. The azimuth carries the comparison because the camera fixes the azimuth of an object well and its range poorly. The detection is measured by the false discovery rate, the precision and the recall,

$$\mathrm{FDR} = \mathbb{E}\left[\frac{|\mathrm{FD}|}{\max(|\mathrm{D}|, 1)}\right], \qquad \text{precision} = 1 - \mathrm{FDR}, \qquad \text{recall} = \frac{|\mathrm{TD}|}{|\mathrm{GT}|}, \tag{8.4}$$

where |D|, |FD| and |TD| count the detections, the false detections (nothing moves there in the camera) and the true detections. The prediction region is measured by its coverage, the share of later measurements $\boldsymbol{z}_{t+\Delta}$ that fall inside the region $\Gamma_{t+\Delta}$ predicted for them,

$$\text{coverage} = \mathbb{P}\big(\boldsymbol{z}_{t+\Delta} \in \Gamma_{t+\Delta}\big) \geq 1 - \epsilon, \tag{8.5}$$

and by its overlap with the camera, the intersection over union (IoU) of the projected region and the camera box of the ball (area) or of the person (image columns). The camera confirms the coverage independently by the share of image positions of the object inside the projected region. The reading is measured by the share of letters of ARTIFACT that are right in a read and by the stable reading, the character-wise majority of the last 15 reads (Section 7.2.2).

One observation gives hope. Every part of the loop has a guarantee of its own: the e-values of Infini-Rex bound the false discovery rate, and conformal calibration bounds the miss rate of a prediction region. If the connections between the parts preserve these guarantees, the loop inherits them.

8.2. Methodology

The method closes the perception-action cycle of a cognitive radar [26]; Chapter 9 draws the cycle for the whole system. The perception half processes one radar look: the range-Doppler map, the detection with error control, the prediction regions, the tracker and the labels of the tracks. The action half chooses the chirp of the next frame from the predicted uncertainty of the tracks. A memory carries information from look to look: the state and covariance of every track, the last 300 conformal scores that calibrate the prediction regions, the adaptive level ϵ_t that follows the misses, the roles and histories of the tracks, and the empty-room echo. Two feedback paths use this memory: the prediction regions weight the detection of the next look, and the predicted covariance steers the choice of the chirp. The four subsections follow this order: the regions themselves, the detection that the regions guide, the chirp that the regions steer, and the tracker that links the methods. Green marks the contributions of this thesis.

8.2.1. KalmanCQR ★

A filter has a best guess of where an object is and a measure of its doubt. Each measurement corrects the guess by a fraction of the surprise, and the doubt shrinks. The doubt is reliable only if the motion model is right, and a swinging ball is not a constant-velocity object. KalmanCQR therefore keeps the filter for the

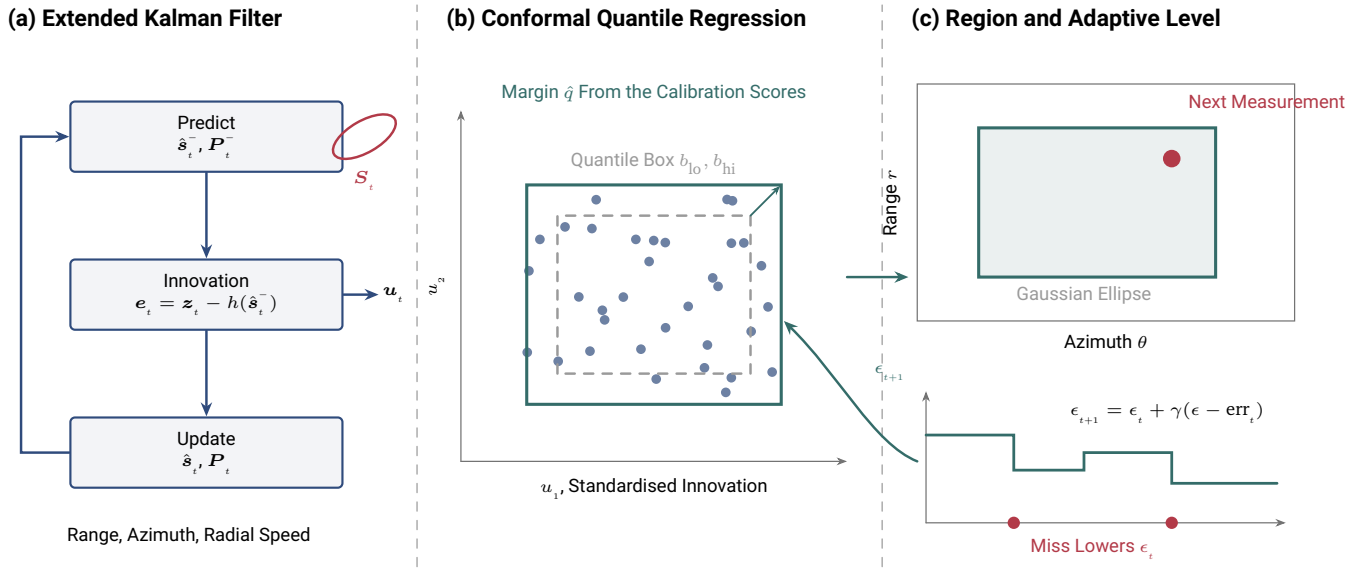


Figure 8.2.: KalmanCQR. (a) The extended Kalman filter predicts the state, forms the innovation from the measurement and updates; its predicted covariance $\boldsymbol{S}_t$ spans an ellipse. (b) Quantile regression bounds the standardised innovations by a box, and conformal calibration widens the box by the margin $\hat{q}$. (c) The widened box becomes the prediction region of the next measurement, and the adaptive level lowers ϵ_t after every miss. Green marks the contributions of this thesis.

guess and measures the doubt from the surprises themselves (Figure 8.2): it standardises the past surprises, learns how large they usually are, and adds a margin so that the region is wrong at most once in ten times.

The Extended Kalman Filter (EKF) [50], [88] carries the state $\boldsymbol{s} = [x, y, v_x, v_y]^\top$, the position and velocity in the radar plane. The motion model assumes a constant velocity over the step Δ and treats every deviation as random acceleration of standard deviation $\sigma_a = 4.9\,\mathrm{m/s}^2$ (the largest acceleration of a pendulum bob is of the order of $g/2$),

$$\boldsymbol{s}_{t+\Delta} = \boldsymbol{F}\boldsymbol{s}_t + \boldsymbol{w}_t, \qquad \boldsymbol{F} = \begin{bmatrix} \boldsymbol{I}_2 & \Delta \boldsymbol{I}_2 \\ \boldsymbol{0} & \boldsymbol{I}_2 \end{bmatrix}, \qquad \boldsymbol{w}_t \sim (\boldsymbol{0}, \boldsymbol{Q}), \tag{8.6}$$

where position grows by velocity times Δ and $\boldsymbol{Q}$ is the covariance that σ_a adds to position and velocity. The radar measures range, azimuth and radial speed, which are nonlinear functions of the state,

$$\boldsymbol{z} = h(\boldsymbol{s}) + \boldsymbol{\nu}, \qquad h(\boldsymbol{s}) = \begin{bmatrix} \sqrt{x^2 + y^2} \\ \operatorname{atan2}(y, x) \\ (x v_x + y v_y)/\sqrt{x^2 + y^2} \end{bmatrix}, \tag{8.7}$$

with measurement noise $\boldsymbol{\nu}$ of covariance $\boldsymbol{R}_\nu$ given by the resolution of the radar. The EKF linearises h with its Jacobian $\boldsymbol{H}$ at the predicted state. The prediction and the update read

$$\hat{\boldsymbol{s}}_t^- = \boldsymbol{F}\hat{\boldsymbol{s}}_{t-\Delta}, \qquad \boldsymbol{P}_t^- = \boldsymbol{F}\boldsymbol{P}_{t-\Delta}\boldsymbol{F}^\top + \boldsymbol{Q}, \tag{8.8}$$

$$\boldsymbol{e}_t = \boldsymbol{z}_t - h(\hat{\boldsymbol{s}}_t^-), \qquad \boldsymbol{S}_t = \boldsymbol{H}\boldsymbol{P}_t^-\boldsymbol{H}^\top + \boldsymbol{R}_\nu, \tag{8.9}$$

$$\hat{\boldsymbol{s}}_t = \hat{\boldsymbol{s}}_t^- + \boldsymbol{K}_t\boldsymbol{e}_t, \qquad \boldsymbol{P}_t = (\boldsymbol{I} - \boldsymbol{K}_t\boldsymbol{H})\,\boldsymbol{P}_t^-, \qquad \boldsymbol{K}_t = \boldsymbol{P}_t^-\boldsymbol{H}^\top\boldsymbol{S}_t^{-1}, \tag{8.10}$$

where $\boldsymbol{e}_t$ is the innovation, the surprise of the measurement, $\boldsymbol{S}_t$ its predicted covariance and $\boldsymbol{K}_t$ the Kalman gain that sets which fraction of the surprise enters the guess. If the model were exact, an ellipse of $\boldsymbol{S}_t$ would hold the next measurement with a known probability. A swinging ball and a walking person violate the model, so the innovations exceed $\boldsymbol{S}_t$, and the filter is overconfident.

Quantile regression models a conditional quantile of a variable, and conformalized quantile regression [52] corrects the fitted quantiles on a calibration set so that the region holds a new value with probability $1-\epsilon$ whenever the new score and the calibration scores are exchangeable (Section 2.2.5). KalmanCQR applies it to the standardised innovations $u_{t,d} = e_{t,d}/\sigma_{t,d}$, with $\sigma_{t,d} = \sqrt{[\boldsymbol{S}_t]_{dd}}$ for range, azimuth and radial speed ($d \in \{1,2,3\}$). The last 300 innovations of all tracks are split alternately into a fitting half and a calibration half. On the fitting half, the lower and upper quantiles of u_d at the levels $\epsilon/2$ and $1-\epsilon/2$ give the box $[b_{\mathrm{lo},d}, b_{\mathrm{hi},d}]$, which holds most standardised innovations. On the calibration half, every innovation receives the score

$$s_i = \max_d \max\left(b_{\mathrm{lo},d} - u_{i,d},\; u_{i,d} - b_{\mathrm{hi},d}\right), \tag{8.11}$$

which is positive if the innovation leaves the box in any dimension. With $\hat{q}$ the $\lceil (1-\epsilon_t)(n_\mathrm{c}+1)\rceil$-th smallest of the n_c calibration scores, the margin $\hat{q}$ widens the box, and the prediction region of the next measurement is

$$\Gamma_{t+\Delta} = \prod_{d=1}^{3}\left[\hat{z}_d + \left(b_{\mathrm{lo},d} - \hat{q}\right)\sigma_d,\; \hat{z}_d + \left(b_{\mathrm{hi},d} + \hat{q}\right)\sigma_d\right], \qquad \hat{\boldsymbol{z}} = h(\hat{\boldsymbol{s}}^{-}_{t+\Delta}). \tag{8.12}$$

The region is centred on the predicted measurement, and its width follows the predicted standard deviation σ_d, so that it widens when the filter is uncertain. As long as fewer calibration scores exist than the quantile needs, the region falls back to the Gaussian box $\hat{z}_d \pm z_{1-\epsilon/6}\,\sigma_d$. Regions for 0.5 and 1 s ahead follow in the same way from innovations at these horizons.

Proposition 8.1 (Coverage)**.** *If the calibration scores and the score of the next measurement are exchangeable, then* $\mathbb{P}(\boldsymbol{z}_{t+\Delta} \in \Gamma_{t+\Delta}) \geq 1-\epsilon_t$.

The proof is in Appendix B.4.1.

The proposition states the central point: the coverage holds whether or not the motion model is right, because the margin is learned from the model's own errors. Motion that changes when a person turns breaks exact exchangeability, and adaptive conformal inference [53] protects the coverage in this case. After every measurement it records a miss, $\mathrm{err}_t = \mathbb{1}\{\boldsymbol{z}_t \notin \Gamma_t\}$, and updates the level,

$$\epsilon_{t+1} = \epsilon_t + \gamma\left(\epsilon - \mathrm{err}_t\right), \qquad \gamma = 0.02. \tag{8.13}$$

A miss lowers the level and widens the next region, and a hit raises it slightly. For this update, the share of misses over T steps differs from ϵ by at most $(\max(\epsilon_1, 1-\epsilon_1)+\gamma)/(\gamma T)$ without any assumption on the data [53]; the implementation keeps ϵ_t between 10^{-3} and 0.5, so this bound applies as long as the clipping does not act. Algorithm 1 lists one step of the loop, in which the innovations of the measurements feed back into the regions.

8.2.2. Track-Guided Detection with FDR Control ★

Every look forms the range-Doppler map of the 16 virtual channels with a range FFT and a Doppler FFT (Section 2.2.1), so that a moving object appears in its own Doppler cell, separated from the static room at

Algorithm 1 *KalmanCQR for One Track at One Look ($\star$).*

1. **Input**: state $\hat{\boldsymbol{s}}_{t-\Delta}$, covariance $\boldsymbol{P}_{t-\Delta}$, memory of the last 300 standardised innovations, level ϵ_t, target ϵ, step γ.
2. **Predict** $\hat{\boldsymbol{s}}_t^-$ and $\boldsymbol{P}_t^-$ by Eq. (8.8); **set** $\hat{\boldsymbol{z}} = h(\hat{\boldsymbol{s}}_t^-)$, $\boldsymbol{S}_t$ by Eq. (8.9) and $\sigma_d = \sqrt{[\boldsymbol{S}_t]_{dd}}$.
3. **Split** the memory into a fitting half and a calibration half; **compute** $b_{\mathrm{lo},d}$ and $b_{\mathrm{hi},d}$ from the fitting half and the scores of Eq. (8.11) from the calibration half; **find** $\hat{q}$.
4. **Output** the region Γ_t of Eq. (8.12), and **gate** the detections of the look with it.
5. **If** a detection $\boldsymbol{z}_t$ is associated with the track:
   5.1. **Compute** the innovation $\boldsymbol{e}_t$ and **update** $\hat{\boldsymbol{s}}_t$ and $\boldsymbol{P}_t$ by Eq. (8.10).
   5.2. **Store** $\boldsymbol{u}_t = \boldsymbol{e}_t/\boldsymbol{\sigma}_t$ in the memory; **set** $\mathrm{err}_t = \mathbb{1}\{\boldsymbol{z}_t \notin \Gamma_t\}$ and ϵ_{t+1} by Eq. (8.13).
6. **Output**: $\hat{\boldsymbol{s}}_t$, $\boldsymbol{P}_t$, the region for the next look and for 0.5 and 1 s ahead.

zero speed; the empty room, recorded before the experiment, is subtracted where this lowers the power of a cell. An OS-CFAR detector proposes up to 24 cells per look, those inside the predicted regions of the tracks first. Infini-Rex for Radar (Section 5.2.4) computes an e-value for every direction of every proposed cell, so that a look poses m hypotheses.

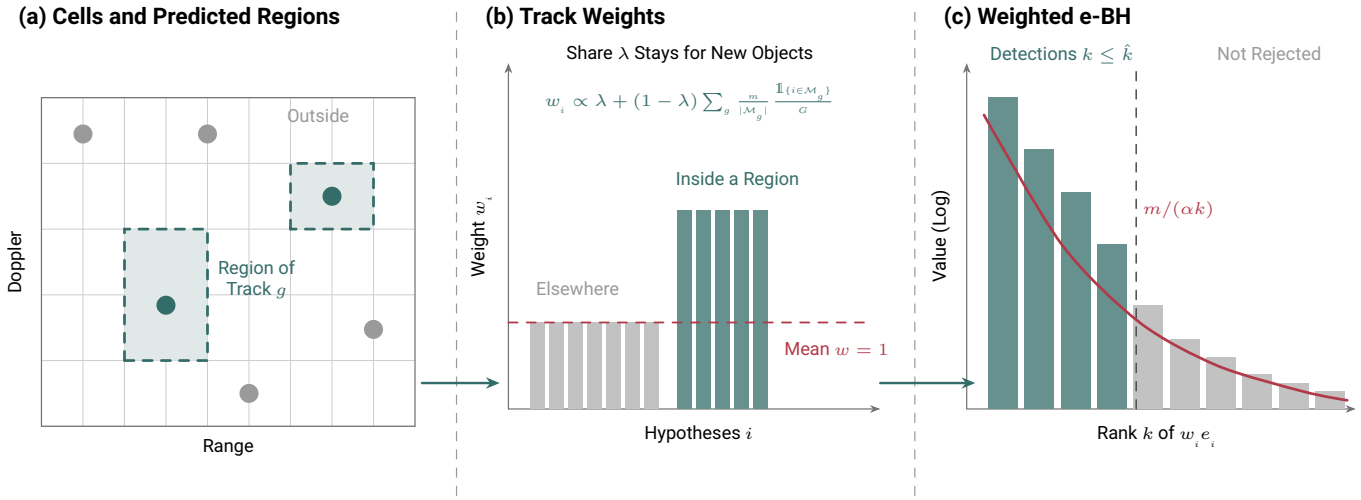


Figure 8.3.: Track-guided detection. (a) The proposed cells in the range-Doppler map; the predicted regions of the tracks mark the cells where a detection is expected. (b) Cells inside a region receive a larger weight, cells elsewhere a smaller one, and the mean weight stays one. (c) The weighted e-BH rejects the hypotheses whose weighted e-value stays above the threshold $m/(\alpha k)$. Green marks the contributions of this thesis.

A detection inside the predicted region of a track is more likely than one elsewhere, so a cell in a region should need less evidence than a cell in empty space. Weighted multiple testing turns this idea into power [160], [161]. Every hypothesis i receives a weight $w_i \geq 0$ with

$$w_i \propto \lambda + (1-\lambda)\sum_g \frac{m}{|\mathcal{M}_g|}\frac{\mathbb{1}\{i \in \mathcal{M}_g\}}{G}, \qquad \sum_{i=1}^{m} w_i = m, \tag{8.14}$$

where $\mathcal{M}_g$ is the set of hypotheses inside the predicted region of track g, G is the number of tracks, and λ is the estimated share of detections that start a new track, so that new objects keep a share of the weight. The weighted e-BH applies Eq. (5.9) to the products $w_i e_i$ and rejects the $\hat{k}$ largest, with $\hat{k} = \max\{k\colon (we)_{[k]} \geq m/(\alpha k)\}$.

Proposition 8.2 (FDR control with track weights)**.** *Let the weights be fixed by the past frames, with* $w_i \geq 0$ *and* $\sum_i w_i = m$*, and let every null e-value satisfy* $\mathbb{E}[e_i \mid \text{past}] \leq 1$*. Then the weighted e-BH has* FDR $\leq \alpha$ *in every look.*

The proof is in Appendix B.4.1.

The weights move the threshold and not the error budget: a cell inside a predicted region needs a smaller e-value to be detected, a cell elsewhere needs a larger one, and the expected share of false detections stays below α. The proposition inherits the assumption of Proposition 5.2 for the e-values.

8.2.3. Cognitive Chirp Choice and Design ★

The chirp choice joins a design and a rule. The design gives the radar a choice without a cost: every frame of 105 ms contains the three sub-frames of Section 8.1, loaded into the radar once (Figure 8.4). The fast sub-frame has a shorter pause between chirps, which raises the maximum unambiguous speed from 3.8 to 5.2 m/s at a coarser velocity step (Section 2.1.5). The radar transmits all sub-frames and decides only which of them it processes, so that the switch costs no blind time. Each processed sub-frame is a look.

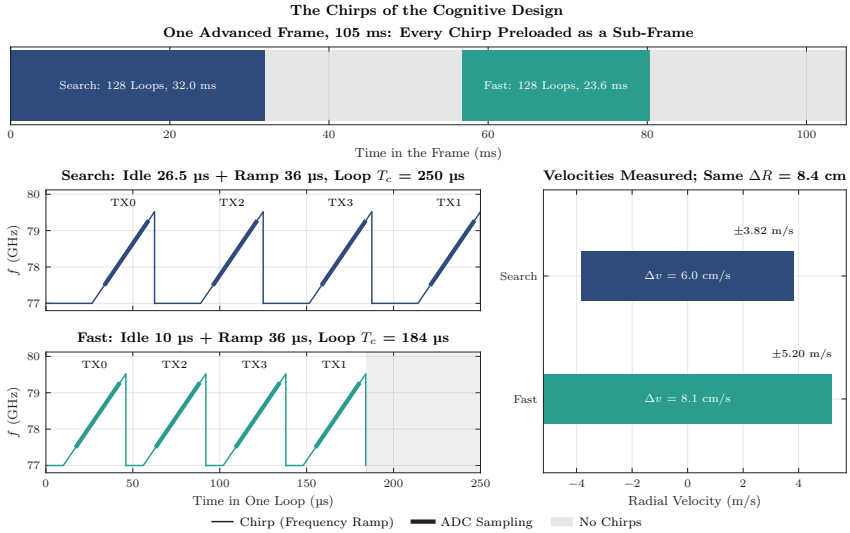


Figure 8.4.: The chirp plan of one frame. The search chirp and the fast chirp are transmitted in every frame of 105 ms; the fast chirp has a shorter pause and reaches a higher maximum speed at a coarser velocity step. The planner chooses which looks the processor works on.

The rule is simple: an extra look is worth its time if it removes enough doubt about the tracks. The principle is to choose the waveform by the tracking error that it will cause [65], [66]. The doubt about a track is the entropy of its Gaussian position estimate [91], averaged over the horizon $H = 1$ s,

$$U = \frac{1}{H} \int_0^H \frac{1}{2} \log_2 \Big((2\pi e)^2 \det \boldsymbol{P}_{\text{pos}}(\tau) \Big) \, \mathrm{d}\tau, \tag{8.15}$$

where $\boldsymbol{P}_{\text{pos}}(\tau)$ is the 2×2 position block of the predicted covariance. After the search look, every plan is scored by U: the covariance is propagated with Eq. (8.8) and updated with Eq. (8.10) at the looks that the plan processes and whose chirp can measure the track, that is, whose maximum speed exceeds the radial speed of the track. No measurement enters this prediction, only covariances, and the time that each look takes to process is included, so a plan whose result arrives too late gains nothing. An extra look is chosen only if it lowers U by at least one bit, which halves the area of the predicted position uncertainty, since U falls by one bit when $\det \boldsymbol{P}_{\text{pos}}$ halves. The rule is sound for two reasons: it uses only the quantities that the filter already predicts, and it counts the processing time as part of the cost, so that it cannot choose a plan that the radar cannot afford.

8.2.4. Multi-Object Tracker

The tracker links the three methods (Figure 8.5). It receives the detections of Infini-Rex, assigns each a role [51], passes the associated measurements to KalmanCQR, and returns the regions that gate the next association and weight the next detection. A detection becomes the *core* measurement of a track if it lies in or near the prediction region of the track and within the distance that the object can have moved since its last update; every track takes at most one core. Further detections close to a moving track become its *parts*, such as the arms and legs of a person, which update the size of the object but not its position. A detection that is a multipath copy of a stronger detection is a *ghost* (Section 2.1.1), and a detection that the empty room explains is dropped. All remaining detections start new tracks. A new track is confirmed after three updates within its first four looks, and a track without an update for one second is deleted.

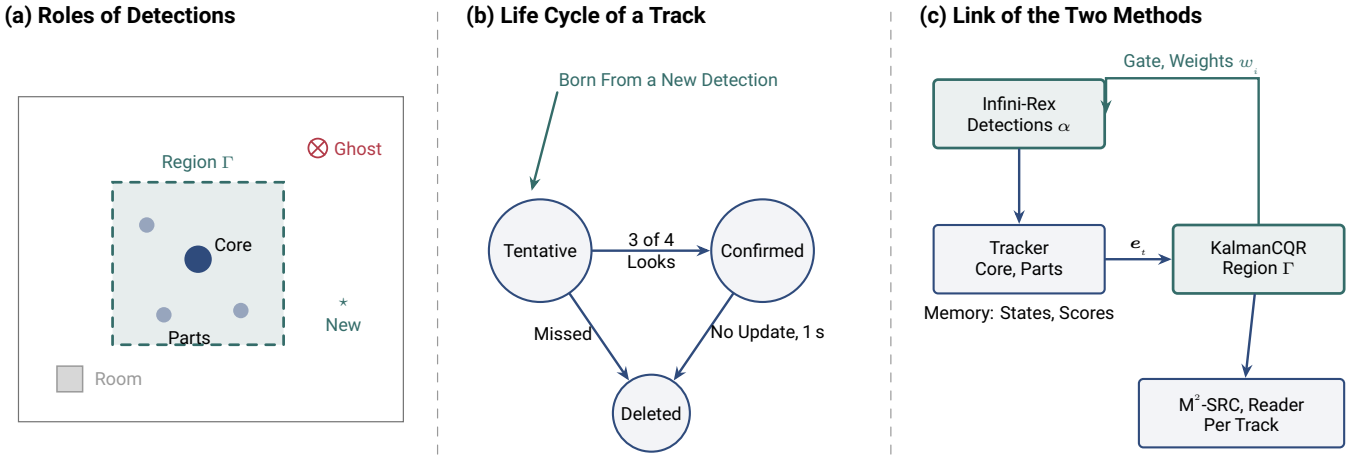


Figure 8.5.: The multi-object tracker. (a) The roles of the detections around the region of a track. (b) The life cycle of a track. (c) The tracker links the detections of Infini-Rex with KalmanCQR: the innovations pass to the filter, and the regions return as gates and weights. Green marks the contributions of this thesis.

Every confirmed track is classified by M^2-SRC (Section 6.2.2) on the window of its last 1.6 s at the range and direction of the track, and the label is the majority of its windows. MetricSight (Section 7.2.2) reads the word at the position and scale that the track predicts, which also checks the track independently. Both methods enter unchanged.

8.3. Results

8.3.1. Tracking Performance

Figure 8.6 compares the two radars on their recordings. The cognitive choice raises the median of all four tracking scores per segment, and the identity score gains most: the median IDF1 climbs from 74.9 to 91.7 %. The scores of the cognitive choice also vary more between segments, whereas the fixed chirp stays within a narrow band. The identity switches show the same picture. The fixed chirp switches identities in every segment, the cognitive choice in few segments, and over the whole recording the count falls from 13 to 8.

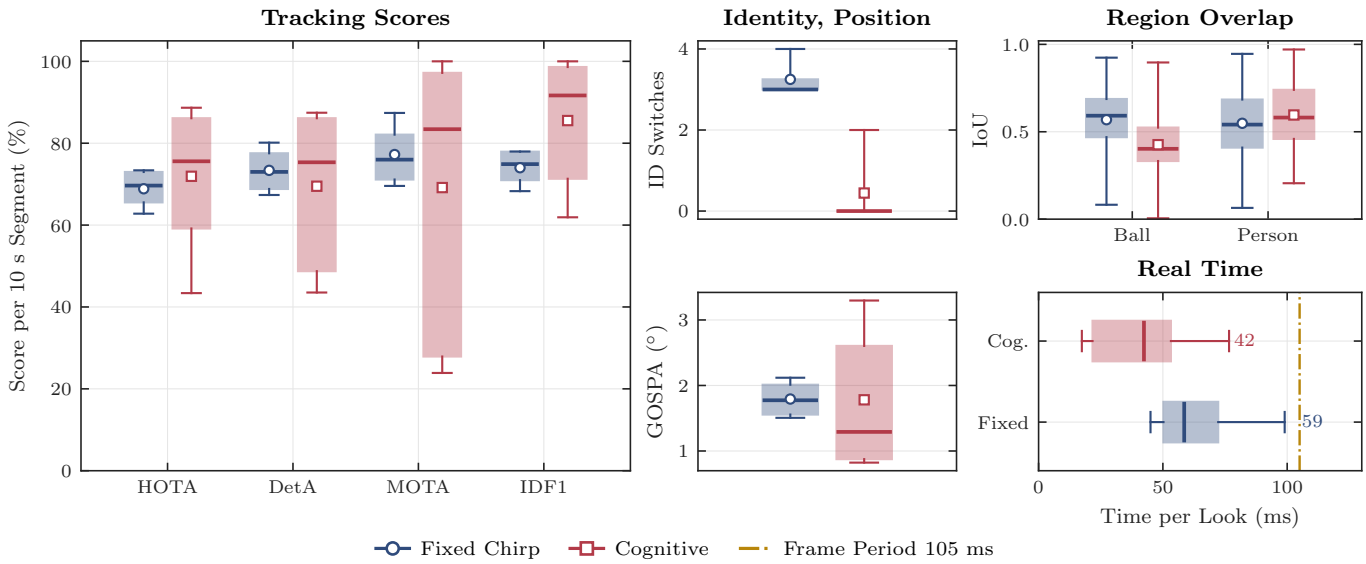


Figure 8.6.: Tracking performance of the fixed chirp and the cognitive choice. Candles per 10 s segment of a recording (box: quartiles, whiskers: range, line: median, marker: mean) for HOTA, DetA, MOTA and IDF1, for the identity switches and for the GOSPA distance; candles of the IoU between the prediction region and the camera box per frame; distribution of the processing time per look (box: quartiles, whiskers: 5 % and 95 %, number: median) against the frame period of 105 ms.

The position error does not change. The GOSPA distance is 1.8° for both radars, so both place the objects equally well. The overlap of the prediction region with the camera box depends on the object: for the ball it falls from a median IoU of 0.59 to 0.40, for the person it stays near 0.55.

The whole-recording scores show one cost. The MOTA per look, which counts every false track point, falls from 76.9 to 66.3 %; the cognitive recording contains 361 false track points against 60 (Appendix C.4.2).

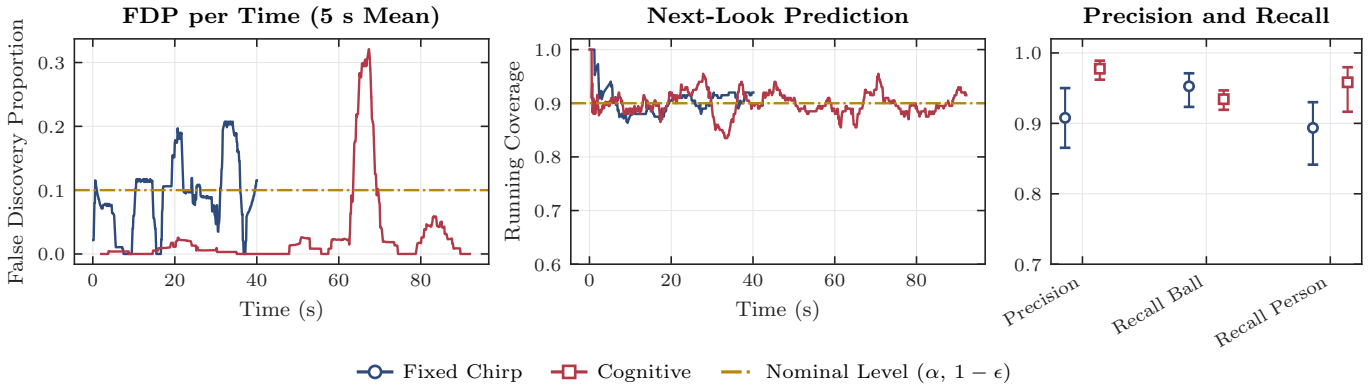
Both radars run in real time. A look takes a median of 59 ms with the fixed chirp and 42 ms with the cognitive choice against a frame of 105 ms, and 95 % of the looks finish within the frame; rare looks of 213 and 286 ms exceed it.

The replay of the cognitive recording isolates the effect of the chirp choice. Doubling the look rate from 9.5 to 18.9 per second lifts the live HOTA from 49.9 to 58.5 %, and taking every look yields the same result (58.3 %). The policies are listed in Appendix C.4.2.

8.3.2. Statistical Guarantees

Figure 8.7 shows that the detection keeps its error rate under the chosen level. The false discovery rate of the fixed chirp, 0.092, lies slightly below $\alpha = 0.1$, and its interval contains the level; that of the cognitive choice, 0.022, lies far below it. The proportion in single windows of 5 s leaves the level for short periods, in 39 % of the fixed recording and in 5 % of the cognitive recording. Precision and recall are high for both radars: the precision exceeds 0.9, and the recall of the ball and the person is 0.89 or more.

The prediction regions hold the next measurement in 92 % of the looks, above the nominal 90 %, and the regions for 0.5 and 1 s ahead reach the same level. The running coverage fluctuates between 0.84 and 0.96 and does not drift away from the level.



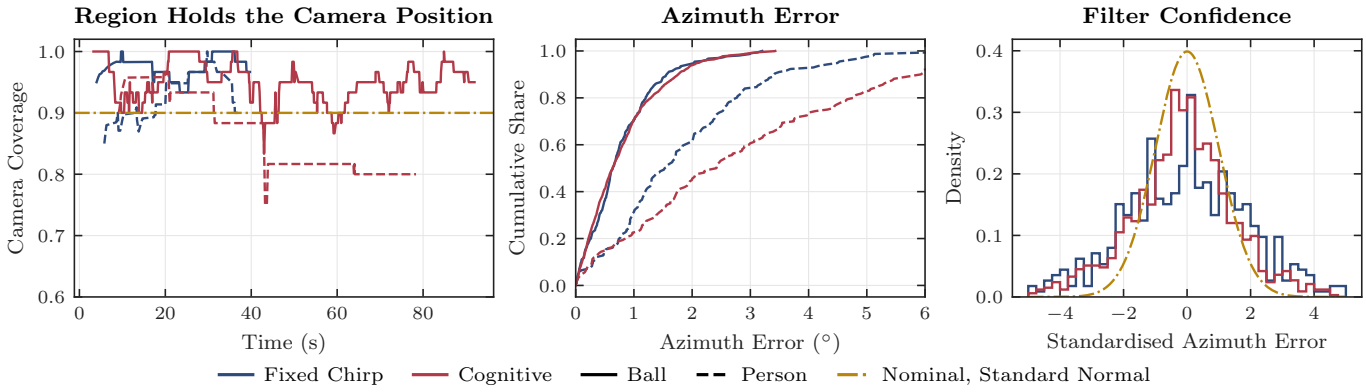
Chirps	FDR	95 % CI	$\Delta\alpha$	Coverage	95 % CI	$\Delta(1-\epsilon)$	0.5 s	1 s
Fixed Chirp	0.092	0.050–0.135	-0.008	0.920	0.897–0.938	+0.020	0.900	0.915
Cognitive	0.022	0.011–0.038	-0.078	0.915	0.904–0.925	+0.015	0.945	0.915

Figure 8.7.: Guarantees measured by the radar. Left: false discovery proportion of the detection over time (5 s mean) against $\alpha = 0.1$. Centre: running coverage of the next-look prediction (last 200 predictions) against $1-\epsilon = 0.9$. Right: precision and recall with their 95 % intervals. Table: false discovery rate and coverage with their 95 % intervals and the deviation Δ from the nominal level, and the coverage of the regions for 0.5 and 1 s ahead.

The camera confirms the regions independently (Figure 8.8). The ball lies inside the projected region in 95 to 97 % of the frames and the person in 87 to 92 %; the value of the person for the cognitive radar lies below 90 %, but its interval contains 90 %. The median azimuth error of the ball stays below 1° for both radars, and that of the person is larger (1.6 and 2.4°). The standardised error is wider than the standard normal density, with a root-mean-square of 1.6 to 4.2 instead of one: this is the overconfidence of the plain filter that the calibration must remove.

8.3.3. Scene Understanding

Figure 8.9 shows the cognitive recording in space and time. The axes are the sideways position, the forward position and the time. The yellow track is the ball. It swings from side to side in an even rhythm along the time axis, which is the uniform motion of a pendulum, and it stays one continuous track. The person appears as green and grey tracks that scatter widely: the tracks jump over long distances in range and in direction,



Chirps	Object	n	Coverage	95 % CI	$\Delta(1-\epsilon)$	Median $\lvert\Delta\theta\rvert$ (°)	90 % (°)	RMS z
Fixed Chirp	Ball	301	0.973	0.948–0.986	+0.073	0.64	1.56	1.77
Fixed Chirp	Person	147	0.918	0.863–0.953	+0.018	1.55	3.41	2.84
Cognitive	Ball	1198	0.948	0.934–0.959	+0.048	0.63	1.76	1.55
Cognitive	Person	160	0.869	0.808–0.913	-0.031	2.42	5.93	4.22

Figure 8.8.: Guarantees confirmed by the camera. Left: running share of the camera positions of the ball and the person inside the projected region (60 frames) against 0.9. Centre: cumulative distribution of the azimuth error against the camera. Right: histogram of the standardised azimuth error against the standard normal density. Table: camera coverage with its 95 % interval and the deviation Δ from 0.9, and the azimuth error.

and the tracks of the person end and restart many times (six identities, the longest holding one third of the person's looks). The translucent wedges around the tracks are the prediction regions of KalmanCQR. Each wedge marks the range and direction in which the next measurement of the track is expected, with a probability of 90 %. The wedges around the ball are compact and follow its swing, and the wedges around the person are large and irregular.

Table 8.1 lists what each radar recognises. Both radars track the ball and the person in most looks. The cognitive radar labels material and motion correctly for both objects in 99 % of the looks; the fixed radar labels the person correctly but the ball as waterlike in every look and as articulated in 30 % of its looks. The reading recovers few letters of the word with either radar (6 and 9 %), but the stable reading collects five of eight letters with the cognitive radar against one with the fixed radar.

Table 8.1.: Scene understanding of the two radars: share of looks (%) in which the ball and the person are tracked, with the correct material label (ball dielectric, person waterlike) and the correct motion label (ball rigid, person articulated), and the reading of the word on the ball.

	Tracked (%)		Material (%)		Motion (%)		Reading		
Chirps	Ball	Person	Ball	Person	Ball	Person	Reads	Letters (%)	Stable
Fixed Chirp	96.2	83.0	0.0	100.0	70.0	100.0	214	5.8	1 of 8
Cognitive	90.6	95.2	99.4	98.8	100.0	98.8	477	9.0	5 of 8

The window counts of the classifier show one weakness: of 135 dielectric windows, 48 (36 %) receive the

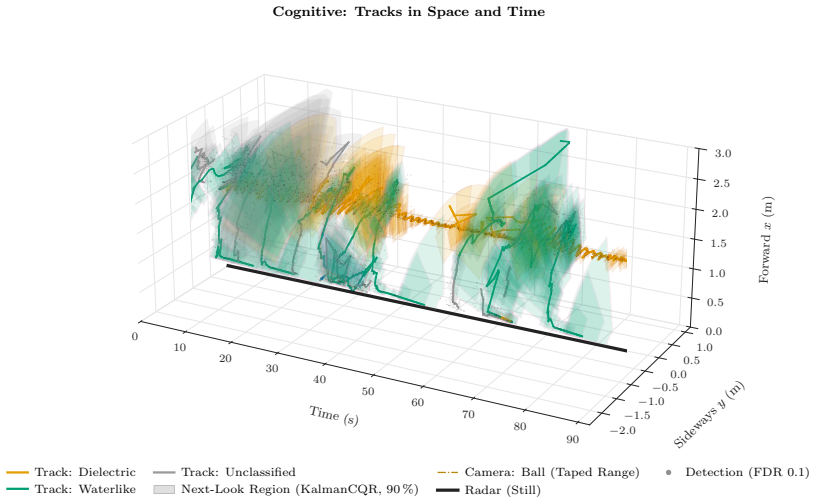


Figure 8.9.: Scene understanding in the cognitive recording. The radar stands at the lower edge, the tracks run along the time axis, the translucent volumes are the next-look prediction regions of KalmanCQR, the dash-dotted line is the camera ground truth of the ball, and the dots are the detections at a false discovery rate of 0.1.

label waterlike.

8.4. Discussion

The results point in two directions. The cognitive choice brings more false track points (361 against 60), the classifier mistakes many dielectric windows for waterlike, and the reading recovers few letters of the ball. In the other direction, the detection stays below its error level, the prediction regions hold the next measurement in 92 % of the looks, as promised, and the cognitive choice raises the live HOTA on one recording from 49.9 to 58.5 %. The question is whether a radar that adapts its own measurement keeps its statistical guarantees, and whether the adaptation improves the track or only adds looks.

8.4.1. Tracking Performance

The cognitive choice keeps the identities of the objects better, but it does not locate the objects better. The identity switches fall and the IDF1 rises, while the GOSPA distance stays the same (Figure 8.6). Both facts follow from the look rate. A radar that looks twice as often sees an object after half the time, and an object that moves with the speed v travels the distance $v\,\Delta t$ in the time Δt, so it travels half as far between two looks. The tracker then links the right detection to the right track more easily. The position error, in contrast, depends on the angular resolution of the array (Section 2.2.1), and the chirp does not change the array. The fast chirp adds a second benefit. It measures radial speeds up to 5.2 m/s without aliasing, whereas a faster motion appears at the wrong speed with the search chirp (Section 2.1.5).

The replay on one recording shows where the gain comes from. The policy that takes every look gains as much as the cognitive choice, so the extra looks always paid off in this scene, and the measured gain belongs to the look rate. The choice differs from the policy of every look only under load: it declines an extra look

whose processing would delay the result. This recording contains no such situation, and the entropy rule of Eq. (8.15) has shown only that it does no harm so far. A scene with many tracks or a slower processor would test the choice itself.

The gain costs false track points, and the data explain only part of the cost. The MOTA counts every false track point, and the radar tracks the person in many looks in which the camera does not see the person (the person is in view in 48 % of the fixed frames and 13 % of the cognitive frames). A correct radar track of an unseen person therefore counts as false. This reading is untested, and the data do not show the source of the false track points. The wide spread of the segment scores of the cognitive radar points to segments with different content, and which segments fail is open. The overlap of the ball regions with the camera box falls while the camera coverage stays high, so the regions still hold the ball but fit its box less tightly; whether the regions grew or the ball moved faster is open as well.

Real time holds on average and not for every look. A look takes 40 to 56 % of the frame, and the classifier is the largest stage with 27 to 31 ms of the median time. The rare looks that exceed the frame delay their own results, and the live scores count this delay. In summary, the cognitive choice improves the identities at no loss in position, and that this recording shows the benefit of looking more often rather than the benefit of choosing when to look.

8.4.2. Statistical Guarantees

A prediction is neither true nor false until a later measurement meets it, and the later measurement must be trustworthy. In this chapter, the measurements that test the regions come from Infini-Rex with a bounded share of false detections, so a region that holds the next detection has been tested against a measurement that is false in at most one case in ten, the level α. The camera adds a test that does not depend on the radar. Both tests agree (Figures 8.7 and 8.8).

The detection keeps its bound inside the tracker, as Proposition 8.2 requires. The proposition promises an upper bound for the expected share of false detections, not equality, so the cognitive recording may lie far below the level. The data do not show whether the track weights or the scene cause this gap. The bound also holds for the expectation over all looks, not for every window of five seconds. This explains why the windowed proportion leaves the level for short periods, and it means that a user who needs a bound for every window needs a different procedure.

The coverage of the prediction regions holds although the filter alone would fail. The filter assumes a constant velocity, but a swinging ball and a walking person do not move this way, so the real errors are wider than the Gaussian errors that the filter predicts (Figure 8.8, right). Ellipses of the predicted covariance would therefore be too small. Conformal calibration repairs this without a correct motion model, because it learns the margin from the errors that the filter has actually made (Proposition 8.1). The method of Gibbs and Candès, which adapts the level after every miss, protects the coverage when the errors are not exchangeable [53], and the test of the cognitive recording rejects exchangeability (KS $p = 0.00$). This is the answer to the part of the research question that asks for honesty about the future when the model is wrong. One value stays below the level: the camera coverage of the person with the cognitive choice is 0.869. It rests on 160 frames, and its interval, 0.808 to 0.913, contains 0.9, so the data neither confirm nor reject a failure there.

8.4.3. Scene Understanding

The two moving objects behave differently because their motion differs. The ball is a rigid body on a string, and a pendulum repeats its swing, so the errors of the filter stay small and regular and the prediction regions stay compact. The person moves with arms and legs that return different echoes from step to step, and the strongest echo jumps from one limb to another. The tracks of the person therefore jump and break, and the regions grow wide to cover the jumps. The data show this contrast but not its exact cause, since a walk out of the camera view could also end a track.

The classification shows one weakness of the method: dielectric objects are mistaken for waterlike ones. Both materials reflect the wave by the same law and differ in their permittivity (Eq. (2.6)), and Chapter 6 found the same confusion, with a recall of 79 and 46 % for the dielectric windows. This means that the echo of a dielectric object lies close to the echo of water in the feature space of that chapter. In the fixed recording, the ball receives the wrong label in every look, and in the cognitive recording the label is right. The chirp choice is therefore not the cause of the error. The label depends on the recording more than on the chirp, and the cause of the fixed-recording label is untested. A wrong material label must not steer a decision, so the label needs a confidence value that the loop can use.

The reading fails for a ball in motion. MetricSight reads a ball that swings across the line of sight far worse than a static board (82 to 93 % of the letters, Chapter 7), because the motion blurs the letters and the crop follows the predicted position and not the true one. The stable reading collects the reads over time and recovers most of the word with the cognitive radar. This shows that the tracker supplies the position and the scale for the reader, and it also shows that reading a moving object remains open.

8.4.4. Findings and Scope

The research question asked whether a radar can state honestly how little it knows about the future position of an object, even when its motion model is wrong, and whether it can use this statement to improve its next measurement. Both parts have an answer. The prediction regions of KalmanCQR hold the next measurement in 92 % of the looks, as stated, although the Kalman filter alone is overconfident by a factor of up to four. The entropy of the predicted covariance selects the chirp. The choice doubles the look rate to 18.9 per second within the frame, raises the live IDF1 from 62.3 to 72.1 % and lowers the identity switches from 13 to 8. The second layer named in the introduction, an error that spreads from the prediction into the measurement and back, did not appear: the false discovery rate and the coverage stayed at their levels with the fixed chirp and with the choice.

This result matters because earlier cognitive radars choose the waveform by the predicted error covariance and report lower tracking errors [65], [66], but the cited works report neither a false discovery rate nor a coverage. However, a radar that acts on its own prediction needs to know whether the prediction can be trusted. The loop of this chapter closes this gap: the detection bounds its false reports, the prediction states its miss rate, the action reads the doubt of the prediction to choose the next measurement, and an independent camera confirms the statements. The limits are as clear: two recordings of one scene, a gain that comes from the look rate, a reading that recovers only a fraction of the letters, a material label that confuses dielectric with waterlike, and an exchangeability assumption that the cognitive recording violates. A radar that states its doubt and spends its next look where the doubt is largest kept its statements true while it acted on them, and this shows that a guarantee can steer an action.

9. Scene Analysis

A radar and a camera ride on a rail through a laboratory, and an obstacle blocks their path. Farther ahead, a ball swings on a string. A person walks through the room and may step into the path at any moment. A phone flashes without warning and blinds the camera for a while. The next step of the person and the flash are hidden variables of the scene: no single measurement reveals them, yet they decide whether the platform completes its drive safely. Only a system that analyses the whole scene while it moves can recover them. Scene analysis states what is where, what it is made of, how it moves and how it changes the view of the sensors. The research questions concern what radar and camera know together about a scene that neither knows alone, and whether a platform that moves can keep the guarantees of its perception while it turns them into safe decisions. The scene (Section 9.1) explains what a scene is and places a platform on a rail between obstacles, a ball, a person and a light. The methodology (Section 9.2) builds the world model, the planner and the perception-action cycle that joins all earlier methods. The results (Section 9.3) follow the movement of the platform, the quality of its scene understanding and the stress test of all methods together, and the discussion (Section 9.4) weighs what the platform achieves and where the metal laboratory stops it.

9.1. Scene: The Obstacle Course

9.1.1. Scene Analysis and the Spatial Envelope ★

A scene is the situation in which a sensor works: the objects around it, their materials and motions, the light, and the empty space between them. Scene analysis turns measurements into statements about this situation. It matters because the meaning of a measurement depends on the scene. The same echo can come from an obstacle on a rail or from a harmless wall, and only the scene tells which. A simple, controlled robot ignores this and reacts to one signal. An autonomous system cannot, because the quantities that decide its safety, such as an approaching person or a blinding light, are hidden in the scene and not in a single measurement.

The literature describes a scene in two ways. The first lists its objects with their positions and kinds, which the Scene Panoptic Quality (PQ) of Section 9.1.2 scores. The second, the spatial envelope of Oliva and Torralba, recognises a scene without segmenting a single object [162]. They describe the space of a scene by a few perceptual properties, among them its openness, its roughness and its expansion, and they estimate these properties from the energy of oriented filters on a coarse grid of the image. A scene is then a space with a layout and not a list of things.

However, the envelope exists for the camera only. This thesis translates it to two sensors. The camera envelope keeps the filter energy per orientation and scale on the grid, as defined. The radar envelope measures what the radar sees of the space: the free share of its view, which parallels openness, and the median range of its detections, which parallels the depth of the scene. Every radar point also takes the colour of the camera pixel that it falls on (Section 7.2.1), so that both sensors describe one space (Figure 9.1). The envelope thus

becomes a coarse, cross-modal way to give a scene a meaning. It motivates the scene analysis of this chapter, and it is a personal contribution to the idea.

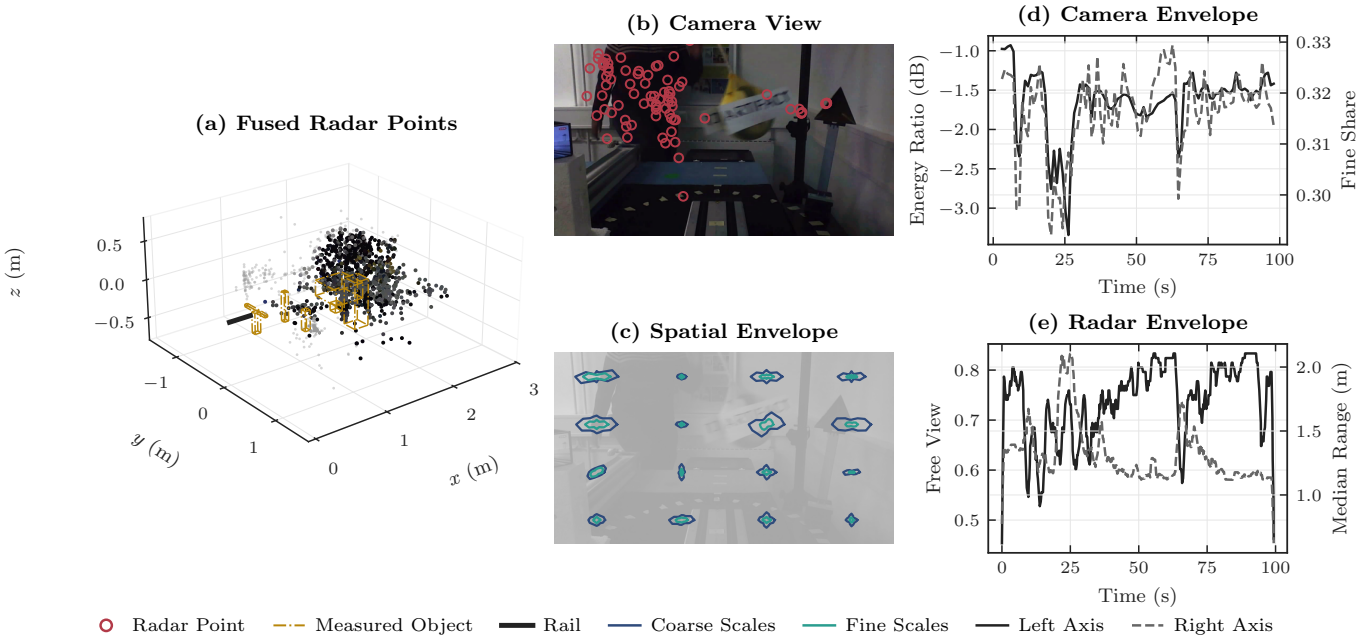


Figure 9.1.: Spatial envelope of camera and radar in an experimental run. (a) Radar points in the colour of their camera pixels, with the measured objects and the rail. (b) Camera view with the projected radar points. (c) Camera filter energy per orientation on a grid. (d,e) Envelope of the camera and of the radar along the run.

9.1.2. Problem Definition

Figure 9.2 shows the scene of the experiment. A corner reflector on a metal plate blocks the path of the platform. This scene is the culmination of all experiments of this thesis. The platform must detect several objects at once, classify their material and motion, read the text on the ball, detect a light, track the ball and a walking person, avoid the obstacle in its path and stay safe by waiting whenever the person may disturb the scene. The three-axis motor stage of Section 4.4 carries the radar of Section 4.1.1 and the camera of Section 4.3 through the scene, and the acquisition of Section 4.5 delivers every frame to the processing in real time.

The obstacle lies 0.5 m ahead. One metre ahead, a phone on its stand serves as the light source, and 1.5 m ahead a metal tripod holds the ball with the printed word ARTIFACT, which swings as a pendulum (Chapter 8). A bottle of water stands next to the tripod. A person walks through the scene and enters the path of the platform six times, and the flash of the phone is on for 14 s. The rail, metal tables and the walls of the laboratory surround everything. The targets differ strongly. The corner reflector returns a strong, point-like echo. The metal plate returns almost none, because a flat surface mirrors the wave away from the radar (Section 2.1.1). The bottle holds water, the ball is a dielectric body that swings, the person moves with articulated limbs, and the light has no echo at all. The room moves past the radar while the platform drives, and this adds the one demand that no earlier experiment had. The platform starts at rest for three seconds

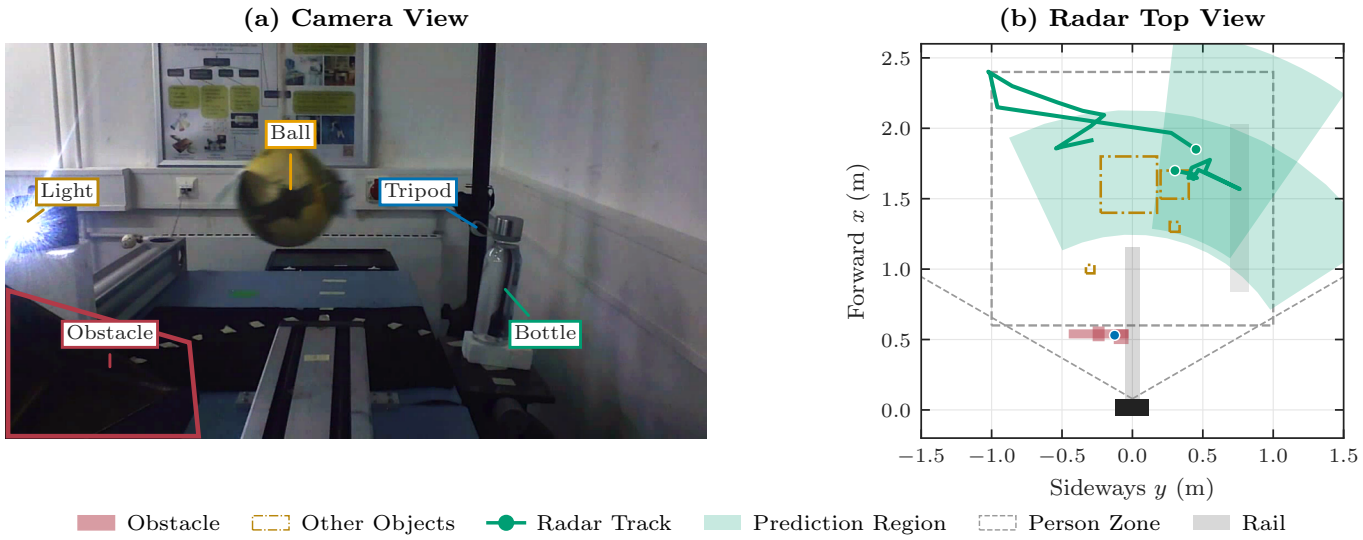


Figure 9.2.: The scene of the experiment. (a) Camera view with the main objects; the red outline marks the obstacle in the path. (b) Radar top view at the instant of the camera view: the measured objects with the obstacle emphasised, the radar tracks of the ball and the person with their prediction regions, the zone in which the person walks, and the rail.

and then drives forward. At every look, it must report every object in view with its position, kind and material, and it must decide whether to drive, to wait or to step aside.

Given the radar frames, the camera frames and the pose of the platform, find the objects of the scene as boxes in a fixed frame with their kind, material and motion, find the light, and decide the next action of the platform. The solution is subject to a false discovery rate of at most $\alpha = 0.1$ per look, a prediction coverage of at least $1 - \epsilon = 0.9$, a gap of 2 cm to every object, and no passage through a person. The solution is measured by the Scene Panoptic Quality below, by the false discovery rate and the coverage of Section 8.1, by the share of person entries after which the platform waits, and by the detection of the light.

The Scene PQ asks at every look which true objects the platform reported, where it placed them and which kind it gave them. It adapts the panoptic quality of image segmentation [69] to boxes on the floor plane. A reported object o and a true object g are compared by

$$s(o, g) = \begin{cases} \max\left(0,\ 1 - \delta(o, g)/c\right) & \text{if } o \text{ has the kind of } g, \\ 0 & \text{otherwise,} \end{cases} \tag{9.1}$$

where $\delta(o, g)$ is the gap between the two boxes, zero if they overlap, and $c = 0.5\,\text{m}$. The Hungarian algorithm pairs reported and true objects with the largest total similarity, and a pair with $s \geq 0.5$ is a true positive. The Scene PQ is

$$\text{Scene PQ} = \underbrace{\frac{\sum_{(o,g)\in \text{TP}} s(o, g)}{|\text{TP}|}}_{\text{SQ}} \cdot \underbrace{\frac{|\text{TP}|}{|\text{TP}| + \frac{1}{2}|\text{FP}| + \frac{1}{2}|\text{FN}|}}_{\text{RQ}}, \tag{9.2}$$

where FP are reported objects without a partner and FN true objects without one. The localisation quality SQ tells how well the found objects are placed, and the recognition quality RQ tells how many objects are

found, with false and missed objects weighted by one half. The score is one only if every object is found, named and placed, and nothing else is reported. A strict version counts a pair only if the material and the person label are also right.

The experimental run lasts 117 s and holds 686 radar looks and 3 378 camera frames. The positions of the objects were estimated by hand, and the person entries and the light were marked on the camera video. The run was recorded raw and processed again by the final method exactly as in the run. The scene scores come from this second pass. The movement of the platform and its decisions are those of the live run, because only the live run moved the rail. Appendix C.5.1 lists the three runs and gives the pooled results. The code of this experiment is in the folder `obstacle/` of the code base (Appendix D).

One observation gives hope. Every part of this problem has an answer in the earlier chapters: detection with a bound on false discoveries, tracks with prediction regions, labels for material and motion, and a camera that reads and detects light. What the platform lacks is a common description of where things are and what they may do next. A world model built from the pose of the platform can provide it.

9.2. Methodology

Three methods turn measurements into actions. The world model places every measurement in one frame and keeps what the platform knows. The planner turns the world model into a decision. The perception-action cycle joins both with the methods of the earlier chapters and closes the loop through the motion of the platform. Green marks the contributions of this thesis.

9.2.1. World Model and Platform Pose ★

A moving platform needs one frame in which all measurements meet (Figure 9.3). The world frame has its origin at the antenna when the rail is at its home position, with x pointing forward along the rail and y to the side. The step counters of the motor controller give the pose of the platform $\boldsymbol{p}_{\mathrm{rail}}$ [144], and each pose is stamped with the time at which it was read, so that it matches the radar frame. A detection at range r and azimuth θ then lies at

$$\boldsymbol{p}_{\mathrm{w}} = \boldsymbol{p}_{\mathrm{rail}} + r\,[\cos\theta, \sin\theta]^{\top}. \tag{9.3}$$

The empty room of Chapter 8 no longer stands still in front of a moving radar. Before the runs, the platform therefore drives the empty path once and records what the radar sees from every rail position. This reference replaces the empty room. A detection counts as a change only if its power exceeds the reference power at the same rail position and range, corrected for the r^4 law of Section 2.1.1. Static objects that the same detector finds in at least half of the reference frames become known parts of the room. The world model then holds three kinds of boxes. A static object is a box around its detections. A known room object is a box from the reference. A mover is a box around the positions that it can reach before the platform can react. The reaction time is measured, not assumed,

$$t_{\mathrm{resp}} = t_{\mathrm{frame}} + t_{\mathrm{proc}} + t_{\mathrm{cmd}} + t_{\mathrm{stop}}, \tag{9.4}$$

where t_{frame} is the duration of a frame, t_{proc} the processing time, t_{cmd} the time to send a command and t_{stop} the time the rail needs to come to rest. The reach of a mover over the horizon $h = t_{\mathrm{resp}}$ is the conformal

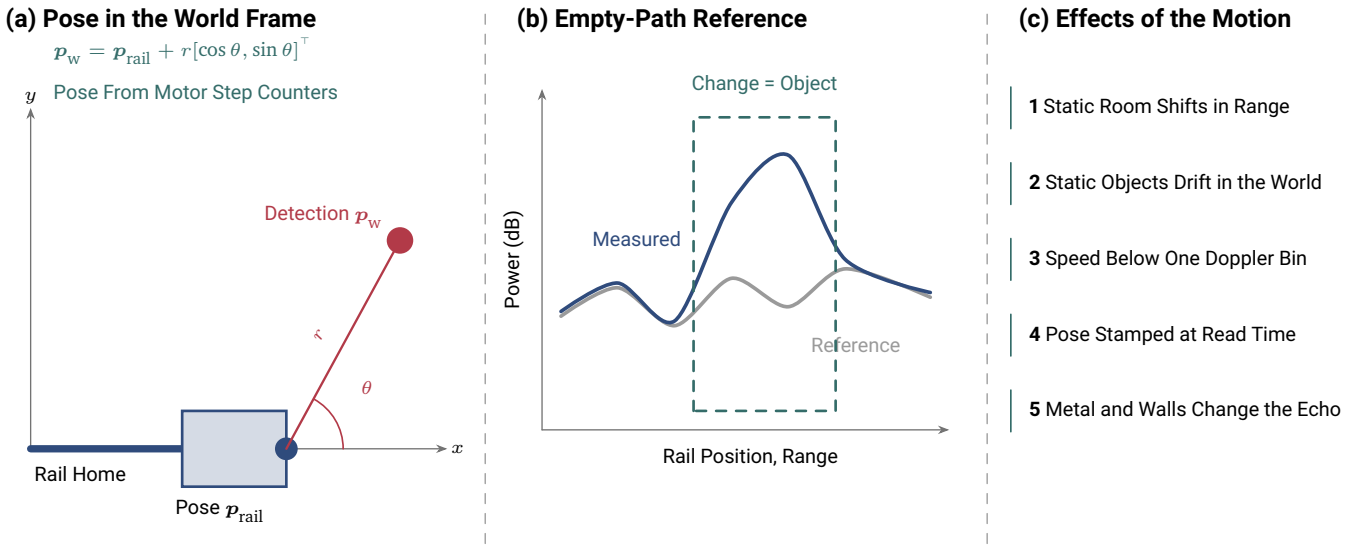


Figure 9.3.: World model and platform pose. (a) The pose from the step counters and a detection at range r and azimuth θ place every measurement in the world frame. (b) The reference run of the empty path gives the power at every rail position and range; a detection is a change if its power exceeds the reference. (c) The five effects of the motion that the discussion follows. Green marks the contributions of this thesis.

quantile of the displacements that the object itself has shown over intervals of length h, in the way of the regions of Section 8.2.1, and it is bounded by the speed of a walking person,

$$R_h = \min\left(v_{\max}\, h,\; Q_{1-\epsilon_{\text{plan}}}\left(\|\boldsymbol{p}(\tau + h) - \boldsymbol{p}(\tau)\|\right)\right), \qquad v_{\max} = 1.5\,\text{m/s}, \quad \epsilon_{\text{plan}} = 0.05, \tag{9.5}$$

where $Q_{1-\epsilon_{\text{plan}}}$ is the empirical quantile over all times τ.

The motion of the platform has five effects. First, the static room shifts in range while the platform drives, so the world model compares with the reference at the same rail position instead of subtracting a fixed background. Second, a static object keeps its world position, because the pose enters the prediction of every track; its measured position still drifts slightly from look to look. Third, the speed of 32 mm/s lies below one velocity step of 6 cm/s of the search chirp, so a slowly moving object cannot be told from a static one by its Doppler shift. Fourth, the pose and the radar frame are read at slightly different times, which the time stamp corrects only in part. Fifth, large metal surfaces and walls change their echo with every position, and the model has no term for this effect. It bounds the effect only through the reference. The first and fourth effects are corrected, the second and fifth are bounded, and the third is accepted.

Three existing ideas meet here. Airborne radar uses the known motion of the platform to separate movers from the ground [8], knowledge-aided radar stores prior knowledge of the environment [68], [123], and radar odometry estimates the pose from the echo of the surroundings [126]. However, the pose on a rail is known, so no estimation is needed, and stored knowledge can be tested as a change detection with a bound on false discoveries [163]. This thesis adds a reference of the empty path that is tested at the level α, and a reach of movers that comes from measured displacements instead of an assumed speed.

9.2.2. Planner ★

The planner applies three rules at every look (Figure 9.4). If a mover can reach the platform within t_{resp}, the platform stops and waits until nothing has been within reach for one second. If a static object blocks the path, the platform stops one range cell before it and searches sideways and in height, on a grid of 1 cm, for an offset that keeps a gap of 2 cm to every box; after the step, it looks for one second from the new position and then drives on. The platform returns to the centre line when the path ahead is free again. A person in the path always makes the platform wait, and the platform never steps around a person. These rules do not prove safety, because a vehicle cannot prove its safety by driving a limited distance [21]. They make every decision traceable to a measured region and a measured reaction time.

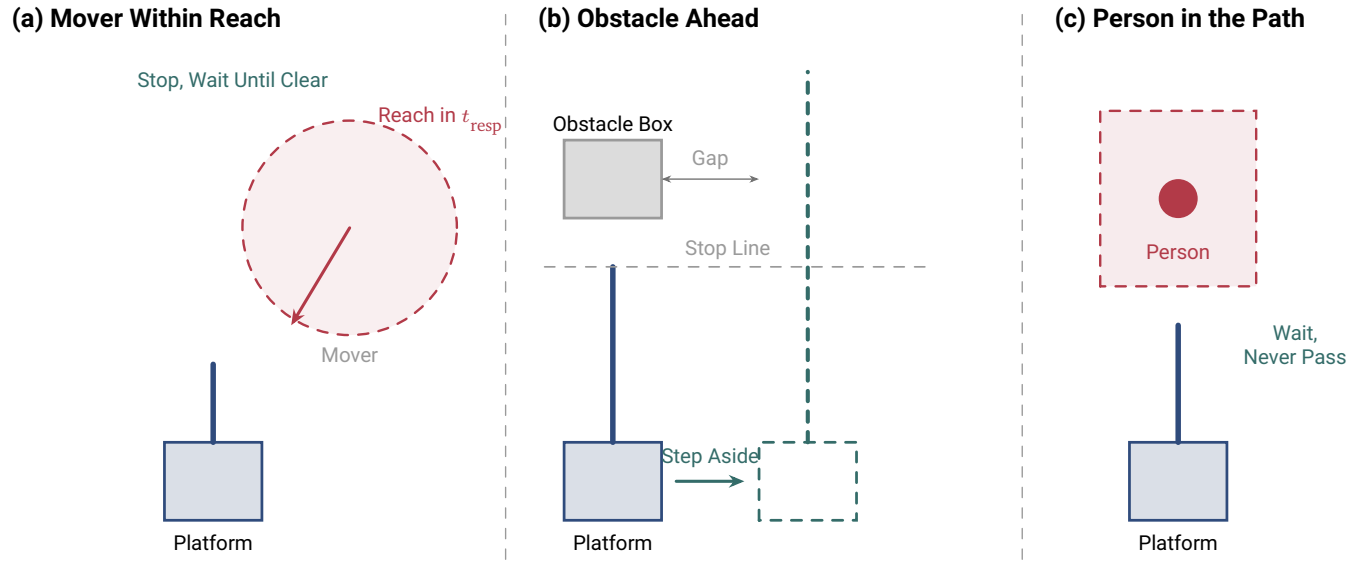


Figure 9.4.: The planner in top view. (a) A mover that can reach the platform within the reaction time stops it. (b) An obstacle ahead makes the platform stop at the stop line and step aside to an offset that keeps a gap to every box. (c) A person in the path makes the platform wait; it never passes. Green marks the contributions of this thesis.

9.2.3. Perception-Action Cycle ★

The perception-action cycle joins all methods of this thesis in one loop (Figure 9.5). Haykin names the loop of a cognitive radar: the radar senses the environment, learns from it and acts so that the next measurement improves [26]. Here, the loop contains the whole platform. Perception runs on the left arc. The radar and the camera sense the scene, and the rail reports the pose. The ego motion stage places the measurements in the world frame and compares them with the empty-path reference. Infini-Rex with the weighted e-BH of Section 8.2.2 detects objects within its error budget. The tracker and KalmanCQR of Section 8.2.1 follow the objects and state where each will be next. M^2-SRC labels material and motion, MetricSight reads, and the light monitor looks for new saturated areas, in place of the per-pixel history of Section 7.2.3, which assumes a camera at rest.

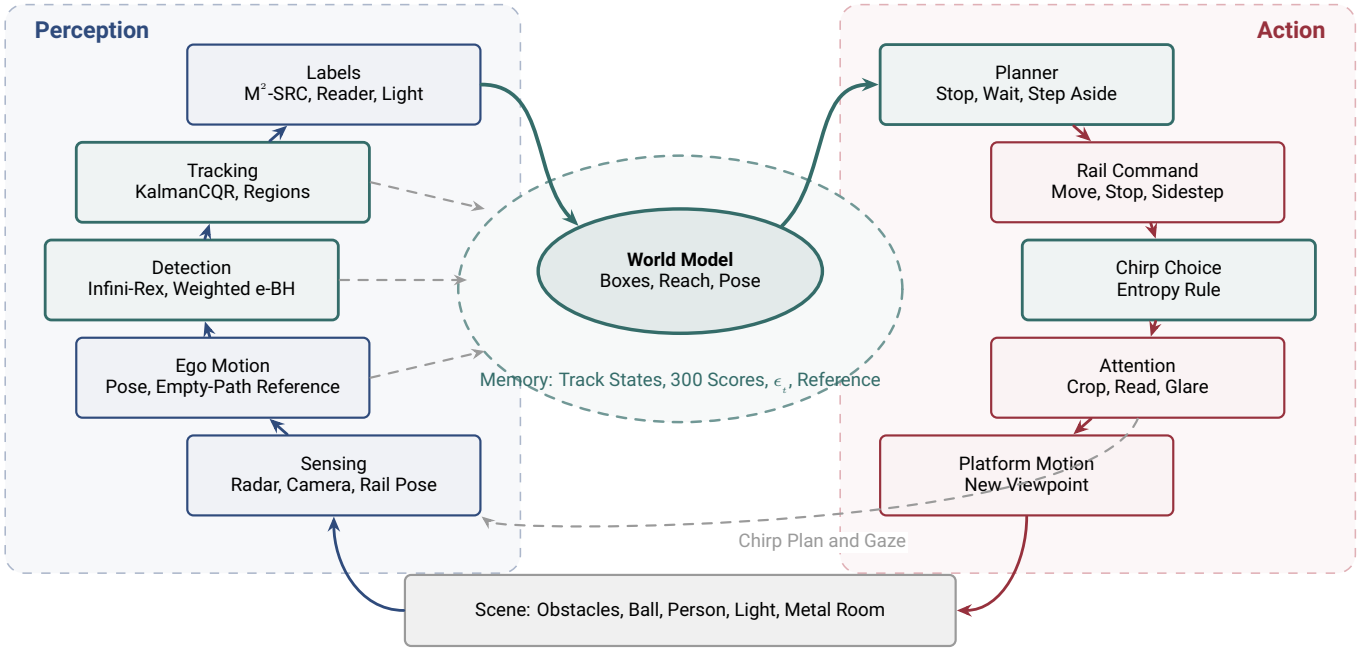


Figure 9.5.: The perception-action cycle of the platform. Perception (left) turns the echoes, the images and the pose into detections, tracks and labels. They update the world model in the centre, whose memory holds the track states, the last 300 scores, the level ϵ_t and the reference. Action (right) turns the world model into a decision, a rail command, a chirp plan and a gaze of the camera. The motion of the platform changes the viewpoint and closes the loop through the scene. Green marks the contributions of this thesis.

The world model is the centre of the cycle. It collects all results in one frame and keeps the memory that carries information from look to look. Action runs on the right arc. The planner decides, the rail executes the command, the chirp choice of Section 8.2.3 selects the next look, and the camera attends to the object that matters most: first an object the platform may collide with, then a person, then the ball. The motion of the platform is not a decision. It is the consequence of a command, and it changes what radar and camera see next. This closes the loop through the scene itself.

Existing cognitive radars close a loop around the radar alone, by choosing the waveform (Section 3.4). The cycle of this thesis closes it around a moving platform and carries a statement through every stage: a bound on false discoveries in the detection, a coverage in the prediction, a measured reach in the decision. A look takes about 0.2 s, a decision follows every look, and the mission lasts 114 s.

9.3. Results

9.3.1. Platform Movement

Figure 9.6 shows how the platform moved in the experimental run. It stands still for three seconds, approaches the obstacle and then advances in short sections that alternate with long waits. In total it travels

1.21 m at a median speed of 32 mm/s and stops 19 times. The rail needs a median of 0.30 s to come to rest after a stop command. It spends 62 of 114 s in the wait state and 33 s approaching.

The obstacle enters the world model as a wide box. At 6 s, the box of the corner reflector reaches 1 m in depth and 25 cm across (left panel, dashed). At 7.2 s the planner decides to avoid it and commands a step of 50 mm to the side. The step reaches 28 mm before a wait interrupts it, and the rail holds this offset until 18.5 s. In the meantime the box shrinks to 24 cm in depth and 8 cm across (solid). At 18.5 s the planner returns the head to the centre line. Its footprint then passes the box with a gap of 3 cm, and the platform continues to 1.15 m.

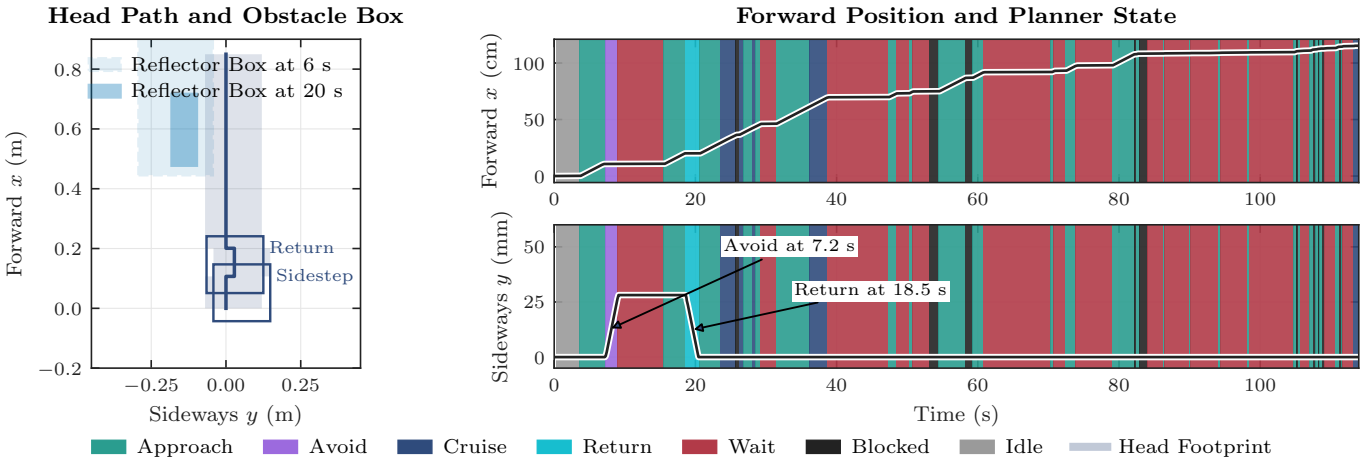


Figure 9.6.: Movement of the platform in the experimental run. Left: top view of the head footprint along the rail and the box that the world model held for the corner reflector at 6 and 20 s. Right: forward position and sideways position of the antenna over time, on the states of the planner; the arrows mark the decision to avoid and the return to the centre line.

9.3.2. Scene Panoptic Quality

Figure 9.7 shows the Scene PQ of the experimental run next to the view from above. The score is 59.6 % over the run, with a localisation quality of 96.9 % and a recognition quality of 61.6 %. When the platform finds an object, it places it well, but it finds only 52 % of the true objects: 1 470 of 2 852 object looks. It reports 453 objects that have no true partner. The strict score, which also requires the right material and person label, is 40.8 %.

The score changes during the run (upper left). It starts at 44 % in the first ten seconds, rises to 72 % between 60 and 100 s and falls to 60 % at the end. The lower left panel shows the parts of the score. The platform finds two to three objects per look and reports at most one object per look that does not exist. The number of missed objects falls from five per look at the start to fewer than one between 60 and 100 s.

The top view shows the tracks of the whole run. A blue track of a metal object ends at the corner reflector, 0.5 m ahead. Between 1.4 and 2.8 m ahead, the tracks of the ball and the person overlap in a dense tangle that covers the whole width of the view. The person produces many tracks that jump over long distances. Per object, the platform finds 58 % of the obstacle looks, 62 % of the ball looks and 56 % of the person looks, and it finds no light source, since a flash returns no radar echo. Pairs of true objects closer than 0.6 m occur

2 656 times over the looks; the platform separates 16 % of them, merges 22 % into one box and misses the rest.

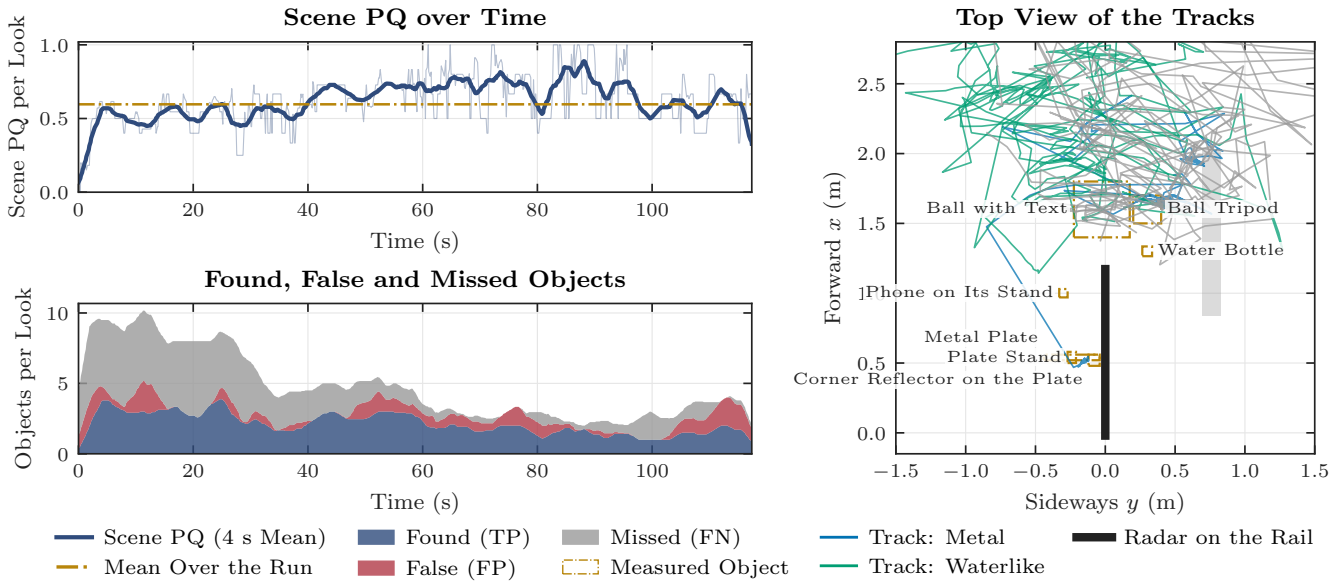


Figure 9.7.: Scene PQ and the top view of the tracks in the experimental run. Upper left: Scene PQ per look and its mean over about 4 s against the mean of the run. Lower left: found (TP), false (FP) and missed (FN) objects per look. Right: the confirmed radar tracks of the whole run, coloured by their material label, with the measured objects (dash-dotted) and the radar on the rail.

9.3.3. Multi-Modal Scene Analysis

Figure 9.8 and Table 9.1 bring all methods together. The top strip shows the decisions of the platform, the person entries and the light over the run. The platform waits for most of the time and drives in short sections. It waited after five of the six person entries, a median of 0.22 s after the entry; the first entry, from 3.1 to 7.6 s, fell into the approach to the obstacle, and no wait began during it. Of its 16 waits, six began during a person entry. The camera detected the annotated light at once, with a delay of 0.00 s, and it reported six light onsets, two of them inside the annotated event.

The statistical statements hold in part. The coverage of the prediction regions is 0.89 to 0.90, and every interval contains the nominal 0.9. The false discovery rate is 0.161 with an interval from 0.128 to 0.192, above the level of 0.1.

The labels are partly right. The material label is right in 61 % and the motion label in 86 % of the labelled looks, and 72 and 67 % of the looks carry a material and a motion label. The material is right for 68 % of the obstacle looks, for none of the ball looks and for 80 % of the person looks; the person label is right in 26 % of the person looks. The reading recovers 2.4 % of the letters. The live run needs a median of 163 ms to process a look, and the result is available 273 ms after the frame.

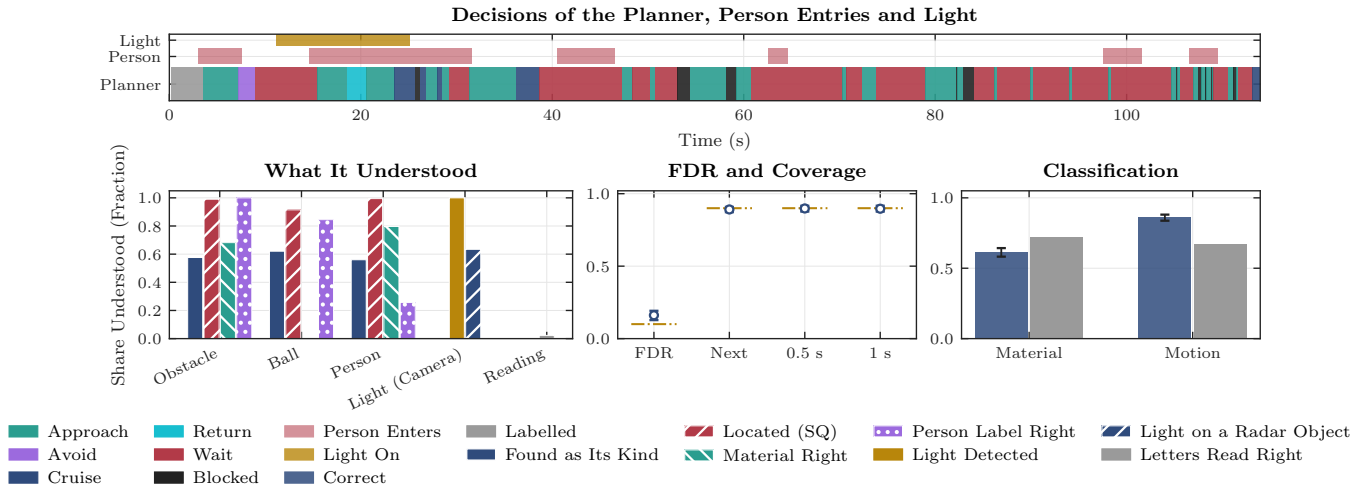


Figure 9.8.: Multi-modal scene analysis in the experimental run. Top: states of the planner, person entries and light over time. Lower left: share of looks in which each object kind was found, located, labelled with its material and its person label, and the detection of the light and the reading. Lower centre: false discovery rate against $\alpha = 0.1$ and coverage of the next look, 0.5 and 1 s ahead against 0.9, with 95 % intervals. Lower right: share of right material and motion labels and share of looks that carry a label.

9.4. Discussion

The results allow two readings. The platform acted well. It avoided the obstacle, waited after five of six person entries within a quarter of a second, kept the coverage of its prediction regions at the nominal level while it moved, and detected the light at once. It also understood its scene only in part. The Scene PQ is 60 %, the false discovery rate of 0.16 lies above the level of 0.1, the reading failed, and the platform waited for more than half of the run. The question is whether the platform acts well because it understands enough of its scene, or whether its actions succeeded although the understanding is incomplete.

9.4.1. Platform Movement

The sidestep followed by a return shows a planner that corrects itself. At 6 s the world model held one wide box for the reflector, and the head of the platform overlapped it by 3 cm, so the planner stepped aside. At 20 s the box was narrow, the head passed it with a gap of 3 cm, and the planner returned. The decision was right at both times, given the evidence of the time. The cause of the wide box is untested. Two explanations are possible: the reflector merged with neighbouring echoes, since the tracker merges objects within 0.6 m, or the first look resolved the direction coarsely. The Rayleigh cell of the array is 9.6° (Chapter 5), which yields a cross-range cell of 17 cm at 1 m and 8 cm at 0.5 m, so a platform that comes closer sees an object more sharply.

Two facts limit this success. First, the radar sees the corner reflector but not the plate under it. A flat plate mirrors the wave away from the radar (Section 2.1.1), so the world model holds the reflector and the plate edges, and a radar-only planner treats the plate surface as free space. The camera sees the plate, but it does

Table 9.1.: All metrics of the multi-modal scene analysis in the experimental run. The first block comes from the recorded data processed by the final method, the second from the live run of the platform.

Stage	Quantity	Value
Recorded data, processed by the final method		
Detection	False discovery rate ($\alpha = 0.1$)	0.161 (0.128–0.192)
	True and false detections per look	5.6 and 1.2
Prediction	Coverage, next look ($1 - \epsilon = 0.9$)	0.891 (0.872–0.909)
	Coverage, 0.5 s and 1 s ahead	0.898 and 0.896
Scene	Scene PQ, localisation SQ, recognition RQ	59.6, 96.9 and 61.6 %
	Scene PQ with kind, material and person label	40.8 %
Objects	Obstacles, ball, person found as their kind	58, 62 and 56 %
	Lamp (radar only)	0 %
Classes	Material and motion label right	61.3 and 86.0 %
	Looks with a material and a motion label	72 and 67 %
Light	Annotated light detected, delay	1 of 1, 0.00 s
	Light onsets reported, inside the event	6, 2
Reading	Letters right in all reads	2.4 %
Live run of the platform		
Planner	Decision at the obstacle (oracle)	avoid (avoid)
	Waits begun, of them during a person entry	16, 6
	Person entries that made it wait	5 of 6
	Median time from entry to wait	0.22 s
	Time in the state wait	62 s of 114 s
Time	Median processing time, frame to result	163 ms, 273 ms

not enter the boxes of the world model in this thesis. Second, the gap that the planner asks for, 2 cm, is smaller than the drift of the static objects in the world model, whose median over the run is 56 mm. The gap protects the platform only as far as the boxes are right. This finding matters beyond the example. The pose of the platform is known exactly, in contrast to radar odometry, which estimates it from the echo [126], and yet the static objects still move in the world model. The remaining error comes from the measurement, not from the pose.

9.4.2. Scene Panoptic Quality

The score of 60 % combines a high and a low value, and the two tell different stories. A localisation quality of 97 % says that the boxes of found objects are placed well. A recognition quality of 62 % says that many true objects are missed (1 382 of 2 852 object looks) and some are invented (453 reports). Three causes explain most of the loss, and each can be checked in the results.

The first cause lies in the truth and not in the method. The light source is part of the true objects, but a flash of light has no radar echo, so the radar cannot find it. It accounts for 12 % of the true looks (348 of 2 852). The camera finds it, which shows what the two sensors know together (Section 9.4.3). The second cause is resolution. The platform separates 16 % of the close pairs and merges 22 % of them into one box. The three parts of the obstacle, the reflector, the plate and its stand, lie within 0.16 m of each other, and the cross-range cell is 8 cm at 0.5 m and larger farther away, so the radar cannot always tell them apart. The third cause is the metal around the ball and the person. The ball hangs 0.4 m from its metal tripod. Both fall into the same range cell, and in the recordings the echo of the ball is about 25 dB weaker than that of the tripod, which

leaves the ball close to the level of the sidelobes of its neighbour. The person produces many tracks that jump over long distances, because the echo of the limbs moves from one strong cell to the next. Only 26 % of the person looks carry the person label, which needs the right material, the right motion and a track that lives long enough. This explains the fall from 60 to 41 % in the strict score.

The rise of the score during the run needs care. The number of missed objects falls from five to fewer than one per look, but the platform also drives forward, and objects leave the field of view and the scored area. The curve therefore compares different object sets at different times, and it does not show learning. A single score also hides much. The Scene PQ of the classical chain, which detects with OS-CFAR, takes the beam peaks as directions, clusters them with DBSCAN and tracks them with the nearest neighbour, is 29 % on the same three runs against 56 % for the method (Appendix C.5.2), which supports the choice of an error-controlled detector. The scores of the literature are not comparable in value: CAFuser reaches a panoptic quality of 59.7 on MUSES [130], but it scores pixels of segments on a large labelled data set, whereas the Scene PQ scores boxes of eight objects in one room. Such methods also draw on denser sensors like LiDAR, while the radar here must separate strong reflectors that stand within 16 cm of each other.

9.4.3. Multi-Modal Scene Analysis

The prediction regions kept their coverage on a moving platform, and the detection did not keep its bound. The two methods differ in one respect. KalmanCQR learns its margin from its own innovations (Proposition 8.1), so a platform motion that the model does not know changes the innovations, and the margin follows. The weighted e-BH cannot adapt in this way. It needs a fair race between null directions and dummies (Assumption 5.1), and in a room of metal the echoes of walls and tables are deterministic and repeat at every look. Chapter 5 found the same mechanism with a static radar, where the false discovery rate rose to 0.13 once the room contributed echoes that the answer key omitted. In Chapter 8, the same detection stayed below the level in the same laboratory. The difference here is the motion and a scene with more metal, and the data do not separate the two. The answer to the part of the research question about guarantees is therefore mixed: the statement about the future survives the motion, and the statement about false detections survives only in part.

The long waits show how the quality of the labels governs the efficiency of the action. The planner waits for any mover that could reach the platform within the reaction time, and a mover that is not classified is treated like a person. Ten of the 16 waits began without a person entry, mostly for such unclassified movers. The platform waited for 62 of 114 s while it advanced 1.2 m. The reaction time itself is short: the rail stops in 0.30 s and a look is processed in 163 ms. The caution comes from the reach rule and from the weak person label, not from the hardware. Better labels would shorten the waiting, and this links the weakness of the material classification of Chapter 6 directly to the speed of the platform. The cost of the caution is time, and the benefit is that the platform waited after five of six person entries. No wait began during the first entry, while the platform approached the obstacle, and the data do not show whether this person could have reached the platform within the reaction time.

The light shows what radar and camera know together. The radar cannot see the flash, and the camera detected it at once. In 63 % of the lit frames the light lay on a radar object, so the radar can supply the range and scale of the source, which the camera cannot measure. The same monitor also reported four onsets that were not the annotated light. A moving camera sees new saturated areas whenever a bright surface enters its view, and shiny metal is plentiful in this laboratory; this explanation is untested. The reading failed for a different reason. The camera attends first to the object that the platform may collide with and to the person,

so the ball rarely received a crop, and the reader was built for a board at rest (Chapter 7). The platform chose safety over reading, as intended.

Finally, the platform ran slower than the camera. The live run skipped 466 of 1 021 frames that arrived while the processor was busy, because a look needed a median of 163 ms, and a result was available 273 ms after its frame. The planner decided on results of this age. The implementation is in Python, and the limit is one of engineering and not of the method.

9.4.4. Findings and Scope

The two research questions receive a partial and encouraging answer. Radar and camera know together what neither knows alone: the camera detects the light that has no echo, the radar supplies its range and scale, and the radar finds the obstacle that the camera cannot measure. The guarantees of perception survive the motion of the platform where they rest on the prediction, with a coverage of 0.89 to 0.90, and they survive only in part where they rest on the detection, with a false discovery rate of 0.16 against 0.1. The platform turns what it knows into safe decisions: it avoided the obstacle, it waited after five of six entries of the person, and every decision can be traced to a measured region and a measured reaction time.

This finding matters because the systems compared in Section 3.6 learn from large labelled data sets, report one score and state nothing about the false objects, and none of them carries the errors of its perception into the action of its platform. However, the platform of this chapter carries a statement through every stage: a bound on false discoveries in the detection, a coverage in the prediction, a measured reach in the decision. This is a new property of a cognitive system, and a laboratory full of metal is a hard place to test it. Metal mirrors, scatters and hides objects, and an open road or the air holds fewer such surfaces. The limits are as clear as the success: one run of one scene with hand-estimated positions, a perception that scores differently when run live (a Scene PQ of 51 % and a false discovery rate of 0.21 in the live pass against 60 % and 0.16 in the replay), a plate that the radar cannot see, and a platform that waits for more than half of its time. The platform did not understand everything it met, but it kept its statements true where they were tested, and it acted on them.

10. Conclusion

10.1. Conclusion

Galileo made his discoveries by accepting only what repeated observation confirmed. This thesis follows his discipline at a far smaller scale: a radar and a camera observe, a known truth checks each measurement, and only findings that withstand doubt remain as bounded truths. In a world that drowns in information and thirsts for knowledge, such truths turn a flood of data into knowledge. The central research question concerns whether a system can distill overwhelming information into a minimal set of bounded truths and reconstruct a true scene through evolving beliefs and cognitive decisions.

Detection (RQ1): in a single snapshot, Infini-Rex for Radar of the T-Rex family finds 94 % of the reflectors at an FDR of 0.02, whereas MUSIC and ESPRIT report 68 % false directions. A guarantee on the false share therefore survives a measured radar signal. With 128 averaged snapshots, however, room echoes raise the FDR to 0.13, above the set level. Hence, the guarantee weakens where echoes add structure.

Material and Motion (RQ2): the sparse classifier M^2-SRC names the motion in 98.1 % and the material in 94.6 % of the windows of an unseen session. An echo alone thus carries the motion and the material of an object, provided the training data covers it.

Radar and Camera (RQ3): the radar tells the camera where to look and how large a word appears. With this knowledge, MetricSight reads text with a character error rate (CER) of 2.1 %, against 13.6 % for a larger neural network. With a bright light source, the radar names it in 94 % of the lit frames, and the camera removes the glare around it. Over three such takes, the removal lowers the error of MetricSight from 32.2 to 26.5 %. Each sensor therefore supplies what the other lacks, so a small reader outperforms a larger network.

Tracking (RQ4): KalmanCQR surrounds the prediction of a Kalman filter with a region that holds 92 % of the next measurements at a stated level of 90 %. The system thus states how far its prediction deserves trust. A cognitive chirp choice raises the higher order tracking accuracy (HOTA) from 49.9 to 58.5 % on one recording, yet the doubled look rate accounts for the gain. Accordingly, the stated trust is earned, whereas the chirp choice itself remains untested.

On the moving platform (RQ5), a planner turns the radar-camera scene into decisions: it avoided the obstacle and waited after five of six entries of a person. Over three runs, its Scene Panoptic Quality is 56 % against 29 % for the classical chain of threshold detection. Its FDR of 0.16, however, exceeds the set level of 0.1. Consequently, a controlled error rate in the detection pays off in the scene, though not at the set level.

The system distills overwhelming information into a few bounded truths and reconstructs a scene from them. In contrast to the classical chain, it scores almost twice as high, although its radar senses far less detail than the LiDAR of related work. Logged recordings replay identically, so every result is reproducible. Ultimately the designed cognitive radar-vision scene analyser is capable of turning many measurements into fewer descisions in a diverse range of tasks.

10.2. Avenues for Future Exploration

Every gap that the experiments leave open defines a more ambitious project. The ideas of this thesis hold in one laboratory on one platform, a controlled setting. Autonomous driving, robotics, aerospace and search and rescue, in contrast, need them in the open world. Each project below names a limit, proposes a remedy and states an expectation.

Open Space and Larger Data. The detection guarantee broke where room echoes accumulated, and the material classifier lost more than twenty points of accuracy when a take was withheld. Recordings in free space, away from walls and corners, let fewer echoes reach the radar. FDR could be increased, since less anomalies would affect measurements. More takes, additional objects and more targets per scene enlarge the data. They allow finer material labels and reveal where the selector breaks.

Beam Steering and Cognitive Control of the Wave. The radar forms its beams in processing only. The loop therefore chooses the chirp but not the direction of the energy. Custom firmware offers transmit beam steering as a new action that concentrates the energy where the doubt is largest. The expected gain is a higher signal-to-noise ratio in that sector.

Beyond One Dummy and One Snapshot: InfiniT-Rex. Infini-Rex stops at the first dummy, $T = 1$, because only this choice has a closed form. It also averages the snapshots, which gained nothing in the room. InfiniT-Rex extends the closed form to $T > 1$, so that one path selects several directions. In simulation, the FDR of MUSIC falls from 0.74 to 0.04 with 128 snapshots. A joint treatment of the snapshots promises the same diversity for the selectors.

Joint Estimation and Classification. M^2-SRC classifies after the detection. Its labels stem from the design of the problem, since a bottle of water or a person is partly dielectric. Consequently, 54 % of the dielectric windows were called waterlike when a take was withheld. Labels that follow the echo, such as bands of the effective permittivity, avoid this overlap. A biconvex formulation of detection and classification lets each task correct the other. These changes promise a stronger dielectric class and a faster classification. The chirp choice then reduces the doubt about the class,

$$U_{\text{total}} = U + \eta\, H, \qquad H = -\sum_{c} p_c \log_2 p_c, \tag{10.1}$$

with U the position entropy of Eq. (8.15), p_c the class share of the code and η a weight.

Radar, Camera and Further Sensors. The alignment error reached 71 pixels on the phone flash, and the track of the person broke into six identities. Mutual checks between camera and radar offer an online correction, and a vision tracker with appearance features bridges identity gaps. A turning sensor pair adds rotation, and additional sensors could add depth. A radar taught by the camera is expected to learn to see like a camera. The spatial envelope of the scene motivates this idea.

Networks, Surfaces and Other Fields. Networked radars with integrated communication and sensing, as envisaged for 6G, offer a route to one shared scene with one guarantee. A reconfigurable intelligent surface is a panel of electrically tunable elements that steers the beam without moving parts. It offers a replacement for the plate of the modular radome. The proposed methods have potential application in other fields presupposing high-dimensional and high-frequency settings: finance with FPGAs, or genome wide studies to even particle physics. Keeping the interdisciplanry nature of the methods and replicating the system opens the doors to explore avenues of the future.

A. Supplementary Information: Implementation

This appendix supplements Chapter 4. It details the four hardware blocks, documents the design and the printing of the housing, describes the laboratory and the measured antenna patterns, and leads from the first cable to the first radar cube.

A.1. Hardware

Figure A.1 shows the four hardware blocks, and Table A.1 lists their key specifications [133], [136], [139], [144]. The radar board carries the AWR2944P chip behind a patch array of four transmit and four receive antennas. Its micro-USB port J10 provides the serial interface for commands and for the flashing of firmware, three jumpers select the boot mode, and a reset button restarts the chip. The capture card receives the raw samples over four LVDS lanes and sends them as UDP packets over Gigabit Ethernet. The camera delivers 1280×720 pixels from its CMOS sensor as M-JPEG over the USB video class. The controller TMCM-310 addresses the three stepper drives of the rail with nine-byte commands of the TMCL protocol.

Table A.1.: Key specifications of the components.

Component	Key specifications
Radar	Texas Instruments AWR2944P evaluation module; 77 to 81 GHz; 4 transmit and 4 receive antennas, 16 virtual channels in time-division multiplexing, 12 in the azimuth row; raw samples as real values; out-of-box demonstration, software development kit 04.07.02.01; 12 V supply with more than 2.5 A
Capture card	Texas Instruments DCA1000EVM; LVDS input, Gigabit Ethernet output; 5 V supply with at most 700 mA
Camera	ISY IW-1000-1; CMOS sensor; USB video class; 1280×720 pixels, M-JPEG
Motor stage	isel ball screw drives; forward axis LES 6, 1 390 mm; sideways axis LES 4, 290 mm; vertical module; stepper motors
Controller	Trinamic TMCM-310; 8 to 34 V; 0.3 to 1.1 A per coil; microsteps up to 16; RS-232 at 9 600 baud
Computer	Lenovo IdeaPad Flex 5; AMD Ryzen 5 5500U, 8 GB; Windows 11; the Ethernet adapter of the capture card on a USB 3 port

A.2. Computer-Aided Design ★

The housing was designed in Autodesk Fusion around a model of the evaluation module, so that every opening, mount and clearance could be checked before printing. All parts were printed on an Original Prusa XL with five toolheads and sliced in PrusaSlicer (Figure A.2). The housing consists of the chassis, the back plate that mounts the chassis on the sensor head, the lid with the engraved name ARTIFACT, and two radome

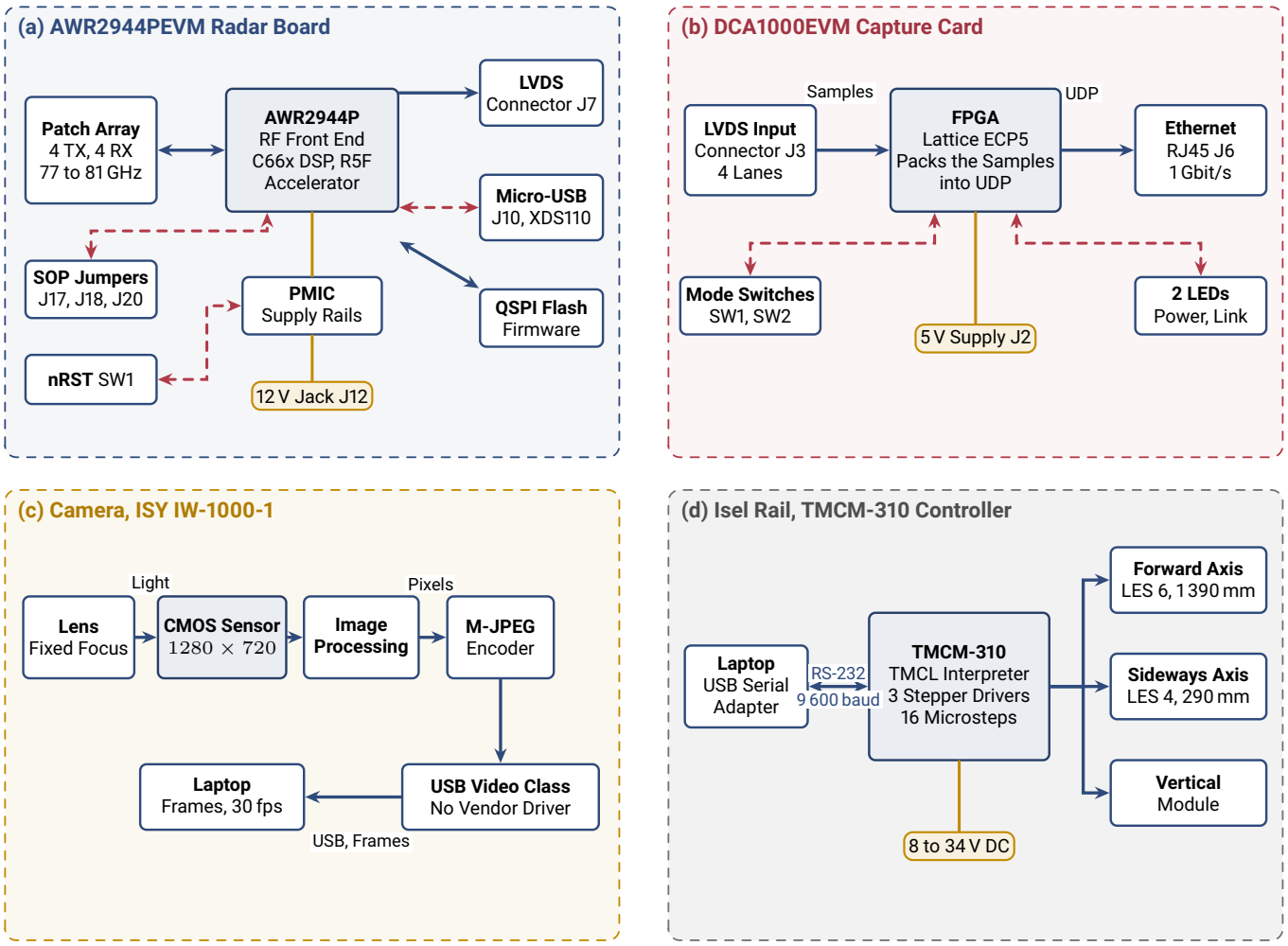


Figure A.1.: Schematics of the four hardware blocks of ARTIFACT: (a) the radar board, (b) the capture card, (c) the camera and (d) the motor controller with its three drives. Solid arrows carry data, dashed arrows carry commands, and gold lines carry power.

plates, one of PETG and one of PLA (Figure A.3). The window of the lid takes an exchangeable plate, so that materials and thicknesses can be tested without printing the housing again (Section 4.2).

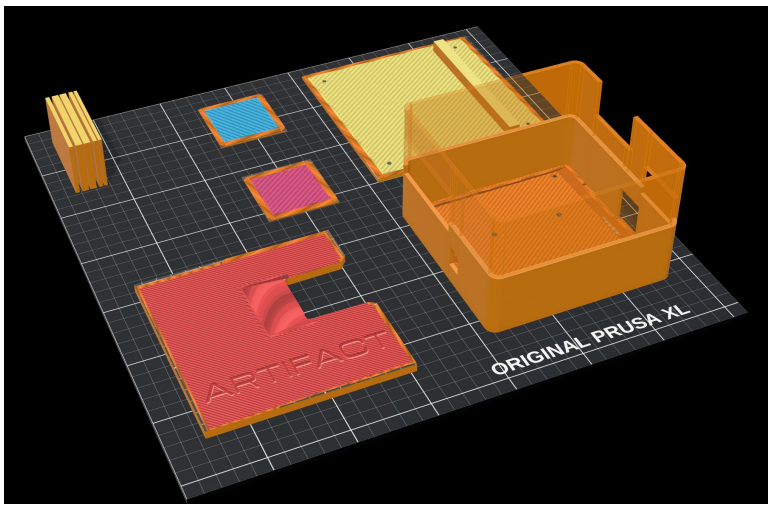


Figure A.2.: The housing in the Original Prusa XL while the print is in progress.

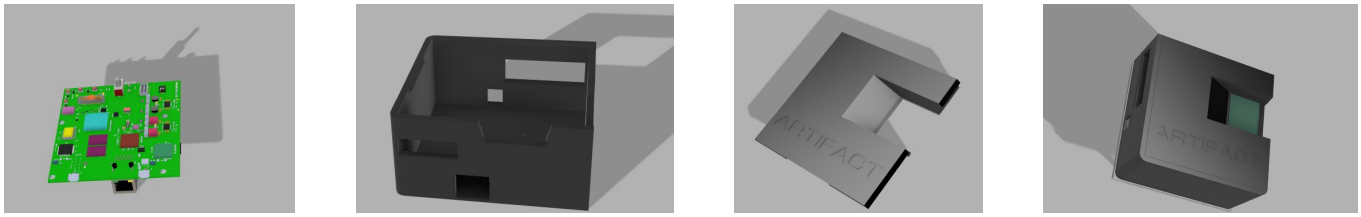


Figure A.3.: Computer-aided design of the housing, from left to right: model of the evaluation module, chassis, lid with the window and the engraved name, and the assembled housing.

Table A.2 lists the dimensions. The radome slot has the footprint of the plates plus the clearance of the fit, and it ends in a funnel that reaches about 9.4 mm below the front face of the lid. This depth fixes the air gap between the plate and the antennas, so that the gap of the design (Section 4.2) is the gap of the print.

Table A.2.: Dimensions of the printed parts.

Part	Dimensions	Remark
Chassis	126 × 126 mm inside, 57.5 mm high	The walls add their thickness
Back plate	136 × 136 mm, 5 mm thick	Mounts the chassis on the sensor head
Lid	125 to 128 mm across	Engraved name; funnel to the radome slot
Radome plates	44 × 44 mm; PETG 1.14 mm, PLA 1.08 mm	Thickness of half a wavelength in the material
Radome slot	Footprint of the plates plus the fit; about 9.4 mm behind the front face	Fixes the air gap to the antennas

A.3. Laboratory Conditions

All recordings were made in one laboratory. Figure A.4 shows the empty room as the radar sees it: the take without objects that every recording subtracts as background. The wall at the back of the room, with its framed posters, returns the broad band of echoes at about 2.85 m, and a second wall closes the room on the right. The metal rail next to the optical rail appears as a strong line at about -0.55 m, and the table with the platform stands at the origin. All of these echoes are real and lie outside every ground truth. They bound the evaluation region of Chapter 5, and they are the likely source of the room echoes that raise the false discovery rate in Chapters 5 and 9. Table A.3 lists the geometry of the experiments.

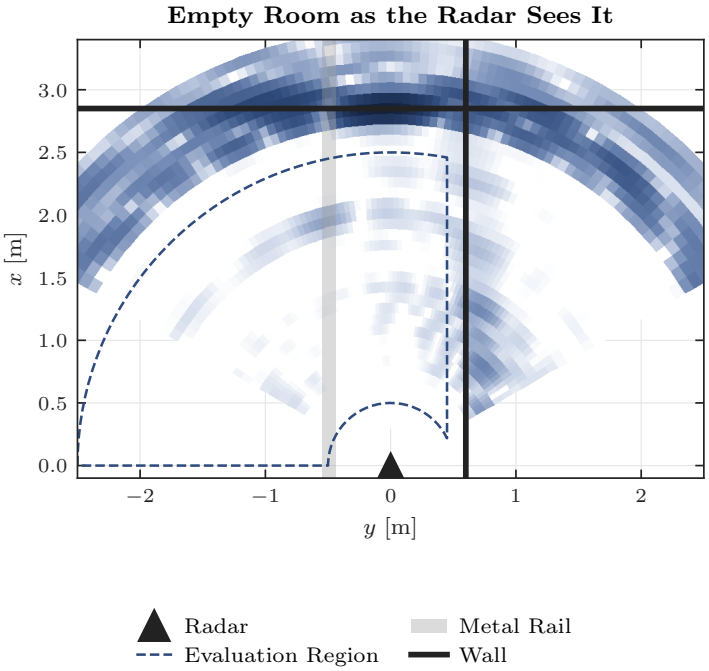


Figure A.4.: The empty room as the radar sees it, seen from above: the echoes of the back wall with its framed posters, the metal rail beside the optical rail and the walls. The dashed line bounds the evaluation region of the detection experiment.

Table A.3.: Geometry of the experiments in the laboratory, relative to the antenna at its home position.

Quantity	Meaning	Value
Minimum range	Closer echoes are dominated by leakage between transmitter and receiver	0.25 to 0.5 m
Evaluation region of Chapter 5	Range from the radar and lateral extent, bounded by the wall and the metal rail	≤ 2.5 m, ≤ 0.45 m
Region of the object in Chapter 6	Range and azimuth of the object window	0.3 to 1.6 m, $\pm 25^\circ$
World of Chapter 9	Forward limit and lateral limits from the centre line of the rail	2.6 m, ± 1 m
Travel of the platform	Forward and sideways travel of the antenna from home	1.24 m, ± 0.06 m

A.4. Antenna and Radome Patterns ★

A corner reflector stood 0.17 m in front of the radar, and the radar turned from -90° to $+90^\circ$. The energy of each of the 16 channels in the range bin of the reflector was averaged into bins of 5° and referred to the largest reading of the open antenna. Three data sets were recorded: the open antenna (170 frames), the antenna behind the PLA plate (124 frames) and behind the PETG plate (161 frames). Table A.4 compares the patterns, and Figure A.5 shows every channel.

Three effects make the measurement approximate. First, the 12-channel row has an aperture of about 21 mm, so that its far field begins at $2D^2/\lambda \approx 0.24$ m, and a reflector at 0.17 m lies in the near field. Second, the noise floor differs between the data sets (-58, -55 and -50 dB), and so does the number of frames; both move the maximum of a noisy pattern and with it the beam widths. Third, the plates differ in thickness (1.14 and 1.08 mm) and in permittivity, and the printed thickness deviates from its design value. The measurement therefore resolves differences of about 1 dB, but it cannot assign them to absorption, thickness or geometry.

Table A.4.: Comparison of the patterns: peak, beam widths at -3 and -10 dB, difference to the open antenna in three sectors and spread between the channels.

Data set	Peak (dB)	BW_{-3} (°)	BW_{-10} (°)	$\Delta\ \lvert\theta\rvert \le 30^\circ$ mean / max	$\Delta\ \lvert\theta\rvert \le 60^\circ$ mean / max	$\Delta\ \lvert\theta\rvert > 60^\circ$ mean	Spread (dB)	Noise (dB)
0 dB: highest 5° bin of the open antenna; Δ: difference to it per bin								
No Radome (Air)	+0.0	12	57	+0.0 / 0.0	+0.0 / 0.0	+0.0	2.8	-58
PLA Radome	-2.4	26	59	-2.2 / 9.8	-3.7 / 14.6	-4.1	2.4	-55
PETG Radome	-2.2	35	83	-0.5 / 7.0	+0.2 / 7.0	-0.7	2.0	-50

A.5. Radar Tutorial and Diagnosis ★

A.5.1. Start-Up

The order of the steps matters, because the radar board decides its mode at power-up (Figure A.6).

1. Plug the 12 V supply into J12 of the radar board and press the reset button nRST (SW1) once, as the manufacturer recommends for a reliable boot [133]. Three LEDs glow: the 12 V supply (D12), the 5 V rail (D16) and the reset indicator (D13). A fourth LED appears when the USB cable of the debugging connection is attached.
2. Plug the 5 V supply into J2 of the capture card. Two LEDs glow.
3. Link the two boards first, with the blue LVDS ribbon cable from J7 of the radar board to J3 of the capture card.
4. Add the other cables: the micro-USB cable from J10 to the laptop for the serial interface, the Ethernet cable from J6 to the adapter on a USB 3 port, and the camera.
5. Run `check_setup.py` of the code base. It tests the serial ports, the answer of the command line to `sensorStop`, the address of the laptop, the handshake with the capture card and the free data port.

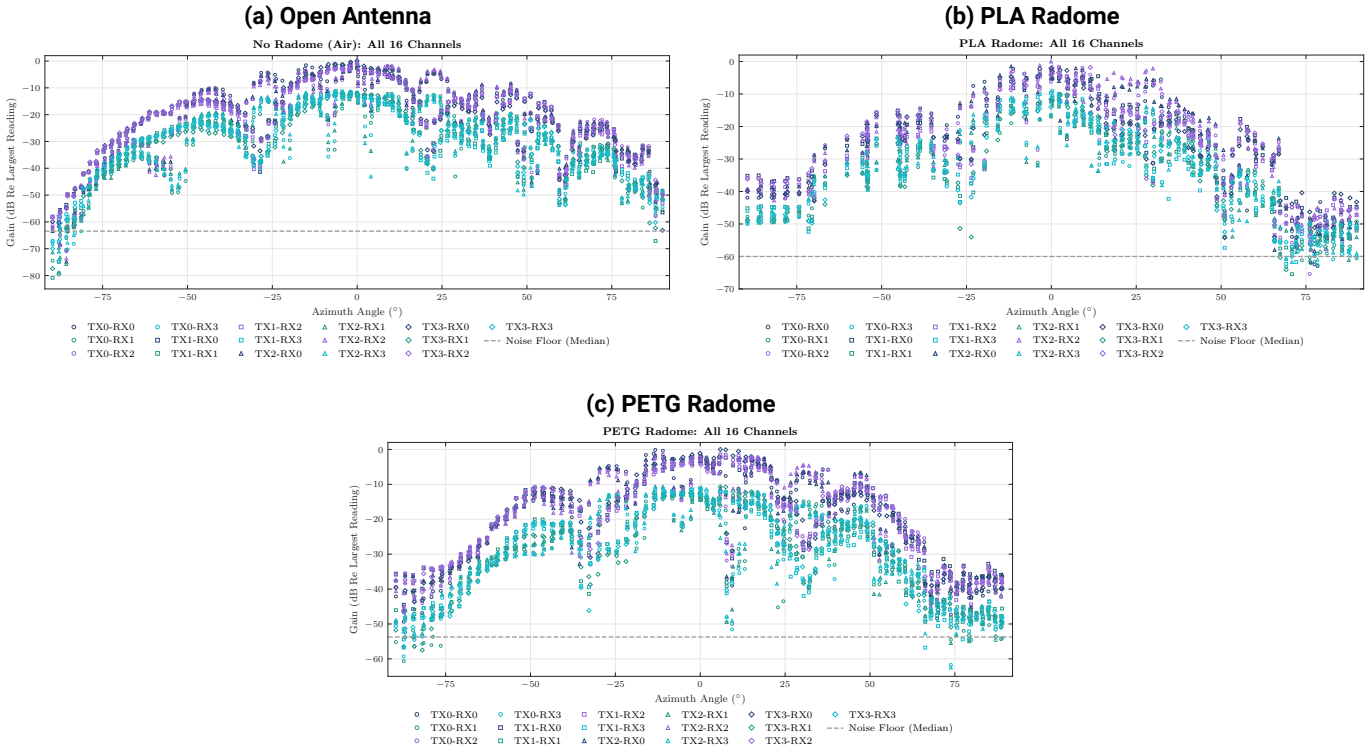


Figure A.5.: Antenna pattern of the 16 channels over the azimuth angle, one marker for each transmitter-receiver pair, relative to the largest reading of each data set: (a) the open antenna, (b) the antenna behind the PLA plate and (c) behind the PETG plate.

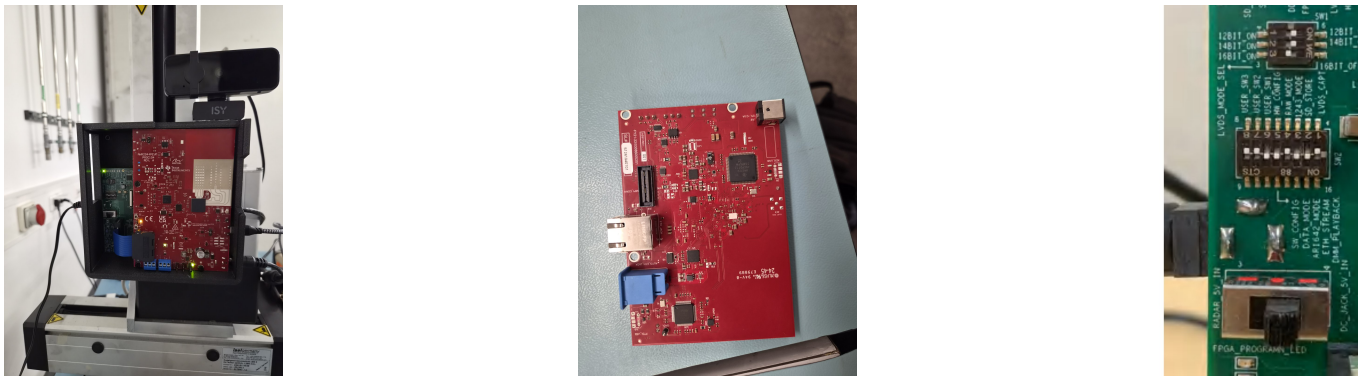
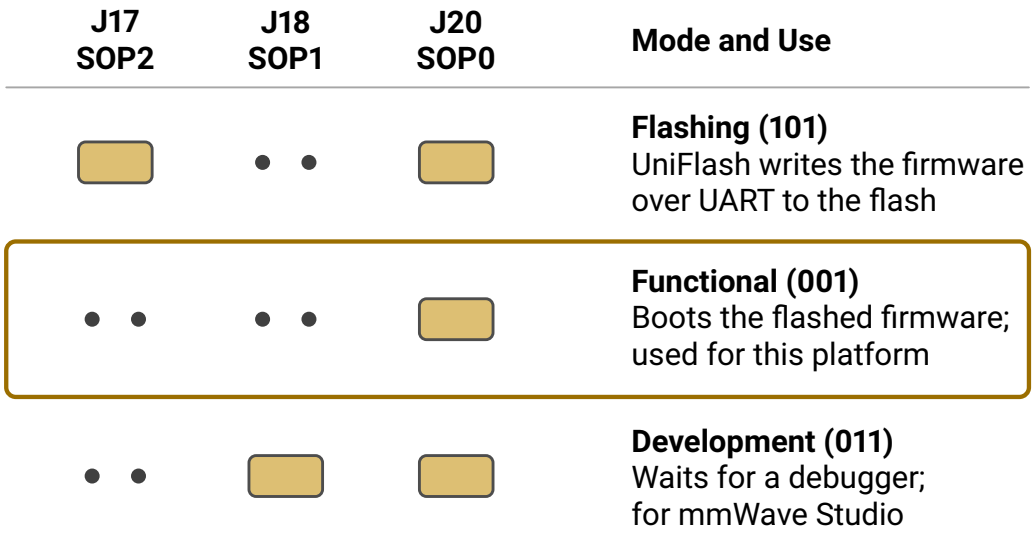


Figure A.6.: The radar board in its chassis with its LEDs lit (left), the capture card (centre) and the switch settings of the capture card for use with the radar board (right, adapted from Texas Instruments [133]).

A.5.2. Operating Modes

The three sense-on-power jumpers J17, J18 and J20 set the state of the lines SOP2, SOP1 and SOP0 while the chip boots (Figure A.7); the lines are read only at boot, so a change takes effect after a press of nRST [133]. In the flashing mode, the UniFlash tool of Texas Instruments writes the firmware of the software development kit to the flash memory through the UART of the board; it needs the COM port of the XDS110 interface and the binary of the kit. In the functional mode, the chip boots the firmware that was flashed, and the platform uses this mode: the configuration of the radar arrives as commands over the serial port. In the development mode, the chip waits for a debugger, which is the mode for mmWave Studio. If the board does not answer, a press of nRST restores the boot state and is the first remedy.



Cap on the pins: closed jumper, 1. No cap: open jumper, 0. The lines are sensed only at boot, so press nRST after a change.

Figure A.7.: Sense-on-power jumpers of the radar board, from left to right J17 (SOP2), J18 (SOP1) and J20 (SOP0), for the three modes. The highlighted row is the mode of the platform.

A.5.3. Diagnosis

Three problems cost most of the time during the setup, and the code base contains a check for each. First, the firmware accepts a command of the configuration only in a specific form. A program that sends the configuration and reads the answers with a long timeout waits for 30 s, while a reader that returns as soon as the line has been quiet for a short time needs 6 s; `probe_cli.py` shows which form of a command the firmware accepts. Second, the raw samples of the AWR2944 are real values in time-division multiplexing, and the settings of complex output that exist for older chips cause the capture card to receive no data at all. Third, the capture card loses packets if its Ethernet adapter sits on a USB 2 port, which is why the adapter belongs on a USB 3 port.

A.5.4. From the Configuration to the Point Cloud

The radar tutorial of the code base leads from the cable to the point cloud and runs on recordings without hardware. The configuration of the radar is a text file of commands that resets the radar, sets the channels and the format of the samples, defines the chirp and its repetition in a frame, and starts the frames. The raw samples of one chirp and their spectrum (Figure A.8) come first. The samples of the 16 virtual channels, the 128 chirps and the 256 samples then form the radar cube, and Fourier transforms over the three axes give range, speed and direction (Figure A.8). The tutorial also reads the point cloud that the chip computes itself; the processing of the experiments replaces it by the methods of this thesis. On synthetic targets of known range, speed and direction, the processing finds every target within one speed bin and 0.3°, and on a recording its output equals that of the loader of the detection experiment.

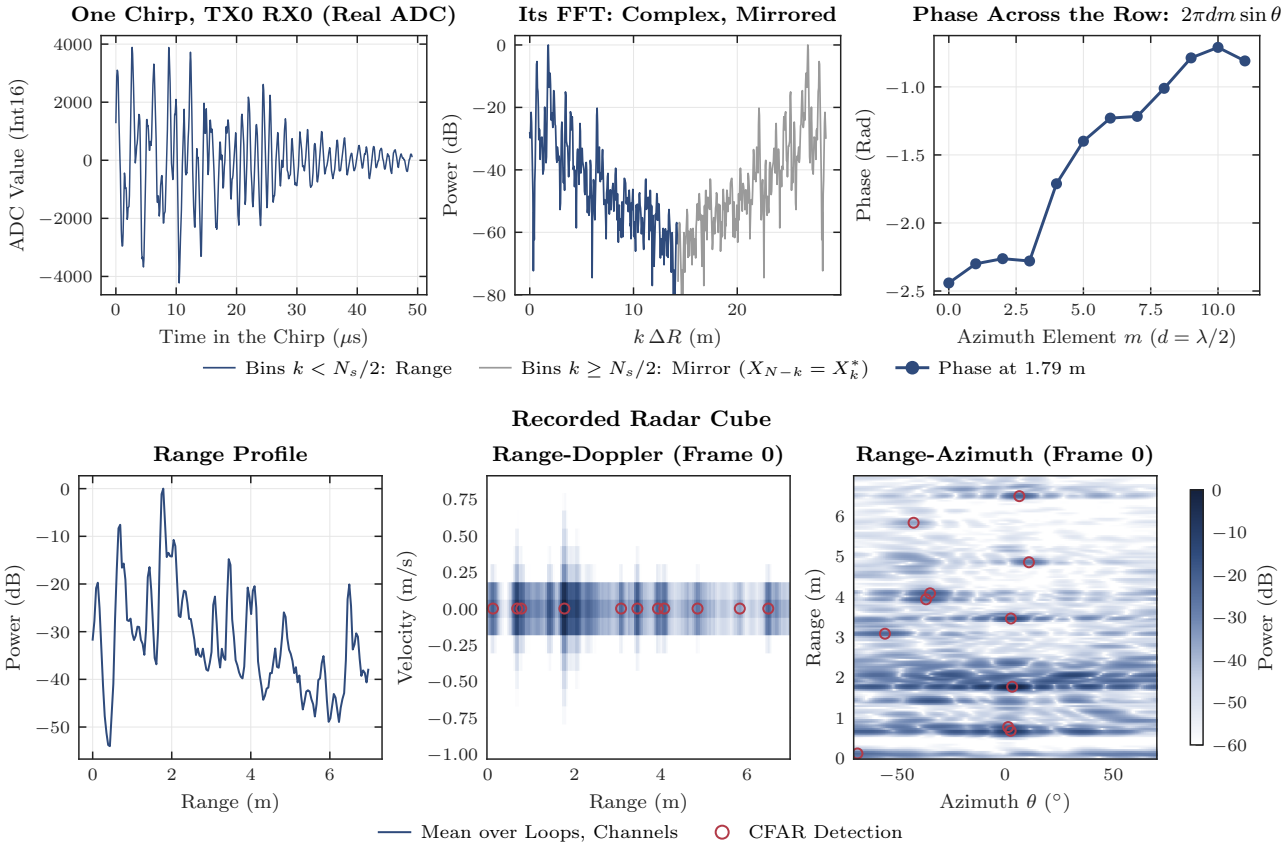


Figure A.8.: Raw samples of the analog-to-digital converter for one chirp and their spectrum (top), and the radar cube of one frame with the maps that follow from it (bottom).

Table B.1.: Methods of the experiment, the directions on which each decides and the guarantee that it claims.

Method	Directions	Error Guarantee
Bartlett, Capon, MUSIC, ESPRIT	spectrum peaks, 83-point grid	none
MLP	83 scores	none
CA- and OS-CFAR	83 cells	P_{fa} per cell
BH, BY, Fixed-X knockoff+	coarse grid, 11 points	FDR $\leq \alpha$
Model-X knockoff+, T-Rex, CT-Rex, Infini-Rex	fine grid, 83 points	FDR $\leq \alpha$
Infini-Rex for Radar	continuous, from 83 points	FDR $\leq \alpha$

B. Supplementary Information: Proofs and Derivations

This appendix collects every proof, derivation, definition and algorithm that the chapters cite, in the order of the chapters. The first section proves the control of the false discovery rate for the selectors of this thesis, including Infini-Rex and its variant for radar.

B.1. Detection and Estimation

B.1.1. Range Estimators

Three estimators refine the range within a gate. The bin centre r_b is the simplest estimate, with an error of up to half a bin. A parabola through the logarithmic powers P_{b-1}, P_b, P_{b+1} of the peak bin and its neighbours places the peak at

$$\hat{r} = \left(b + \frac{1}{2} \frac{P_{b-1} - P_{b+1}}{P_{b-1} - 2P_b + P_{b+1}} \right) \Delta r. \tag{B.1}$$

The Chirp Z-Transform (CZT) evaluates the same spectrum on a fine grid of 257 frequencies within one bin on either side of the peak and takes the maximum [164]. The CZT costs more than the parabola, but it does not assume a parabolic peak shape.

B.1.2. Additional Methods

Table B.1 lists all methods of the experiment, and Algorithm 2 states the test of Benjamini and Hochberg.

Algorithm 2 *Benjamini–Hochberg and Benjamini–Yekutieli on the Coarse Grid.*

1. **Input**: $\bar{y}$, $\boldsymbol{X}_{\mathrm{N}}$, noise variance σ^2/K, level α, variant BH or BY.
2. **Compute** $\hat{\boldsymbol{\beta}}$ and $\mathrm{Var}(\hat{\beta}_j)$ by Eq. (5.5); **set** $T_j = |\hat{\beta}_j|^2/\mathrm{Var}(\hat{\beta}_j)$ and $p_j = e^{-T_j}$ for every direction.
3. **Sort** the p-values, $p_{(1)} \le \cdots \le p_{(p_{\mathrm{N}})}$. **If** the variant is BY, **set** $\alpha \leftarrow \alpha/\sum_{i=1}^{p_{\mathrm{N}}} 1/i$.
4. **Find** $\hat{k} = \max\{k : p_{(k)} \le k\alpha/p_{\mathrm{N}}\}$, with $\hat{k} = 0$ if none qualifies.
5. **Output**: the directions of the $\hat{k}$ smallest p-values.

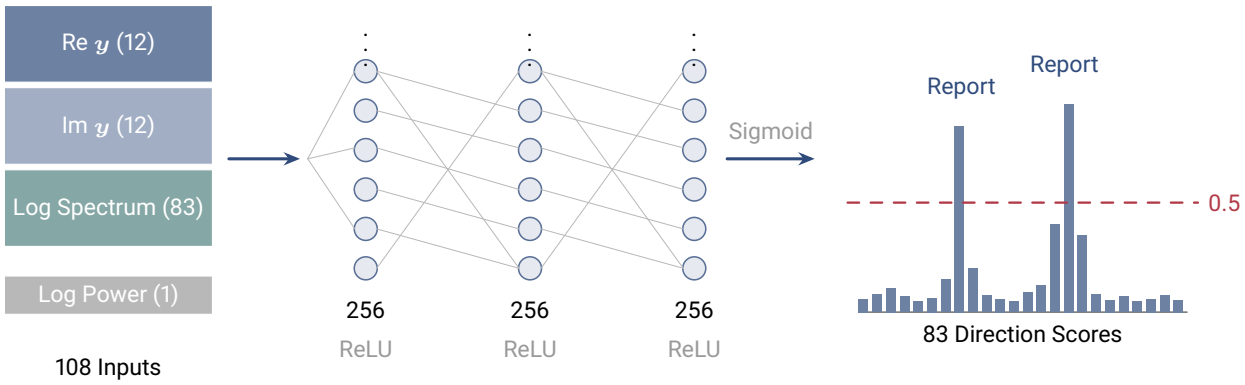


Figure B.1.: The multilayer perceptron: 108 inputs, three hidden layers of 256 ReLU units and 83 sigmoid scores; peaks above 0.5 are reported.

Learned baseline. The learned baseline takes one idea of Liu and colleagues, a network that scores every direction of a grid [118], in its simplest form, a Multilayer Perceptron (MLP) (Figure B.1). Its 108 inputs are the real and imaginary parts of the snapshot, rotated to the phase of the strongest beam and normalised, the logarithm of the beam spectrum on the 83 directions and the logarithm of the power. Three hidden layers of 256 units with ReLU activation follow, and a sigmoid turns 83 outputs into scores. A direction is reported if it is a peak with a score above 0.5. The network was trained on 200 000 simulated gates of an ideal array with up to three reflectors; it knows only the conditions of its training and gives no error guarantee.

Infini-Rex. Algorithm 3 states Infini-Rex and its variant for radar for one range gate.

CT-Rex. Algorithm 4 states CT-Rex for one range gate.

B.1.3. Proofs and Derivations ★

The e-BH rule.

Proposition B.1 (e-BH). *Let $e_1, \ldots, e_p \ge 0$ be e-values with $\mathbb{E}[e_j] \le 1$ for every null index $j \in \mathcal{H}_0$, let $e_{[1]} \ge \cdots \ge e_{[p]}$ be their order, and let $\hat{k} = \max\{k\colon e_{[k]} \ge p/(\alpha k)\}$, with $\hat{k} = 0$ for an empty set. The rule that selects the $\hat{k}$ largest e-values has* FDR $\le \alpha\, p_0/p$ *for any dependence between the e-values, where $p_0 = |\mathcal{H}_0|$.*

Proof. Idea. A selected index has a large e-value, so its selection indicator is at most a multiple of its e-value, and a null e-value has mean at most one.

Algorithm 3 *Infini-Rex and Infini-Rex for Radar ($\star$) for One Range Gate.*

1. **Input**: $\bar{\boldsymbol{y}}$, $\boldsymbol{X}$ with unit-norm columns, level α, set $\mathcal{L}$; ($\star$) channel noise levels σ_m^2, each the median of the squared range bins beyond the minimum range over all snapshots, divided by $\ln 2$.
2. ($\star$) **Whiten** $\bar{\boldsymbol{y}}$ and $\boldsymbol{a}(u)$ with $\boldsymbol{W} = \operatorname{diag}(\sigma_m^{-1})$ scaled to unit mean.
3. **Initialise** $\boldsymbol{r}_0 \leftarrow \bar{\boldsymbol{y}}$, $S_0 \leftarrow 0$, $\mathcal{A} \leftarrow \emptyset$, $s_{\max} \leftarrow \lfloor n/2 \rfloor$.
4. **For** $t = 1, \ldots, s_{\max}$ **do**:
   - 4.1. **Select** $j_t = \arg\max_{j \notin \mathcal{A}} |\boldsymbol{x}_j^H \boldsymbol{r}_{t-1}|^2$ and **set** $\mathcal{A} \leftarrow \mathcal{A} \cup \{j_t\}$. ($\star$) **Orthogonalise** $\boldsymbol{x}_{j_t}$ against the refined steering vectors.
   - 4.2. **Compute** ρ_t by Eq. (5.6), $P_t = (1-\rho_t)^{n-1}$ and $S_t = S_{t-1} + P_t$, then e_{j_t} by Eq. (5.8). **If** $e_{j_t} < p/(\alpha s_{\max})$, **stop**.
   - 4.3. ($\star$) **Refine** every selected direction by Eq. (5.10) until no direction moves.
   - 4.4. **Update** $\boldsymbol{r}_t$ as the part of $\bar{\boldsymbol{y}}$ orthogonal to the selected (refined) columns.
5. **Set** $e_j \leftarrow 0$ for $j \notin \mathcal{A}$, **find** $\hat{k}$ by Eq. (5.9); ($\star$) **merge** directions closer than one grid step.
6. **Output**: the directions of the first $\hat{k}$ steps (refined for $\star$).

Algorithm 4 *CT-Rex for One Range Gate.*

1. **Input**: $\bar{\boldsymbol{y}}$, $\boldsymbol{X}$, level α, forward solver, $K_{\text{rex}} = 20$, voting levels $v \in \{0.5, 0.55, \ldots, 0.95\}$.
2. **For** $\ell = 1, \ldots, 10$ **do**, and **leave the loop** as soon as $\widehat{\text{FDP}}_{1,L}(0.75) \le \alpha$: **set** $L = \ell p$; **for** every run k **draw** $\boldsymbol{D}_k \in \mathbb{C}^{n \times L}$ with independent standard complex Gaussian entries, **normalise** its columns and **run** the solver on $[\boldsymbol{X}\ \boldsymbol{D}_k]$ until one dummy has entered; **compute** $\Phi_{1,L}$ and $\widehat{\text{FDP}}_{1,L}(v)$.
3. **Set** $T \leftarrow 1$. **While** $\widehat{\text{FDP}}_{T,L}(0.95) \le \alpha$ and $T < \lceil n/2 \rceil$ **do**: **set** $T \leftarrow T+1$, **continue** every run until T dummies have entered and **recompute** $\Phi_{T,L}$ and $\widehat{\text{FDP}}_{T,L}(v)$.
4. **Solve** $(v^\star, T^\star) = \arg\max |\{j : \Phi_{T,L}(j) > v\}|$ subject to $\widehat{\text{FDP}}_{T,L}(v) \le \alpha$ over all voting levels and $T < T_{\text{stop}}$.
5. **Output**: $\hat{\mathcal{A}} = \{j : \Phi_{T,L}(j) > v^\star\}$.

1. For $\hat{k} = 0$, no discovery is made. For $\hat{k} \ge 1$, every selected j has $e_j \ge e_{[\hat{k}]} \ge p/(\alpha\hat{k})$, that is, $\mathbb{1}\{j \text{ selected}\} \le \alpha\hat{k}\, e_j/p$.
2. Summing over the nulls gives
$$\text{FDP} = \frac{1}{\hat{k}} \sum_{j \in \mathcal{H}_0} \mathbb{1}\{j \text{ selected}\} \le \frac{\alpha}{p} \sum_{j \in \mathcal{H}_0} e_j. \tag{B.2}$$
3. The expectation gives FDR $\le (\alpha/p) \sum_{j \in \mathcal{H}_0} \mathbb{E}[e_j] \le \alpha\, p_0/p$. No step uses the dependence between the e-values [95, Theorem 2].

□

Proof of the FDR control of Infini-Rex.

Proof. Idea. The e-value of a column is $L+1$ times a lower bound on the probability that the column beats L dummies. In a fair race, this probability is at most $1/(L+1)$, so the e-value has mean at most one.

1. *The path is fixed by the data.* Condition on the measurement $\bar{\boldsymbol{y}}$. As long as no dummy has entered, the columns $j_1, j_2, \ldots$ and the shares $\rho_1, \rho_2, \ldots$ depend on $\bar{\boldsymbol{y}}$ alone, because a dummy that has not entered leaves the residual unchanged.

2. *One dummy.* At step τ, a fixed dummy beats the column with the probability P_τ of Eq. (5.7). By the union bound, it beats at least one of the steps $1, \ldots, t$ with a probability of at most $S_t = \sum_{\tau \le t} P_\tau$; the steps need not be independent.

3. *L dummies.* The dummies are independent, so column j_t is selected before all of them with a probability of at least $(1 - S_t)_+^L$. Let $A_j(L)$ be the event that a null column j is selected before all L dummies. If j enters at step t, then $(1 - S_t)_+^L \le \mathbb{P}(A_j(L) \mid \bar{\boldsymbol{y}})$; if j never enters, its e-value is zero.

4. *The e-value has mean at most one.* By Assumption 5.1, $\mathbb{E}\big[(L+1)(1-S_t)_+^L\big] \le (L+1)\,\mathbb{P}(A_j(L)) \le 1$. The average over $\mathcal{L}$ keeps the bound, so $\mathbb{E}[e_j] \le 1$, and an early stop of the path, which sets further e-values to zero, keeps them valid.

5. *Selection.* Proposition B.1 gives FDR $\le \alpha\, p_0/p$.

□

Distribution of the dummy share. Eq. (5.7) rests on one fact: for $\boldsymbol{d} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{I}_n)$ and a fixed $\boldsymbol{r} \neq \boldsymbol{0}$, the share $c = |\boldsymbol{d}^H \boldsymbol{r}|^2 / (\|\boldsymbol{d}\|_2^2 \|\boldsymbol{r}\|_2^2)$ follows a Beta$(1, n-1)$ law.

1. The law of $\boldsymbol{d}$ is invariant under unitary maps, so $\boldsymbol{r}$ can be rotated onto the first axis, and $c = |d_1|^2 / \sum_{m=1}^n |d_m|^2$.

2. The $|d_m|^2$ are independent and exponentially distributed with mean one, that is, Gamma distributed with shape one. The ratio of one of them to the sum of itself and $n-1$ others is Beta$(1, n-1)$, with the survival function

$$\mathbb{P}(c \ge \rho) = \int_\rho^1 (n-1)(1-x)^{n-2}\,\mathrm{d}x = (1-\rho)^{n-1}. \tag{B.3}$$

The race of Infini-Rex for Radar. In Infini-Rex for Radar, the path runs on the continuous steering manifold. At step t, the grid column with the largest correlation to the residual enters, the angles of all entered columns are refined, and the residual $\boldsymbol{r}_t$ is what the refined steering vectors leave. The share of the step is measured on the grid column after the removal of its projection onto the refined steering vectors, $\rho_t = |\tilde{\boldsymbol{g}}^H \boldsymbol{r}_{t-1}|^2 / \|\boldsymbol{r}_{t-1}\|_2^2$ with $\|\tilde{\boldsymbol{g}}\|_2 = 1$. A dummy $\boldsymbol{d} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{I}_n)$ is scored in its raw form, $|\boldsymbol{d}^H \boldsymbol{r}_{t-1}|^2 / (\|\boldsymbol{d}\|_2^2 \|\boldsymbol{r}_{t-1}\|_2^2)$, and it beats step t if its share reaches ρ_t. This defines the race of the radar variant.

Proposition B.2 (FDR control of Infini-Rex for Radar)**.** *Let Assumption 5.1 hold for the race of Infini-Rex for Radar. Then the e-values of Eq.* (5.8) *with $P_t = (1-\rho_t)^{n-1}$ satisfy $\mathbb{E}[e_j] \le 1$ for every null direction, and the selection of Eq.* (5.9) *has* FDR $\le \alpha\, p_0/p$.

Proof. Idea. The proof of Proposition 5.2 needs three facts about the race, and each holds here.

1. *The path is fixed by the data.* The refinement and the least-squares residual use $\bar{\boldsymbol{y}}$ only, and a dummy that has not won does not enter the residual.

2. *The dummy law is exact.* Given $\bar{\boldsymbol{y}}$, the residual $\boldsymbol{r}_{\tau-1}$ is fixed and the dummy is independent of it. By Eq. (B.3), the raw share is Beta$(1, n-1)$, so the dummy beats step τ with the probability $P_\tau = (1-\rho_\tau)^{n-1}$.

3. *The union bound applies.* With the independence of the dummies, it gives $(1 - S_t)_+^L \le \mathbb{P}(A_j(L) \mid \bar{\boldsymbol{y}})$ for a column j that enters at step t.

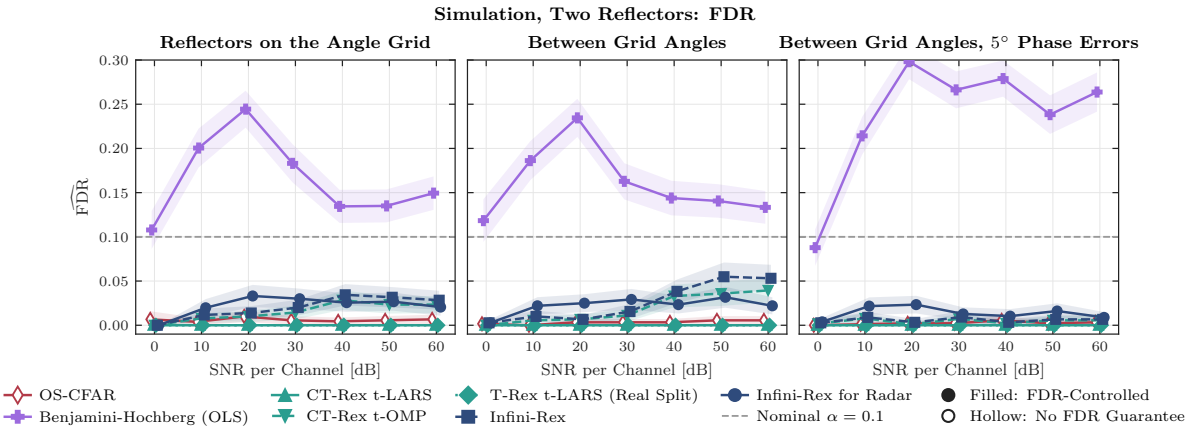


Figure B.2.: Achieved FDR for two simulated reflectors at $\alpha = 0.1$ against the SNR per channel: on the grid (left), between grid directions (centre) and between grid directions with 5° phase errors per channel (right). Filled markers denote FDR-controlled methods; every point averages 300 trials.

The remaining steps are those of the proof of Proposition 5.2, with Assumption 5.1 for the radar race, and Proposition B.1 ends the proof. □

Scope of the assumption.

- *First step: exact.* The share is the raw correlation, and under white noise a fixed null direction and a dummy are exchangeable with the same Beta$(1, n-1)$ law, so a null direction beats L dummies with a probability of at most $1/(L+1)$.
- *Later steps: a condition on the data.* The share is measured after the projection, so the experiments test the assumption. In the simulation of two reflectors (three models, SNRs of 10, 20, 30 and 50 dB, 300 trials each), the achieved FDR at the level of 0.1 stays between 0.013 and 0.033. On the measured session, it is 0.021 with one snapshot and 0.134 with 128 snapshots, where the excess lies in room echoes (Chapter 5).
- *Sensitivity check.* The script `radar_dummy_check.py` scores the dummy with the law Beta$(1, n-t)$ of the projected column, so that $P_t = (1-\rho_t)^{n-t}$. The FDR falls to 0.000 to 0.010 in the simulation and to 0.011 and 0.107 on the measured session, the TPR stays the same, and the angle error rises from 0.45° to 0.52°, below the offset of up to 1.2° of the tape marks.

Figure B.2 shows the simulation.

The z-test of BH. Under $H_{0,j}$ and white Gaussian noise, $\hat{\beta}_j$ of Eq. (5.5) is circularly symmetric complex Gaussian with variance $\mathrm{Var}(\hat{\beta}_j)$, so $|\hat{\beta}_j|^2/\mathrm{Var}(\hat{\beta}_j)$ is the sum of the squares of two independent real Gaussians with variance $1/2$ each, which is exponentially distributed with mean one. Hence $\mathbb{P}(T_j \geq t) = e^{-t}$, and $p_j = e^{-T_j}$ is uniformly distributed under the null.

FDR control of BH and BY. Let $p_1, \ldots, p_m$ be the p-values, $\mathcal{H}_0$ the set of true nulls, $m_0 = |\mathcal{H}_0|$ and R the number of rejections. The BH procedure at level a uses the thresholds $t_k = ka/m$ and rejects every $p_i \leq t_{\hat{k}}$, with $\hat{k} = \max\{k\colon p_{(k)} \leq t_k\}$ and no rejection for an empty set. With the counting function $N(t) = \#\{j\colon p_j \leq t\}$, this reads $\hat{k} = \max\{k\colon N(t_k) \geq k\}$, and a hypothesis is rejected if and only if $p_i \leq t_R$ [29].

Proposition B.3 (BH under independence)**.** *If the p-values are independent and every true null satisfies Eq.* (2.51)*, then BH at level* α *obeys* FDR $\leq m_0\alpha/m$.

Proof. Idea. Freeze the other p-values. Then "i is rejected" is a threshold event for p_i alone, and a null passes the threshold $\alpha R/m$ with a probability of at most $\alpha R/m$, which cancels the division by R.

1. *Leave one out.* Fix a true null i, replace p_i by zero, and let $R^{(i)} \geq 1$ be the number of BH rejections of the modified vector; it depends on $(p_j)_{j\neq i}$ only. Its counting function is $N^{(i)}(t) = N(t) + \mathbb{1}\{p_i > t\} \geq N(t)$.

2. *Claim.*
$$i \text{ is rejected } \iff p_i \leq \frac{\alpha R^{(i)}}{m}, \qquad \text{and then } R = R^{(i)}. \tag{B.4}$$
($\Leftarrow$) If $p_i \leq t_{R^{(i)}}$, then $N(t_{R^{(i)}}) = N^{(i)}(t_{R^{(i)}}) \geq R^{(i)}$, so $R \geq R^{(i)}$. For $k > R^{(i)}$, $N(t_k) \leq N^{(i)}(t_k) < k$, so $R \leq R^{(i)}$. Hence $R = R^{(i)}$, and i is rejected because $p_i \leq t_R$. ($\Rightarrow$) If i is rejected, then $p_i \leq t_R \leq t_k$ for all $k \geq R$, so $N^{(i)}(t_k) = N(t_k)$ for $k \geq R$; this gives $R^{(i)} = R$ and $p_i \leq \alpha R^{(i)}/m$.

3. *Expectation.* With Eq. (B.4),
$$\text{FDR} = \sum_{i\in\mathcal{H}_0} \mathbb{E}\left[\frac{\mathbb{1}\{i \text{ rejected}\}}{R}\right] = \sum_{i\in\mathcal{H}_0} \mathbb{E}\left[\frac{\mathbb{P}\big(p_i \leq \alpha R^{(i)}/m \mid (p_j)_{j\neq i}\big)}{R^{(i)}}\right] \leq \sum_{i\in\mathcal{H}_0} \mathbb{E}\left[\frac{\alpha R^{(i)}/m}{R^{(i)}}\right] = \frac{m_0\,\alpha}{m}, \tag{B.5}$$
where the bound on the conditional probability is Eq. (2.51), valid because p_i is independent of the other p-values.

□

Proposition B.4 (BH under any dependence)**.** *If the true null p-values are uniform on* $[0,1]$*, whatever the joint dependence of all p-values, then BH at the level* $\alpha_{\text{BY}} = \alpha/\sum_{k=1}^{m} 1/k$ *of Eq.* (2.64) *obeys* FDR $\leq m_0\alpha/m$.

Proof. Idea. Cut $[0, \alpha_{\text{BY}}]$ into cells of equal width. A rejection with $R = k$ puts a null into one of the first k cells, and the weight $1/k$ is at most $1/j$ for the cell $j \leq k$. The harmonic sum of these weights is the factor that the level removes.

1. *Cells.* Write $\gamma = \alpha_{\text{BY}}/m$, so that $t_k = k\gamma$, and let $I_j = ((j-1)\gamma, j\gamma]$. For a true null i, a rejection with $R = k$ requires $p_i \leq k\gamma$, that is, $p_i \in I_j$ for one $j \leq k$.

2. *Bound.*
$$\mathbb{E}\left[\frac{\mathbb{1}\{i \text{ rejected}\}}{R}\right] \leq \sum_{k=1}^{m}\sum_{j=1}^{k} \frac{1}{k}\,\mathbb{P}(R = k,\, p_i \in I_j) = \sum_{j=1}^{m}\sum_{k=j}^{m} \frac{1}{k}\,\mathbb{P}(R = k,\, p_i \in I_j) \leq \sum_{j=1}^{m} \frac{1}{j}\,\mathbb{P}(p_i \in I_j) = \gamma\sum_{j=1}^{m}\frac{1}{j} = \frac{\alpha}{m}. \tag{B.6}$$
The second inequality uses $1/k \leq 1/j$ for $k \geq j$ and $\sum_{k\geq j}\mathbb{P}(R = k, p_i \in I_j) \leq \mathbb{P}(p_i \in I_j) = \gamma$.

3. *Sum.* No step uses the dependence between the p-values. Summation over the m_0 true nulls gives the claim [96].

□

FDR control of knockoffs. A knockoff selector computes statistics $W_1, \dots, W_p$, where a large positive W_j is evidence for variable j, and selects $\{j\colon W_j \geq \tau\}$ with the threshold τ of Eq. (2.67), which runs over the nonzero values of $|W_j|$ and selects nothing if no value qualifies. Both constructions of knockoffs establish the same property of the statistics, and the proof of the guarantee uses this property alone.

Assumption B.5 (Sign symmetry of the nulls)**.** Let $\boldsymbol{\xi} \in \{\pm 1\}^p$ be independent of the data, with $\xi_j = +1$ for every non-null variable j and independent fair signs for every null variable. Then $(W_1, \dots, W_p) \overset{d}{=} (\xi_1 W_1, \dots, \xi_p W_p)$.

Proposition B.6 (Knockoff+ from sign symmetry)**.** *Under Assumption B.5, the knockoff+ selector obeys* FDR $\leq \alpha$.

Proof. Idea. Reveal the variables from the smallest magnitude upward. The ratio of positive to negative null statistics is then a supermartingale, and stopping it at the threshold bounds the proportion of false discoveries.

1. *Setup.* Condition on $|W_1|, \dots, |W_p|$ and on the signs of the non-null statistics; by Assumption B.5, the null signs stay independent and fair. Variables with $W_j = 0$ are never selected and are dropped. Order the remaining p' variables by decreasing $|W_j|$, with ties broken by a rule that depends on $|W|$ only. For the first i variables, let N_i^+ and N_i^- count the positive and negative statistics, m_i the nulls, and Y_i the positive statistics among these nulls. The threshold selects the positive statistics among the first i^* variables, where
$$i^* = \max\left\{i\colon 1 + N_i^- \leq \alpha \max(N_i^+, 1)\right\}, \tag{B.7}$$
and nothing if the set is empty, in which case FDP $= 0$.

2. *Bound on the proportion.* At i^*, $\max(N^+, 1) \geq (1 + N^-)/\alpha \geq (1 + m_{i^*} - Y_{i^*})/\alpha$, so
$$\text{FDP} = \frac{Y_{i^*}}{\max(N_{i^*}^+, 1)} \leq \alpha\, M_{i^*}, \qquad M_i = \frac{Y_i}{1 + m_i - Y_i}. \tag{B.8}$$

3. *Supermartingale.* Scan i downwards from p', and let $\mathcal{G}_i$ be generated by $|W|$, the non-null signs, the total $Y_{p'}$ and the null signs above position i. The counts $N_{i'}^{\pm}$ with $i' \geq i$ are $\mathcal{G}_i$-measurable, so i^* is a stopping time of the increasing sequence $\mathcal{G}_{p'} \subset \mathcal{G}_{p'-1} \subset \dots$; if no i qualifies, the scan ends at $i = 0$ with $M_0 = 0$. Given $\mathcal{G}_i$, a null at position i is positive with probability Y_i/m_i. If position i holds a non-null variable, $M_{i-1} = M_i$. If it holds a null, then
$$\mathbb{E}[M_{i-1} \mid \mathcal{G}_i] = \frac{Y_i}{m_i} \cdot \frac{Y_i - 1}{1 + m_i - Y_i} + \frac{m_i - Y_i}{m_i} \cdot \frac{Y_i}{m_i - Y_i} = \frac{Y_i}{1 + m_i - Y_i} = M_i, \tag{B.9}$$
where the second term is absent if $Y_i = m_i$, and then $M_{i-1} = m_i - 1 < M_i$. Hence (M_i) is a supermartingale along the scan.

4. *Optional stopping.* It gives $\mathbb{E}[M_{i^*}] \leq \mathbb{E}[M_{p'}]$. The total $Y_{p'}$ is binomial with $m_{p'}$ trials and success probability $1/2$, and $\binom{m}{k}\frac{k}{m-k+1} = \binom{m}{k-1}$ gives
$$\mathbb{E}\left[\frac{Y_{p'}}{1 + m_{p'} - Y_{p'}}\right] = \sum_{k=1}^{m_{p'}} \binom{m_{p'}}{k-1} 2^{-m_{p'}} = \frac{2^{m_{p'}} - 1}{2^{m_{p'}}} \leq 1. \tag{B.10}$$
With Eq. (B.8), FDR $\leq \alpha$.

□

Proposition B.7 (Fixed-X knockoffs)**.** *Let $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \sigma\boldsymbol{z}$ with $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, $n \geq 2p$, and a fixed design $\boldsymbol{X}$ with unit-norm columns and $\boldsymbol{\Sigma} = \boldsymbol{X}^\top \boldsymbol{X}$. Let the knockoffs $\widetilde{\boldsymbol{X}}$ satisfy $\widetilde{\boldsymbol{X}}^\top \widetilde{\boldsymbol{X}} = \boldsymbol{\Sigma}$ and $\boldsymbol{X}^\top \widetilde{\boldsymbol{X}} = \boldsymbol{\Sigma} - \operatorname{diag}(\boldsymbol{s})$, and let $W_j = f_j(\boldsymbol{G}, [\boldsymbol{X}\ \widetilde{\boldsymbol{X}}]^\top \boldsymbol{y})$ with the Gram matrix $\boldsymbol{G}$ of $[\boldsymbol{X}\ \widetilde{\boldsymbol{X}}]$ and the antisymmetry $W_j \mapsto -W_j$ under the swap of $\boldsymbol{x}_j$ and $\widetilde{\boldsymbol{x}}_j$. Then Assumption B.5 holds, and knockoff+ obeys* FDR $\leq \alpha$.

Proof. Idea. Swapping a null variable with its knockoff leaves the law of everything that W uses unchanged, while W_j changes its sign.

1. *The Gram matrix survives the swap.* Let $[\boldsymbol{X}\ \widetilde{\boldsymbol{X}}]_{\text{swap}(S)}$ exchange $\boldsymbol{x}_j$ and $\widetilde{\boldsymbol{x}}_j$ for $j \in S$, a set of nulls. Its Gram matrix is $\boldsymbol{G}$, because the off-diagonal entries of $\boldsymbol{X}^\top\widetilde{\boldsymbol{X}}$ equal those of $\boldsymbol{\Sigma}$ and the diagonal entries are symmetric.
2. *The law of the inputs survives the swap.* The vector $[\boldsymbol{X}\ \widetilde{\boldsymbol{X}}]_{\text{swap}(S)}^\top\boldsymbol{y}$ is Gaussian with covariance $\sigma^2\boldsymbol{G}$. For $j \in S$, the means $\boldsymbol{x}_j^\top\boldsymbol{X}\boldsymbol{\beta}$ and $\widetilde{\boldsymbol{x}}_j^\top\boldsymbol{X}\boldsymbol{\beta}$ differ only in the term $s_j\beta_j$, which vanishes for a null. So the law of $(\boldsymbol{G}, [\boldsymbol{X}\ \widetilde{\boldsymbol{X}}]^\top\boldsymbol{y})$ does not change.
3. *Sign flips.* W is a function of this pair and flips its sign at every swapped j, so W with flipped signs on S has the law of W. Taking $S = \{j\colon \xi_j = -1\}$ for independent $\boldsymbol{\xi}$ gives Assumption B.5, and Proposition B.6 concludes [30].

□

Proposition B.8 (Model-X knockoffs)**.** *Let $\boldsymbol{X}$ be random, $\boldsymbol{y}$ have any conditional law given $\boldsymbol{X}$, and call j a null if $\boldsymbol{y}$ is independent of $\boldsymbol{x}_j$ given the other variables, so that the law of $\boldsymbol{y}$ given $\boldsymbol{X}$ depends on the non-null columns only. Let the knockoffs satisfy $\boldsymbol{y} \perp\!\!\!\perp \widetilde{\boldsymbol{X}} \mid \boldsymbol{X}$ and $[\boldsymbol{X}\ \widetilde{\boldsymbol{X}}]_{\text{swap}(S)} \overset{d}{=} [\boldsymbol{X}\ \widetilde{\boldsymbol{X}}]$ for every S, and let $W_j = w_j([\boldsymbol{X}\ \widetilde{\boldsymbol{X}}], \boldsymbol{y})$ flip its sign under the swap of $\boldsymbol{x}_j$ and $\widetilde{\boldsymbol{x}}_j$. Then Assumption B.5 holds for any n and p, and knockoff+ obeys* FDR $\leq \alpha$.

Proof. Idea. The response depends on the non-null columns only, and a swap of nulls leaves them unchanged. So the exchangeability of the knockoffs carries over to the pair of columns and response.

1. *The response sees the non-null columns only.* Let S be a set of nulls and $M = [\boldsymbol{X}\ \widetilde{\boldsymbol{X}}]$. By $\boldsymbol{y} \perp\!\!\!\perp \widetilde{\boldsymbol{X}} \mid \boldsymbol{X}$ and the null property, $\boldsymbol{y} \mid M$ has the kernel $\kappa(\boldsymbol{X}_{\mathcal{H}_1}, \cdot)$, where $\boldsymbol{X}_{\mathcal{H}_1}$ holds the non-null columns, and the swap of S leaves these unchanged.
2. *The swap leaves the law unchanged.* For a bounded g and $Q(M) = \int g(M, \boldsymbol{y})\,\kappa(\boldsymbol{X}_{\mathcal{H}_1}, \mathrm{d}\boldsymbol{y})$,
$$\mathbb{E}\, g(M_{\text{swap}(S)}, \boldsymbol{y}) = \mathbb{E}\, Q(M_{\text{swap}(S)}) = \mathbb{E}\, Q(M) = \mathbb{E}\, g(M, \boldsymbol{y}), \tag{B.11}$$
where the middle equality is the exchangeability of the knockoffs. Hence $(M_{\text{swap}(S)}, \boldsymbol{y}) \overset{d}{=} (M, \boldsymbol{y})$.
3. *Sign flips.* Applying w_j shows that W with flipped signs on S has the law of W. Taking $S = \{j\colon \xi_j = -1\}$ for independent $\boldsymbol{\xi}$ gives Assumption B.5, and Proposition B.6 concludes [31].

□

FDR control of the T-Rex selector. The proof follows Machkour [32], [97]. The forward selection adds in each step the variable with the largest absolute sample correlation with the current residual and never removes a variable, as LARS does. Let $\mathcal{Z}$ be the set of null variables and $p_0 = |\mathcal{Z}|$. Let $\Phi_{T,L}$ be the relative occurrence of Eq. (2.69) and $\Delta\Phi_{t,L}(j) = \Phi_{t,L}(j) - \Phi_{t-1,L}(j) \geq 0$ its increment in step t. For a voting level v, let $\widehat{\mathcal{A}}(v) = \{j\colon \Phi_{T,L}(j) > v\}$ be the selection, $R(v) = |\widehat{\mathcal{A}}(v)|$ its size and $V(v) = |\widehat{\mathcal{A}}(v) \cap \mathcal{Z}|$ its number of false discoveries. The deflated relative occurrence and the estimators are

$$\Phi'_{T,L}(j) = \sum_{t=1}^{T}\left(1 - \frac{p - \sum_{q=1}^{p}\Phi_{t,L}(q)}{L-(t-1)} \cdot \frac{1}{\sum_{q\in\widehat{\mathcal{A}}(0.5)}\Delta\Phi_{t,L}(q)}\right)\Delta\Phi_{t,L}(j), \qquad \widehat{V}(v) = \sum_{j\in\widehat{\mathcal{A}}(v)}\big(1 - \Phi'_{T,L}(j)\big), \tag{B.12}$$

with $\widehat{\text{FDP}}(v) = \widehat{V}(v)/\max(R(v), 1)$ and $\widehat{V}'(v) = \widehat{V}(v) - \sum_{j\in\widehat{\mathcal{A}}(v)}(1 - \Phi_{T,L}(j)) \leq \widehat{V}(v)$. The selector uses the voting level $v^* = \inf\{\nu \in [0.5, 1)\colon \widehat{\text{FDP}}(\nu) \leq \alpha\}$, with $v^* = 1$ for an empty set. The proof lets the number K of random experiments tend to infinity, so that averages over experiments equal expectations, and it uses three assumptions on the variables that the dissertation states and verifies numerically.

Assumption B.9 (T-Rex)**.** (i) The variables are standardised, and the null variables are mutually independent and independent of the active variables. (ii) For every $v \in [0.5, 1)$, $V(v)$ is stochastically dominated by the negative hypergeometric law $\mathrm{NHG}(p_0 + L, p_0, T)$, the number of nulls drawn before the T-th dummy when drawing without replacement from p_0 nulls and L dummies. (iii) For large T, the terms of the active variables in $\widehat{V}'$ are negligible, so that, with $\widehat{\mathcal{A}}_0(v) = \widehat{\mathcal{A}}(v) \cap \mathcal{Z}$,

$$\widehat{V}'(v) = \sum_{t=1}^{T} \frac{p_0 - \sum_{q \in \mathcal{Z}} \Phi_{t,L}(q)}{L - (t-1)} \cdot \frac{\sum_{j \in \widehat{\mathcal{A}}_0(v)} \Delta\Phi_{t,L}(j)}{\sum_{q \in \widehat{\mathcal{A}}_0(0.5)} \Delta\Phi_{t,L}(q)}. \tag{B.13}$$

Proposition B.10 (T-Rex selector)**.** *Under Assumption B.9, for $\widehat{V}'(v^*) > 0$ and as $K \to \infty$, the T-Rex selector obeys* $\mathrm{FDR}(v^*, T, L) \leq \alpha$.

Proof. Idea. The estimate $\widehat{V}'$ is built so that the ratio $V/\widehat{V}'$ is a supermartingale when the voting level falls. Stopping it where the estimated proportion reaches α bounds the true proportion.

1. *Expected null counts.* At every forward step, the nulls and dummies that are not yet included are unrelated to the response, so each of them is included with the same probability. Terminating after t dummies is therefore a draw without replacement, and the number of included nulls is $\mathrm{NHG}(p_0 + L, p_0, t)$ with mean $tp_0/(L+1)$. For $K \to \infty$, $\sum_{q \in \mathcal{Z}} \Phi_{t,L}(q) \to tp_0/(L+1)$, and at $v = 0.5$, Eq. (B.13) gives the deterministic value

$$\widehat{V}'(0.5) = \sum_{t=1}^{T} \frac{p_0 - tp_0/(L+1)}{L - (t-1)} = \sum_{t=1}^{T} \frac{p_0}{L+1} = \frac{T\, p_0}{L+1}. \tag{B.14}$$

2. *Thinning.* The relative occurrences of the nulls with $\Phi > 0.5$ are independent and identically distributed. Given $V(0.5)$, the count $V(v)$ is therefore binomial with $V(0.5)$ trials and success probability $\vartheta(v) = \mathbb{P}(\Phi_{T,L}(j_0) > v \mid \Phi_{T,L}(j_0) > 0.5)$, so $\mathbb{E}[V(v) \mid V(0.5)] = \vartheta(v) V(0.5)$. The ratio of two sums of i.i.d. terms with a and b summands has expectation a/b, so Eq. (B.13) yields $\mathbb{E}[\widehat{V}'(v) \mid \widehat{V}'(0.5)] = \vartheta(v)\widehat{V}'(0.5)$.

3. *Supermartingale.* Let $H(v) = V(v)/\widehat{V}'(v)$ and $\mathcal{F}_v = \sigma(\{R(u), V(u), \widehat{V}'(u)\}_{u \geq v})$. Lowering v by the smallest step ϵ^* that adds one variable adds either an active variable, which leaves V, $\widehat{V}'$ and H unchanged, or a null with $V(v - \epsilon^*) = V(v) + 1$. In the second case, Jensen's inequality and step 2 give

$$\mathbb{E}\big[H(v - \epsilon^*) \,\big|\, \mathcal{F}_v\big] \geq \frac{V(v) + 1}{\mathbb{E}[\widehat{V}'(v - \epsilon^*) \mid \widehat{V}'(v)]} = \frac{V(v)\, \vartheta(v - \epsilon^*)/\vartheta(v)}{\widehat{V}'(v)\, \vartheta(v - \epsilon^*)/\vartheta(v)} = H(v), \tag{B.15}$$

where $V(v) + 1 = \mathbb{E}[V(v - \epsilon^*) \mid V(v)] = V(v)\vartheta(v - \epsilon^*)/\vartheta(v)$ is step 2 applied to the null that enters (Lemma 5 of the dissertation). Hence H is a supermartingale along decreasing v.

4. *Optional stopping.* The voting level v^* is a stopping time of $(\mathcal{F}_v)$, and H is bounded, so $\mathbb{E}[H(v^*)] \leq \mathbb{E}[H(0.5)]$. By the choice of v^*, $\widehat{\mathrm{FDP}}(v^*) \leq \alpha$, and $\widehat{V}' \leq \widehat{V}$ gives

$$\mathrm{FDP}(v^*) = \widehat{\mathrm{FDP}}(v^*)\, \frac{V(v^*)}{\widehat{V}(v^*)} \leq \alpha\, H(v^*), \qquad \mathbb{E}[H(v^*)] \leq \frac{\mathbb{E}[V(0.5)]}{\widehat{V}'(0.5)} \leq \frac{Tp_0/(L+1)}{Tp_0/(L+1)} = 1, \tag{B.16}$$

where the last inequality is the domination by $\mathrm{NHG}(p_0 + L, p_0, T)$ with mean $Tp_0/(L+1)$ and Eq. (B.14). Hence $\mathrm{FDR} = \mathbb{E}[\mathrm{FDP}(v^*)] \leq \alpha$.

□

Algorithm 5 *M^2-SRC Training and Classification.*

1. **Input**: training windows $\{\boldsymbol{u}_n\}_{n=1}^{N}$ with labels, features of the new window, $\lambda = 0.05$, 300 iterations.
2. **For** each head, **select** its features, **standardise** every window with the mean and the standard deviation of the training set, **clip** to ± 5, **scale** to unit length and **stack** the vectors as the columns of $\boldsymbol{D}$.
3. **Set** $\boldsymbol{a}_0 = \boldsymbol{z}_1 = \boldsymbol{0}$ and $\tau_1 = 1$. **For** $k = 1, \ldots, 300$ **compute** $\boldsymbol{a}_k$, τ_{k+1} and $\boldsymbol{z}_{k+1}$ by Eq. (6.9); **compute** $\boldsymbol{e}_c = \boldsymbol{y} - \boldsymbol{D}\boldsymbol{a}_{300,c}$ for every class c of the head.
4. **Output**: $\hat{c} = \arg\min_c \|\boldsymbol{e}_c\|_2$, the confidence of Eq. (6.7), the atoms with $\boldsymbol{a}_{300} > \boldsymbol{0}$ and the feature shares of Eq. (6.8).

FDR control of the CT-Rex selector. The CT-Rex selector of Scheidt, Machkour and Muma replaces, in the T-Rex selector, the real forward selection by a terminating complex least angle regression, which adds in each step the variable with the largest modulus $|\boldsymbol{x}_j^H \boldsymbol{r}|$ of the complex correlation with the residual, and the real dummies by independent circularly symmetric complex standard normal columns [47]. The paper states that the guarantee of the T-Rex selector carries over and gives no separate proof. The following reduction shows where the proof of Proposition B.10 uses the data, and why these places hold for complex data.

Proposition B.11 (CT-Rex selector)**.** *Let $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with complex entries and let Assumption B.9 hold for the complex relative occurrences. Then, for $\widehat{V}'(v^*) > 0$ and as $K \to \infty$, the CT-Rex selector obeys* $\mathrm{FDR}(v^*, T, L) \leq \alpha$.

Proof. Idea. The proof of Proposition B.10 touches the data in three places only, and none of them needs real values.

1. *The forward rule.* The selection adds one variable per step by a rule that depends on the variables through their correlation moduli with the residual, and it never removes one. The terminating complex least angle regression has this form.
2. *Equal chances (step 1).* Given the residual, every null that is not yet included and every dummy must have the same probability of reaching the largest modulus. For a dummy $\boldsymbol{d} \sim \mathcal{CN}(\boldsymbol{0}, \boldsymbol{I}_n)$ and a fixed residual $\boldsymbol{r} \neq \boldsymbol{0}$, the unitary invariance of $\boldsymbol{d}$ gives the share $|\boldsymbol{d}^H \boldsymbol{r}|^2 / (\|\boldsymbol{d}\|^2 \|\boldsymbol{r}\|^2) \sim \mathrm{Beta}(1, n-1)$, the same law for all dummies. The law is the same for a circularly symmetric Gaussian null, and, as n grows, for any null with finite variance, by the Lindeberg-Feller argument that the paper cites for the dummy generation. The draw of nulls and dummies is therefore again a draw without replacement, and the number of included nulls is $\mathrm{NHG}(p_0 + L, p_0, t)$.
3. *The remaining steps.* Steps 2 to 4 use only the definitions of Eq. (B.12), Assumption B.9(ii) and (iii), and the law of large numbers over the experiments.

Hence the steps of Proposition B.10 apply unchanged. □

B.2. Material Classification

B.2.1. Proofs and Derivations

Table B.2 defines the fourteen features. Here, $x_f[\kappa]$ is the slow-time signal of the object in frame f, v_f and $\sigma_{v,f}$ follow from Eq. (6.3), b_f is the range bin of the object in frame f, and P and P_0 are the range profiles of the window and of the empty room in the object's direction.

Algorithm. Algorithm 5 states the training and the classification of M^2-SRC.

Table B.2.: The fourteen features of a window.

Feature	Head	Definition
Speed	motion	median of $\lvert v_f \rvert$ over the frames
Speed change	motion	standard deviation of v_f
Speed spread	motion	median of $\sigma_{v,f}$
Pattern change	motion	$\log_{10}(1-\bar{\gamma})$, with $\bar{\gamma}$ the mean coherence of consecutive frames of x_f
Flicker	motion	standard deviation over mean of the amplitude of x_f
Range motion	motion	standard deviation of the range of b_f
Cross-range motion	motion	standard deviation of $r\,u$ at the peak direction near the object
Change	material	$C(b_0, a_0)$ of Eq. (6.2) in dB
Reflectivity	material	ρ of Eq. (6.4) in dB
Back echo	material	strongest power 2 to 8 bins behind the object relative to the object, in dB
Shadow	material	power of the room behind the object relative to the empty room, in dB
Ripple	material	standard deviation of the reflectivity over four sub-bands around a straight line
Spectral slope	material	slope of the reflectivity over the four sub-bands, in dB/GHz
Depth	material	power-weighted standard deviation of the echo around b_0, in metres

Proof of the duality gap.

Proof. Idea. Any dual-feasible point bounds the optimal value from below, and the scaled residual is such a point. The gap to this bound limits the distance of the current value to the optimum.

1. *Lagrangian.* Write the residual as a variable, $\boldsymbol{w} = \boldsymbol{y} - \boldsymbol{D}\boldsymbol{a}$, and attach the multiplier $\boldsymbol{\theta}$ to this constraint: $\frac{1}{2}\|\boldsymbol{w}\|_2^2 + \lambda \mathbf{1}^\top \boldsymbol{a} + \boldsymbol{\theta}^\top(\boldsymbol{y} - \boldsymbol{D}\boldsymbol{a} - \boldsymbol{w})$.

2. *Dual function.* The minimum over $\boldsymbol{w}$ is attained at $\boldsymbol{w} = \boldsymbol{\theta}$. The minimum over $\boldsymbol{a} \geq \mathbf{0}$ is finite only if $\boldsymbol{D}^\top \boldsymbol{\theta} \leq \lambda \mathbf{1}$, and then it is attained at $\boldsymbol{a} = \mathbf{0}$. Hence $g(\boldsymbol{\theta}) = \boldsymbol{\theta}^\top \boldsymbol{y} - \frac{1}{2}\|\boldsymbol{\theta}\|_2^2$ on the set $\boldsymbol{D}^\top \boldsymbol{\theta} \leq \lambda \mathbf{1}$, and weak duality gives $g(\boldsymbol{\theta}) \leq f(\boldsymbol{a}^\star)$ for every $\boldsymbol{\theta}$ in this set.

3. *The scaled residual is feasible.* The scaling factor of the proposition ensures $\max_i (\boldsymbol{D}^\top \boldsymbol{\theta})_i \leq \lambda$, so $\boldsymbol{\theta}$ lies in the set.

4. *Conclusion.* $f(\boldsymbol{a}) - f(\boldsymbol{a}^\star) \leq f(\boldsymbol{a}) - g(\boldsymbol{\theta}) = G(\boldsymbol{a})$, and the lower bound $0 \leq f(\boldsymbol{a}) - f(\boldsymbol{a}^\star)$ holds because $\boldsymbol{a}^\star$ is the minimiser.

□

B.3. Computer Vision

B.3.1. Proofs and Derivations

Metric scale. By the pinhole model, a length w at range r covers $s\,w$ pixels with $s = f/r$. MetricSight cuts a region of $w_{\text{reg}} \times h_{\text{reg}}$ metres centred at the projected radar object and resamples it to $w_{\text{reg}}/\Delta \times h_{\text{reg}}/\Delta$ pixels with $\Delta = 1.5\,\text{mm}$, so one pixel of the metric image covers Δ on the board at every range.

Veil fit. For a fixed exponent q, the model of Eq. (7.4) is linear in (b, a). With the ring radii ρ_i and the ring medians L_i, the least-squares coefficients solve

$$\min_{b,a} \sum_i \left(L_i - b - a\rho_i^{-q} \right)^2 , \tag{B.17}$$

which has the closed-form solution of a linear regression with the regressors 1 and ρ_i^{-q}. The exponent is chosen on a grid from 0.5 to 4 in steps of 0.05, and fits with $a < 0$ are discarded, since a lamp cannot darken the image.

B.4. Radar Tracking

B.4.1. Proofs and Derivations

Proof of Proposition 8.2. *Idea.* The weights move the threshold of each test, and the expectation of the weighted e-values stays below one on average.

1. Let $\mathcal{R}$ be the set of rejected hypotheses. Every rejected i satisfies $w_i e_i \geq m/(\alpha|\mathcal{R}|)$, so $1/|\mathcal{R}| \leq \alpha w_i e_i / m$.
2. With $\mathcal{H}_0$ the set of true nulls, FDP $\leq \frac{\alpha}{m} \sum_{i \in \mathcal{H}_\circ} w_i e_i$.
3. The expectation given the past is at most $\frac{\alpha}{m} \sum_{i \in \mathcal{H}_\circ} w_i \leq \alpha$, and the bound holds for the unconditional expectation as well.

Proof of Proposition 8.1. *Idea.* Under exchangeability, the rank of the new score among all scores is uniform.

1. The next measurement lies in $\Gamma_{t+\Delta}$ exactly when its score is at most $\hat{q}$.
2. The rank of the new score among the $n_{\mathrm{c}} + 1$ scores is uniformly distributed, so it is at most $\lceil (1-\epsilon_t)(n_{\mathrm{c}}+1) \rceil$ with a probability of at least $1 - \epsilon_t$ [52], [64].

Entropy of a plan. For a Gaussian position estimate with covariance $\boldsymbol{P}_{\text{pos}}$, Eq. (2.47) with $d = 2$ gives the entropy in Eq. (8.15). Since $\det(c\boldsymbol{P}_{\text{pos}}) = c^2 \det \boldsymbol{P}_{\text{pos}}$, scaling the uncertainty ellipse by an area factor c changes the entropy by $\log_2 c$ bits; a gain of one bit therefore halves the area. The plans are compared by the time average of this entropy over the horizon, with the prediction of Eq. (8.8) between looks and the update of Eq. (8.10) at the looks of the plan. The covariance update does not depend on the measured values, so the comparison needs no measurement.

B.5. Scene Analysis

B.5.1. Proofs and Derivations

Range of Scene PQ. Every true positive has $0.5 \leq s(o, g) \leq 1$ by the matching rule, so $0.5 \leq \mathrm{SQ} \leq 1$ whenever $|\mathrm{TP}| > 0$. Since $|\mathrm{TP}| \leq |\mathrm{TP}| + \frac{1}{2}(|\mathrm{FP}| + |\mathrm{FN}|)$, the recognition quality satisfies $0 \leq \mathrm{RQ} \leq 1$, with $\mathrm{RQ} = 1$ only if $|\mathrm{FP}| = |\mathrm{FN}| = 0$. The product of the two lies in $[0, 1]$ and equals one only if all objects are found without false reports and every matched pair has $s = 1$, that is, overlapping boxes and the right kind.

Reach of a mover. For a mover, the planner needs the region that the object can reach within the reaction time t_{resp}. The method collects the displacements $\|\boldsymbol{p}(\tau + h) - \boldsymbol{p}(\tau)\|$ of the object over intervals of length $h \geq t_{\text{resp}}$ and takes their conformal quantile at level $1 - \epsilon_{\text{plan}}$, in the same way as Eq. (8.12). Under exchangeability of the displacements, the next displacement stays within this reach with probability at least $1 - \epsilon_{\text{plan}}$ (Proposition 8.1). The physical bound $v_{\max} h$ caps the reach when only few displacements are known.

C. Supplementary Information: Experiments

This appendix describes the data sets, the measured objects and the additional results of the five experiments. Every table and figure is generated by the code base (Appendix D).

C.1. Detection and Estimation

C.1.1. Dataset

The measured session holds ten evaluated takes of 20 frames each: three with one corner reflector, six with two or three, and a take of the empty room, which also provides the background. A calibration take with one large reflector at broadside fixes the phase offsets of the channels. Table C.1 lists the takes with the positions of the reflectors in range r, azimuth θ and elevation φ, Figure C.1 shows where each take was measured, and Figure C.2 shows the reflectors. Table C.2 shows the first rows of the data set: one row per reflector observation, with its taped position, the SNR and the complex values of the channels in its range bin.

Table C.1.: Takes of the measured session.

Take	Role	Reflectors (R, θ, φ)
cal_0deg	calibration	big CR: 2.05 m, $+0.0^\circ$, -0.3°
empty	background	–
multi_close_az	multi	small CR 2: 1.50 m, -2.0°, -4.2°; small CR 1: 1.50 m, $+4.0^\circ$, $+1.5^\circ$
multi_doa	multi	small CR 1: 1.50 m, -25.0°, $+1.5^\circ$; small CR 2: 1.50 m, -2.0°, -4.2°
multi_range	multi	small CR 1: 1.25 m, $+0.0^\circ$, $+1.8^\circ$; small CR 2: 1.50 m, -24.9°, -4.2°; big CR: 2.05 m, -35.0°, -0.3°
multi_range_close	multi	small CR 1: 1.50 m, -25.0°, $+1.5^\circ$; small CR 2: 1.67 m, -1.0°, -3.8°
multi_same_az	multi	small CR 2: 1.50 m, $+0.0^\circ$, -4.2°; small CR 1: 1.67 m, $+0.0^\circ$, $+1.4^\circ$
multi_strong_weak	multi	big CR: 1.55 m, -28.0°, -0.4°; small CR 1: 1.50 m, -2.0°, $+1.5^\circ$
null	null	–
single_m30	single	small CR 1: 2.00 m, -30.0°, $+1.1^\circ$
single_m45	single	small CR 1: 1.80 m, -45.0°, $+1.3^\circ$
single_p10	single	small CR 1: 1.50 m, $+10.0^\circ$, $+1.5^\circ$

C.1.2. Additional Results

Figure C.3 compares all methods of the measured session: the achieved false discovery rate rises with the number of chirps for the methods without a guarantee, and the true positive rate of every method stays nearly

Table C.2.: First rows of `dataset.csv` of the detection experiment (320 rows, 44 columns): one row per reflector observation; dots mark omitted rows and columns.

take	frame	reflector	label	x_m	y_m	z_m	range_m	azimuth_deg	elevation_deg	snr_db	ch1_re	ch1_im	…
multi_close_az	0	0	small CR 2	1.499	-0.052	-0.11	1.5039	-1.98	-4.19	67.7911	21279.2	-24095.8	…
multi_close_az	0	1	small CR 1	1.496	0.105	0.04	1.5002	4.01	1.53	67.7911	21279.2	-24095.8	…
multi_close_az	1	0	small CR 2	1.499	-0.052	-0.11	1.5039	-1.98	-4.19	67.0224	21099.1	-24188.9	…
…	…	…	…	…	…	…	…	…	…	…	…	…	…

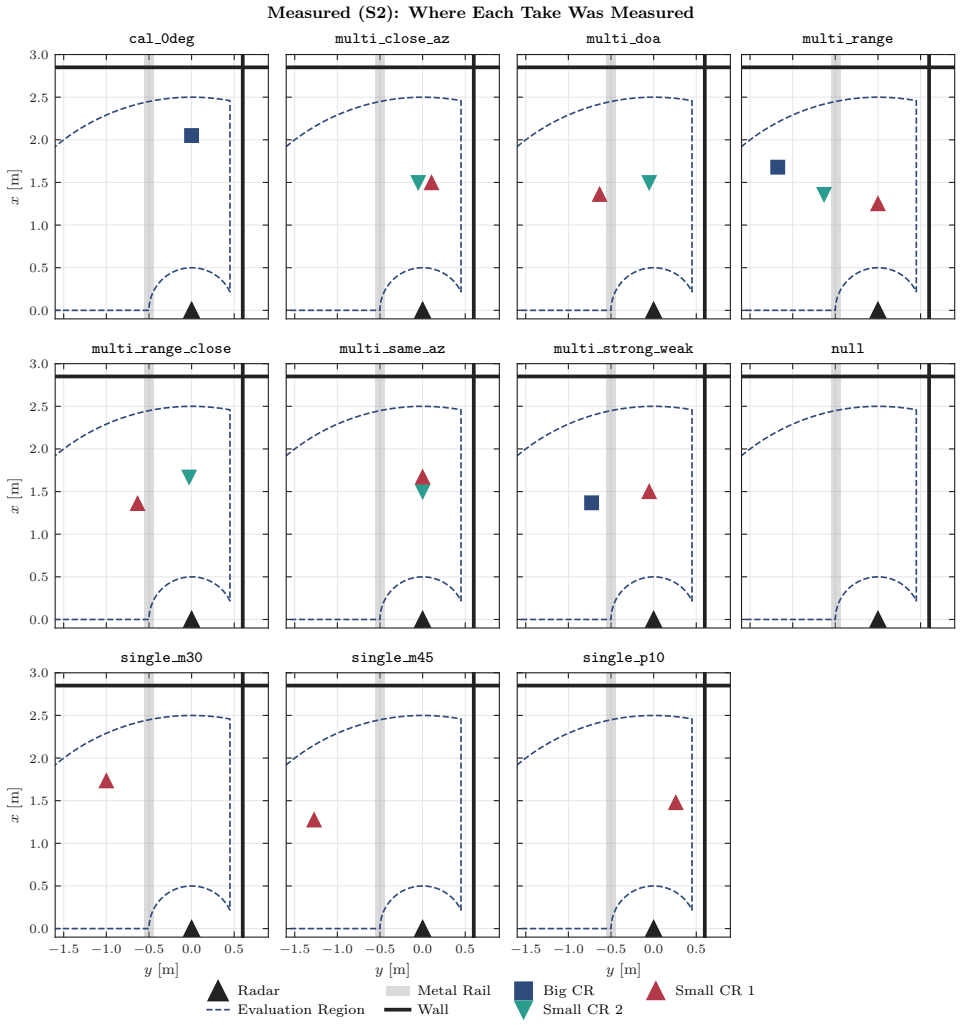


Figure C.1.: Taped positions of the corner reflectors in every take, seen from above, with the evaluation region bounded by the wall and the metal rail.

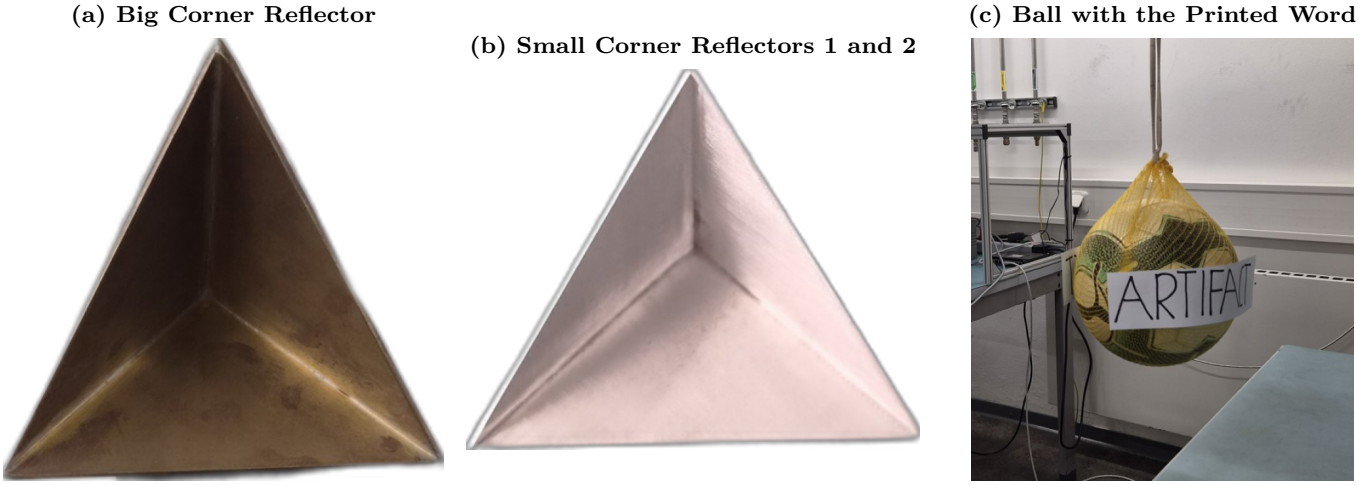


Figure C.2.: Objects of the detection, vision and tracking measurements, with the background removed where the object allows it: (a) the big corner reflector, (b) the small corner reflectors 1 and 2, which are identical, and (c) the ball with the printed word. The phone flash of the vision experiment and the person of the tracking experiment are not shown.

constant. Figure C.4 plots the computation per range gate: the median time and the peak memory of every method.

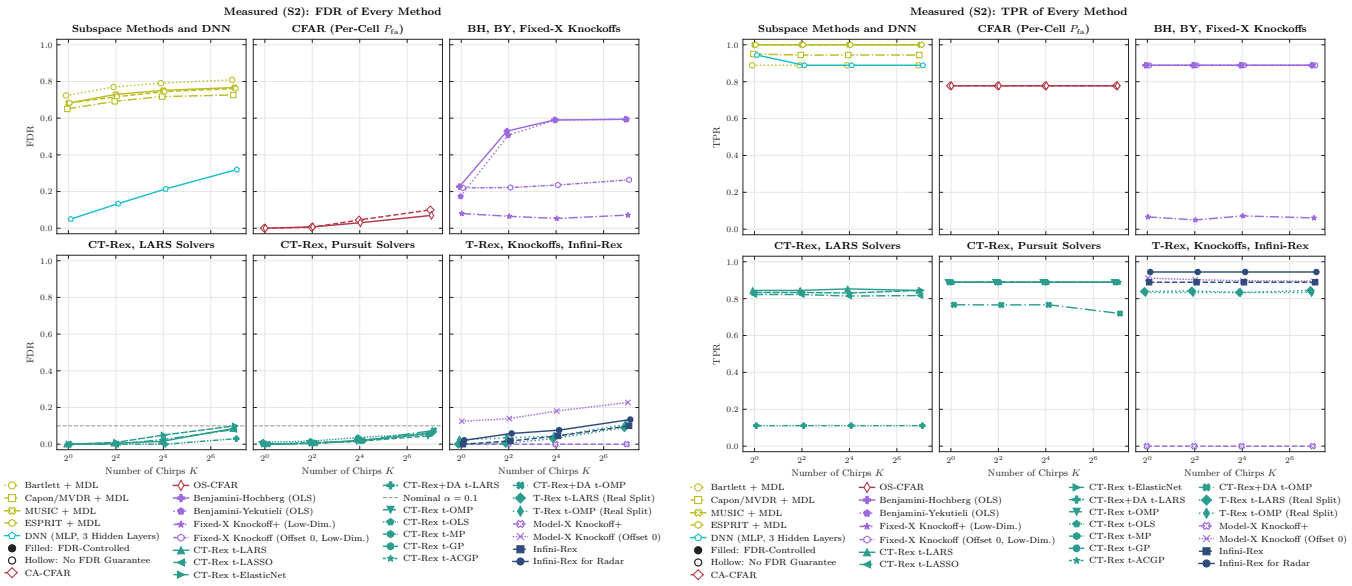


Figure C.3.: All methods on the measured session at $\alpha = 0.1$, grouped by family: achieved FDR (left) and TPR (right) against the number of chirps K.

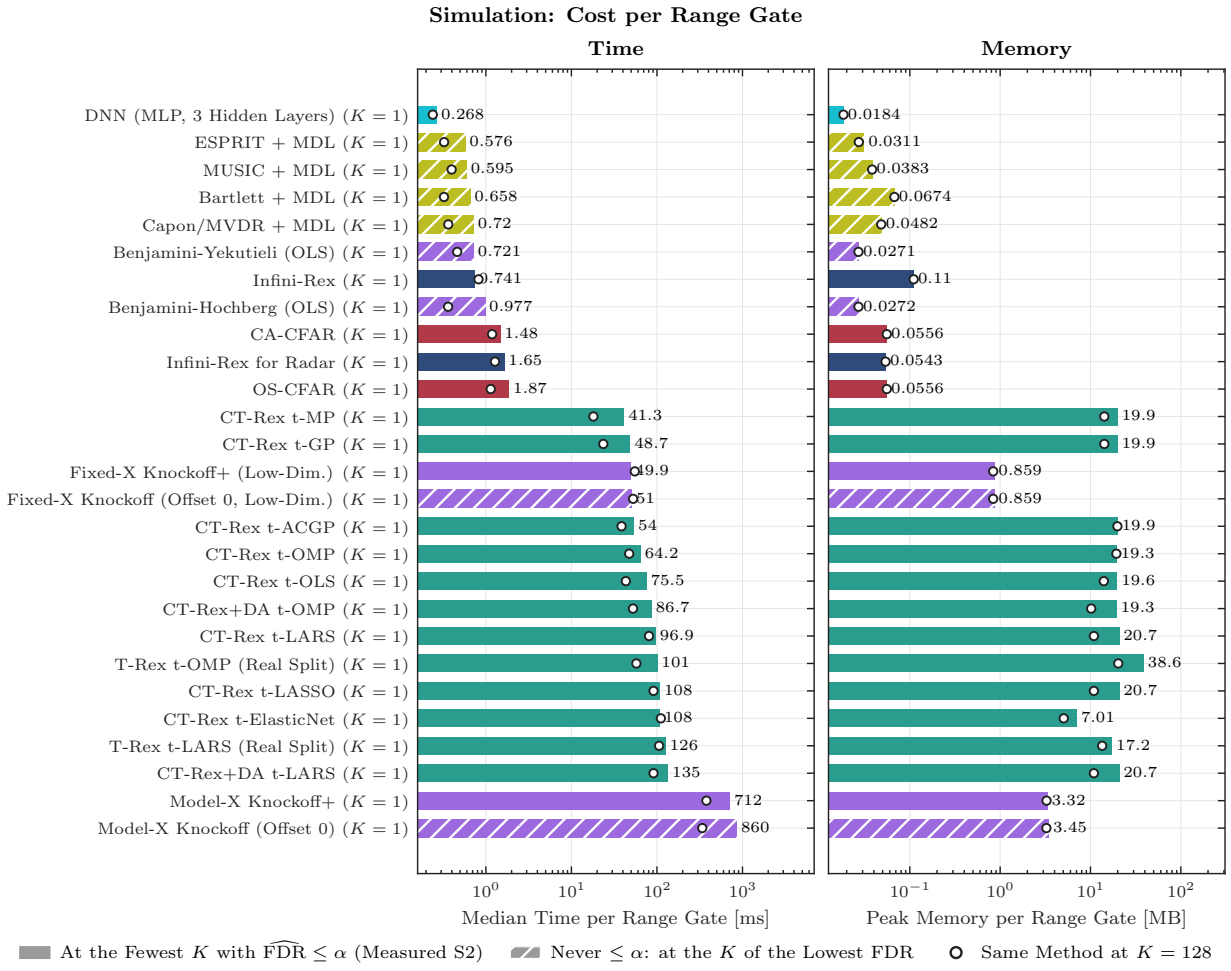


Figure C.4.: Computation per range gate: median time (left) and peak memory (right) of every method on the grid of the experiment.

C.2. Material Classification

C.2.1. Dataset

Figure C.5 shows representative objects of the three materials, and Table C.3 shows the first rows of the data set, with one row per window. From every take, windows of 40 frames (1.6 s) are cut every 10 frames. Session A (34 takes, 935 windows, 705 with an object and 174 empty) and session A2 (8 takes, 232 windows, 162 and 70) train the dictionary, and session B (14 takes, 384 windows, 314 and 70) tests it.

Figure C.5.: Representative objects of the material classification with the background removed. Metal: steel ball (a), brass (b) and aluminium (c) trihedral corner reflectors. Dielectric: red plastic plate (d), plastic box (e) and bag (g). Waterlike: water bottle (f) and, in the full photograph (h), water poured into a perforated bottle as rain.

Table C.3.: First rows of `dataset.csv` of the material classification (1 551 rows, 22 columns): one row per window of 40 frames.

session	take	material	motion	t_start_s	detected	range_m	azimuth_deg	speed_mps	reflectivity_db	…
sessionA	dielectric-rigid/plasticbox_pendulum_take1	dielectric	rigid	0.0	1	0.502	9.0	0.46437	85.8452	…
sessionA	dielectric-rigid/plasticbox_pendulum_take1	dielectric	rigid	0.4	1	0.502	9.0	0.4403	85.90657	…
sessionA	dielectric-rigid/plasticbox_pendulum_take1	dielectric	rigid	0.8	1	0.502	9.0	0.4403	85.90657	…
…	…	…	…	…	…	…	…	…	…	…

C.2.2. Additional Results

Table C.4 gives precision, recall, F1 score and Area Under the Receiver Operating Characteristic Curve (AUROC) per class for leave-one-take-out and for the unseen session B. Figure C.6 shows the micro-Doppler signature of every combination of material and motion, Figure C.8 compares FISTA with the exact solver Clarabel, and Figure C.7 breaks down the time per window of the live classification.

Table C.4.: Per-class results in percent for leave-one-take-out and the unseen session B: precision P, recall R, F1 score and AUROC.

		Leave one take out				Test on B			
Class	n	P	R	F1	AUC	P	R	F1	AUC
motion head									
static	505	98.8	94.3	96.5	99.2	96.4	100.0	98.2	99.9
rigid	161	99.3	93.8	96.5	99.6	100.0	88.9	94.1	99.5
articulated	135	99.3	100.0	99.6	100.0	100.0	100.0	100.0	100.0
distributed	66	64.9	95.5	77.3	99.4	100.0	100.0	100.0	100.0
material head									
metal	242	97.9	78.9	87.4	97.0	97.6	100.0	98.8	100.0
dielectric	269	79.0	46.1	58.2	85.7	100.0	79.0	88.3	99.4
waterlike	356	61.9	89.6	73.2	88.9	91.0	100.0	95.3	99.9

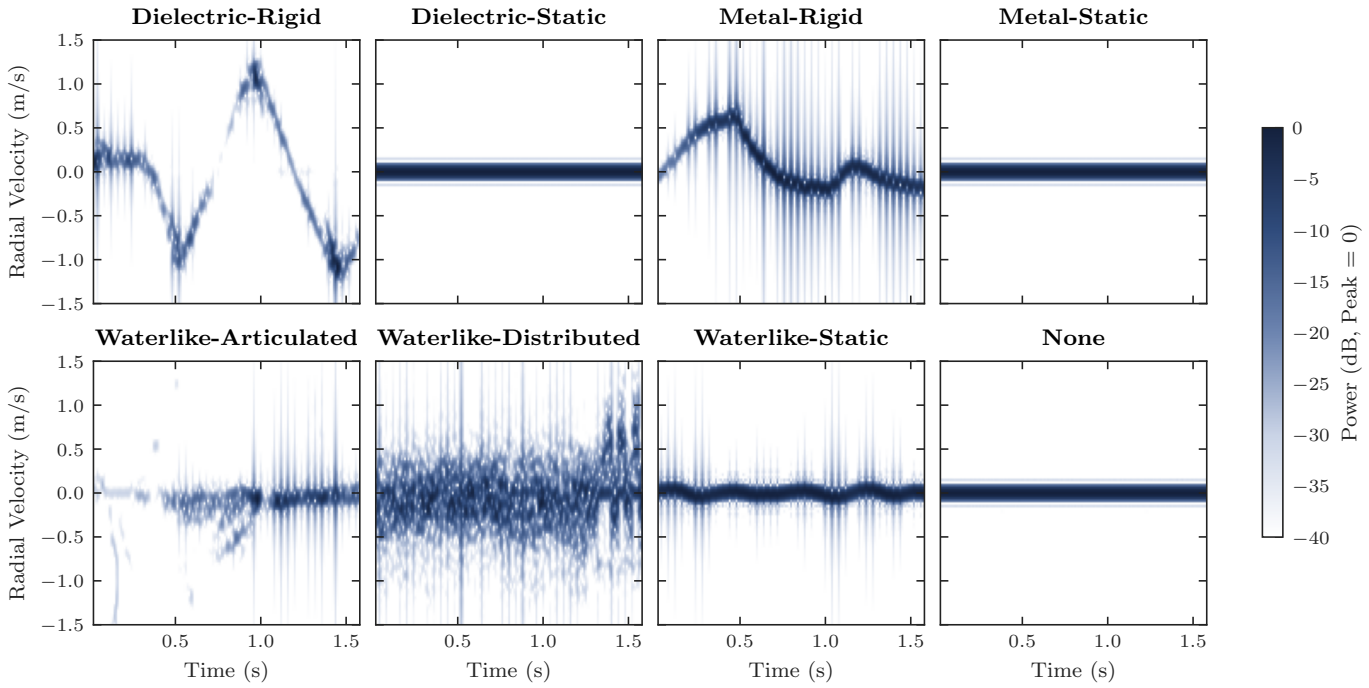


Figure C.6.: Micro-Doppler signatures: radial speed against time over one window for each combination of material and motion. A swinging object traces a single curve, a walking person spreads the speed around the body speed, and falling water fills a broad band.

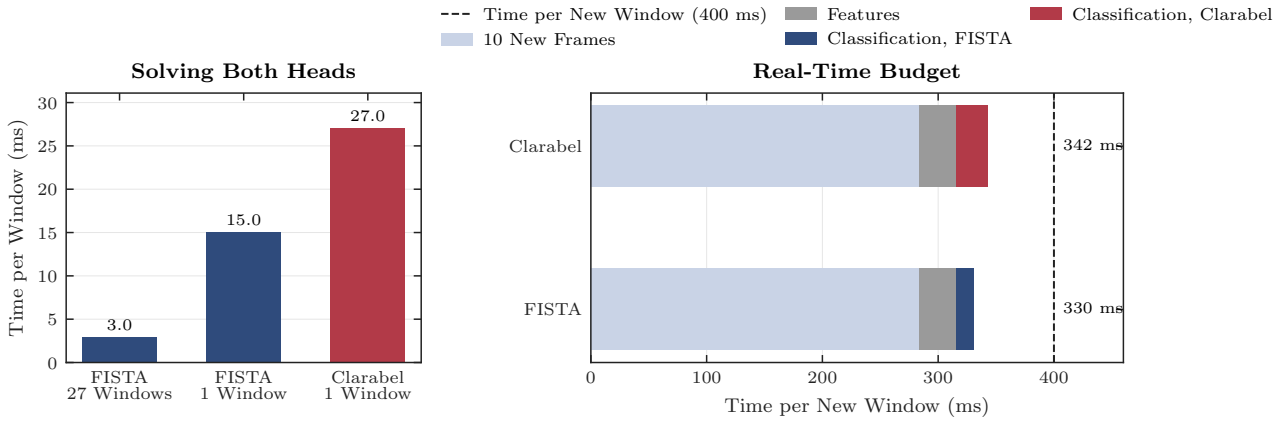


Figure C.7.: Time per window of the live classification: solving both heads with FISTA for 27 windows and for one window, and with Clarabel for one window (left); the share of the 400 ms between two windows that the frame processing, the features and the classification take (right).

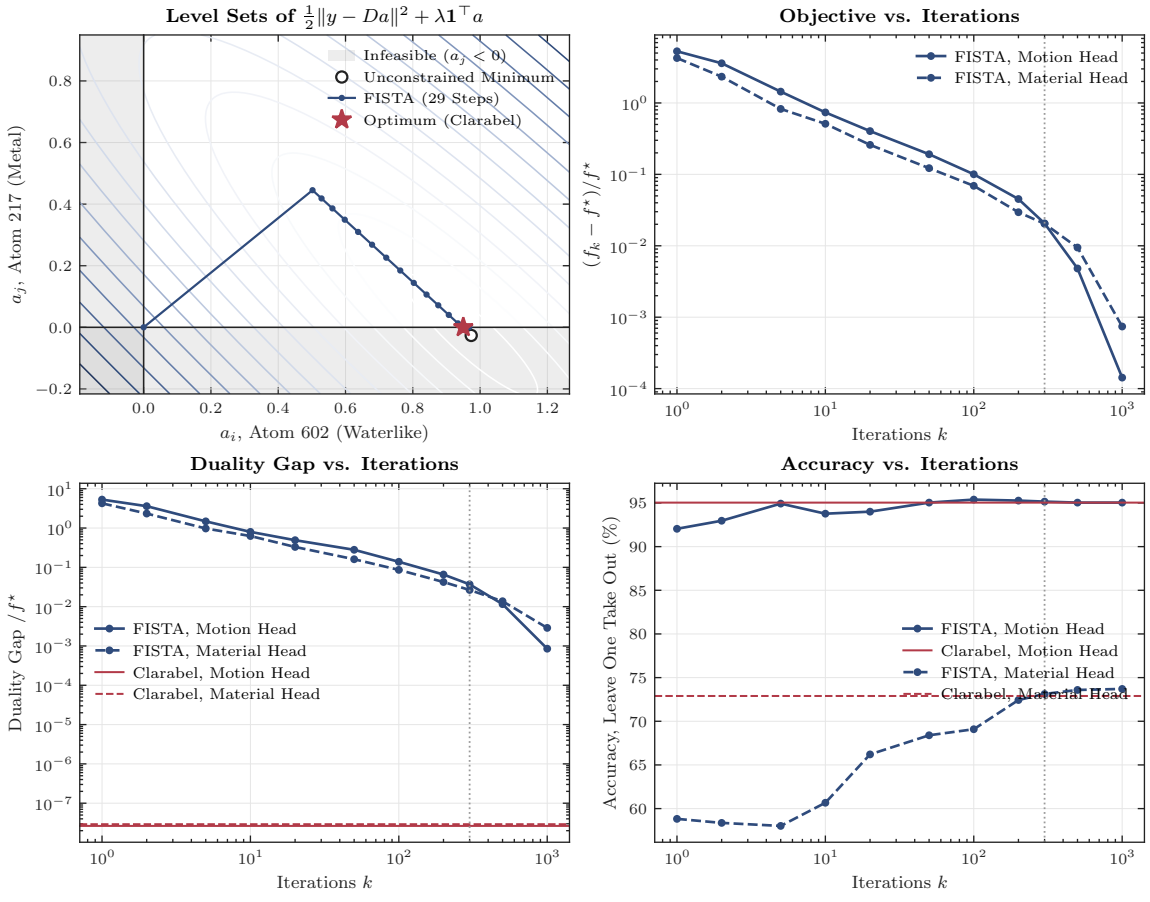


Figure C.8.: Optimisation of the sparse code of one window: level sets of the objective with the path of FISTA (top left), and the relative objective, the duality gap and the accuracy of the labels against the number of iterations for the motion and the material head, with the exact solution of Clarabel as reference. The vertical line marks the 300 iterations of the classifier.

C.3. Computer Vision

C.3.1. Dataset

Table C.5 shows the first rows of the data set for one take with the light, with one row per radar frame. The five takes (`text_085b`, `text_085c`, `light_1` to `light_3`) hold 2 987, 279, 726, 421 and 1 188 frames, and the radar finds the board in 99.8, 96.8, 100.0, 68.9 and 100.0 % of them. The light is on in 433, 215 and 603 frames of the three light takes, a veil is fitted in 402, 215 and 544 of them, and the radar names the light in 1.8, 0.0 and 93.9 % of the lit frames. The board with the printed word hangs on the ball (Figure C.2); two corner reflectors align radar and camera, and the flash of a phone tests the alignment independently.

Table C.5.: First rows of `dataset.csv` of the computer vision experiment (5 601 rows, 21 columns), take `light_1`: one row per radar frame.

take	k	t_s	lamp	text_found	text_range_m	text_azimuth_deg	lamp_radar	source_range_m	source_azimuth_deg	radar_ms	…
light_1	0.0	0.0	0.0	1.0	0.8365	1.059	0.0			18.4	…
light_1	2.0	0.067	1.0	1.0	0.8365	0.904	1.0	0.8365	0.986	30.5	…
light_1	4.0	0.133	1.0	1.0	0.8365	0.904	1.0	0.8365	0.986	30.5	…
…	…	…	…	…	…	…	…	…	…	…	…

C.3.2. Additional Results

Table C.6 gives the character error rate per take with the light, for the raw and the cleaned image, and Figure C.9 shows the reading over time while the light is switched.

Table C.6.: CER in percent per take with the light on, for the raw and the cleaned image, per frame and stable (subscript s).

		MetricSight				CNN			
Take	Frames	Raw	Removed	Raw_s	Removed_s	Raw	Removed	Raw_s	Removed_s
light_1	402	41.4	24.9	10.4	5.5	95.2	93.7	88.5	85.3
light_2	141	10.5	16.0	9.5	14.6	14.0	17.1	12.7	15.9
light_3	544	31.1	30.3	3.9	2.8	96.0	99.0	96.0	96.9
all	1087	32.2	26.5	7.0	5.3	85.1	86.4	82.4	82.1

C.4. Radar Tracking

C.4.1. Dataset

Table C.7 shows the first rows of the camera ground truth, with one row per video frame and object. The two recordings, one with the fixed chirp (40.1 s, 402 radar looks, 1 153 camera frames) and one with the cognitive chirp choice (92.2 s, 1 744 looks, 2 233 frames), show the ball in 84 and 100 % of the frames, the person in 48 and 13 %, and the person at the edge of the view in 22 and 23 %. The ball swings as a pendulum (Figure C.2), and the person walks through the scene. In the cognitive recording, the ball is visible to the camera throughout, while the person spends most of the time outside the view of the camera.

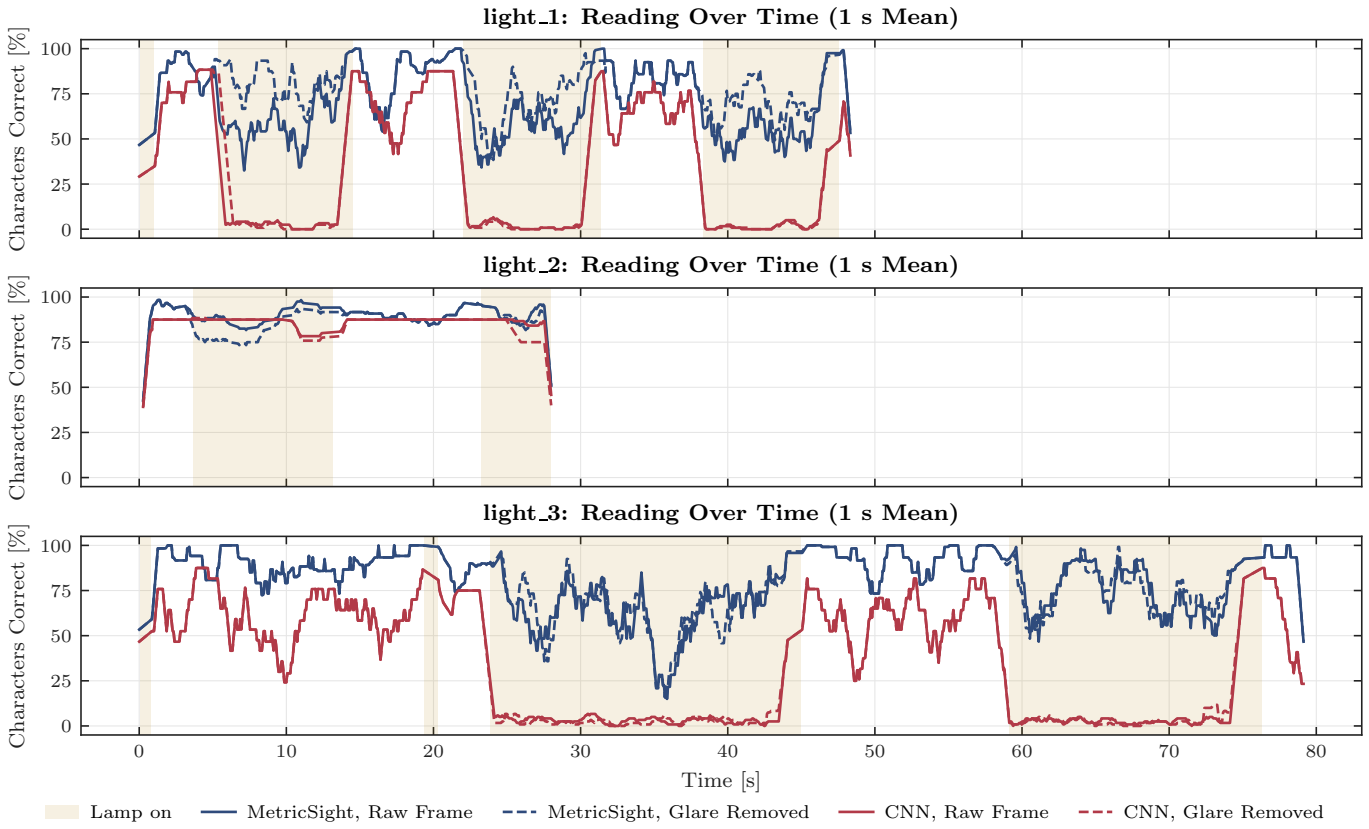


Figure C.9.: Characters read correctly over time in the three takes with the light, for both readers on the raw and the cleaned image; shaded intervals mark the light being on.

Table C.7.: First rows of `dataset.csv` of the tracking experiment (6 772 rows, 12 columns): one row per video frame and object.

take	video_frame	t_s	object	found	partial	u	v	u0	v0	u1	v1
final_fixed_1	0	0.0	ball	1	0	588.0	217.0	512.0	132.0	664.0	302.0
final_fixed_1	0	0.0	person	1	0	288.0	208.0	136.0	0.0	440.0	416.0
final_fixed_1	1	0.0328	ball	1	0	613.0	224.0	538.0	146.0	688.0	302.0
final_fixed_1	1	0.0328	person	1	0	304.0	208.0	144.0	0.0	464.0	416.0
…	…	…	…	…	…	…	…	…	…	…	…

C.4.2. Additional Results

Figure C.10 shows three look policies on the cognitive recording, Table C.8 lists all tracking scores against the camera, and Table C.9 gives the prediction regions: coverage by the next radar measurement and at 0.5 and 1 s, coverage of the camera positions, and tests of exchangeability.

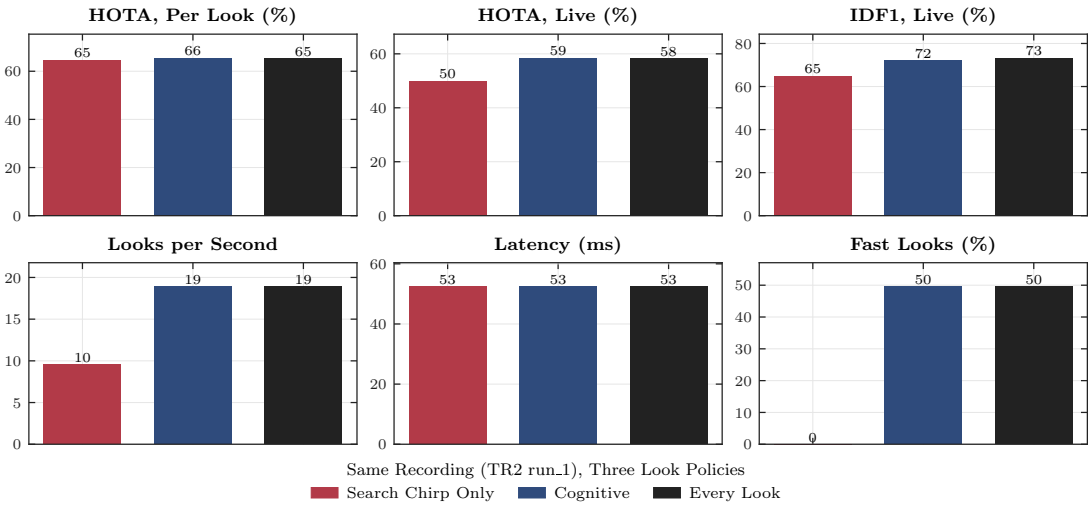


Figure C.10.: Three look policies on the same recording of the cognitive run: tracking scores per look and live, looks per second, latency and share of fast looks.

Table C.8.: All tracking scores in percent against the camera (the lower block in degrees).

	Fixed Chirp		Cognitive	
Metric	Per Look	Live	Per Look	Live
HOTA	64.7	53.1	65.6	58.5
DetA	72.8	60.9	65.0	58.8
AssA	57.6	46.3	66.2	58.3
LocA	89.2	84.2	89.2	86.5
MOTA	76.9	64.6	66.3	56.4
MOTP	88.2	81.8	88.1	85.1
IDF1	66.5	62.3	77.3	72.1
ID switches	13	15	8	10
False positives	60	85	361	231
Misses	44	77	133	107
Ground truth	506	500	1491	799
GOSPA (°)	1.80	2.53	1.86	2.20
localisation RMS (°)	1.45	2.12	1.40	1.71
missed per look	0.12	0.18	0.10	0.12
false per look	0.16	0.20	0.27	0.30
OSPA (°)	1.49	2.14	1.69	1.99
Track switches (T-GOSPA)	25.5	36.5	52.5	51.0
T-GOSPA per look (°)	2.79	3.51	2.76	3.18

Table C.9.: Prediction regions: coverage by the next radar measurement and at 0.5 and 1 s, coverage of the camera positions, and tests of exchangeability of the scores.

	Radar coverage			Camera coverage		IoU median		Exchangeability		
Chirps	Next	0.5 s	1 s	Ball	Person	Ball	Person	KS p	Lag-1	Runs p
Fixed Chirp	0.92	0.90	0.92	0.97	0.92	0.59	0.54	0.06	0.14	0.59
Cognitive	0.92	0.94	0.92	0.95	0.87	0.40	0.58	0.00	0.14	0.37

C.5. Scene Analysis

C.5.1. Dataset

Table C.10 lists the three runs, and Table C.11 shows the first rows of the ground truth, with one row per look and object. Every run contains the ball, the light, the obstacle and the person; the person entered the path between three and seven times per run.

Table C.10.: Runs of the scene experiment: duration, looks, camera frames, objects, person entries, light events, travel and Scene PQ.

Run	Seconds	Radar looks	Camera frames	Objects	Person entries	Light events	Travel (m)	Scene PQ (%)
OB2_run_3	99.3	483	2837	ball, lamp, obstacle, person	3	0	0.29	47.1
OB2_run_4	105.9	653	3039	ball, lamp, obstacle, person	7	0	0.69	61.0
OB2_run_5	117.4	686	3378	ball, lamp, obstacle, person	6	1	1.21	59.6

Table C.11.: First rows of `dataset.csv` of the scene experiment (62 989 rows, 22 columns): one row per look and object.

take	look	t_s	rail_x_m	rail_y_m	light	person	object	kind	label	material	present	x_lo	x_hi	…
replay/OB2_run_3	0	0.0	0.0	0.0	0	0	obstacle:1	obstacle	metal plate	metal	1	0.52	0.56	…
replay/OB2_run_3	0	0.0	0.0	0.0	0	0	obstacle:2	obstacle	plate stand	metal	1	0.5	0.58	…
replay/OB2_run_3	0	0.0	0.0	0.0	0	0	obstacle:3	obstacle	phone on its stand		1	0.97	1.03	…
…	…	…	…	…	…	…	…	…	…	…	…	…	…	…

C.5.2. Additional Results

The results pool the three runs. Table C.12 compares the method with its baselines on the same recordings: with the detector of CT-Rex in place of Infini-Rex, with OS-CFAR in place of Infini-Rex, and with the classical chain of OS-CFAR, beam peaks, DBSCAN and nearest-neighbour tracking. Figure C.11 shows the scene understanding per object and over time, and the separation of close objects.

C.6. Dashboards

Figure C.12 shows every detection method on one measured take, and Figure C.13 the classifier on one window of each class. The live views are part of Chapters 8 and 9.

Table C.12.: The method against its baselines on the same three runs: scene understanding, tracking, detections and classification.

Metric	Method (Infini-Rex)	CT-Rex detector	OS-CFAR detector	Classical chain
Scene PQ (%)	56.2	55.3	48.8	28.7
HOTA (%)	46.1	40.3	41.7	12.4
DetA (%)	40.0	38.0	33.1	16.1
AssA (%)	53.4	42.9	52.9	9.6
MOTA (%)	32.7	27.5	5.5	-21.2
IDF1 (%)	44.0	38.0	36.7	8.2
GOSPA (m)	0.713	0.747	0.850	0.824
ID switches	173	189	198	318
Detections per look on a measured object	5.5	7.0	8.9	20.5
Detections per look on nothing known	1.3	2.2	4.3	25.0
FDR	0.173	0.223	0.319	0.543
Obstacles found as static (%)	51.7	49.8	47.5	4.6
Ball found as a mover (%)	61.9	65.6	69.7	98.8
Person found as a mover or person (%)	56.3	69.8	77.4	94.0
Material right (%)	66.3	77.1	54.9	–
Motion right (%)	85.7	89.6	88.6	–
Person entries that made it wait	15/16	16/16	15/16	16/16

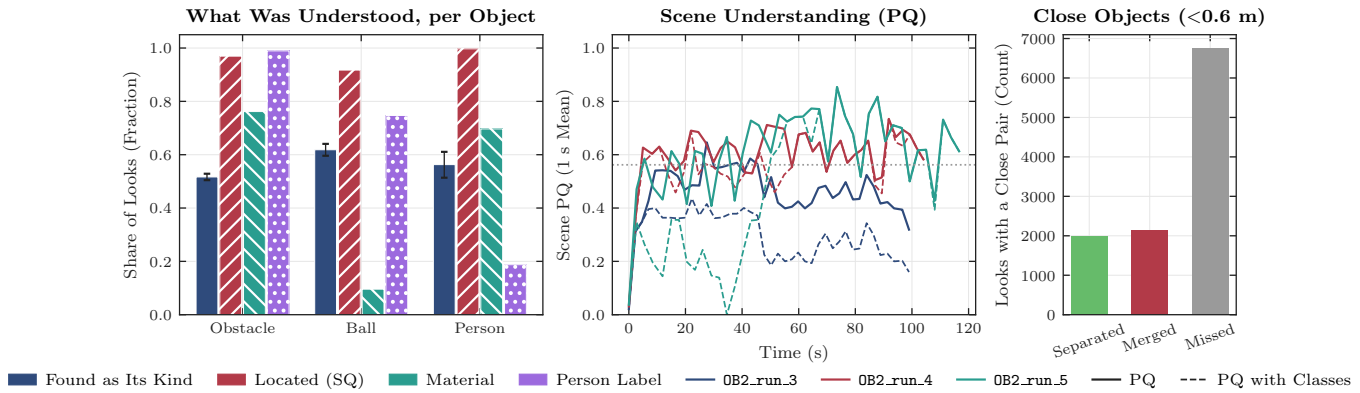


Figure C.11.: Scene understanding per object and over time, and the separation of close objects.

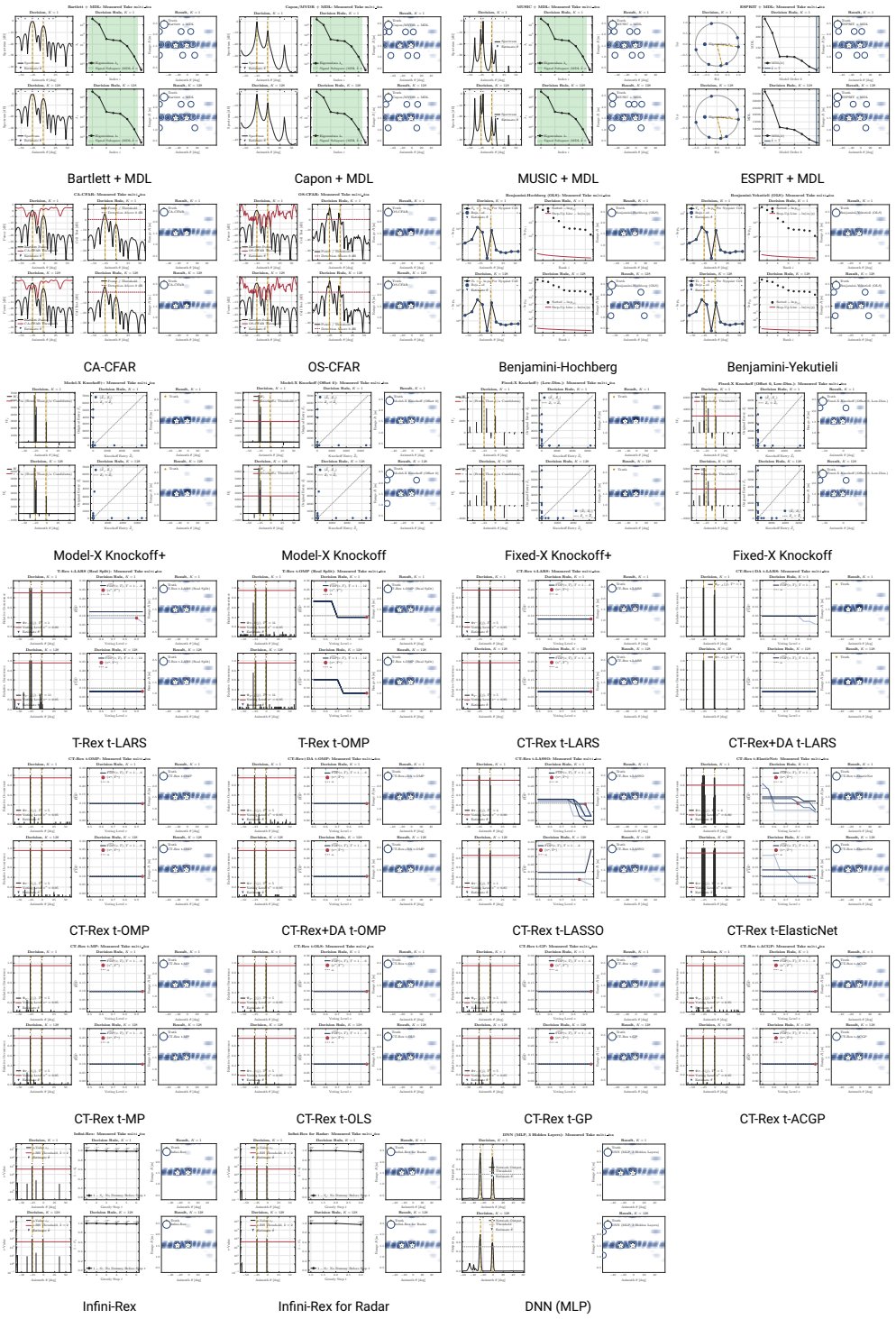


Figure C.12.: Dashboards of all 27 detection methods on one measured take: scores, decision and reports on the range-azimuth map for one and for 128 snapshots.

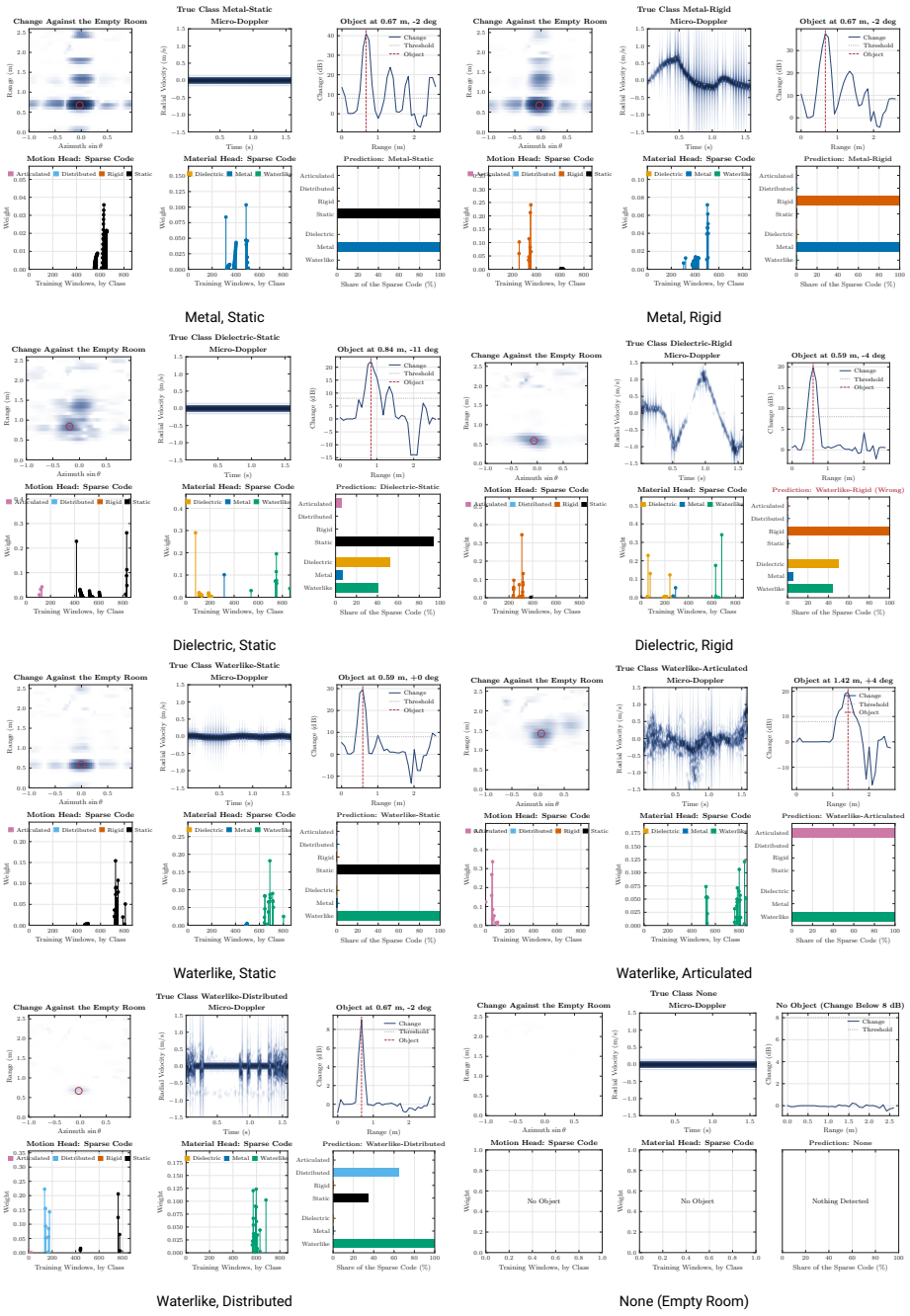


Figure C.13.: Dashboards of the classifier for one window of each class: micro-Doppler, change against the empty room, object profile, sparse codes of both heads and predicted class.

D. The Code Base ★

All methods, experiments and figures of this thesis are open source under the Apache License 2.0 at `https://github.com/itislmn/ARTIFACT`. The code base turns an evaluation module and a capture card into a complete research platform in Python. It configures the radar, receives its raw samples through the capture card, and runs every method of this thesis on them, live or on a recording. No vendor processing tool is involved between the samples and the results.

Every experiment lives in its own folder with the same structure (Table D.1). A single command, `python run_all.py`, recreates all tables and figures of the experiment from its recordings, and a replay runs a method on a recording exactly as it ran live, with the same timing. Methods are shared rather than copied where several experiments use them: the tracking and scene experiments import the selectors of `estimation/` and the classifier of `material_classification/`, so a change in a method reaches every experiment that uses it. Every figure follows one shared style, defined in `visualization/`, and every experiment documents its commands and its main result in a README.

The repository contains the code and the figures of the READMEs, but not the raw recordings of about 35 GB. With the recordings, every number of this thesis can be reproduced; without them, the code still runs live with the hardware of Chapter 4 and the instructions of Appendix A.5.

Table D.1.: Structure of the repository.

Folder	Content
`radar_tutorial/`	Radar Tutorial and Diagnosis (Appendix A.5)
`pattern_measurements/`	Antenna and Radome Patterns (Section 4.2)
`estimation/`	Detection and Estimation (Chapter 5)
`material_classification/`	Material Classification (Chapter 6)
`computer_vision/`	Computer Vision (Chapter 7)
`tracking/`	Radar Tracking (Chapter 8)
`obstacle/`	Scene Analysis (Chapter 9)
`isel/`	Rail Control (Section 4.4)
`visualization/`	Figure Style, Cover and Literature Map
`thesis/`	Thesis

The code base serves six tasks, each with one command.

- **Check the set-up:** `python check_setup.py --cli COM4` in `radar_tutorial/` tests the serial ports, the command line, the capture card and the data port (Appendix A.5).
- **Record raw frames:** `python capture_raw.py --cli COM4 --name corner_1m --seconds 3` stores the frames of the capture card and a log of lost packets.
- **View the radar live:** `python live_raw.py --cli COM4` shows range-azimuth power and the detections of a running radar; `--replay` shows a recording the same way.

- **Reproduce an experiment:** `python run_all.py` in any experiment folder recreates its data set, predictions, tables and figures; `--again` recomputes everything (about 3 h for the detection experiment).
- **Replay a take:** the tracking and scene experiments run their methods on recorded takes exactly as live, with the same timing.
- **Check the proof:** `python radar_dummy_check.py` in `estimation/` compares the two dummy conventions of the radar variant (Appendix B.1.3).

Table D.2 lists the Python packages. The knockoff filters and the T-Rex calibration are implemented in Python after the R packages *knockoff* [148] and *TRexSelector* [165], so that no part of the code base depends on R.

Table D.2.: Requirements of the code base (Python 3.12.7).

Package	Version	Purpose	Reference
NumPy	2.5.3	Arrays and FFTs	[145]
SciPy	1.18.1	Signal Processing and Statistics	[146]
Matplotlib	3.11.2	Figures	[166]
scikit-learn	1.9.1	Lasso Paths	[167]
CVXPY	1.9.3	Convex Optimization	[153]
Clarabel	0.11.1	Interior-Point Solver	[152]
SCS	3.3.1	Conic Solver	[168]
OpenCV	5.0.0	Camera and Video	[169]
Pillow	12.3.0	Images and Fonts	[170]
pySerial	3.5	Radar and Motor Commands	[171]
psutil	7.2.2	Process Priority and Load	[172]
pygrabber	0.2	Camera Capture	[140]
comtypes	1.4.17	Camera Exposure Control	[141]

The last four packages connect the code to the hardware. The camera runs through DirectShow: pygrabber finds the webcam by its name, selects the MJPG format at 1280×720 pixels and delivers the frames from a sample grabber, and the class `IAMCameraControl` declares the DirectShow interface of the camera with comtypes, so that the exposure can be set manually. pySerial sends the configuration and the start and stop commands to the command port of the radar and the commands to the controller of the motor stage. psutil raises the priority of the radar process and reports the processor load and the peak memory of the camera loop and of the recording session.

E. Literature Map

Figure E.1 links each of the 174 cited sources to the chapters that cite it.

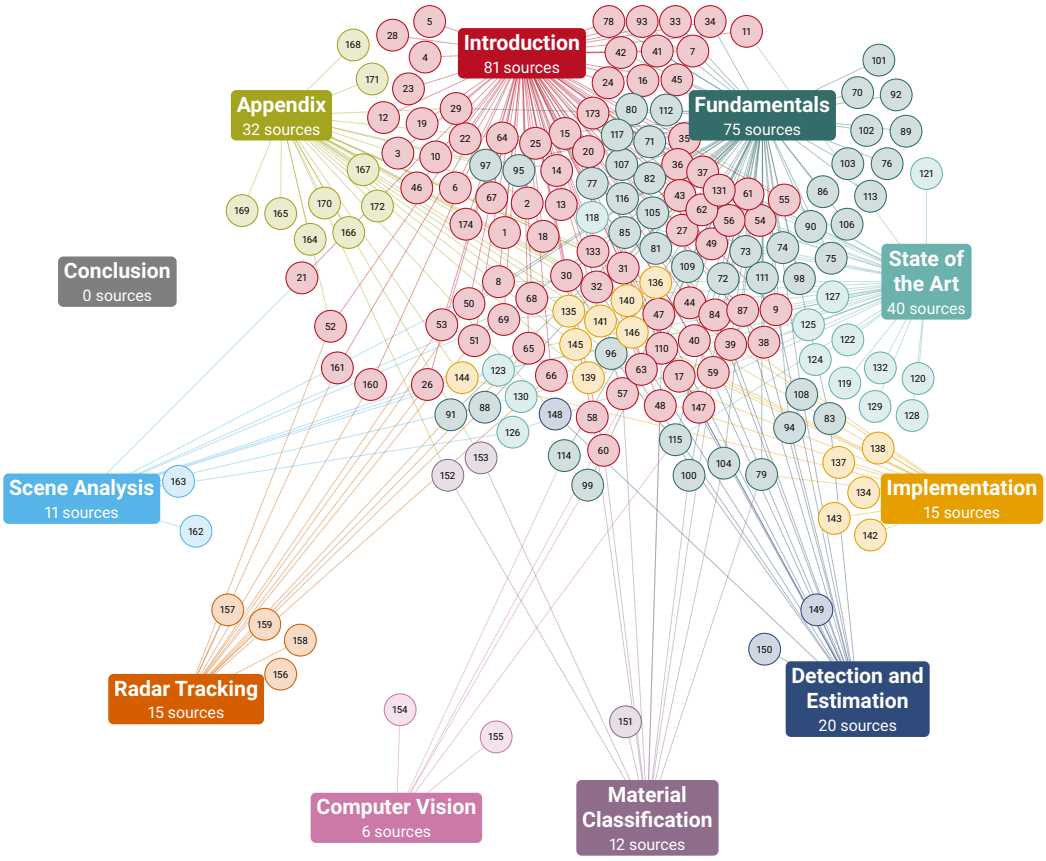


Figure E.1.: Literature map of the thesis. Each coloured box is a chapter, with the number of sources it cites. Each circle is one source, labelled with its number in the references. A circle takes the colour of the first chapter that cites it, and a line joins it to every chapter that cites it (307 links in total). A source near a single box is cited only in that chapter. A source between several boxes connects them, for example a method that is introduced in the fundamentals and used in an experiment.

F. Tools

Table F.1.: Research, development and writing toolchain.

Tool	Version	Purpose
Literature and Notes:		
Google Scholar	Online	Literature Search
Zotero	10.0.6	Reference Management
Obsidian	1.15.0	Literature Notes
Programming:		
Python	3.12.7	Programming Language
Visual Studio Code	1.143.1	Development Environment
Git and GitHub	2.47.0	Version Control
Writing and Images:		
LaTeX (TeX Live)	2025	Typesetting
GIMP	3.2.6	Image Editing
Design and Fabrication:		
Autodesk Fusion	2706.1.14	Computer-Aided Design
PrusaSlicer	2.9.6	Slicing for the Original Prusa XL (5 Toolheads)
Hardware and Firmware:		
TI AWR2944PEVM	—	Radar Sensor [133]
TI DCA1000EVM	—	Data Capture [136]
TI MMWAVE-MCUPLUS-SDK	04.07.02.01	Radar Firmware [135]
ISY IW-1000-1	—	Camera [139]
isel Rail with TMCM-310	—	Motor Stage [144]

G. Declaration on the Use of AI Tools

The use of AI-based tools is declared according to the guidelines “Citing AI Tools” of the University of Basel[1].

Table G.1.: Use of AI-based tools.

AI-Based Tool	Type of Use	Affected Sections of the Thesis
Claude (Anthropic)	Debugging	Code Base
Gemini (Google)	Refinement of Images	All Images Involving Photographs
Gemini (Google)	Literature Search with Google Scholar	Literature of All Chapters
Consensus	Literature Search with Google Scholar	Literature of All Chapters

[1] https://www.unibas.ch/en/Studies/My-Studies/AI-in-learning-and-teaching.html

List of Acronyms and Abbreviations

ACI	Adaptive Conformal Inference
ADC	Analog-to-Digital Converter
AIC	Akaike Information Criterion
AUROC	Area Under the Receiver Operating Characteristic Curve
BH	Benjamini–Hochberg Procedure
BY	Benjamini–Yekutieli Procedure
CA-CFAR	Cell-Averaging Constant False Alarm Rate
CER	Character Error Rate
CFAR	Constant False Alarm Rate
CNN	Convolutional Neural Network
CRNN	Convolutional Recurrent Neural Network
CQR	Conformalized Quantile Regression
CT-Rex	Complex-Valued Terminating-Random Experiments Selector
CZT	Chirp Z-Transform
DoA	Direction of Arrival
e-BH	Benjamini–Hochberg Procedure for E-Values
EKF	Extended Kalman Filter
ESPRIT	Estimation of Signal Parameters via Rotational Invariance Techniques
EVM	Evaluation Module
FDR	False Discovery Rate
FPGA	Field-Programmable Gate Array
DSP	Digital Signal Processor
FFT	Fast Fourier Transform
FISTA	Fast Iterative Shrinkage-Thresholding Algorithm
FMCW	Frequency-Modulated Continuous Wave

FWER Family-Wise Error Rate

GOSPA Generalized Optimal Sub-Pattern Assignment

HOG Histogram of Oriented Gradients

HOTA Higher Order Tracking Accuracy

IDF1 Identification F1 Score

JPEG Joint Photographic Experts Group

LARS Least Angle Regression

LVDS Low-Voltage Differential Signaling

M^2-SRC Material-Motion Sparse Representation Classification

LSTM Long Short-Term Memory

MDL Minimum Description Length

MIMO Multiple-Input Multiple-Output

MJPEG Motion JPEG

MLP Multilayer Perceptron

MOTA Multiple Object Tracking Accuracy

MUSIC Multiple Signal Classification

NOMP Newtonized Orthogonal Matching Pursuit

OMP Orthogonal Matching Pursuit

OS-CFAR Ordered-Statistic Constant False Alarm Rate

PCA Principal Component Analysis

PQ Panoptic Quality

RCS Radar Cross Section

ROC Receiver Operating Characteristic

RMSE Root-Mean-Square Error

RQ Research Question

SFCW Stepped-Frequency Continuous Wave

RMS Root Mean Square

SNR Signal-to-Noise Ratio

SRC Sparse Representation Classification

T-Rex Terminating-Random Experiments Selector

TDM Time-Division Multiplexing

TMCL Trinamic Motion Control Language

TPR True Positive Rate

UDP User Datagram Protocol

UVC USB Video Class

List of Operators & Symbols

Operators

$\sum$	Summation
$\prod$	Product
$(\cdot)^\top$	Transpose
$(\cdot)^H$	Hermitian Transpose
$(\cdot)^{-1}$	Matrix Inverse
$\det(\cdot)$	Determinant
$\text{tr}(\cdot)$	Trace
$\text{diag}(\cdot)$	Diagonal Matrix
$\lvert \cdot \rvert$	Absolute Value
$\lVert \cdot \rVert_1$	ℓ_1-Norm
$\lVert \cdot \rVert_2$	ℓ_2-Norm
$\lceil \cdot \rceil$	Ceiling
$\max(\cdot)$	Maximum
$\arg\min$	Argument of the Minimum
$\arg\max$	Argument of the Maximum
$\exp(\cdot)$	Exponential Function
$\log(\cdot)$	Natural Logarithm
$\log_2(\cdot)$	Binary Logarithm
$\mathbb{E}[\cdot]$	Expectation
$\mathbb{P}(\cdot)$	Probability
$\sim$	Distributed As
$\propto$	Proportional To
$\approx$	Approximately Equal

General

a	Scalar
$\boldsymbol{a}$	Vector
$\boldsymbol{A}$	Matrix
$\mathcal{A}$	Set
$\mathbb{R}$	Real Numbers
$\mathbb{C}$	Complex Numbers
$\mathcal{N}(\mu, \sigma^2)$	Normal Distribution
$\mathcal{CN}(\mathbf{0}, \boldsymbol{I})$	Complex Normal Distribution
$\mathbb{1}\{\cdot\}$	Indicator Function

α	Target Level of the Error Rate

Radar and Signal Model

c	Speed of Light
λ	Wavelength
f_0, B, S	Start Frequency, Bandwidth, Chirp Slope
f_b	Beat Frequency
T_c	Chirp Period of One Transmitter
Δr, Δv	Range and Speed Resolution
r, θ, φ	Range, Azimuth, Elevation
v	Radial Speed
P_t, P_r	Transmitted and Received Power
G_t, G_r	Antenna Gains
σ	Radar Cross Section
ε_r, Γ	Relative Permittivity, Reflection Coefficient
n	Number of Array Channels
K	Number of Chirps
d_m	Position of Channel m in Wavelengths
u	Direction Sine, $u = \sin\theta$
$\boldsymbol{a}(u)$	Steering Vector
Δu	Grid Spacing in Direction Sine
p	Number of Grid Directions
$\boldsymbol{X}$	Steering Matrix, Design Matrix
$\boldsymbol{y}$, $\bar{\boldsymbol{y}}$	Snapshot, Chirp-Averaged Snapshot
$\boldsymbol{\beta}$	Sparse Amplitudes
$\boldsymbol{\varepsilon}$	Noise Vector
$\hat{\boldsymbol{R}}$	Sample Covariance
$\boldsymbol{W}$	Whitening Matrix
$\boldsymbol{r}_t$	Residual After Step t

Detection, Error Control and Selection

H_0, H_1	Null and Alternative Hypothesis
P_fa, P_d	Probability of False Alarm and of Detection
m, m_0	Number of Tests and of True Nulls
V, S, R	False Discoveries, True Discoveries, Discoveries
FDP, FDR	False Discovery Proportion and Rate
TPR	True Positive Rate
p_j, e_j	P-Value, E-Value
$\hat{k}$	Number of Selected Hypotheses
ρ_t	Squared Correlation of Step t
P_t, S_t	Dummy Probability and Its Cumulative Sum
L, T, K_rex	Dummies, Included Dummies, Random Experiments
$\Phi_{T,L}$	Relative Occurrence

$\boldsymbol{D}$, $\boldsymbol{a}^\star$	Dictionary and Sparse Code of M^2-SRC
λ	Weight of the Sparsity Penalty
w_i	Weight of Hypothesis i

State Estimation and Prediction

$\boldsymbol{s}_t$, $\boldsymbol{P}_t$	State and Its Covariance
$\boldsymbol{F}$, $\boldsymbol{Q}$	Transition Matrix and Process Noise
$\boldsymbol{H}$, $\boldsymbol{R}_\nu$	Measurement Jacobian and Noise Covariance
$\boldsymbol{e}_t$, $\boldsymbol{S}_t$, $\boldsymbol{K}_t$	Innovation, Its Covariance, Kalman Gain
Γ_t	Prediction Region
ϵ, ϵ_t, γ	Miscoverage Level, Adaptive Level, Step Size
U	Mean Position Entropy
h	Differential Entropy

Vision and Scene

$\boldsymbol{I}$, H, W	Image, Height, Width in Pixels
(u, v)	Pixel Column and Row (Images)
Y	Luma
f, (u_0, v_0), k_1	Focal Length, Principal Point, Radial Distortion Coefficient
n_w	Width of an Object in Pixels
$\boldsymbol{p}$, $\boldsymbol{q}$	Point in Space, Its Pixel
$\boldsymbol{R}_\mathrm{c}$, $\boldsymbol{c}$	Rotation and Position of the Camera
$\boldsymbol{\vartheta}$, $\boldsymbol{J}$, μ	Camera Parameters, Residual Jacobian, Damping
ρ	Distance From the Lamp in Pixels
$L(\rho)$	Brightness Model of the Veil
N_S, N_D, N_I	Substitutions, Deletions, Insertions
$s(o, g)$	Similarity of Object and Truth
t_resp	Reaction Time of the Platform

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